Applied/Advanced Management Accounting

Part 1

MAC4861/MAC4862
NMA4861/NMA4862
ZMA4861/NMA4862



FJC Benadé
WJ Coetzee
A Combrink
L Crafford
SK du Toit
TJ Matsoma
A Ravat
S Riekert
F Venter
A von Wielligh

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PART 2: FINANCIAL MANAGEMENT



SECTION A: COSTING

QUESTION 1 **40 marks**

GAUTENG RAIN GUTTERS (PTY) LTD **100 marks**

(Source: Exam 2013 MAC4861)

Gauteng Rain Gutters (Pty) Ltd ('GRG') operates in and around Johannesburg installing gutter systems for various customers.

The company purchases rolls of flat pre-painted aluminium-zinc steel sheets which are rolled into seamless continuous gutters on the customer's premises according to the lengths required using mobile gutter profiling machines. (Note: each metre of aluminium-zinc steel sheet is rolled into a metre of gutter.) The current book value of GRG's machines and vehicles is R2,1 million. The gutters are prepared in domestic or industrial size and installed together with downpipes (domestic or industrial size). The installation also requires various sundry materials e.g. UV resistant silicone to seal corners, blind rivets, end caps, etc. The rolls of metal, downpipes and sundry materials are purchased from a local supplier. Customers can also install combinations of domestic and industrial sizes depending on requirements e.g. domestic gutters with industrial downpipes.

The company operates in three main markets:

- Domestic installations – individual home owners
- Commercial installations – commercial and government property owners e.g. factories, shopping centres, schools and hospitals
- Construction projects – building projects for construction companies and developers. Currently 70% of the work (in Rand terms) is done for a JSE listed property development company.

The new financial manager of GRG reported the following for the financial year ended 30 June 2013:

	Domestic	**Commercial**	**Construction**
Average metres of gutter used per installation (domestic & industrial)	60m	145m	975m
Average metres of downpipe used per installation (domestic & industrial)	35m	95m	585m
Average wastage % included in above quantities	3%	4%	3%
Average charge to customer per combined metres of gutter and downpipe used	R125 /m	R95 /m	R85 /m
Average cost per metre:			
- gutter (domestic & industrial)	R35 /m	R42 /m	R37 /m
- downpipe (domestic & industrial)	R40 /m	R48 /m	R43 /m
Number of installations	520	240	63

Customers are billed using a single tariff per metre based on the total number of metres of gutter and downpipe used (irrespective of the actual proportion of gutter and downpipe used). Volume discounts are given if more combined metres are required per installation.

Current installations use the existing skilled work force to their full capacity.

Operating overheads (including salaries and wages) for the 2013 financial year amounted to R9 500 000. GRG has for many years allocated operating overheads to the different markets based on the number of installations per market in order to evaluate the relative profitability of its different markets.

The new financial manager is considering using activity-based costing to allocate operating overheads and after meetings with various managers and staff members identified the following main activity areas:

Activity area	**Cost driver**
Ordering materials	Number of orders
Travelling to and from customers' premises	Kilometres travelled to and from customers' premises
Rolling seamless gutters using mobile machines	Metres of aluminium-zinc steel sheets used
Installing gutters and downpipes	Combined metres of gutter and downpipe used
Other overheads	Working days at customer premises

The operating overheads for the year ended 30 June 2013 were analysed as follows:

Activity area	**Total cost R**
Ordering materials	410 000
Travelling to and from customers' premises	2 150 000
Rolling seamless gutters using mobile machines	2 140 000
Installing gutters and downpipes	4 200 000
Other overheads	600 000
	9 500 000

Other data for the year ended 30 June 2013:

	Domestic	**Commercial**	**Construction**
Average number of orders per installation	3	5	8
Average distance travelled to and from customers' premises per installation	70km	185km	290km
Average number of working days per installation	1 day	2,5 days	5 days

New manufacturing division

GRG has always purchased domestic and industrial size downpipes from a local supplier but is now considering purchasing new machinery that will enable them to manufacture their own downpipes in the sizes and lengths as needed for each installation.

The machinery will have a capacity to manufacture 150 000 metres of downpipe per year (domestic and industrial combined). GRG estimates that its own requirements will be 90 000 metres per year (domestic and industrial combined) for the next two years and 95 000 metres per year for the two years after that. GRG has commissioned market research at a cost of R10 000 and found that the demand from other gutter installation businesses for the first year of operation of the new machinery is likely to be as follows:

Metres of downpipe per year (domestic and industrial combined)	Probability
50 000	50%
60 000	30%
70 000	20%

It is expected that the demand for downpipes (domestic and industrial combined) from the external market will increase by 2% each year for the following four years irrespective of the demand in the first year. Management are however adamant that internal requirements should at all times be satisfied first before external demand. 70% of demand, in terms of quantities, is expected to be for domestic downpipes for both external sales and internal usage.

The external selling price per metre is calculated to be R39 for domestic downpipes and R50 for industrial downpipes. This is based on a 45% mark-up on the actual variable cost of each product. If GRG does not manufacture the downpipes internally they expect to pay R38 per metre for domestic downpipes and R49 for industrial downpipes in the following year after volume discounts obtained from their current supplier.

The new machine is expected to cost R3 000 000 and have a useful life of four years after which it can be sold for R500 000. Fixed costs for this investment, including straight line depreciation over four years (with a residual value of R500 000) and marketing costs of R50 000 per year, are expected to be R1 350 000 per year and will remain at this level for the machinery's useful life. It also includes rent for the small factory required at 5% of the fixed cost. This cost will however increase at 9% per annum (from year 2 onwards), 3% higher than the anticipated inflation rate. Additional working capital of R250 000 will also be required at the start of the new venture.

A wear-and-tear allowance on the new machinery is available on a 40:20:20:20 basis. The current tax rate is 28%. Half of the tax is payable in the year in which the income is produced while the balance is paid in the following year. Any taxable losses arising from this investment can be set-off against profits made by GRG's other activities.
GRG will use a post-tax cost of capital of 15% p.a. to evaluate the investment. (Inflation can be ignored.)

During the past financial year, the management and shareholders were presented with two projects. These were outlined as follows:

1. An agent would acquire a contract from the Angolan government to assist with the renovation of government buildings throughout the country. The contract would have a duration of two years and GRG would be responsible for the guttering aspect. This part of the full contract has a value of US$ 3 000 000, of which 10% would be retained by the agent, and would require the equivalent of the entire construction work force.

2. The Department of Defence has decided to take over the maintenance of their buildings from the Department of Public Works. They are prepared to use GRG for work required in the northern provinces on the basis of cost plus 75% and estimate that the initial phase would take three years. It would require the equivalent of 50% of the domestic and commercial work force.

REQUIRED		Marks	
		Sub-total	**Total**
(a)	Calculate the contribution per installation and contribution per combined metre for the three markets for the year ended 30 June 2013.	(6)	(6)
(b)	Calculate the operating profit margin percentages for the three markets and the company for the year ended 30 June 2013 using the activity-based costing (ABC) information provided. Rand values should be rounded to the nearest Rand. Percentages should be rounded to one decimal.	(17)	(17)
(c)	Identify five challenges that GRG will face in apportioning operating overheads to the activity areas.	(5)	(5)
(d)	Advise the management of GRG as to whether they should go ahead with the new manufacturing investment from a financial perspective. Inflationary adjustments are not required. Rand values *per unit* should be rounded to the nearest cent. Other Rand values should be rounded to the nearest Rand.	(26)	(26)
(e)	Discuss the use of a hurdle rate above WACC in evaluating projects.	(3)	(3)
(f)	Prepare a memorandum to the management of GRG discussing the **qualitative** factors that GRG should consider before investing in the manufacturing division. *Communication skills – layout and structure; appropriate style*	(6) (2)	(8)
(g)	Calculate the break-even value (Rand) for the manufacturing division for the first year of operation as well as the margin of safety percentage.	(5)	(5)
(h)	GRG is considering implementing a transfer pricing strategy between its two divisions in order to motivate each division to maximise its reported profit. The management of GRG favour a policy that each division should be allowed autonomy to set transfer prices and purchase from whomever they choose. - Suggest a suitable minimum and maximum transfer price per metre of domestic and industrial downpipe in the first year of operation of the new division; - Explain how this will change in the fourth year; - Also highlight other factors that GRG should consider in its decision on a transfer price. Assume that domestic and industrial downpipes use the same quantity of manufacturing capacity per metre manufactured. Ignore inflation. *Communication skills – logical argument*	(3) (7) (4) (1)	(15)
(i)	Discuss whether return on investment and residual income are good bases for performance measurement. List and motivate other (non-ratio and / or non-financial) measures that may be used. *Communication skills – logical argument*	(4) (4) (1)	(9)
(j)	List the risks that GRG should consider for each of the proposals presented: Proposal 1 – Angolan contract Proposal 2 – Department of Defence *Communication skills – appropriate style*	(3) (2) (1)	(6)
TOTAL MARKS			**100**

QUESTION 2

SKY (PTY) LTD **100 marks**

(Source: Exam 2012 MAC4862)

Sky (Pty) Ltd ('Sky') is a small company that manufactures electronic components and other products for satellite communication related uses. The company has three factories located in Gauteng and products are sold across South Africa.

Antenna division

The Antenna division manufactures three types of antennas at one of the factories. Each type of antenna is manufactured in batches of 100 units but they are sold as single units. The antenna division uses a just-in-time manufacturing process in terms of satisfying demand for its products, and therefore keeps no inventory of any of the products. Sky has several competitors in the antenna market and can therefore not set its own prices but must accept the market prices. The following management statement shows the results of the antenna division per product and in total for the year ended 31 August 2012:

Product	**Alpha**	**Nebula**	**Terra**	**Total**
Number of units sold	10 000	13 000	8 000	31 000
Machine hours	20 000	19 500	8 000	47 500
Direct labour hours	5 000	13 000	8 000	26 000
	R	**R**	**R**	**R**
Sales	1 300 000	2 275 000	2 120 000	5 695 000
Direct materials	(450 000)	(910 000)	(940 000)	(2 300 000)
Direct labour	(400 000)	(1 040 000)	(640 000)	(2 080 000)
Overhead costs	(400 000)	(390 000)	(160 000)	(950 000)
Operating profit / (Loss)	50 000	(65 000)	380 000	365 000
Finance costs				(97 900)
Net profit before tax				267 100

Total divisional controllable net assets R890 000

The manager of the antenna division is concerned about the loss being made by the Nebula antenna and has investigated the products and their costs and found the following:

1. The number of units sold of each product is completely independent of the sales of the other products.

2. Labour is to be considered a variable cost.

3. A budgeted blanket absorption rate of R20 per machine hour has been applied.

4. A detailed analysis of the overhead cost shows that part of it is driven by the number of machine hours used, part is driven by the number of batches produced and part of the costs are product specific fixed overheads that would be avoided if the product were discontinued. The rest are general fixed overhead costs that would only be avoided if the factory closed. The details are as follows:

	R
Machine hour related	190 000
Batch related	155 000
Product specific fixed overhead	590 000
- Alpha	330 000
- Nebula	75 000
- Terra	185 000
General fixed overhead – allocated	15 000
Total	950 000

Converter division and Processor division

The Converter division manufactures only one type of component at one of the factories, namely, analogue to digital converters. It sells these converters to external customers and also to the processor division. The Processor division manufactures digital signal processors. The following statement shows the performance of each division for the year ended 31 August 2012:

	Converter division R'000	**Processor division R'000**
Sales	50 400	125 050
Variable cost	(35 000)	(95 100)
Contribution	15 400	29 950
Fixed manufacturing costs	(9 240)	(17 500)
Operating profit	6 160	12 450
Finance costs	(1 335)	(2 780)
Net profit before tax	4 825	9 670
Total divisional controllable net assets	12 125	32 860

During the year ended 31 August 2012, the Converter division operated at its full capacity of 140 000 units. The transfer of 70 000 units to the Processor division satisfied that division's total demand for converters. However the external demand was not satisfied. A further 42 000 converters could have been sold to external customers by the Converter division at the current market price of R387,50.

The current policy of Sky is that internal sales should be transferred at their opportunity at marginal or variable cost and where appropriate including an opportunity cost. Therefore, during the year, some converters were transferred to the Processor division at the market price and some were transferred at variable cost.

The fixed manufacturing cost of the Converter division includes allocated fixed overhead cost of R4 900 000, of which R3 206 000 relates to the depreciation of the existing manufacturing equipment, included in the divisional assets stated above. On 31 August 2012, this equipment had a book value of R2 061 000, but will only realise a nominal amount on the sale thereof. The allocated fixed cost of the Processing division amounted to R4 235 000 for the year ending 31 August 2012.

The Converter division is considering investing in new equipment which will reduce its unit variable costs by 20% in current terms and increase its capacity to 160 000 units per annum for each of the next seven (7) years. The capital cost of the investment is R30,8m payable in full upfront. The equipment will have no tax value after five (5) years. Sky and its divisional managers evaluate investments using net present value (NPV) with a 12% cost of capital over a five (5) year term. Once the board of Sky approves an investment proposal, the assets so acquired are deemed to be controllable divisional assets. The same cost of capital is used for divisional performance measurement purposes. The installation period of the new equipment will be six (6) weeks.

Due to the government's communication strategy, it is estimated that there will be huge demand for digital converters in future and that the external annual demand for the next seven (7) years will continue to be a minimum of 112 000 converters at R387,50 each (in current prices). Sky company management will however insist that the internal annual demand of 70 000 converters to be transferred to the Processor division must be satisfied in full.

The factory operations of Sky closes down for three (3) weeks during the December/January festive season each year, compulsory closure. The relevant factory building can currently not accommodate new equipment and no rental space is available in the immediate vicinity.

Processor division bid

The management of Sky noted with great interest the announcement that the major part of the SKA (Square Kilometre Array) telescope had been awarded to Africa. The future plans include, amongst others, the building of about three thousand mid-frequency dishes, between 2016 and 2023 with the highest concentration thereof being in the Northern Cape, South Africa. The marketing manager identified an opportunity to gain involvement in the project by putting in a bid for ten (10) high-speed digital signal processors needed for the current 64-dish Karoo Array Telescope (known as MeerKAT) which will be integrated into phase 1 of the SKA. The marketing manager intends pricing the bid using a relevant costing approach.

The marketing manager held a two-hour meeting with a representative of MeerKAT and obtained the following information pertaining to the bid for the ten processors. The marketing manager is paid an annual salary equivalent to R1 400 per 8-hour day.

- 110kg of material A will be needed. Material A is regularly used by the processor division and there are currently 200kg in inventory which were bought at a cost of R22 per kg. They have a resale value of R20,50 per kg and their current replacement cost is R22,50 per kg.

- 30 litres of material B will be required. This will have to be purchased for the contract because it is not normally used by the processor division. The minimum order quantity from the supplier is 40 litres at a cost of R19 per litre. The processor division does not expect to have any use for any of this material that remains after this contract is completed.

- 500 components will be required. The purchase price is R8 per component.

- A total of 235 direct labour hours will be required. The current wage rate for the relevant grade of direct labour is R25 per hour. The processor division currently only has 75 direct labour hours of spare capacity at this grade. These employees are permanent employees. The additional hours would need to be obtained by either (i) overtime at a total cost of R32 per hour, or (ii) recruiting temporary staff at a cost of R27 per hour. However, if temporary staff is used they will not be as experienced as the processor division's existing workers and will require 10 hours supervision by an existing supervisor who would be paid overtime at a cost of R40 per hour for this work.
- 25 machine hours will be required. The machine to be used is already leased at a weekly leasing cost of R1 500 with a capacity of 40 hours per week. The machine has sufficient available capacity for the contract to be completed. The variable running cost of the machine is R15 per hour.

- The processor division absorbs its fixed overhead costs using an absorption rate of R30 per direct labour hour.

Sky Foundation

The Sky Foundation ('the Foundation'), a non-profit organisation, was established by Sky to coordinate the social responsibility initiatives of the company. The Sky Foundation obtains all its funding from Sky. Sky also donated inventory to the Foundation for training purposes. Ms Matlala is the CEO of the Foundation. Amongst other initiatives, she is supportive of poor people and learners growing vegetables in gardens at their homes and schools for their own use and possibly to earn an income for themselves. She identified a need for fertiliser for these gardens and therefore acquired a small plant to produce the fertiliser. Unemployed people are also trained at the plant in technical skills and there is a small laboratory where visiting school groups are introduced to chemistry.

The plant uses a standard costing system for the manufacture of the fertiliser. Direct labour is considered a variable cost. Fixed overheads are absorbed on a machine hour basis. Fractions of kilograms can be produced. Up until the end of the 2012 financial year the Foundation produced only one type of fertiliser, namely, Green.

The Foundation does not keep any inventory of production raw materials, work-in-progress or finished goods.

The Foundation decided to add two new types of fertiliser from the start of the new financial year namely, Grow and Nutri to the existing product (Green). The Foundation's accountant has prepared the following budget for the year ending 31 August 2013 based on needs identified in the areas where the Foundation operates:

	Green	**Grow**	**Nutri**	**Total**
	Kg	**Kg**	**Kg**	**Kg**
Need	7 300	5 000	3 000	15 300
	R	**R**	**R**	**R**
Material G	16 425	22 500	10 800	49 725
Material R	17 520	–	10 800	28 320
Material O	28 105	17 500	6 300	51 905
Direct labour	21 900	30 000	13 500	65 400
Fixed overhead	29 200	15 000	18 000	62 200
Total cost	113 150	85 000	59 400	257 550

The standard quantities **per kilogram** produced have been set as follows:

	Green	**Grow**	**Nutri**
Material G (kg)	0,25	0,50	0,40
Material R (kg)	0,20	–	0,30
Material O (kg)	0,55	0,50	0,30
Direct labour (hours)	0,10	0,20	0,15
Machine hours	0,20	0,15	0,30

At the start of the financial year it was discovered that the production plant would need urgent repairs and will therefore need to immediately be shut down for three (3) weeks. This would result in the loss of 180 machine hours for the financial year compare to the original/existing budget.

At the same time a fire broke out at the production plant of the main producer of Material G and their entire inventory was destroyed. Ms Matlala managed to obtain contracts to purchase Material G from other suppliers at the same price but could only secure 5 600 kg for the financial year.

The Foundation also received notification that due to environmental regulations and limitations on gaseous emissions, the production plant is not allowed to produce more than the following maximum quantities of each product:

Green:	10 000 kg
Grow:	7 000 kg
Nutri:	2 800 kg

In the light of these circumstances Ms Matlala felt that the Foundation's planning should be revised to take these events into account.

The Foundation has already committed to donating the following quantities to communities and schools for the forthcoming year:

Green:	4 000 kg
Grow:	500 kg
Nutri:	1 500 kg

REQUIRED		**Marks**
(a)	Prepare a statement in columnar format for the Antenna division that is more useful for decision making than the profit statement presented in the scenario. Your statement should also show the current total profit for the business.	(8)
(b)	Explain to the Board of Sky why your statement is suitable for decision making.	(4)
(c)	Advise the Board of Sky which, if any, of its three antenna products should be discontinued in order to maximise its company profits.	(4)
(d)	Calculate the break even volume (in batches) for product Alpha to break even at a "product level".	(4)
(e)	Discuss the benefits that the Antenna division should gain from operating a JIT purchasing system for materials	(4)
(f)	Based on measurement by residual income (RI) and one other relevant measure, determine which division performed best for the year ended 31 August 2012. List all other information that will assist you in making a more informed decision regarding their performance.	(9)
(g)	Prepare an analysis of the sales made by the Converter division that clearly shows, in units and in Rand-terms, the internal and external sales made during the year ended 31 August 2012.	(3)
(h)	Discuss the fairness of the current transfer pricing policy and the effect of possible changes in external demand on the profits of the Converter division, assuming the current transfer pricing policy continues.	(6)
(i)	Assuming that the current transfer pricing policy continues (i) Evaluate the proposed investment from the perspective of the manager of the Converter division. (ii) Evaluate the proposed investment from the perspective of Sky company. **Note:** Ignore inflation and taxation.	(8) (4)
(j)	Assume the new converter equipment will be acquired. Calculate the impact of this on the product cost per unit for 2013, from the Convertor division's perspective.	(4)
(k)	Assume approval will be given just prior to the 2012 financial year end to proceed with the acquisition of the new equipment for the Converter division. Advise management regarding what the strategy in this regard should be and show any supporting calculations.	(5)
(l)	Calculate the price that the Processor division should bid for the high speed processor contract. Clearly show the relevant cost value for each of the items. Also explain each relevant value you have included and why the values you have excluded are not relevant.	(14)
(m)	Calculate the revised production mix of fertilisers that will maximise the use of the resources for the Foundation for the forthcoming year ending 31 August 2013. Show total kilograms to be produced for each product.	(15)
(n)	The Foundation is registered as a Public Benefit Organisation (PBO) in terms of Section 30(3) of the Income Tax Act. Discuss the normal tax implications of donating inventory from the perspective of the Foundation and the Company.	(5)
Presentation Language Logic		(1) (1) (1)
TOTAL MARKS		**100**

QUESTION 3

CORLEONIE'S (PTY) LTD **100 marks**

(Source: Exam 2012 MAC4862 adapted)

Corleonie's was started by an Italian couple, Alberto and Gaby Corleonie, who immigrated to South Africa to start an Italian take-away. After struggling for years in the very competitive market, Alberto decided to use his grandfather's lasagna recipe and convert it into a microwavable meal, which can be sold to larger outlet stores such as Spar, Checkers and Pick n Pay. Corleonie's started off with the Julio, a meat lasagna but the Carrie, a vegetarian lasagna (named after his daughter) was recently introduced by Alberto as market research showed that career women with children, are looking for a healthy meal that is easy to prepare. Corleonie's currently operates a small delicatessen in Brooklyn and supplies a few independent grocers on a cash on delivery basis.

Even though it is an age-old recipe, production takes place at a small fairly automated plant. The production process starts with the mixing of ingredients and the kneading of the dough. The two lasagnas are then split into two processes, the meat and the vegetable process. The existing costing system is not very sophisticated as Corleonie's grew a bit too fast for the systems. The costs include direct materials (dough, butter, mince, vegetables etcetera). As Alberto (with the help of his engineering friend) was able to convert the process into a fairly automated process, fixed production overheads are quite high. Alberto is however confident, that should capacity increase, Corleonie's plant will be able to handle it.

Extracts from the budget for the year ended 30 September 2012 and the actual results per the management accounts for the year are set out below:

CORLEONIE'S (PTY) LTD
BUDGETED AND ACTUAL RESULTS FOR THE YEAR ENDED 30 SEPTEMBER 2012

	Notes	Budget R	Actual R
Revenue	1	4 015 000	4 032 000
Less: Cost of Sales		(2 225 837)	(2 272 871)
Opening inventory	2	297 000	297 000
Production cost	3	2 037 300	2 083 015
		2 334 300	2 380 015
Closing inventory	4	(108 463)	(107 144)
Gross Profit		1 789 163	1 759 129
Selling and administrative cost	5	(1 750 000)	(1 755 000)
Net profit before tax		39 163	4 129

Notes:

1. Budgeted selling prices are R50 for Julio and R45 for Carrie, but Alberto believed that he could keep up with the budgeted sales volume (65 000 for Julio and 17 000 for Carrie) if he increased the selling price of the Carrie by R1 and was proved correct by the actual outcome. The Julio is currently producing at 80% capacity and the Carrie at 40%.

2. Opening inventory at 1 October 2011 consisted of 8000 units of Julio and 3 000 units of Carrie at an actual cost of R216 000 and R81 000 respectively. A fixed manufacturing expense per lasagne was R16 for the 2011 financial year. No material inventories of raw materials were held on this date.

3. Alberto budgeted for 60 000 Julios and 15 000 Carries to be manufactured during the year. Actual production showed 59 000 Julios and 16 000 Carries were manufactured.

4. Corleonie's uses the Weighted Average method of inventory valuation and the closing inventory values on 30 September 2012 are considered to be correct.

5. Selling and administrative cost consist of 60% fixed cost and 40% variable cost and a 6:1 ratio between Julio and Carrie. Variable cost was R35 000 more than budgeted.

Additional information:

- Flour, butter and dough are bought premixed from a local supplier. The budgeted cost per kg mix is R14. One Julio requires 220g and one Carrie 200g of the mix. Total actual cost of the mix for the year was R230 040 (R191 160 for Julio) after Alberto was able to negotiate a price of R13,50 per kg.

- The actual mince cost for the Julio lasagna was R283 200. Alberto was very concerned, because although the mince price remained at the budgeted R20 per kilogram this was R43 200 more than that was budgeted.

- Currently the vegetable portions that fill the Carrie are the same quantity that fills a Cornish pie. Alberto budgeted R2 per Carrie lasagna. The actual cost was R30 400.

- As the manufacturing process is fairly automated, only six (6) unskilled workers are required. As they are willing to work for the minimum stipulated wage, Alberto decided to keep them on. The workers are split in two groups, four (4) workers allocated for the Julio and the rest for the Carrie. Alberto budgeted for 9 000 hours at the minimum rate of R15 per hour. Alberto reviewed the timesheets and discovered that it takes on average 6 minutes for a Julio and 12 minutes for a Carrie to be processed. This is based on a 125 hour month per worker. Corleonie's were part of the striking members in a nationwide strike, the outcome of which forced Alberto to increase their hourly rate by 50c. Gaby is in control of the workers, and she allowed an extra 2 250 hours that Alberto did not budget. The Julio workers clocked 8 856 hours in total and the Carrie workers 1 200 hours each.

- Budgeted variable manufacturing overheads were calculated at R2,50 per lasagna. Actual variable manufacturing overheads amounted to R180 000.

- The lasagne is wrapped in a disposable labelled plastic holder which was budgeted for at 24c per holder. Due to increasing pressure from the public to go "green", the supplier offered a discount of 4 cents should Corleonie's continue with the order and pay cash, which Alberto accepted.

- Fixed manufacturing overheads of R1 200 000 were budgeted. Currently these overheads are allocated based on produced quantities. Actual fixed manufacturing overheads amounted to R1 170 000.

Activity-based costing

Alberto heard of activity-based costing (ABC). He did some research on his own to familiarize himself with ABC. He applied some basic techniques to observe and record how the two products (Julio and Carrie) differed in using the production processes. The following activity areas and associated costs were identified:

Actual fixed overhead cost for each activity area:

Activity area	**Note**	**Total cost**
		R
Set-up costs	6	120 000
Ordering of materials	7	59 700
Handling of materials	8	770 300
Quality control	9	220 000
		1 170 000

Notes:

6. Alberto's friend, the engineer, was in charge of the set-ups of the automated plant. Alberto put his friend on a monthly retainer. He has to set-up the machines based on production projections for that month. At the end of the year he had to do twice as many set-ups for Julio as for the Carrie. He completed 36 set-ups during the year.

7. Materials are ordered once a month for both the Julio and the Carrie.

8. Every unit of Julio and Carrie is handled during the manufacturing process. The Julio is handled twice, before it is put in the process and when it is put in the plastic container. The Carrie is handled three times; it follows the same process as for the Julio, but an additional activity, when the vegetables are "stirred/mixed" halfway through the process is required.

9. Alberto is in charge of the quality control inspections. He decided to allocate R220 000 of his directors remuneration to his time spent on quality control. He inspects the Julio process every nine (9) labour hours and the Carrie process every six (6) labour hours.

Expansion

Alberto's brother, Luca, recently came to visit the Corleonies. He was very impressed with the growth of Alberto's company. He hinted that he would also like to get involved, especially if he can use his grandfather's recipe. Alberto discussed the idea with Gaby, but was surprised to learn that Gaby was not very keen on the idea; Gaby instead suggested franchising to Alberto as the only option.

Special order

Alberto is considering whether or not to accept a once-off order in the next financial year for the supermarkets (Big & Small Bites) situated in the Kruger National Park. This however would require Corleonie's to produce a third product. This third product will be a Julio with some vegetables added. Alberto decided to call it the Romeos. The premix flour, butter and dough will work perfectly for the new Romeos. Alberto is unsure how many grams of filling will be necessary per Romeo, but an estimate of 220g seems appropriate. The mince required is half of what is necessary for a Julio, and the remainder will be vegetables. Two workers from the Julio process agreed to work on the Romeo line at an overtime rate of R20 per hour. Alberto believes in good training, and has monthly training sessions at a cost of R6 000 per session. In one of these sessions, Alberto was able to train the two workers to complete a Romeo in the same time as a Carrie. Alberto felt that an additional R20 000 of his remuneration be allocated to inspections for this order. Big & Small Bites requires 7 500 Romeos for this order.

The materials in inventory are as follows:

	Notes	Quantity
Flour, butter and dough (kg)	10	300
Mince (kg)	11	150
Vegetables (kg)	12	200
Plastic holders (units)	13	6 000

Notes:

10. The current purchase price is R14 per kg, and if not used soon, will have to be disposed of at a cost of R1,00 per kg.

11. The inventory cost is still R20 per kg, and if not used for the Romeos, will be used in the current process. The new contract price for mince is R25/kg provided 100 kg is delivered at a time.

12. The current price is R1,10 per kg, and if not used for the Romeo will be used in the Carrie.

13. The inventory on hand cost R0,20 each. The Romeo will require a new labeled plastic holder at R0,30 each. The supplier is only prepared to supply 10 000 holders per order.

REQUIRED		MARKS
(a)	Draft a Budgeted Statement of Comprehensive Income for the year ended 30 September 2012 for both lasagnas applying direct costing principles. A total column is **not** required.	(15)
(b)	Indicate why you would prefer a variable costing system to an absorption costing system from a management accounting perspective.	(5)
(c)	Determine the variances for: ○ The dough premix ○ Mince ○ Selling and administrative cost And briefly comment thereon (variance calculations 8 marks and comment 8 marks)	(16)
(d)	Critically discuss and show the impact of operating at full capacity on Corleonie's budgeted fixed manufacturing expenses per product and for the business (where processing time is used as an allocation basis). Any change in the calculation process should be clearly motivated.	(6)
(e)	Calculate the budgeted break-even point in Rand for Corleonie's.	(4)
(f)	Calculate the actual fixed overhead manufacturing cost per unit for Carrie and Julio, based on the **current overhead allocation** method and the fixed manufacturing cost per unit, should Alberto decide to **implement ABC**.	(12)
(g)	How would the ABC information assist Corleonie's Ltd to make better decisions?	(4)
(h)	Explain how ABC will affect the break-even points of the two products and then discuss the viability of Carrie.	(8)
(i)	Discuss the reasons why Gaby should be sceptical about the idea of Luca getting involved with Corleonie's.	(6)
(j)	List the benefits and possible downfalls of franchising.	(6)
(k)	Calculate the minimum selling price per Romeo for the special order from Big and Small Bites.	(10)
(l)	Corporate clients indicated to Alberto that should his products be accepted, their normal business practice is to take 75 days from delivery before payment is made. List all issues that Corleonie's should consider before venturing into corporate sales.	(5)
Presentation Language Logic		(1) (1) (1)
Total		**100**

QUESTION 4 (MAC4862) **100 marks**

ANANI GROUP

(Source: Supplementary exam 2012 MAC4862 adapted)

The Anani Group owns several home lifestyle stores which are in part supplied by their own factories operating in a separate division. The remainder of their products are imported from a Scandinavian business and are paid for in US dollar. In each store, consultations if needed are undertaken by specialists, who also visit potential customers in their homes, using specialist software to help customers realise their lifestyle objectives. Customers visit the stores to make their selections from the wide range of self-assembly goods offered, after which sales staff collect payment and raise a purchase order. Customers then collect their goods from the warehouse, using the purchase order as authority for collection to take place. Administrative staff process purchase orders and also arrange consultations.

The Anani Group operates an absorption costing system, and costs other than the cost of goods sold are apportioned on the basis of sales floor area.

Results for the trading departments in Anani's stores for the last three months are as follows:

	Departments			
	Breakfast Nook R'000	**Bedrooms R'000**	**Living Rooms R'000**	**Total R'000**
Sales	21 000	11 250	44 000	76 250
Cost of goods sold	(6 300)	(3 750)	(17 600)	27 650
Other costs	(13 025)	(8 250)	(11 470)	32 745
Profit	1 675	(750)	14 930	15 855
Average inventory held (in R'000)	1 475	950	2 550	
Target inventory turn	4 times	3 times	5 times	

The management accountant of Anani is concerned that the Bedrooms department of the stores has been showing a loss for some time, and is considering a proposal to close the Bedrooms department in order to concentrate on the more profitable Breakfast Nook and Living Rooms departments. The required return for the trading divisions and individual departments are 25% on total assets under the control of the respective divisions excluding cash balances which are transferred to the group treasury division on a daily basis. Group treasury pays for all purchases and uses the interest earned to set head office charges off against. He has found that other costs for the last three months are made up of:

	R'000
Sales staff salaries	6 644
Consultation staff salaries	2 496
Warehouse staff wages	3 055
Administrative staff salaries	3 050
General overheads (electricity, rates, etc.)	17 500
	32 745

Salaries, wages and general overheads are considered to be of a fixed nature, except for 30% of general overheads which vary with the total cost of sales.

He has also collected the following information for the last three months:

	Departments		
	Breakfast Nook	**Bedrooms**	**Living Rooms**
Number of items sold	1 000	1 500	4 000
Customer purchase orders	1 000	900	2 500
Floor area (square metres)	16 400	10 000	14 400
Number of consultations	798	200	250

The following pairing of cost drivers to cost pools has been suggested by a competent consultant:

Activity	**Driver**
Sales staff salaries	Customer purchase orders
Consultation staff salaries	Number of consultations
Warehouse staff wages	Items sold
Administration staff salaries	Customer purchase orders plus consultations
General overheads	Floor space

The management accountant believes that he can use this information to review the department's performance in the last three months from an activity-based costing (ABC) perspective.

Lindela factory

The Lindela factory produces a fairly basic product which can be used in all the lifestyle departments, as it is easily assimilated into other units. The factory currently operates at 60% of its normal capacity.

The current sales split of the factory is 80% to the life-style departments and 20% to external customers. No export sales are made.

The standard cost per unit for **external sales** is as follows:

	Notes	**R**
Raw materials :	(1)	
A 0,15m^3 @ R3 500/ m^3		525,00
B 12 units @ R3,20/unit		38,40
C 1 litre @ R21,60/litre		21,60
Labour and variable overhead		
36 min @ R40/labour hour		24,00
Packaging	(2)	56,00
Fixed overhead	(3)	60,00
		725,00
Sales price per unit		1 160,00

Notes:

(1) Raw materials B and C are easily sourced and no material inventories are held. Material A, Mukwa, bought in Zambia and Zimbabwe, cannot be obtained at short notice. The factory manager in the past insisted that eight (8) weeks' inventory of Mukwa based on current production, be held at any point in time.
(2) For internal sales to the stores division, more rudimentary packaging is used at a cost of R26 per unit.
(3) The fixed overhead is based on normal capacity of the Lindela factory of 27 000 labour hours per working year of 50 weeks.

A design specialist has performed a market research study and found that a small design change which will add R45 to the current material cost will enable the Lindela factory to increase the current sales to

- Local external customers by ⅔ (two thirds)
- Life style departments by 25%
- Generate export sales to France and Germany of 10% of the normal capacity. The export sales will be denominated in Euro at a discount of 10% on current local sales prices.

A subsequent price decrease of 5% on the current local external price and internal transfer price will increase the new local sales volumes (internal and external) by a further 10%.

The current policy of the group is to use market prices as the transfer prices between divisions/subsidiaries, adjusted for internal savings.

Naldnak factory

The Naldnak factory in particular has struggled for the last few years and management is concerned about closure and job losses. They are negotiating with the Kiash family to set up a joint venture (JV) which will entail an expanded operation. The expansion will require additional funding of R35 million. As part of this exercise, two options are being considered:

Option A

- A 5-year R27 million loan from the Industrial Development Corporation (IDC) at a fixed rate of 7% per annum. The cost of preparing the proposal in the format which the IDC requires will be 5% of the loan amount payable immediately. The remainder of the required funding will be provided by the joint venture participants.

- In acquiring the loan, the factory will qualify for a grant from the Department of Trade and Industry (DTI) amounting to 40% of the loan. The grant will be payable to the JV in five (5) equal instalments at the end of each year of the loan, upon certification that the JV complied with DTI conditions.

- The IDC will require that the annual DTI grant payment paid to the JV be paid directly to them, which they will off-set against the loan. This will be the only repayment made on the loan, except for the payment referred to hereunder.

- Final settlement of the loan will take place at the end of the 5-year period by means of a single bullet repayment.

Option B

- An equivalent Euro-denominated loan at 1,5% per annum, repayable in full at the end of a 5-year period. The cost of preparing this proposal will be €100 000.

- Should the JV obtain the Euro-denominated loan, they will not qualify for the DTI grant detailed above.

- The current €/R exchange rate is R10,80 = €1.

- The loan will allow the JV to obtain preferential status for exports to all European Union countries.

REQUIRED		Marks
(a)	Prepare a profit statement for the last three months for the Bedrooms department using the activity based costing method.	(12)
(b)	Evaluate and discuss the proposal to close the Bedrooms department. Your evaluation should include a break-even calculation for this department, as well as the calculation of other relevant indicators. [Calculation 10, discussion 6]	(16)
(c)	Discuss the strategic issues that the management of the Anani Group should consider with regard to • Customers • Consultants	(9)
(d)	The Bedrooms department management, after consultation with the affected parties, has proposed that the salaries for sales staff and consultants be cut by 50% and be replaced by a sales commission of 4% of sales. Determine the sales level at which the Department and staff will be better off.	(4)
(e)	Explain the impact of the current production level on the Lindela factory's cost structure, if it is assumed that all of the current production is sold externally.	(8)
(f)	Explain the strategic impact and impact on inventory management on Lindela, of holding eight (8) weeks inventory of Mukwa.	(8)
(g)	Determine the production level of the Lindela factory after both the design change and the price reduction had been affected and show the sales prices to be charged to the different customer categories.	(8)
(h)	Calculate the mark-up percentage on units transferred from the Lindela factory to other group departments or divisions, based on the adjusted standard costs and pricing.	(6)
(i)	For the Naldnak factory, calculate and show the following in respect of options A and B: • The bullet payments at the end of the term • The cash available for the project • The forward €/R exchange rate applicable for a year 5 payment, at which management would be indifferent between options A and B. *Ignore tax for all calculations.*	(10) (2) (2)
(j)	List the strategic, financial and operational issues that the management of Naldnak should consider in (i) evaluating and comparing financing options A and B. (ii) the operations of the Joint Venture.	(7) (5)
Logic, layout an legibility		(3)
Total		**100**

QUESTION 5 (MAC4861) **100 marks**

ANANI GROUP

(Source: Supplementary exam 2012 MAC4861)

The Anani Group owns several home lifestyle stores which are in part supplied by their own factories operating in a separate division. The remainder of their products are imported from a Scandinavian business and are paid for in US dollar. In each store, consultations if needed, are undertaken by specialists, who also visit potential customers in their homes, using specialist software to help customers realise their lifestyle objectives. Customers visit the stores to make their selections from the wide range of self-assembly goods offered, after which sales staff collect payment and raise a purchase order. Customers then collect their goods from the warehouse, using the purchase order as authority for collection to take place. Administrative staff process purchase orders and also arrange consultations.

The Anani Group operates an absorption costing system, and costs other than the cost of goods sold are apportioned on the basis of sales floor area.

Results for the trading departments in Anani's stores for the last three months are as follows:

	Departments			
	Breakfast Nook	**Bedrooms**	**Living Rooms**	**Total**
	R000	**R000**	**R000**	**R000**
Sales	21 000	11 250	44 000	76 250
Cost of goods sold	6 300	3 750	17 600	27 650
Other costs	13 025	8 250	11 470	32 745
Profit	1 675	(750)	14 930	15 855
Average inventory held (in R'000)	1 475	950	2 550	
Target inventory turn	4 times	3 times	5 times	

The management accountant of Anani is concerned that the Bedrooms department of the stores has been showing a loss for some time, and is considering a proposal to close the Bedrooms department in order to concentrate on the more profitable Breakfast Nook and Living Rooms departments. The required return for the trading divisions and individual departments are 25% on total assets under the control of the respective divisions excluding cash balances which are transferred to the group treasury division on a daily basis. Group treasury pays for all purchases and uses the interest earned to set head office charges off against. He has found that other costs for the last three months are made up of:

	R000	**Number of Employees**
Sales staff salaries	6 644	12
Consultation staff salaries	2 496	4
Warehouse staff wages	3 055	6
Administrative staff salaries	3 050	4
General overheads (electricity, rates, etc.)	17 500	
	32 745	

Salaries, wages and general overheads are considered to be of a fixed nature, except for 30% of general overheads which vary with the total cost of sales.

He has also collected the following information for the last three months:

	Breakfast Nook	Departments Bedrooms	Living Rooms
Number of items sold	1 000	1 500	4 000
Customer purchase orders	1 000	900	2 500
Floor area (square metres)	16 400	10 000	14 400
Number of consultations	798	200	250

The following pairing of cost drivers to cost pools has been suggested by a competent consultant:

Activity	**Driver**
Sales staff salaries	Customer purchase orders
Consultation staff salaries	Number of consultations
Warehouse staff wages	Items sold
Administration staff salaries	Customer purchase orders plus consultations
General overheads	Floor space

The management accountant believes that he can use this information to review the department's performance in the last three months from an activity-based costing (ABC) perspective.

Lindela factory

The Lindela factory produces a fairly basic product which can be used in all the lifestyle departments, as it is easily assimilated into other units. The factory currently operates at 60% of its normal capacity.

The current sales split of the factory is 80% to the life-style departments and 20% to external customers. No export sales are made.

During a five week period in October 2012, Lindela manufactured 2 300 units, 2 000 of which were delivered to the retail section and 300 to external customers. In respect of the production, the following costs were incurred:

- Material A: 370m^3 issued with an average cost of R3 600/m^3
- Material C: 2 280 litres @ R21,50/litre
- Packaging: External cost R17 000
  Internal cost R51 000
  R68 000
- Fixed overhead paid R274 000
- Other costs incurred were as scheduled.

The standard cost per unit for **external sales** is as follows:

	Notes	**R**
Raw material :	(1)	
A 0,15m^3 @ R3 500/ m^3		525,00
B 12 units @ R3,20/unit		38,40
C 1 litre @ R21,60/litre		21,60
Labour and variable overhead		
0,6 hours @ R40/labour hour		24,00
Packaging	(2)	56,00
Fixed overhead	(3)	60,00
		725,00
Sales price per unit		1 160,00

Notes:

(1) Raw materials B and C are easily sourced and no material inventories are held. Material A, Mukwa bought in Zambia and Zimbabwe, cannot be obtained at short notice. The factory manager in the past insisted that eight (8) weeks' inventory of Mukwa based on current production, be held at any point in time.

(2) For internal sales to the stores division, more rudimentary packaging is used at a cost of R26 per unit.

(3) The fixed overhead is based on normal capacity of the Lindela factory of 27 000 labour hours per working year of 50 weeks.

A design specialist has performed a market research study and found that a small design change which will add R45 to the current material cost, will enable the Lindela factory to increase the current sales to

- Local external customers by ⅔ (two thirds)
- Life style departments by 25%
- Generate export sales to France and Germany of 10% of the normal capacity. The export sales will be denominated in Euro at a discount of 10% on current local sales prices.

A subsequent price decrease of 5% on the current local external price and internal transfer price will increase all local sales volumes (internal and external) by a further 10%.

The current policy of the group is to use market prices as the transfer prices between divisions/subsidiaries, adjusted for internal savings.

Naldnak factory

The Naldnak factory in particular has struggled for the last few years and management is concerned about closure and job losses. They are negotiating with the Kiash family to set up a joint venture (JV) which will entail an expanded operation. The expansion will require additional funding R35 million. As part of this exercise, two options are being considered:

Option A

- A 5-year R27 million loan from the Industrial Development Corporation (IDC) at a fixed rate of 7% per annum. The cost of preparing the proposal in the format which the IDC requires will be 5% of the loan payable immediately. The remainder of the required funding will be provided by the joint venture participants.
- In acquiring the loan, the factory will qualify for a grant from the Department of Trade and Industry (DTI) amounting to 40% of the loan and The grant will be payable to the JV in five (5) equal instalments at the end of each year of the loan, upon certification that the JV complied with DTI conditions.
- The IDC will require that the annual DTI grant payment paid to the JV be paid directly to them, which they will off-set against the loan. This will be the only repayment made on the loan, except for the payment referred to hereunder.
- Final settlement of the loan will take place at the end of the 5-year period by means of a single bullet repayment.

Option B

- An equivalent Euro-denominated loan at 1,5% per annum, repayable in full at the end of a 5-year period. The cost of preparing this proposal will be €100 000.
- Should the JV obtain the Euro-denominated loan, they will not qualify for the DTI grant detailed above.
- The current €/R exchange rate is R10,80 = €1.
- The loan will allow the JV to obtain preferential status for exports to all European Union countries.

REQUIRED		Marks
(a)	Prepare a profit statement for the last three using months for the Bedrooms department using the activity based costing method.	(12)
(b)	Discuss the advantages and disadvantages that may arise for the Anani Group from introducing activity-based costing in its stores.	(10)
(c)	Evaluate and discuss the proposal to close the Bedrooms department. Your evaluation should include a break-even calculation for this department as well as the calculation of other relevant indicators. [Calculation 10, discussion 6]	(16)
(d)	Calculate the following for October 2012: • Sales revenue for the month • Variances incurred in respect of materials A, C and packaging • Fixed overhead variances – state clearly whether the efficiency variance represents an under-or-over-recovery	 (2) (6) (5)
(e)	List and motivate which of the variances calculated in (d) above should be investigated by management.	(7)
(f)	Elaborate on the strategic impact of holding 8 (eight) weeks inventory of Mukwa and the effect thereof on inventory management for Lindela.	(6)
(g)	Determine the production level of the Lindela factory after both the design change and the price reduction had been affected and show the sales prices to be charged to the different customer categories.	(8)
(h)	Calculate the mark-up percentage on units transferred from the Lindela factory to other group departments or divisions, based on the adjusted standard costs and pricing.	(6)
(i)	For the Naldnak factory, calculate and show the following in respect of options A and B: • The bullet payments at the end of the term • The cash available for the project • The forward €/R exchange rate applicable for year 5 payment at which management would be indifferent between options A and B.	 (10) (2) (2)

QUESTION 6

FILMSTARS (PTY) LTD **100 marks**

(Source: Exam 2011 TOE408W (MAC4862))

Filmstars (Pty) Ltd is an entertainment company with its primary focus being the African continent. The company manufactures and distributes DVD's and related products as well as distributing the rights relating to movies produced by a variety of producers. They also sell the movies to cinema chains for viewing in South Africa and other African countries.

Both the company's manufacturing and distribution divisions have experienced difficulties recently due to falling demand as a result of the economic recession, as well as difficulty experienced in sourcing sufficient movies to maintain profitability.

MOVIE AND DVD BUSINESS SEGMENT

DVD Manufacturing and distribution division

New technology has provided an opportunity to produce a new type of DVD which entails utilising its existing facilities. Filmstars (Pty) Ltd anticipates that this will improve its market share. It has already paid R1 million towards development expenditure, market research and a feasibility study for DVD production as well as R50 000 on a market research survey to determine the potential for the DVD production business.

Sales of 165 000 DVD's per annum for five years at a selling price of R40 per DVD are anticipated.

Estimated operating costs, at today's prices, are as follows:

Current cost per DVD:	**R**
Material cost	16
Direct labour	4
Variable overhead	3
Fixed overhead (allocated)	3
Distribution	4

The existing facilities will need to be redesigned and set-up at a cost of R4 million after allowing for all the relevant tax relief. The current tax rate is 28%.

Filmstars (Pty) Ltd's shareholders require a nominal return of 14% per annum after tax, which includes an allowance for inflation of 5,56% per annum.

Movie production division

Filmstars (Pty) Ltd's board has expressed its concern regarding the increasing uncertainty about the potential success or failure of independently produced movies. None the less it was decided at a recent board meeting, to commission movies in its own right and to invest in such productions.

One option is to purchase an existing company at a cost of R800 000. Cinema chains are adopting a policy of only buying movies from large international movie companies, and while Filmstars (Pty) Ltd understands that more movies may fail than succeed, they believe this part of their business will still be profitable.

Data has been collected covering a five year period. A target of producing three movies a year has been set.

Sales in year one will probably be:

Sales (R million)	**Probability**
8	0,2
10	0,4
14	0,3
20	0,1

Sales, in terms of value, are expected to grow at an annual rate of 5% from year two onwards.

Costs are as follows:

	R'000
Purchase of film-making company	800
Annual legal and professional costs	40
Annual operating lease payments (office equipment)	24
Studio and set hire per movie (incurred annually)	360
Camera or specialist equipment hire per movie (incurred annually)	80
Technical staff wages per movie (incurred annually)	1 040
Screenplay per movie (incurred annually)	100
Actors' salaries per movie (incurred annually)	1 400
Costume and wardrobe hire per movie (incurred annually)	120
Set design and painting per movie (incurred annually)	300
Annual wages of non-production staff	120

Additional information gathered

- All the above expenses are deductible in terms of section 24F (film allowances) of the Income Tax Act.
- Tax is payable one year in arrears and the tax benefits of any assessed losses can be taken into account as they arise, when performing net present value (NPV) calculations.
- Wages for technical and non-production staff as well as actors' salaries are expected to rise by 7% per annum from the beginning of year 2 onwards.
- Studio hire costs will be subject to an increase of 25% at the beginning of year three and 10% per annum thereafter.
- Screenplay costs per film are expected to rise by 12% per annum from the beginning of year 2 onwards, due to a shortage of skilled writers.
- The new business will occupy office accommodation that is currently being rented out for R40 000 per annum, with no escalation provided for in the rental contract.
- Other costs not specified above will remain stable as cost savings due to efficiencies and technological advances will counter inflationary increases.

MANUFACTURING SEGMENT

The manufacturing segment focuses on the manufacturing of specialised film industry equipment, decoders and wiring sets. Each of the three product ranges occupies dedicated space in their Brits factory.

Decoders

Although the employees of this section boast excellent credentials in their field and are upbeat about future prospects in the market, 70% spare capacity currently exists. The hard working sales manager has recently won a contract to manufacture and supply 6 000 decoders to a new pay-to-view TV station, Hot TV, during the following year.

The price of this decoder is critical as Hot TV subsidizes the decoder price to entice users to subscribe to their services. Hot TV believes there is a direct positive correlation between decoder sales and subscriptions and would like to use this as a growth tool. Hot TV is likely to order additional decoders in the event of higher demand. The sales manager has asked for your help in setting the right price for the order and has supplied you with the following information:

- Sales have been budgeted at R3 627 000 at full capacity for the coming year with a budgeted contribution margin of 35%. Without special orders, capacity utilisation is likely to remain at 30%.

- The production manager has set the overhead recovery rate at R38,50 per machine hour. Fixed production overheads include labour, depreciation and rent. The budgeted machine hours have been determined as being 34 500 hours for the year.

- Due to highly skilled technical staff, management is confident that production will derive an 85% cumulative average learning rate during the manufacturing of the required 6 000 units. The time taken to manufacture the first batch of 1 500 units was 5 463 machine hours. The order will be manufactured in batches of 1 500 units. All decoders manufactured after the special order will take the same average time per unit as that of the special order.

- Variable costs will amount to

Materials	R123,65 per unit
Production overheads	R24,60 per machine hour

- This project will incur additional fixed costs of R28 500 per annum.

- Hot TV would like to earn a 50% mark-up on their cost, but has consciously made a decision to subsidize the decoder price to grow their fledgling business. Hot TV has provided you with the following schedule of their projected sales volumes at different price levels:

Price per decoder R	**Projected Volume**
305	9 200
320	8 800
340	8 200
360	7 500
380	6 800
400	6 000

Wiring sets

A wire set consists of the wiring to connect to different devices as well as the dish connection. This enables the independent installers to cut down substantially on their installation time. After the wiring is put together, colour coded plastic is moulded onto it. Products are inspected before being released for distribution and a 10% loss is considered normal and acceptable. A completed set is sold for R90 while a rejected set is sold for R10 to a welfare society, which recovers the wiring components. The proceeds from the rejected sets are set off against the normal loss incurred. No material inventories of accepted or rejected sets are held at month ends.

For budget purposes, the following cost structure has been prepared:

Monthly volume (sets)	**Total overheads** [Note 1] **(R)**
400 – 500	11 760 – 12 225
600 – 1 000	22 490 – 24 350
1 100 – 1 500	36 115 – 37 950
1 600 – 1 700	42 340 – 42 805

[Note 1] The range of the budgeted total overheads indicates the total overhead costs on a monthly basis for the budgeted volume of wire sets indicated above.

The fixed overhead rate for each production month is calculated using the average quantity of the range, in which production fell during that month.

The following actual figures were recorded in this division during September 2011:

- 1 320 wire sets at a cost of R19 800 (the cost being the material component of the sets only) were sent for inspection. The rejection rate was as expected.
- Labour cost incurred was R17 754.
- Variable overhead incurred was as per budget.
- The fixed overhead actually incurred was R28 248.

REQUIRED		Marks
(a)	Assess the financial feasibility of producing the DVD's in **real terms**, computing both the net present value (NPV) and the internal rate of return (IRR). For this assessment, **assume no delay in respect of the tax cash flows exists**.	(12)
(b)	Determine the values for the: (i) price (ii) Volume at which the DVD project's NPV becomes zero. Discuss your results and suggest appropriate management action.	 (4) (4) (2)
(c)	Calculate the NPV of the proposed investment in the movie production company. (Workings to be in R'000)	(20)
(d)	Distinguish between specific (diversifiable) and market (non-diversifiable) risk with specific reference to Filmstars (Pty) Ltd's activities.	(4)
(e)	Making use of the risk measure – coefficient of variation – comment on the risk associated with the company's anticipated sales of produced films for the current year.	(10)
(f)	Briefly explain how the tax treatment of capital purchases can affect an investment decision.	(2)
(g)	Calculate the budgeted breakeven point in **Rand** for Filmstars (Pty) Ltd's decoder section.	(2)
(h)	**Address the following in respect of the order received by** Filmstars (Pty) Ltd for 6 000 decoders to be supplied to Hot TV: • Calculate the minimum price at which the decoders could be supplied. • Calculate the price(s) at which the decoders could be supplied on the assumption that the decoder division does not want to make a loss during the ensuing year. • Make a final recommendation to management regarding a possible course of future action.	 (6) (15) (4)
(i)	Briefly explain whether the subsidization process used by Hot TV is reasonable and sustainable from a business perspective.	(5)
(j)	Calculate the: (i) Fixed overhead over or under allocated and the (ii) Normal loss incurred during September 2011 for/by the wiring set division. Do calculations for this section to 3 decimals where necessary.	 (6) (4)

Formulae

Fisher: $1 + \text{nominal rate \%} = (1 + \text{real rate \%}) \times (1 + \text{inflation rate \%})$

Discount factor: $\frac{1}{(1 + r)^n}$

Coefficient of variation: $\frac{\text{Standard deviation}}{\text{Expected value}}$

QUESTION 7

SMILE (PTY) LTD **100 marks**

(Source: Exam 2011 DIPAC26 (MAC4861))

Please note that the standard of the DIPAC26 questions is not always comparable to MAC4861 standards but that these questions can be used for revision purposes for MAC4861 as they test basic principles and concepts that are necessary for MAC4861/2.

Smile (Pty) Ltd (Smile) is a small company manufacturing toothpaste for sensitive teeth.

In the week preceding the current date, the management accountant of Smile resigned with only 24 hours of notice in order to take up an appointment in London. You have been engaged as a consultant to assist the financial director while his replacement is being sought. The company's year end is 30 September and the management accountant, before his departure, had been preparing the income statement for management purposes for the year ended 30 September 2011, as well as the budget for the 2012 financial year.

Smile currently manufactures two types of fluoride toothpaste for sensitive teeth namely 'Mint' (regular) and 'Citrus' (with an added antibacterial agent). The toothpaste is sold in 100ml tubes packaged in a box. The basic mix of ingredients for the two types of toothpaste is the same. The same quantity of flavouring (mint or citrus) is added to each tube of toothpaste and costs the same per millilitre. The antibacterial agent is added in tiny quantities and does not affect the volume of the final product.

You obtained the following information:

1. No opening or closing inventory of raw materials and packaging are kept. Completed products are recorded on the first-in-first-out basis.

   The quantities for the year were as follows:

	Mint Tubes	Citrus Tubes	Total Tubes
Budget			
Opening inventory	7 000	6 000	13 000
Closing inventory	12 000	4 000	16 000
Sales	120 000	62 000	182 000
Actual			
Opening inventory	7 000	6 000	13 000
Closing inventory	6 000	8 000	14 000
Sales	135 000	57 000	192 000

2. Under normal operating conditions 5% of the direct materials (basic ingredients and flavouring) purchased go to waste during the production process. This is an unavoidable part of the process. The actual cost of direct materials for the 2011 financial year was R42,75 per litre. No abnormal losses occurred during this year.

3. The company allocates fixed manufacturing overheads to products using machine hours. The **budgeted** machine hours were 1 750 hours for Mint and 960 hours for Citrus. The budgeted fixed manufacturing overheads for 2011 were R600 000 and actual overheads were R620 000.

4. Selling and distribution costs for the 2010 (previous) financial year amounted to R135 360 for Mint and R111 234 for Citrus for sales of 132 000 tubes of Mint and 55 800 tubes of Citrus respectively. Selling and distribution costs for the current year amounted to R135 900 for Mint and R111 510 for Citrus. The total fixed selling and distribution costs for each product line remained the same for these two years and are included in the amounts given. The variable selling and distribution costs per tube for each product also remained the same for the two years.

5. Budgeted costs per tube for the 2011 year were as follows:

	Mint R	**Citrus R**
Direct materials (basic ingredients and flavouring)	4,00	4,00
Antibacterial agent	–	0,50
Packaging – tubes	0,40	0,40
Packaging – boxes	0,15	0,15
Direct labour	0,60	0,80
Variable overhead	2,00	2,20

6. The following variances between **total original budgeted costs** and **total actual costs** for the 2011 year were calculated by the management accountant:

	Mint R		**Citrus R**	
Direct materials (basic ingredients and flavouring)	Refer Note 2			
Antibacterial agent	–		2 270	F
Packaging – tubes	8 960	A	1 960	A
Packaging – boxes	2 670	F	1 920	F
Direct labour	25 500	A	5 100	A
Variable overhead	42 120	A	750	A

F = Favourable; A = Adverse

These variances relate to the total production of Mint and Citrus during the 2011 year.

7. The actual selling prices during the 2011 year were R16,20 per packaged tube of Mint and R20,00 per packaged tube of Citrus.

8. Administrative costs do not vary with production and amounted to R420 000 for the 2011 year.

The company is considering whether or not to accept a once-off order for 30 000 tubes of children's' toothpaste (strawberry flavour). The tubes will contain 50ml of toothpaste. The costs per tube will be as follows:

	Strawberry R
Direct materials (basic ingredients and flavouring, including normal spoilage)	2,00
Packaging – tubes	0,25
Packaging – boxes	0,12
Direct labour	0,55
Variable overhead	1,80

An additional manufacturing machine will have to be rented at a cost of R75 000 to accommodate the order. In addition, incremental fixed manufacturing overheads will amount to R35 000 and delivery costs will be R7 000.

While preparing the 2012 budget the management accountant investigated the possible use of Activity Based Costing. He analysed the budgeted fixed manufacturing overheads for the 2012 year as follows:

	Total R
Store costs	180 000
Set-up costs	120 000
Power	105 000
Depreciation	255 000
	660 000

Further budgeted details for the 2012 year are as follows:

	Mint	Citrus
Production (tubes)	145 000	62 000
Machine hours	1 950	1 050
Batch size (tubes)	2 500	2 000
Orders per batch	4	5
Set-ups per batch	1	2
Power (kJ) per batch	50	45

It was found that store costs are driven by the number of orders.

The management accountant had gathered the following information regarding the sales budget for the 2012 financial year:

- The company's current market share of the market for toothpaste for sensitive teeth (for Mint and Citrus combined) is 6%.
- The market is expected to grow by 1% during the 2012 year.
- The company is expected to maintain its market share. However, if additional marketing costs are incurred the company could increase its market share to 6,5%.
- The current selling price per tube can be increased by 6,5% for Mint and 6,0% for Citrus.
- The new sales mix is expected to be 71% Mint and 29% Citrus.

REQUIRED			Marks
Round Rands to the nearest cent (two decimals) and all other figures (e.g. percentages and hours) to three decimals.			
(a)	(i)	How many litres of direct material were purchased (and used) in total to produce the actual quantity of toothpaste manufactured during the 2011 year?	(3)
	(ii)	What is the actual direct material cost per tube of toothpaste, per product line?	(1)
(b)	(i)	Calculate the fixed overhead recovery rate for the company using machine hours as the allocation base.	(1)
	(ii)	Calculate the amount of overhead that was budgeted to be allocated to each product, **per tube,** in 2011.	(5)
	(iii)	Calculate the over- or under-recovery of fixed manufacturing overheads for the company. State whether the calculated amount represents an over- or under-recovery of fixed overheads.	(2)
	(iv)	Calculate the fixed manufacturing overhead expenditure variance. State whether the variance is favourable or adverse.	(1)
(c)	(i)	Calculate the actual total fixed selling and distribution cost per product line for the 2011 financial year.	(5)
	(ii)	Calculate the actual contribution per tube per product for the 2011 financial year, as well as the actual contribution percentage per tube per product.	(18)
	(iii)	Calculate the actual number of tubes of each product line that Smile had to sell for the 2011 financial year in order to break even.	(7)
	(iv)	Calculate the total percentage margin of safety relating to the actual results for the 2011 year and comment on your answer. Also give advice to management on how the results of the ratio may be improved. (Calculation = 3 marks; Comments / advice = 4 marks)	(7)
(d)	(i)	Calculate the minimum selling price per tube of Strawberry toothpaste that Smile will have to ask to break even on the once-off order.	(5)
	(ii)	Discuss the conditions that must generally apply for such a special pricing order to be acceptable.	(3)
	(iii)	Discuss the qualitative factors to be considered.	(4)
(e)	(i)	Calculate the 2012 budgeted fixed manufacturing overhead cost per tube for both Mint and Citrus using Activity Based Costing (ABC). Round total costs to the nearest Rand.	(10)
	(ii)	Assume the 2012 budgeted fixed manufacturing overhead cost using traditional absorption costing is R2,96 and R3,73 per tube of Mint and Citrus respectively. Compare the traditional and ABC 2012 budgeted fixed manufacturing overhead costs per tube and briefly comment on the impact of ABC on the cost per tube which could have pricing and profit consequences.	(5)
(f)	A foreign investor is interested in taking up a majority share in Smile. The investor's initial investigation, however, indicated that the current return on capital achieved by Smile is not sufficient to make an investment worthwhile. The investor has asked management to consider improving the profitability of the company in 2012 by either increasing the selling price of the products by 10% or reducing the size of the tubes by 10%. The financial director has asked you to discuss the possible consequences of both these proposals. Note that calculations are NOT required. (10% increase in selling price = 4 marks; 10% reduction in size = 4 marks)		(8)
(g)	Define incremental and zero-based budgeting and summarise the main differences between the two budgeting techniques.		(5)
(h)	Calculate the expected sales (in Rand) for the 2012 sales budget for both the Mint and Citrus product lines. Assume that the company will incur the additional marketing cost and that Mint and Citrus will be the only two products sold in the 2012 year.		(7)
(i)	The managing director has heard of the residual income approach for purposes of evaluating performance. Briefly explain what the residual income approach is and name one advantage of the residual income approach.		(3)

QUESTION 8

EWORX LTD **100 marks**

(Source: Supplementary exam 2011 TOE408W (MAC4862))

A. Background information relating to Eworx Ltd

Eworx Ltd is a diversified company listed on the AltX section of the JSE. The company develops and maintains software products, manufactures telecommunication equipment, performs data monitoring services for clients worldwide and operates other related businesses.

Management has extracted the following financial information for the past two years:

Year ended 30 June	**2011 R'000**	**2010 R'000**
Operating profit	28 994	27 618
Finance charges	906	1 272
Profit before tax	28 088	26 346
Taxation	7 865	5 796
Net profit	20 223	20 550
Total assets	98 826	92 494
Total equity	87 441	82 921

B. Hardware installation

This division sources applicable hardware for clients based on a review of client needs, installs the hardware and then monitors the maintenance process thereof. Management accounts for this division show the following for the past two years:

	2011 R'000	**2010 R'000**
Revenue generated	9 764	10 841
Labour costs	(8 313)	(7 697)
Variable costs	(1 457)	(1 301)
Rental of office space	(420)	(382)
Other fixed costs	(920)	(860)
Head office charge	(1 260)	(1 230)
Net loss	(2 606)	(629)

You have been requested to brief the senior management team on the implications of the possible closure of this division and have obtained the following data:

(i) The revenue (mainly fee based) includes a maintenance monitoring fee of R732 000 in 2011 and R759 000 in 2010. The monitoring fees are fixed and are based on a 3-year contract entered into with the client upon installation of the hardware and are currently estimated at R548 000 for the 2012 financial year.

(ii) Labour is considered a fixed cost. 15% of the employees, with an equivalent 25% of the Rand cost, can be employed elsewhere in the company. Retrenchment costs for the remainder will amount to R1 198 000, based on fair labour practice plus a sweetener.

(iii) Variable costs are incurred during the evaluation and installation process.

(iv) The office space currently occupied by this division can be taken over by one of the other divisions as Eworx Ltd currently experiences an acute shortage of office space.

(v) Other fixed costs refer to depreciation of equipment and vehicles, 40% of which will not be taken up directly by other divisions in the event of closure.

(vi) The head office charge consists of an applied cost recovery of R720 000 for 2011 (2010: R720 000) and the target profit for 2011, obtained by applying the weighted average cost of capital (WACC) of Eworx Ltd - 15% - on the division's assets, of R3,6 million.

C. Telecommunication division

- The Telecommunication division has been in business for a number of years producing telecommunication equipment. The division has been approached by a local manufacturer of in-vehicle satellite navigation systems with a view to forming a strategic alliance. The proposal is that one of the components, which is currently used in the Telecoms end products, will be purchased by the partner for use in one of its products, for a three-year period.

- The partner has suggested that the price for the component should be calculated on a target costing basis. The proposal is to deduct an agreed profit margin and the processing cost of the partner, from the market price of the component to derive a transfer price for the component.

- The managing director has suggested that the division should take this opportunity to expand its production capacity and change the production structure to allow for this. He has suggested that two product lines be formed. One of these should manufacture the relevant component, and the other should manufacture its own final products. Some of the output of the component manufacturing line will be transferred to the end-product manufacturing line and some will be sold to the partner.

- He has noted that if this structure is adopted, it will be necessary to implement an appropriate system of transfer pricing. In addition it is proposed to pay a bonus to production managers. This will be calculated as 4% of the amount by which divisional profits exceed 5% of revenue.

- Two possible transfer pricing systems, namely a market-based and a cost-based system respectively, have been suggested. In the market-based system, internal transfers will be charged at the price paid by the external customer (the partner). In the cost-based system, internal transfers will be charged at variable cost plus 70%.

- One of the senior managers has questioned the need for what he has referred to as "another cost line-item". He argues that there will be no benefit to the division as the income of one product line will be a cost to the other, having a neutralising effect on the division as a whole.

You have been requested to brief the senior management team on the implications of establishing a transfer pricing system. You have obtained the following data:

(i) The selling price of the satellite component will be R156,00.
(ii) The agreed margin of the partner will be 30%.
(iii) The partner's costs of processing will be R25,00 per unit.
(iv) The variable cost of manufacturing is R51,00 per unit.
(v) Additional fixed costs for the component line will be R2,7 million per annum.
(vi) The capacity of the component line will be 450 000 machine hours per annum.
(vii) Each component requires 2,5 hours of machine time.
(viii) The expected demand of the external partner the first year of operation is 60 000 units which will increase to 80 000 units per annum thereafter.
(ix) Internal transfers will absorb 60 000 components in the first year of production and will remain stable thereafter. A condition of setting up the component line is that internal usage will at all times be satisfied.
(x) From year 2 onwards, up to 50 000 components per annum may be exported to the USA at a fixed US $ price of $12 per component for an initial 3-year period, with a current exchange rate of $1 = R7,20.
(xi) The expansion will be funded from internal sources.

D. Client service

One of the services provided by Eworx Ltd. is to offer customers an on-line booking service for packaged holidays, tickets for events and flights, under the trading name Fastbook.com.

The results for the current year (2011) are:

	Holidays	**Tickets**	**Flights**
	R'000	**R'000**	**R'000**
Sales revenue	7 685	3 770	3 045
Direct costs	(5 870)	(2 670)	(2 260)
Share of central costs	(1 157)	(845)	(767)
Profit	658	255	18

The direct costs are made up of the cost of buying the various services from the providers and other costs incurred by Fastbook.com to operate and promote this line of business. With the exception of advertising costs of R2 972 900, the direct costs are considered to be variable. The price charged to the customer is calculated by adding a standard mark-up to the cost of buying the service from the provider.

The analysis of advertising costs is:

Holidays	**Tickets**	**Flights**
R'000	**R'000**	**R'000**
1 766,21	773,69	433,00

The central costs, which are considered to be fixed, are allocated to the lines of business on the basis of the usage of the central computer system, at a rate of R65 per processing hour.

Management is considering whether to undertake a further investment in the booking system. They have received a quote of R819 000 for the upgrade, which will have a useful life of three years and will increase processing capacity by 13%. They estimate that the following cost increases will occur in the forthcoming year:

	R
Advertising:	
Holidays	160 000
Tickets	55 000
Flights	35 000
Central costs before system upgrade	215 000

You have been requested to investigate:

(i) The profitability of each line of business, based on the contribution per processing hour.
(ii) The total revenue which must be generated in the incoming year to increase the company's profit by 10%, and
(iii) Whether the booking system upgrade will provide sufficient processing hours to generate the revenue required to achieve the profit target.

For the purposes of your investigation, management do not anticipate any change in either:

- The proportion of total revenue which is generated by each line of business or
- The contribution per processing hour for each line of business.

Note to candidates:

Pay particular attention to the exact wording of the required section, and the number of marks allocated. Answers should be carefully planned before being executed and clearly numbered. The length and extent of answers should relate to the allocated number of marks.

REQUIRED		Marks
(a)	Calculate the following: (i) The Return on Investment (ROI) for Eworx Ltd for both 2011 and 2010. (ii) The Return on Equity (ROE) for Eworx Ltd for 2011. (iii) The Residual Income (RI) for Eworx Ltd for 2011.	 (2) (1) (2)
(b)	Discuss, with specific reference to the stated business activities of Eworx Ltd, whether you consider Return on Investment to be a fair tool or not, for the measurement of management performance.	(5)
(c)	Explain Economic Value Added (EVA) in basic common terminology and then indicate in respect of each of the following whether it may be considered for the application of Economic Value Added by Eworx Ltd: • Development of software packages. • Manufacturing of electronic components. • Data monitoring services.	(9)
(d)	Calculate the break-even point for the Hardware Installation division for 2011.	(7)
(e)	Advise management whether, based on the divisional information presented, they should close the Hardware Installation division or not. State clearly all issues management should consider during this process.	(10)
(f)	Calculate the bonus that may be payable to managers of the telecommunication division in year one with an internal transfer price based on a market-based approach.	(7)
(g)	Use the cost-based approach for year 2 and calculate the profit generated by the component product line. For this purpose you may assume that due to contractual arrangements, no price or cost increases will take place.	(8)
(h)	Briefly explain the impact on Eworx Ltd of funding the new component product line from internal sources.	(8)
(i)	List and explain briefly the risks that Eworx Ltd should consider in terms of the new component product line.	(10)
(j)	Calculate the following in respect of the Client Service division: (i) The profitability of each line of business based on the contribution per processing hour. (ii) The total revenue that Fastbook.com should generate to meet the target of a 10% increase in profit in the 2012 financial year. (iii) Whether the upgraded system will provide sufficient processing hours to generate the revenue required to achieve the profit target as identified in j(ii) above.	 (12) (7) (12)
TOTAL		**100**

QUESTION 9

MATOJENI LTD **100 marks**

(Source: Supplementary exam 2011 DIPAC26 (MAC4861))

Please note that the standard of the DIPAC26 questions is not always comparable to MAC4861 standards but that these questions can be used for revision purposes for MAC4861 as they test basic principles and concepts that are necessary for MAC4861/2.

THIS QUESTION HAS THREE (3) UNRELATED PARTS

PART A **[35]**

Month 1

Matojeni Ltd processes and sells brown rice. It buys unprocessed rice seeds and then, using a relatively simple process, removes the outer husk of the rice to produce the brown rice. This means that there is substantial loss of weight in the process. The market for the purchase of seeds and the sales of brown rice has been, and is expected to be, stable. Matojeni Ltd uses a variance analysis system to monitor its performance.

There has been some concern about the interpretation of the variances that have been calculated in month 1.

1. The purchasing manager is adamant, despite criticism from the production director, that he has purchased wisely and saved the company thousands of Rands in purchase costs by buying the required quantity of cheaper seeds from a new supplier.
2. The production director is upset at being criticised for increasing the wage rates for month 1; he feels the decision was the right one, considering all the implications of the increase. Morale was poor and he felt he had to do something about it.
3. The maintenance manager feels that saving R8 000 on fixed overhead has helped the profitability of the business. He argues that the machines' annual maintenance can wait for another month without a problem as the machines have been running well.

The variances for month 1 are as follows:

		R
Material price	Favourable	48 000
Material usage	Adverse	52 000
Labour rate	Adverse	15 000
Labour efficiency	Favourable	18 000
Labour idle time	Adverse	12 000
Variable overhead expenditure	Adverse	18 000
Variable overhead efficiency	Favourable	30 000
Fixed overhead expenditure	Favourable	8 000
Sales price	Adverse	85 000
Sales volume	Adverse	21 000

Matojeni Ltd uses labour hours to absorb the variable overhead

Month 2

<u>In month 2 the following standard data applies:</u>

Standard costs for 1 tonne of brown rice:

- 1,4 tonnes of rice seeds are needed at a cost of R60 per tonne
- It takes 2 labour hours of work to produce 1 tonne of brown rice and labour is normally paid R20 per hour.
- 2 hours of variable overhead at a cost of R30 per hour
- The standard selling price is R240 per tonne
- The standard contribution is R56 per tonne

Budget information for month 2 is:

- Fixed costs were budgeted at R210 000 for the month
- Budgeted production and sales were 8 400 tonnes

The actual results for month 2 were as follows:

- Actual production and sales were 8 000 tonnes
- 12 000 tonnes of rice seeds were bought and used costing R660 000
- 15 800 labour hours were paid for costing R303 360
- 15 000 labour hours were worked
- Variable production overhead cost R480 000
- Fixed costs were R200 000
- Sales revenue achieved was R1 800 000

PART B **[30]**

- Telematics Ltd has been in business for a number of years producing telecommunication equipment. The company has been approached by a manufacturer of in-car satellite navigation systems with a view to forming a strategic alliance. The proposal is that one of the components, which is currently used in its end products, would be purchased by the partner for use in one of its products.
- The partner has suggested that the price for the components should be calculated on a target costing basis. The proposal is to deduct an agreed profit margin and further processing costs from the market price for the navigation system to derive the price for the component.

- The managing director has suggested that Telematics should take this opportunity, expand the company's production capacity and change the structure to allow for this. He has suggested that two divisions should be formed. One of these would manufacture the component, and the other would manufacture the final products. Some of the output of the component manufacturing division would be transferred to the end-product manufacturing division and some would be sold to the partner.

- He has noted that if this structure is adopted, it will be necessary to implement an appropriate system of transfer pricing. In addition it is proposed to pay a bonus to divisional managers. This will be calculated as 4% of the amount by which divisional profits exceed 5% of revenue.

- Two possible transfer pricing systems, market-based and cost-based, have been suggested. In the market-based system, internal transfers would be charged at the price paid by the external customer. In the cost-based system, internal transfers would be charged at variable cost plus 70%.

- One of the senior managers has questioned the need for what he has referred to as 'another cost heading'. He argues that there will be no benefit to Telematics as the income of one division will be a cost to the other, leading to a neutral effect on the company as a whole.

You have been asked to brief the senior management team on the implications of establishing a transfer pricing system.

You have obtained the following data:

(i) The selling price of the satellite navigation system will be R78.
(ii) The agreed margin will be 25%.
(iii) The customer's costs of further processing will be R12,50 per unit.
(iv) The variable cost of manufacturing is R26,80 per unit.
(v) Fixed costs for the component division will be R2,7 million per annum.
(vi) The expanded capacity of the component division will be 450 000 machine hours per annum.
(vii) Each unit requires 2,5 hours of machine time.
(viii) The expected external demand in the first year of operation is 60 000 units.
(ix) The remaining production capacity will be utilised by internal transfers.

PART C **[35]**

Lastminute.com offers customers an on-line booking service for packaged holidays, tickets for events, and flights. The results for the current year are:

	Holidays R000	Tickets R000	Flights R000
Sales revenue	7 685	3 770	3 045
Direct costs	(5 870)	(2 670)	(2 260)
Share of central costs	(1 157)	(845)	(767)
Profit	658	255	18

The direct costs are made up of the cost of buying the various services from the providers and other costs incurred by Lastminute.com to operate and promote that line of business. With the exception of advertising costs of R2 972 900, the direct costs are considered to be variable. The price charged to the customer is calculated by adding a standard mark up to the cost of buying the service from the provider.

The analysis of the advertising costs is:

Holidays R000	Tickets R000	Flights R000
1 766,21	773,69	433,00

The central costs, which are considered to be fixed, are allocated to the lines of business on the basis of the usage of the central computer system, at a rate of R65 per processing hour.

The directors are considering whether to undertake a further investment in the booking system. They have received a quote of R819 000 for the upgrade, which will have a useful life of three years and will increase processing capacity by 13%. They estimate that the following cost increases will occur in the incoming year:

	R
Advertising:	
Holidays	160 000
Tickets	55 000
Flights	35 000
Central costs before system upgrade	215 000

The directors have asked you to investigate:

1. The profitability of each line of business, based on the contribution per processing hour.
2. The total revenue which must be generated in the incoming year to increase the company's profit by 10%; and
3. Whether the booking system upgrade will provide sufficient processing hours to generate the revenue required to achieve the profit target.

For the purposes of your investigation, the directors do not anticipate any change in either:

1. The proportion of total revenue which is generated by each line of business; or

2. The contribution per processing hour for each line of business.

PART A

REQUIRED		Marks
(a)	Using the variances and other relevant information above comment on the performance of the following managers in **month 1**: (i) The purchasing manager (ii) The production director (iii) The maintenance manager	(10)
	Note : You are not required to comment on the performance of the business or its managers in month 2.	
(b)	Calculate the variances for **month 2** in as much detail as the information above allows.	(10)
(c)	Reconcile the budget profit to the actual profit using marginal costing principles.	(15)
Total for Part A		**35**

PART B

REQUIRED		Marks
(a)	A good transfer pricing system should avoid sub-optimal decision making by divisional managers if overall objectives of the organisation are to be attained**.** **List** five objectives of a transfer pricing system.	(5)
(b)	Assess the argument that a transfer pricing system will provide no benefits to Telematics, and discuss the conflicts which the company must resolve if a transfer pricing system is introduced.	(5)
(c)	Calculate the bonus which would be payable to the manager of the component division in the first year of divisionalisation in each of the proposed transfer pricing systems.	(15)
(d)	Recommend the basis for calculating a transfer price which is likely to be appropriate for use by Telematics (No computations are necessary here).	(5)
Total for Part B		**30**

PART C

REQUIRED		Marks
(a)	**Calculate:** (i) The profitability of each line of business, based on the contribution per processing hour (ii) The total revenue which Lastminute.com must generate to meet the directors' target of a 10% increase in profit in the incoming year. (iii) Whether the upgraded system will provide sufficient processing hours to generate the revenue required to achieve the profit target (Only computations are required here).	(14) (8) (9)
(b)	Briefly discuss the limitations of your analysis in (a).	(4)
Total for Part C		**35**

TOTAL	**100**

QUESTION 10 **100 marks**

JUG AND BREW GROUP LIMITED

Jug and Brew Group Limited is a producer of a variety of high grade coffee blends. The firm buys coffee beans from mainly East, Southern and the horn of Africa (mainly Ethiopian Highlands). The coffee beans are then roasted, blended and packaged in one kilogram bags for sale to specialist coffee houses around the world and the Group's 13 own coffee shops. The major raw material in the coffee blends is the beans and the company's automated roasting, blending and packaging processes' have substantial overheads. Since the company is highly automated it uses very little direct labour.

Coffee blends from the mainly Ethiopian Highlands are very popular with American and European consumers and sell in large volumes while blends from Zimbabwe and Malawi sell in moderate volumes. The company has a mixed pricing strategy. In markets where demand is inelastic, the company prices its coffee blends at cost plus a mark up of 25% and in those markets where demand is highly elastic demand based pricing is applied. Group coffee shops pay market prices for their purchases.

The company's master budget for the coming year (2011) includes estimated fixed production overheads of R22 000 000 and the company assigns production overheads to products on the basis of direct labour hours. The expected direct labour cost is R2 400 000 and the budgeted direct labour hours are 50 000 hours. The labour cost is considered a variable cost. Based on the sales budget of R87,43 million for 2011 (all sales take place on a credit basis) and the material requirement budget, the company will during the forthcoming year purchase and use R34,1 million of raw materials in the form of coffee beans. Purchases of beans are paid for in US dollar ($), contracted for at the beginning of the year. The budgeted spot exchange rate for 2011 is $1 = R7,80. The budgeted debtors outstanding days is 47 and the budgeted raw material inventory turn is 12 times.

The budgeted direct costs for one - kilogram bags of two of the company's coffee products are as follows:

Per 1 kilogram bag:	**Ethiopian Roast R**	**Zimbabwean Roast R**
Direct materials	14,50	12,90
Direct labour (standard of 0,01 hours per bag)	0,48	0,48
Packaging bag	3,02	2,72

Note: one kilogram of coffee beans yields one kilogram of roasted and blended coffee.

The company's marketing manager, Mr Singh, who has just returned from a short course at Harvard in the USA, is of the belief that the company's traditional costing system may be providing misleading cost information related to the products. While at Harvard, he learnt of alternative methods of assigning manufacturing overheads to products and has quickly gathered the following information which relates to the manufacturing of **all the various products** of the company:

Cost Pool	**Measure**	**Yearly Activity of all products**	**2011 Budgeted Cost R**
Purchasing	Purchase Orders	5 000 Orders	5 600 000
Material Handling	Number of set-ups	2 500 setups	1 930 000
Quality Control	Number of batches	1 250 Batches	900 000
Roasting	Roasting Hours	237 500 Hours	10 450 000
Blending	Blending Hours	80 000 Hours	1 920 000
Packaging process	Packaging Hours	75 000 Hours	1 200 000
Total manufacturing overhead			**22 000 000**

Purchasing, materials handling and quality control have some labour content, whilst the other activities are automated.

Other costs:

- Sales, marketing and distribution costs amount to 3% of turnover.
- Administrative cash costs of the Jug and Brew Group amount to R13 200 000 per annum.
- Group depreciation, not included in the figures above, amounts to R6 000 000 per annum.

Data regarding the expected production for the **two selected products** is given below:

	Ethiopian Roast	**Zimbabwean Roast**
Forecast sales	80 000 kg	40 000 kg
Budgeted sales price (R/kg)	35,00	33,50
Budgeted production	80 000 kg	30 000 kg
Batch size	5 000 kg	3 000 kg
Setups	2 per batch	1 per batch
Purchase order size	20 000 kg	5 000 kg
Roasting time per 100 kg	1,5 hours	1,5 hours
Blending time per 100 kg	0,5 hours	0,4 hours
Packaging time per 100 kg	0,3 hours	0,3 hours
Opening inventory	5 000 kg	20 000 kg
Opening inventor (R/kg)	17,40	15,87

Note: Jug and Bean Ltd does not hold any inventory of unroasted coffee beans at the beginning or end of their financial year.

Orders

- One of the group's coffee shops, situated in Nelspruit, is experiencing profitability and liquidity problems. The management of this coffee shop has requested that a transfer price of R20,20 / kg for Ethiopian roast and R18,30 /kg for Zimbabwean roast for their 2011 orders of 3 000 kg and 2 000 kg respectively, be considered.

- A UK firm has approached Jug and Brew Group Ltd with a request to quote for a special order of 50 000 kg of the Zimbabwean roast where they will pay for the transport cost. The UK buyer has sent an e-mail to Mr Singh, stating that '… should a favourable price be quoted, follow up orders can be expected. In addition, we offer you an all-expenses paid trip to the UK to watch two year-end test matches between the Springboks and home unions of your choice'. This e-mail has inadvertently also landed in the managing director's inbox. Mr Singh has mentioned the possibility of the special order and future orders at a management meeting, but did not disclose the offer of the overseas trip.

- The production manager, Ms Ramphele, investigated and advises that the additional 50 000 kg Zimbabwean beans can be acquired at $1,80 per kg all inclusive. The budgeted forex rate will apply. Overtime of time and a half will be required for the specific production period. The order can be run as 2 batches of 25 000 kg each. Special packaging, which will be used as a substitute for the existing packaging, at R1,70 per 1 kg bag will also be required. No changes in any machine related costs are expected to arise from the execution of this order.

New capacity

- Ms Ramphele has presented the management committee with the following proposal: That an out-of-use neighbouring premises be converted to a factory manufacturing two lines of coffee machines for brewing coffee, for which the following information was gathered:

	Basic	**Advanced**
Sales price per unit	R1 200	R3 500
Quantity year 1 – units	3 000	1 000
Sales growth per annum (%)	20	10
Period of growth (years)	2	3
Manufacturing cost (R)	900	2 275
Including depreciation of capital assets used in production (R)	220	535
Number of unit returns per annum	160	30
Guarantee cost per return* (R)	800	2 500
Distribution and marketing cost (% of turnover)	4	6

The proposal is based on research results which cost R350 000, which have already been incurred. The premises will require an immediate refurbishment of R0,6 million and the manufacturing equipment will cost R7,5 million which will be purchased immediately. The equipment will have a total manufacturing life of 5 years, but will require a technological upgrade costing an estimated R1 million after three years. At the end of it's lifespan it will have a value of R0,3 million. The equipment provider has offered financing at an all-in cost of 14%. Jug and Brew Group Ltd has a tax rate of 28% and a real weighted average cost of capital of 18%. SA Revenue Services will allow a 40/20/20/20 wear and tear allowance on the equipment purchased.

Sufficient other taxable income is generated to absorb any tax losses resulting from this expansion project. Tax is paid in the year incurred.

Raw materials of R1,1 million will be required upfront, which will be exhausted at the end of the 5-year period.

* The guarantee cost of returns is paid out one year after the sale and return of the product.

Financial

- The financial director, Mr Adam, has observed that futures on coffee beans for close-out 12 months from now are trading at 30% above the current budget prices, due to anticipated market shortages. The forward rate in the currency market for the R/$ exchange rate at the end of 2011 is R8,50 = $1.

- Mr Adam has proposed that ten of the company owned coffee shops be converted to franchise operations and that the franchise channel be actively pursued under the following conditions:

  - Set up cost - an average R1,5 million payable by franchisee
  - Initial fee of R80 000 paid to Jug and Brew Group Ltd
  - Management service fee of 6% of turnover
  - Marketing fee of 2% of turnover
  - Operations cost of 45% of turnover-cost of products sold in the franchise operation and acquired solely from Jug and Brew Group Ltd.

REQUIRED		Marks
(a)	Using direct labour hours as the base for assigning manufacturing overhead cost to products, undertake the following: (i) Calculate the predetermined overhead rate that will be used during 2011 year. (ii) Calculate the unit product cost of one kilogram each of the **Ethiopian and Zimbabwean roast.**	 (1) (2)
(b)	Using activity based costing as a basis for assigning manufacturing overhead cost to products, undertake the following: (i) Calculate the total amount of manufacturing overhead cost assigned to the **Ethiopian and Zimbabwean roasts** for the year. (ii) Using the data calculated under **b(i)** above, compute the overhead cost per kg of the **Ethiopian and Zimbabwean roasts.** (Work to 2 decimal places). (iii) Prepare an activity based absorption costing statement of comprehensive income for the **Ethiopian and Zimbabwean roasts** respectively for 2011. Clearly indicate the most profitable roast.	 (19) (2) (14)
(c)	Briefly describe the accounting treatment for a change in inventory valuation should Jug and Brew Group Ltd value inventory using activity based costing principles.	(3)
(d)	Briefly discuss the ethical issues arising from the e-mail received by Mr Singh. List any additional information that you may require to reach a final decision.	(9)
(e)	Calculate the cash flow impact if the budgeted debtors days change to 40 and the inventory turn changes to 15 times. Calculations may be rounded to R'000 and to be based on 365 days per annum where relevant.	(5)
(f)	Explain the impact of accepting a transfer price to the Nelspruit coffee shop on Jug and Brew Group Ltd. Your explanation should be supported by relevant calculations and make a clear recommendation.	(8)
(g)	Calculate a price for the UK special order, incorporating the current gross profit margin based on the cost calculated in **(a)(ii)** above.	(6)
(h)	Evaluate Ms Ramphele's proposal to expand Jug and Brew Group Ltd's product range. Show all calculations clearly. **You may round to 1 decimal of R'000**. Your evaluation needs to be a quantitative analysis only.	(19)
(i)	Explain the impact of Mr Adam's observations regarding the coffee bean futures and the R/$ exchange rate on Jug and Brew Group Ltd's *current (2011) budget* and indicate how this impact may be countered. Ignore the potential special order being considered when answering this part of the question.	(6)
(j)	Explain the impact of the proposed franchise operation on Jug and Brew Group Ltd.	(6)

QUESTION 11

SWEET ROAST LIMITED **80 marks**

(Source: Exam 2010 DIPAC26 (MAC4861))



Please note that the standard of the DIPAC26 questions is not always comparable to MAC4861 standards but that these questions can be used for revision purposes for MAC4861 as they test basic principles and concepts that are necessary for MAC4861/2.

THIS QUESTION HAS THREE (3) UNRELATED PARTS

PART A **[34]**

Sweet Roast Limited is a producer of a variety of high grade coffee blends. The firm buys coffee beans mainly from the Ethiopian Highlands. The coffee beans are then roasted, blended and packaged in one kilogram bags for sale to specialist coffee houses around the world. The major raw material in the coffee blends is the beans. The company's automated roasting, blending and packaging processes result in substantial overheads. Since the company is highly automated it uses very little direct labour.

Coffee blends from the mainly Ethiopian Highlands are very popular and sell in large volumes while blends from Tanzania and Malawi sell in moderate volumes. The company has a mixed pricing strategy.

The company's budget for the coming year includes estimated production overheads of R2 200 000. The company assigns production overheads to products on the basis of direct labour hours. The expected direct labour cost is R600 000 and the budgeted directed labour hours are 50 000. Based on the sales budget and the material requirement budget, the company will, during the forthcoming year, purchase and use R5 000 000 of raw materials, mostly in the form of coffee beans.

The budgeted costs for direct materials and direct labour for one-kilogram bags for two of the company's coffee products are as follows:

	Ethiopian Roast R	**Malawi Roast R**
Direct materials (per 1 kg bag)	4,50	2,90
Direct labour (per 1 kg bag) (0,02 Hours per bag)	0,24	0,24

The company's marketing manager is of the belief that the company's traditional costing system may be providing misleading cost information related to the products. She has learnt of alternative methods of assigning manufacturing overheads to products and has collected the following information for the **total production** in the company:

Cost Pool	Measure	Yearly Activity of all products	Budgeted Annual Cost R
Purchasing	Purchase orders	2 000 orders	560 000
Material handling	Number of set-ups	1 000 setups	193 000
Quality control	Number of batches	500 batches	90 000
Roasting	Roasting hours	95 000 hours	1 045 000
Blending	Blending hours	32 000 hours	192 000
Packaging	Packaging hours	24 000 hours	120 000
Total manufacturing overhead			**2 200 000**

Data regarding the expected production for the **two selected products** is given below:

	Ethiopian Roast	Malawi Roast
Forecast sales (and production quantities)	80 000 kg	4 000 kg
Batch size	5 000 kg	500 kg
Setups	2 per batch	2 per batch
Purchase order size	20 000 kg	500 kg
Roasting, blending and packaging time per 100 kg	2,3 hours	2,3 hours

The company holds no inventory of unroasted coffee beans at either the beginning or the end of the financial year.

REQUIRED		Marks
(a)	Using direct labour hours as the base for assigning manufacturing overhead cost to products, undertake the following:	
	(i) Calculate the predetermined overhead recovery rate that will be used during the forthcoming year.	(1)
	(ii) Calculate the unit product cost of one kilogram each of the **Ethiopian and Malawi roast** products.	(3)
(b)	Using activity based costing as a basis for assigning manufacturing overhead cost to products, undertake the following:	
	(i) Calculate the total amount of manufacturing overhead cost that should be assigned to the total production of each of the roast blends for the year.	(19)
	(ii) Using the data calculated under **b(i)** above, compute the overhead cost per kilogram of each of the **Ethiopian and Malawi roasts.** (Work to 2 decimal places)	(3)
	(iv) Calculate the unit product cost of one kilogram each of the **Ethiopian and Malawi roast** products.	(3)
(c)	In bullet point form, prepare short notes for the marketing manager on your findings under (a) and (b) above and clearly indicate the implications to the company of using direct labour as the base for assigning manufacturing overheads to products.	(5)

PART B **[32]**

The following information is available for a firm producing and selling a single product:

Budgeted costs **(at normal activity):**

	R
Direct materials and labour	26 400 000
Variable production overhead	4 800 000
Fixed production overhead	14 400 000
Variable selling and administration overhead	2 400 000
Fixed selling and administration overhead	9 600 000

- The overhead absorption rates are based upon normal activity of 240 000 units per period.
- During the period just ended 260 000 units of product were produced, and 230 000 units were sold at R300 per unit.
- At the beginning of the period 40 000 units were in stock. These were valued at the budgeted costs shown above.
- Actual costs incurred were as per budget.
- Any under lower-recovery of overheads is set-off against cost of sales.

REQUIRED		**Marks**
(a)	Based on normal activity, calculate the unit product cost under both the absorption and marginal costing methods.	(3)
(b)	Calculate the fixed production overhead absorbed during the period, and the extent of any under/over absorption, if an absorption costing method is employed.	(2)
(c)	Calculate the operating profit for the period using absorption costing.	(9)
(d)	Calculate the operating profit for the period using marginal costing.	(9)
(e)	Reconcile the profit figures which you have calculated under (c) and (d) above.	(2)
(f)	State the situation in which the profit figures calculated under both absorption costing and marginal costing would be the same.	(1)
(g)	List the arguments (advantages) put forward for the use of absorption and marginal costing systems respectively.	(6)

PART C **[14]**

Phumelela Engineering Ltd is a company based in Mbombela. It tenders for various size contracts in most parts of South Africa.

It has recently successfully tendered for a Local Authority contract in Mangaung in the Free State province. Work is due to begin in January next year. The price that the company submitted with which it won its bid was R432 000 000.

The company has also been requested to undertake another contract in Bela Bela in the Limpopo province. The price that has been offered for this work is R528 000 000. However, due to staffing problems, the two contracts cannot both be carried out at the same time. Phumelela has to choose which is the most financially rewarding to the company.

Phumelela is able to withdraw from the contract in Mangaung, thanks to a clause that was written into the contract before it was signed. However, if this is done the company would have to notify the Local Authority of its intention by October this year and pay a penalty clause of R42 000 000.

The following estimates relating to the costs and revenue of each contract have been compiled:

		Mangaung R'000	**Bela Bela R'000**
Materials:	Material A (in stock, original cost)	32 400	
	Material B (in stock, original cost)		37 200
Committed orders at original cost for material A		45 600	
Further material A required (current cost)		90 000	
Further material C required (current cost)			106 800
Direct labour wages from local agency		129 000	165 000
Plant on site: depreciation		14 400	19 200
Site foremen		51 000	51 000
Temporary accommodation for foremen		10 200	8 400
Interest cost at 8%		7 680	9 600
Total contract costs		380 280	397 200
Central overheads from headquarters		19 014	19 860
Total costs		399 294	417 060
Contract price		432 000	528 000
Anticipated profit		32 706	110 940

Additional information is as follows:

1. Material A does not have a resale value outside of the company, as it is not used commonly in the engineering industry. It could be used as a substitute for another material which currently costs 10% less than the original cost of material A.

2. The price for material B has increased to twice the amount at which it was first purchased. Material B can be resold at its current market price, in which case the costs of disposal are 12% of the selling price. If not sold, any material B could be used on any future contracts as it does not deteriorate.

3. The plant that would be required is common to both contracts. If it were used on the Bela Bela contract, all of the plant would be required due to the nature of the work to be undertaken. However, if the Mangaung contract goes ahead, less plant will be required. The plant that is not used will be rented to a neighbouring business for a total rental of R9 000 000 during the period of the contract.

4. The interest charge of 8% relates to a notional charge for working capital that may be required during the period of the contract. The customer would pay the company progress payments according to the work completed.

5. Both contracts would run for a 12 month period.

6. Site foremen are paid an annual salary. This does not vary according to the amount of work undertaken. They will however, be expected to stay in temporary accommodation near the contract site. The accommodation costs have been estimated based on past experience.

7. The costs of the central headquarters consist of the salaries of the personnel who work in the administration and payroll departments, which amount to R162 000 000 in total. The headquarters supervise approximately 15 contracts at any one time.

8. It is not expected that labour will be in short supply.

REQUIRED		**Marks**
(a)	Making use of the principles of relevant costing, prepare a statement that compares the relevant net benefits of the two contracts in Rand terms.	(14)

QUESTION 12

OOSCAR LTD **90 marks**

(Source: Supplementary exam 2010 TOE408W (MAC4862))

Ooscar Ltd is a manufacturer in the automotive industry and listed on the JSE Securities Exchange SA. Ooscar Ltd manufactures parts and components which are either sold and delivered directly to motor car manufacturers through the Manufacturing Division or sold to the retail and post sales market through the Retail Division. The Retail Division receives the branded Ooscar products from the Manufacturing Division at full cost plus 15%.

Ooscar Ltd has a target debt : equity ratio of 2:1. The latest share price is R35,00 and a dividend of 105 cents per share was paid. The past growth rate in dividends was 12% per annum, but management considers 8% per annum to be more realistic for the foreseeable future due to the current economic turmoil. Debt consists of debentures of R1 000 each, which will be redeemable in 3 years time at R1 050 each. Interest on the debentures is payable at a nominal rate of 11,5% per annum. Similar instruments currently trade in the market at 9,25%. Ooscar Ltd has an effective tax rate of 20%.

The head office management accountant has listed the following items for discussion at the management meeting of 29 November 2010:

Manufacturing Division

Statement of comprehensive income – 10 months to 31 October 2010

		Note	R'000
Revenue			310 600
Cost of sales			168 640
Opening inventory		1	50 000
Production:			
Materials and components			75 490
Labour	- skilled	2	21 000
	- unskilled		10 750
Overhead	- variable		7 250
	- fixed		27 300
Packaging			6 510
Closing inventory		3	(29 660)
Fixed costs	- underrecovered		(20 700)
	- expenditure variance		(600)
Gross profit			120 660
Administrative costs		4	(40 300)
Other operational costs			(24 100)
Finance costs		5	(39 100)
Profit before tax			17 160

Notes

1. Opening inventory consists of:

Variable manufacturing costs (R'000)	30 000
Fixed manufacturing costs (R'000)	20 000
	50 000

   Actual fixed manufacturing costs were close to budget and as the factory was running at 98% of capacity, no adjustment was made for the underrecovery of fixed costs.

2. Skilled labour is considered a fixed cost, whilst unskilled labour is a variable cost.

3. Closing inventory consists of:

Variable manufacturing costs (R'000)	20 000
Fixed manufacturing costs (R'000)	9 660
	29 660

   The factory has been running at 70% of capacity during the current 10-month period, resulting in the under recovery per the statement of comprehensive income.

4. Administrative costs are considered fixed, whilst other operating costs are variable.

5. Finance costs relate to the debenture costs dealt with earlier. The manufacturing division carries 80% of the debenture costs.

Retail Division

Turnover to date amounts to R142 000 000, all sales being on credit. The current credit policy allows debtors 60 days to pay from the date of sale. The debtors outstanding days on 31 October 2010 amounted to 71 days.

Due to slow cash inflow, the management of this division has recommended that the credit policy be changed to:

- Cash sales will receive a 7% cash discount.
- The credit terms - 30 days will be allowed from date of sale.
- Credit sales outstanding after 30 days – interest will be levied at a fixed interest rate of 12% pa in respect of the period that amounts are in arrears.
- Credit sales being paid within 12 days from date of sale will receive a 3% settlement discount.

A survey conducted by them indicates that the response to the changed credit policy will be:

- 10% of sales (based on value) will now buy on a cash basis.
- 15% of credit sale customers (based on value) will take up the settlement discount offer.
- 65% of credit sale customers (based on value) will pay on the 30th day after purchase.
- 17% of credit sale customers (based on value) will pay on the 60th day after purchase .
- 3% of credit sale customers (based on value) will become bad debt, up from the current 1,5%.

The new credit policy will necessitate a new debtors clerk being appointed, costing R210 000 per annum and incremental IT costs of R10 000 per month being incurred. Ooscar Ltd can invest their cash resources at 7,5% per annum.

Contract offer

Ooscar Ltd has received an offer from a major vehicle manufacturer to enter into a contract with them, linked to a new 3 year export programme that they have negotiated. The contract calls for the annual supply of sets of seats: 3 000 sets in the initial year and increasing by 300 sets per annum thereafter. 70% of the sets will be covered by material and 30% by leather.

The cost accountant has prepared the following cost estimate:

	Note	Material	Leather
Set of frames (@ R4 000)	1	1	1
Cover	2	10 m^2	9 m^2
Labour - skilled	3	2 hours	4 hours
- unskilled		3 hours	3 hours
Variable overhead	4	2 hours	4 hours
Sales price per set (R)		R9 000	R12 000

Notes

1. A set of frames are fully upholstered seats and benches (back seats) without the final cover.

2. A material called 'Plastitex' will be used for the material cover. The manufacturing division has 20 000 m^2 of this material in inventory, acquired at R90 per m^2. If not used on this contract, it can be sold at R85 per m^2 or used as a substitute material in the manufacture of body panels where the current material used costs R100 per m^2. New supplies of Plastitex will cost R110 per m^2.

   The division has 17 000 m^2 of leather on hand bought from a liquidated competitor at R100 per m^2. If not used on the new contract, it can be resold at a net price of R140 per m^2. New leather will cost in the region of R180/m^2.

3. Skilled labour currently gets R45 per hour. For purposes of the contract a rate of R50 per hour was used to incorporate a possible increase as there is a permanent shortage of these skills (refer also note 5). These labourers will have to be moved from manufacturing lines where they are currently responsible for a contribution of R10 per labour hour. It is anticipated that these labourers will be able to upgrade the skills of enough unskilled colleagues to such an acceptable level that they can be redeployed after two years. Unskilled labour, of which there is never any shortage, is paid R15 per hour.

4. Variable overhead is usually closely related to actual costs and are costed at R20 per direct skilled labour hour.

5. The offered contract makes provision for an annual price increase of 8% to cover increased costs. The cost accountant has therefore not increased any prices or costs, as the relative positions will remain unchanged. He has however noted that this, 'inflation' of 8% should also be incorporated into the weighted average cost of capital.

6. Special new equipment with a useful life of 4 years will be required for this contract. The equipment will cost R20 million and will require a relatively small amount for an upgrade to extend its useful life. Ooscar Ltd provides depreciation over the useful life of their equipment, although SARS will allow wear and tear on a 40%/20%/20%/20% basis. Carrying value has been found to be a close approximation of realizable value.

7. The equipment will be installed in a section of their factory which is currently not used. The cost to clear this section and prepare it for the new equipment will be R500 000.

8. Incremental fixed costs will be R7,7 million per annum, including depreciation of the new equipment and factory rental of R1,2m allocated to this project.

9. Should the contract offer receive an extension on their export programme, there is a high probability that Ooscar Ltd will remain involved, provided that there are no major problems during the three year period.

10. Ooscar Ltd is in a tax-paying position.

REQUIRED		**Marks**
(a)	Calculate the current value of a R1 000 debenture.	(2)
(b)	Calculate Ooscar Ltd's weighted average cost of capital.	(5)
(c)	Restate the Manufacturing Division's statement of comprehensive income as at 31 October 2010 on a variable costing basis, including any notes that management should consider.	(10)
(d)	Calculate the Manufacturing Division's break-even point on 31 October 2010.	(3)
(e)	Calculate the impact of the proposed credit policy on the Retail Division's accounts receivable amount as at 31 October 2010.	(10)
(f)	Calculate the cost impact of implementing the new credit policy. You may assume for this purpose that the policy was implemented on 1 January 2010.	(11)
(g)	Determine whether Ooscar Ltd should accept the new contract offer. You may assume a hurdle WACC of 11% for this purpose.	(35)
(h)	After signing the new contract, the National Union of Metalworkers organized a fierce strike which resulted in skilled workers being granted pay of R55 per hour and unskilled workers R17 per hour. Assuming that 7 500 skilled labour hours and 9 600 unskilled labour hours will actually be worked in the first year of the new contract, calculate the relevant standard cost variances.	(4)
(i)	Briefly list the risks that the management of Ooscar Ltd should consider in running the business.	(10)

Note: Where applicable you may round to one decimal of R'000 and use 360 days for relevant calculations. Discount factors should be rounded to three decimals.

QUESTION 13 **80 Marks**

DREAMY CREAMY (PTY) LTD

(Source: Supplementary exam 2010 DIPAC26 (MAC4861))

Mr Farmer owns a large herd of cattle on a farm near Ermelo in Mpumalanga. On the farm is also a small dairy factory where drinking yoghurts are manufactured. As the farm has ample underground water, part of the farm is used to produce sods of lawn grass which are sold in the Johannesburg/Pretoria metropolis.

The yoghurt business

Dreamy Creamy (Pty) Ltd (Dreamy Creamy) manufactures a drinking yoghurt-juice blend which is available in two flavours. Both products are sold locally.

Budgeted production information for the year ended 31 October 2010

	Strawberry Surprise R	Grape Treat R	Note
Sales price	7,90	7,40	1
Yoghurt	1,60	2,00	2
Apple Juice	0,18	0,05	3
Grape Juice	0,16	0,30	3
Strawberry juice	0,35	0,06	3
Flavourants and colourants	0,50	0,50	4
Bottles	0,45	0,45	5
Labels	0,75	0,75	6
Manufacturing overheads	0,80	0,80	7
Allocated head office cost	0,20	0,20	8

Actual information for the year ended 31 October 2010

1. The completed products are sold in 340 ml yoghurt-juice bottles. The content of each consists only of liquid ingredients and no losses occur during the process of manufacturing. The actual sales prices of both products were 6% higher than budget.

   Actual production

Strawberry surprise	500 000 units
Grape treat	270 000 units

   Actual sales

Strawberry surprise	480 000 units
Grape treat	270 000 units

   There was no opening stock of either of the products. Actual production was equal to budgeted production.

   Any closing inventory is held at actual cost.

2. Dreamy Creamy has an annual fixed contract with Dairy Do CC, a supplier of yoghurt. Dairy Do CC gave Dreamy Creamy a discount of 10% on the budgeted yoghurt cost because Dreamy Creamy paid cash on delivery.

3. Juice is bought from Fruity Tutti Juices (Pty) Ltd. The budgeted cost was equal to the actual cost of juices in 2010. Due to the easy obsolescence of the fruit juices, required quantities are purchased on demand (there is no opening or closing juice stock).

4. Flavourants and colourants are bought from a local supplier. He charged R0,20 more per unit of colourant than was budgeted, passing on some of his cost increases.

5. The cost of bottles per the budget was equal to the 2009 actual bottle costs because the accountant was not sure when he set the budget whether or not the price of bottles would increase. Bottles actually did cost 5% more in 2010.

6. The budgeted cost of labels was equal to the actual cost of labels. In addition however, the supplier charged a R2 000 delivery charge for one overnight delivery when Dreamy Creamy needed labels in a hurry. This cost should be split equally between the two products and is regarded as a once off cost.

7. Manufacturing overheads consist only of fixed manufacturing costs. Dreamy Creamy takes total budgeted manufacturing overheads and allocates it evenly to all units based on normal capacity. Normal capacity is 770 000 units. Total budgeted *and* actual manufacturing overheads for the Strawberry Surprise product line were R400 000 and for Grape Treat product line they were R216 000.

8. Head office costs are based on the volume of units produced and are allocated to all the farms' products.

9. Both products use all fruit juice types. The difference in final products is achieved through different combinations of liquids. The manufacturing process is machine intensive and as such there are no significant labour costs in the manufacturing process.

10. The company makes use of an absorption costing system.

Special order

Baby Bear (Pty) Ltd is a company based in Pretoria, Gauteng that specialises in the distribution of various baby food products. They have approached Dreamy Creamy to supply them with a special range of toddler drinking yoghurts from 1 November 2010.

Dreamy Creamy is required to supply a similar version of the Strawberry Surprise product to the toddler food market. The order will run until 31 May 2011. Mr Farmer's production assistant has put together the following cost information:

The request is for 2 500 units per month at a special price of R5 per unit. The additional production capacity requirements can be met by tweaking the current production process slightly. This will not impact on current production levels.

The following conditions were set:

- A new bottle in the shape of a bear's face which will cost R0,50 per unit.

- A new label for the product will be required as this new product for toddlers will be called Strawberry Baby Bear. These labels will cost R0,60 per unit.

- The flavourants and colourants will be reduced because it is a product for toddlers. The flavourants' cost for Strawberry Baby Bear will decrease by R0,20 compared to the cost of the normal product.

- Manufacturing overheads are expected to increase by R0,30 per unit because of the stricter quality control procedures required for a toddler product during the manufacturing process.

- Costs not listed above will remain unchanged from the 2010 actual costs.

The milk business

As part of the herd of cattle, Mr Farmer has a herd of milk cows totalling 1 600 head. On any given day 800 cows will be milked. Cows receiving power feeds deliver 30 litres of milk per day and those on grazing 18 litres per day. Approximately 60% of the cows in milk will be fed power feeds. These feeds cost an average of R10,00 per cow per day. Currently Mr Farmer sells the milk from his farm to a large national producer of dairy products at an average price of R2,60 per litre. The base overall cost of a litre of milk is R2,00 excluding the power feed costs. You may assume that milking takes place, on average, on 360 days in a year.

Mr Farmer has discussed the current low return received on investment with a number of local dairy farmers. Their conclusion was that they require their own pasteurization and bottling facility to improve their margins and to ensure sustainable farming. To end this end they have agreed to form a co-operative which would operate as Ermelo Dairy Producers (EDP).

A consultant has collected the following project information for EDP:

- Capital cost of the facility: R65 million at the start of year one and R25 million at the start of year two. The supplier of the equipment will arrange financing for the entire facility at 12% per annum over a five year period.
- Four delivery and collection cooling trucks at a cost of R300 000 each, will be acquired at the time when the processing facility commences with production.
- Capacity of the plant: 300 000 litres of milk per day, of which 50 000 litres will be processed and delivered in bulk to a fruit and vegetable retail group (F & V group) and the remainder will be used to fill one litre plastic bags for delivery to contracted retailers.
- Retailers will pay R4,10 per litre bag of milk and the F & V group will pay R3,50 per litre delivered.
- Suppliers of milk will be paid R2,30 per litre of milk and will share in the net profit according to litres delivered.
- Processing costs per litre, all cash, will be:

	R
Bag (where appropriate)	0,60
Labour	0,15
Fixed overhead	0,25
Collection and storage of milk	0,05
Delivery	0,10
Management of facility	0,05

- The management of the facility will be undertaken by Mr Farmer who will also make the land available for the facility. This will not affect Mr Farmer's current activities or the other production or harvesting activities taking place on his farm.
- Processing will take place for 360 days in the year, as is the industry norm. Processing in year one will be for only 180 days, following the completion of the plant. The fixed overhead cost will also only be incurred for 180 days in year one.

- The plant will have a useful life of five years from date of commencement of processing.
- SARS will allow a 40%/30%/30% write off on the plant and equipment and 25% per annum on the cooling vehicles from the date of acquisition of the vehicles.
- EDP should use a 28% tax rate and 18% weighted average cost of capital (WACC) for evaluation purposes.
- A conservative view will be taken that the plant and equipment will have no realizable value after five years from date of commencement of processing.
- For the purposes of this evaluation, you may assume that there won't be any price increases during the five years of processing.
- The co-operative will not engage in any other activities and will earn no other profits.

The lawn grass business

Lawn grass is sold at R6/m^2 with all-in production costs of R2/m^2. Delivery and sales costs amount to R1/m^2.

Forecasts for normal sales of 300 000 m^2 per annum, from 1 November 2010 onwards, show the following two possible scenarios and their related probabilities:

	Poor sales: Normal – 20%	**Normal sales**	**High sales: Normal + 10%**
Expected by Mr Farmer based on historical trends	0,20	0,65	0,15
Market forecast	0,45	0,50	0,05

- Market stagnancy in the building of houses, office parks and recreational areas is regularly in the news. Mr Farmer will therefore consider both scenarios separately in his planning for the year.
- In spring and summer months (6 months) sales are three times the quantity per month that is sold in the remainder of the year.

REQUIRED		Marks
(a)	Calculate the **budgeted** breakeven point in units for each of the Strawberry Surprise and Grape Treat product lines.	(9)
(b)	Prepare, in a column format, the **actual** statement of comprehensive income up to earnings before interest and tax, for the year ended 31 October 2010, for each product separately. A total column is not required. Where actual prices or costs are different from budgeted prices or costs, also show the actual price or cost per unit. (Round unit prices and costs to the nearest cent).	(20)
(c)	Indicate, with the necessary supporting calculations and motivation whether or not Dreamy Creamy should accept Baby Bear (Pty) Ltd's enquiry as a **special order**. [Calculation (7), Motivation (3)]	(10)
(d)	Evaluate the milk processing proposal and make a recommendation to the dairy farmers on its feasibility. (Round to one decimal of R'000 throughout.)	(25)
(e)	Assume that Mr Farmer will either deliver to the national dairy producer *or* participate as supplier and manager of EDP, once processing commences. Calculate the income differential per day between these two options. You may ignore the profit share that he would receive as a supplier to the co-operative, as well as taxation and the time value of money. (Round to the nearest Rand.)	(8)
(f)	Calculate the expected income for December 2010 from the lawn grass business under the two given scenarios. (Round to the nearest m^2 and the nearest Rand.)	(8)

QUESTION 14

TUNING TONIES; FELINE PIXELS; TEDDY **40 marks**

(Source: Test 1 2013 MAC4862)

PART 1 **15 marks**

Tuning Tonies (Pty) Ltd's main line of business is the tuning of sports and performance cars. It is managed and owned by two cousins, Tony Mendez and Tony Cabrera. The tuning is done by qualified technicians on a contract basis. Tony Mendez is considering an opportunity to lease expensive machinery that will result in much less technician hours required by the company.

The following monthly information is available for the two scenarios:

	Manual R	**Automated R**
Fees received	420 000	420 000
Equipment lease paid	55 350	154 980
Labour, paid on hourly basis	108 000	63 000
Maintenance contract	18 750	28 800
Parts bought	69 000	38 880
Rent paid for premises	29 800	29 800
Variable overheads	34 000	13 500

PART 2 **8 marks**

Feline Pixels (Pty) Ltd provides the following information concerning its two product lines:

	Product A	**Product B**
Number of units produced and sold	550	110
Machine setup hours per production run	7	17,5
Product units manufactured per production run	110	55
Direct labour hours per unit	2	2

Total budgeted machine setup-related cost – R42 000

PART 3 **17 marks**

Teddy Ltd manufactures a range of similar teddy bears. Teddy Ltd is considering the following proposals for changes to the design of the teddy bears:

(1) Reduce some of the decorations on the teddy bear e.g. stitching

(2) Instead of using glass eyes, use plastic eyes (two eyes per teddy bear)

(3) Replace the synthetic material that is currently used for filling the teddy bears with scrap material left over from the manufacturing of the body.

The following information has been obtained regarding the proposed changes:

(a) The current glass eyes cost R180 per hundred while the plastic eyes will cost R140 per hundred. The plastic eyes are however not as strong as the glass eyes and are more likely to be damaged during the manufacturing process. It is estimated damaged and broken plastic eyes will be 10% of the quantity issued from stores compared to 5% of issues of glass eyes at present.

(b) Currently the synthetic filling material is bought for R720 per tonne. One tonne is sufficient for filling 2 000 teddy bears.

(c) Before the scrap material can be used to fill the teddy bears it will have to be shredded. This will cost R0,50 per teddy bear. There will be sufficient scrap material available for filling all the bears. The scrap material is currently disposed of at a cost of R1 000 per month.

(d) It is expected that the reduction of the decorations on the teddy bear will reduce the attractiveness of the teddy bear and will result in an estimated drop in sales of 10% from the current level. It will also reduce production costs by R6,00 per teddy bear. It is not expected that the change in eyes or filling material will negatively affect sales volume.

(e) 300 000 teddy bears are currently manufactured and sold per year. Apportioned fixed costs amount to R4 050 000 per year, which will not be impacted by the changes considered. The net profit per teddy bear at the current sales level is R30.

PART 1

REQUIRED		**Marks**
(a)	Calculate the monthly break-even in sales value for the manual and automated scenarios;	(5)
(b)	Calculate the margin of safety for both scenarios;	(5)
(c)	Evaluate the two tuning methods and advise the company on which method to choose.	(5)
Total		15

PART 2

(d)	Calculate how much machine setup-related cost is allocated to each of A and B's product lines if Feline Pixels (Pty) Ltd uses a traditional cost allocation system based on direct labour hours. Calculate the fixed overhead cost per unit for Product A and B;	(2)
(e)	Calculate how much machine setup-related cost is allocated to each of A and B's product lines if Feline Pixels (Pty) Ltd uses activity-based costing (ABC). Calculate the fixed overhead cost per unit for Product A and B;	(2)
(f)	Indicate if overhead allocation to A and B's product lines is evidence of product diversity or volume diversity.	(1)
(g)	Explain how ABC will affect the break-even sales value of the two product lines.	(3)
Total		8

PART 3

(h)	Assuming that all three proposals are implemented, prepare an analysis which shows the estimated effect on annual profit and which enables management to check whether the proposals combined will achieve an annual target profit increase of R250 000. The proposals for plastic eyes and the change in filling should be evaluated after the decoration reduction proposal has been evaluated.	(12)
(i)	Calculate the percentage reduction in sales due to the decoration reduction at which the implementation of all three design change proposals combined would result in the same total profit from the teddy bears as that earned before the implementation of the design changes.	(5)
Total		17
		(Drury - adapted)

QUESTION 15

MADIBSA 17 marks

(Source: Extract from Test 1 2013 MAC4861)

Information for MAC4861 only:

PART 3 **17 marks**

Madibsa (Pty) Ltd manufactures two types of cigars in its factory, the (Product Winnie and Product Graca. The following relates to product Winnie:

	R/unit
Selling price	90
Direct material	30
Direct labour	15
Variable overhead	10
Commission on sales	4%

Budgeted fixed overheads are R1 000 000, split evenly between products (product Winnie and product Graca), which reflects the manufacturing time required.

The annual maximum budgeted production quantities are:
- Winnie 50 000 units and Graca 25 000 units.

Completed cigars of Winnie are packaged 20 to a carton, essentially required to ensure undamaged distribution and sales. Each carton costs R24.
During the financial year (2013) 49 000 units of Winnie were manufactured, of which 47,500 were sold at the budgeted price. There was no opening inventory. No overtime is currently allowed.
Actual overhead costs incurred and related to product Winnie amounted to R480 000 in the financial year (2013). Actual variable costs approximated the budgeted variable cost per cigar.
After some market research by Madibsa they found some interesting data. Should Madibsa lower their selling price by 5% the sales volume would increase by 10%.

PART 3 (MAC4861 only)

REQUIRED

(h)	Prepare an absorption costing Statement of Comprehensive Income for product Winnie.	(8)
(i)	What would the profit be should a variable costing system be used.	(2)
(j)	Explain why budgeted overhead rates are used instead of actual overheads when determining product cost.	(2)
(k)	Indicate whether you would expect any difference between the profit under absorption costing and variable costing if a company uses a “just-in-time” inventory system.	(2)
(l)	What would the net profit be should Madibsa lower the selling price of product Winnie with 5%.	(3)
Total		17
		(LG Rayburn - adapted)

QUESTION 16

P&S LTD; SAVE LTD **40 marks**

(Source: Test 2 2013 MAC4862)

This question has two independent parts:

PART A

P & S Ltd manufactures floor polish and liquid soaps for commercial clients. Both product lines are sold in 10 litre containers. The company is critically aware of their trading environment and uses a standard costing system, which is reviewed every six months. The last review was done on 1 March 2013 for the six months ending 28 August 2013.

Floor polish division

The following standard cost was set for one 10-litre container of polish:

	Standard Cost R	**Unit Cost R**
Direct material:		
• 5 litre policloro	R3,60/litre	18,00
• 3,5 litre oxotone	R2,20/litre	7,70
• 4 litre lather	R0,60/litre	2,40
	R21,875/hour	5.25
Direct labour: 0,24 productive hours		
Variable overhead: 0,24 hours		14,88
Fixed overhead: 0,24 hours		8,40
Container		5,10
Total standard cost per unit		61,73
Sales price per unit		90,00

Budgeted idle time of 4% was considered in detailing the above standard cost card. P & S allocates both variable and fixed overhead costs to production on the basis of productive labour hours. The budgeted fixed manufacturing overhead for the six months is R10 320 000. Manufacturing activities and incurring of production cost take place evenly throughout a year.

The actual costs incurred during March 2013 amounted to:

Purchase of direct materials:

- Policloro 1 100 000 litre @ R3,80/litre
- Oxotone 725 000 litre @ R2,30/litre
- Lather 830 000 litre @ R0,56/litre

Labour: 53 800 clocked hours @ R21,50 / hour (49 920 worked hours)
Variable overhead: R3 072 800
Fixed overhead: R1 689 200
Containers: 212 000 containers @ R5,10 each

The actual production for March 2013 was 210 000 units. All containers purchased were issued. 1 000 000 litres of Policloro, 790 000 litres of Oxotone and 850 000 litres of Lather were issued to production.

Soap division

The soap division manufactures three distinct liquid products. As part of the standard and budget setting process, the capacities of the product lines have to be considered. The following information is available for the six month period from 1 March 2013.

Unit standard costs and prices

Products	**LD R**	**SO R**	**AP R**
Direct materials	60,00	40,00	172,50
Direct labour	20,00	10,00	35,00
Factory overhead	25,00	12,50	43,75
Selling and admin costs	45,00	22,50	67,50
Selling price	200,00	100,00	300,00

Monthly demand

Product units	50 000	18 000	38 000

Capacity and machine time (all hours)

Products	**LD**	**SO**	**AP**	**Monthly capacity**
Mixing	½	¼	¼	15 000 hours
Processing and finishing	¼	⅛	¾	30 000 hours

The monthly factory overheads are estimated at R1 385 000 of which 40% varies directly with labour costs. The monthly selling and admin costs are estimated at R2 400 000. One third of these costs vary directly with sales.

PART B

Save Limited operates a countrywide chain of shops, which sells clothing, food and household items. In 2010 the company started to incur losses. The general consensus was that the company was losing strategic direction. The business formula that had proved successful in the earlier years was no longer proving effective. A new Chief Executive Officer from Australia was appointed to turn the company around.

She implemented a three-pronged strategy:

- Remove levels in the hierarchy;
- Decentralise the organisation; and
- Focus on the core competencies of the business.

The core competencies were identified as essentially buying and selling, and from this analysis the philosophy of outsourcing was developed. The argument supporting this is that the core activities have to be world class and supporting activities should follow suit. As it is extremely difficult to achieve both yourself (it requires different focuses), it was decided to outsource the non-core activities. E.g. distribution was outsourced and had been reduced in size from 30 staff required to 5. Even though very high standards were required (e.g. certain foods would have to be kept in a cold-chain and never be exposed to temperatures exceeding $5^{o}C$ from the time of packing to sale), this area had been outsourced successfully.

Other activities followed this success, e.g. quality control and packaging.

Detail

The proviso for outsourcing is always that improvements be made in the specific area and that it should not be implemented just for the sake of it.

In relation to Information and Communication Technology (ICT), the feeling of senior management was that ICT was performing reasonably well in an operational sense, but not really delivering its full potential to the business. Projects within the ICT department had been run on a relatively informal manner and projects were tending to overrun budgets. During the previous year an investment of R5 million was made in order to upgrade personal computer systems.

Save Ltd decided to investigate the possibility of outsourcing all ITC activities from 1 March 2013.

Investigation into the possible outsourcing of ICT

A shortlist of 4 possible vendors were compiled. These vendors each made a presentation to the management of Save Ltd. Following this, PCC Ltd (PCC) was selected as it was felt that this company best understood the philosophy of Save Ltd, it had the resources and understood the future development needs.

Further negotiations with PCC were then entered into. Detailed requirements could be presented to PCC as Save Ltd had already, during the previous 2 years, prepared a detailed listing of requirements.

Offer by PCC

A three year contract with a fixed price of R12 500 000 per year. PCC agreed to take on 24 of the 30 ICT personnel, maintaining the terms of conditions they had with Save Ltd. (Average salary of ICT personnel is R480 000 per staff member per year).

Additional information provided by the financial director and human resources manager

- Barry Childs (one of the existing 30 ICT personnel) agreed on early retirement as he wants to further his writing career. If he does retire early, Save Ltd would have to pay an extra R1 000 000 lump-sum into the pension fund.

- Five ICT personnel members can be retained by Save Ltd to help administer the planned ICT outsourcing contract.

- The building housing the ICT department was on a 4 year lease (from 1 March 2012) and the company was committed to an annual rental of R500 000 per year for that period. This building could be sublet if ICT was outsourced, generating R240 000 in the second year, R360 000 in the third and R480 000 in the fourth (final) year.

- The resale value of the computer equipment purchased in the previous year is R1 500 000.

- Annual overheads of the ICT department are R1 440 000 per year. 60% of the overhead varies with staff members, the remaining 40% is a share of central overhead charges. Assume that this amount would remain constant.

- All relevant personnel provisionally agreed to the planned changes.

- It is estimated that the value of money would depreciate by 8% per year.

- Average salary increases are estimated at 7% per year.

- A contract manager (new appointee) would have to be appointed by Save Ltd, with an expected salary of R750 000 per year.

- It is assumed that tax be paid one year after the relevant income/expenditure flow. Assume that the tax rate remains at 28% per annum.

REQUIRED		**MARKS**
(a)	For the Floor polish division, calculate the • Material • Labour Variances for March 2013 in as much detail as possible.	(8)
(b)	Prepare the journal entries to reflect the • Use of direct material • Payment of labour wages	(5)
(c)	For the Soap division, determine the order in which the products should be manufactured to maximise contribution.	(7)
(d)	Prepare an incremental costing analysis to determine whether Save Ltd should proceed with the outsourcing of their ICT division.	(15)
(e)	Briefly list other factors that Save Ltd should consider before deciding on the outsourcing route.	(5)
TOTAL		**40**

QUESTION 17

P&S LTD **40 marks**

(Source: Test 2 2013 MAC4861)

P & S Ltd manufactures floor polish and liquid soaps for commercial clients. Both product lines are sold in 10 litre containers. The company is critically aware of their trading environment and uses a standard costing system, which is reviewed every six months. The last review was done on 1 March 2013 for the six months ending 28 February 2013.

Floor polish division

The following standard cost was set for one 10-litre container of polish:

	Standard Cost R	**Unit Cost R**
Direct material:		
• 5 litre policloro	R3,60/litre	18,00
• 3,5 litre oxotone	R2,20/litre	7,70
• 4 litre lather	R0,60/litre	2,40
Direct labour: 0,24 productive hours		5,25
Variable overhead: 0,24 hours		14,88
Fixed overhead: 0,24 hours		8,40
Container		5,10
Total standard cost per unit		61,73
Sales price per unit		90,00

Budgeted idle time of 4% was considered in detailing the above standard cost card. P & S allocates both variable and fixed overhead costs to production on the basis of productive labour hours. The budgeted fixed manufacturing overhead for the six months is R10 320 000. Manufacturing activities and incurring of production cost take place evenly throughout a year.

The actual costs incurred during March 2013 amounted to:

Purchase of direct materials:

- Policloro 1 100 000 litre @ R3,80/litre
- Oxotone 725 000 litre @ R2,30/litre
- Lather 830 000 litre @ R0,56/litre

Labour: 53 800 clocked hours @ R21,50 / hour (49 920 worked hours)
Variable overhead: R3 072 800
Fixed overhead: R1 689 200
Containers: 212 000 containers @ R5,10 each

The actual production for March 2013 was 210 000 units. All containers purchased were issued. 1 000 000 litres of Policloro, 790 000 litres of Oxotone and 850 000 litres of Lather were issued to production.

Soap division

The soap division manufactures three distinct liquid products. As part of the standard and budget setting process, the capacities of the product lines have to be considered. The following information is available for the six month period from 1 March 2013.
Unit standard costs and prices

Products	**LD R**	**SO R**	**AP R**
Direct materials	60,00	40,00	172,50
Direct labour	20,00	10,00	35,00
Factory overhead	25,00	12,50	43,75
Selling and admin costs	45,00	22,50	67,50
Selling price	200,00	100,00	300,00

Monthly demand

Product units	50 000	18 000	38 000

Capacity and machine time (all hours)

Products	**LD**	**SO**	**AP**	**Monthly capacity**
Mixing	½	¼	¼	15 000 hours
Processing and finishing	¼	⅛	¾	30 000 hours

The monthly factory overheads are estimated at R1 385 000 of which 40% varies directly with labour costs. The monthly selling and admin costs are estimated at R2 400 000. One third of these costs vary directly with sales.

Performance

The Floor polish and Soap divisions have reported the following summarised financial data for the six months ended 28 February 2013:

	FP R'000	**Soap R'000**	**Group Total R'000**
Sales	101 180	132 320	233 500
Variable costs	65 520	89 220	154 740
Contribution	35 660	43 100	78 760
Fixed costs	9 560	13 500	23 060
Operating profit	26 100	29 600	55 700
Head office fixed costs			10 034
Interest on long-term debt			9 146
Profit before tax			36 520
Income tax (at 28%)			9 945
Profit for the period			26 575

			MAC4861 MAC4862
Assets			
Non-current assets	135 520	142 905	278 425
Current assets	10 380	15 845	26 225
Total assets	145 900	158 750	304 650
Equity and liabilities			
Total equity			133 280
Long-term borrowing			152 000
Current liabilities	2 030	17 340	19 370
			304 650

P & S Ltd defines residual income (RI) for each division as operating profit minus a required return of 16% on the total assets of the division. The long-term borrows of the group are at present not allocated to the two divisions.

REQUIRED		Marks
(a)	For the Floor polish division, calculate the • Material (price, quantity) • Labour (rate, idle time, efficiency) • Fixed overhead (expenditure, volume) • Container (price, quantity) Variances for March 2013.	(9)
(b)	Calculate mark-up percentage for one unit of floor polish.	(2)
(c)	Briefly discuss the advantages of reviewing the standard cost of products on a six monthly basis.	(3)
(d)	For the Soap division, use the budgeted demand to calculate the break-even point in Rand for this division.	(8)
(e)	For the Soap division, determine the order in which the products should be manufactured to maximise contribution and explain briefly what method will be used to solve the production conundrum.	(8)
(f)	Evaluate the performance of the two divisions for the six months ended 28 February 2013. List the other information that you deem necessary to make a more informed assessment.	(10)
TOTAL		40

QUESTION 18

SOUTH LTD **40 marks**

(Source: Test 1 2012 MAC4861 and MAC4862)

Note: MAC4861 and MAC4862 shared certain information and questions (Required) for Test 1. The information and required sections for both MAC4861 and MAC4862 are included below.

Information for MAC4861 and MAC4862:

South Ltd (South) manufactures two types of industrial cleaning agents, Super Clean and Quick Clean.

The following were the standard direct labour hours per litre and budgeted production quantities for a 4-week period:

	Super Clean	**Quick Clean**
Standard direct labour hours per product litre	0,40 hours	0,56 hours
Budgeted production quantities	11 000 litres	6 800 litres

During the entire 4-week period 55 workers were directly employed in the production process and all worked a standard 40-hour week. The standard wage rate for all these workers was R20,00 per hour.

One of the materials used in the manufacture of both products is material Bravo. The following actual information for the 4-week period is available:

Production quantities:

Super Clean	10 800 litres
Quick Clean	8 200 litres
Direct wages paid	R177 400
Material Bravo purchased	14 460 litres costing R26 190
Material Bravo price variance	R4 176 Favourable
Material Bravo usage (Super Clean)	10 285 litres
Material Bravo usage variance (Super Clean)	R3 227,70 Adverse

The following budgeted annual revenue and costs per litre for the year ended 31 March 2012 are available:

	Super Clean R	**Quick Clean R**
Selling price	29,20	13,50
Variable costs	16,10	7,20
Fixed costs:		
Product specific	4,90	3,50
General	4,60	4,60

Unit fixed costs are based on the following annual sales and production volumes for the year ended 31 March 2012:

Super Clean	143 000 litres
Quick Clean	88 400 litres

Information for MAC4861 only:

Budgeted annual overheads specific to Super Clean and Quick Clean are as follows:

		R
Material purchase and storage		52 000
Setting up of machines		195 000
Production		645 000
Quality control		118 100
	Total	1 010 100

Additional information regarding overheads:

- The products are manufactured by machines and takes on average three machine hours for every Super Clean *production unit* and two machine hours for every Quick Clean *production unit*. A production unit equates to 200 litres of product for both products.
- It takes twice as long to set up the machine for Super Clean than for Quick Clean.
- 125 orders were budgeted for the year (100 for Super Clean and 25 for Quick Clean)
- 300 hours were budgeted for setting up the machines.
- An inspection is done after every 100 machine hours of production.

Information for MAC4862 only:

The company is considering whether or not to accept a once-off order in the next financial year for 20 000 litres of a special cleaning liquid for the Department of Health for use in their clinics. The costs per litre will be as follows:

	R
Variable costs (excluding Antibact – refer below)	5,40
Fixed costs:	
Product specific	3,90
General	4,80

Not included in the variable costs above is material Antibact. The special cleaning liquid requires 0,25 litres of Antibact per product litre. The material in inventory (4 000 litres) is not used for the current products and if not used soon will have to be disposed of at a cost of R0,50 per litre. It was originally purchased at R4,00 per litre. The current purchase price of Antibact is R4,70 per litre.

In addition to the above the company will have to hire a temporary staff member for the period of the order to handle the additional administration. This will cost R15 500.

REQUIRED		Marks
MAC4861 and MAC4862		
(a)	Calculate the direct labour variances for the given 4-week period.	(6)
(b)	Calculate the standard purchase price for material Bravo for Super Clean for the 4-week period.	(2)
(c)	Calculate the standard usage of material Bravo per litre of Super Clean produced.	(3)
(d)	Calculate the break-even point sales (Rand) for South Ltd based on the budgeted revenue mix.	(10)
(e)	Calculate the number of litres of Quick Clean at the break-even point determined in (d) above.	(3)
MAC4861 only		
(f)	Calculate the cost driver rate per activity pool using activity based costing (ABC). (Round of to two decimals).	(6)
(g)	Calculate the product specific overhead cost per litre, for Super Clean and Quick Clean using the driver rates calculated in (f) above.	(7)
(h)	Briefly explain the impact of the ABC approach on the allocation of the two products' manufacturing overhead cost.	(3)
MAC4862 only		
(f)	Comment on the viability of the production and sales of product Quick Clean only.	(6)
(g)	Calculate the minimum selling price per litre of the special cleaning liquid that South Ltd will have to ask to break even on the once-off order. Briefly comment on the choice of relevant costs.	(10)

(Drury adapted)

QUESTION 19

MR JOHNSON **40 marks**

(Source: Test 2 2012 MAC4861)

THIS QUESTION CONSISTS OF TWO (2) INDEPENDENT PARTS

PART A **31 Marks**

Mr Johnson trades as a chandler (who sells ship supplies) at the Knysna Waterfront. His profit in this business during the year ended 31 March 2012, still depressed by the prevailing economic conditions, was R480 000. Mr Johnson also undertakes occasional contracts to build seagoing yachts and is considering the price at which to bid for a contract to build the 'Blue Blood' for Mr CV Pedume. The yacht will have to be delivered in a year's time. He has no other contract in hand, or under consideration, for the next few months.

Mr Johnson intends to hire Ms Q Aplon, an unqualified accounting practitioner, to undertake his bookkeeping and other paperwork at a cost of R80 000. This will enable him to spend 25% of his time on the contract.

He will also have to employ an independent supervisor on the contract at a cost of R440 000 as well as two highly skilled artisans at a cost of R352 000 each. These costs include Mr Johnson's normal apportionment of the fixed overheads of his business at the rate of 10% of labour cost.

During spells of bad weather in winter, Mr Johnson can redeploy one of the artisans for the equivalent of up to three months fulltime in maintenance and painting work in the chandler's business. He will use materials costing R40 000. Mr Johnson already had two all inclusive quotations from jobbing builders for this maintenance and painting work, one for R100 000 and the other for R140 000, the work to start immediately.

The equipment which will be used on the Blue Blood contract was bought nine (9) years ago for R840 000. Depreciation has been written off on a straight- line basis assuming a ten (10) -year life and scrap value of R40 000. The current replacement cost of similar new equipment is R2 400 000. This is expected to increase to R2 640 000 in one year's time. Mr Johnson has recently been offered R240 000 for the equipment and considers that in a year's time he will have little difficulty in obtaining R120 000 for it. The equipment is useful to Mr Johnson only for occasional contract work.

In order to build the Blue Blood Mr Johnson will need six types of specified material, as follows:

No of Units			Price per unit (R)		
Material code	**In inventory**	**Needed for contract**	**Purchase price of inventory items**	**Current purchase price**	**Current resale value**
			R	**R**	**R**
B	10	100	440	1 200	800
L	110	100	800	360	400
U	-	10	-	2 400	-
E	10	20	1 600	1 200	800
O	5 000	500	72	80	100
D	100	300	360	800	400

Materials L and O are sold regularly in the chandler's business. Material B can be sold to a local sculptor, if not used for the contract. Materials B and O can also be used for other purposes, such as property maintenance. Mr Johnson has no other use for materials E and D, the inventories of which are obsolete. The Blue Blood will be built in a building held on a lease with four years remaining at a fixed rental of R200 000 per annum. It will occupy half of this building's space, which is useful to Mr Johnson only for contract work. Mr Johnson anticipates that the direct expenses of the contract, other than those noted above, will be R260 000.

Mr Johnson has also recently been offered a one-year appointment at a fee of £50 000 to manage a boat-building firm on the Isle of Wight. If he accepted the offer, he will be unable to take on the contract to build the Blue Blood or any other contract. He will have to employ a manager to run the chandler's business at an annual cost (including all insurances) of R400 000, and will incur additional personal living costs abroad of £7 200. The current R1/£ foreign exchange rate is R12 = £1.

Junior Mercury, Mr Johnson's nephew who has recently graduated with a B Com degree, has calculated the minimum price that Mr Johnson should charge for the contract in order to break even:

	R'000	**R'000**
Salary of Q Aplon		80,0
Supervision cost		396,0
Cost of craftsmen		633,6
Cost of equipment		840,0
Material costs:		
B (10 x R800) + (90 x R1 200)	116	
L (100 x R360)	36	
U (10 x R2 400)	24	
E (10 x R1 600) + (10 x R1 200)	28	
O (500 x R0,80)	40	
D (100 x R0,360) + (200 x R800)	196	440,0
Other direct expenses		260,0
Owner's opportunity cost		600,0
Minimum contract price		3 249,6

PART B **9 Marks**

Bee (Pty) Ltd is a small electronics entity based in Wynberg that manufactures two products, namely Mac and Ruz. The manufacturing process takes place in machine and assembly centres respectively. The production schedule of each product is as follows:

	Mac	**Ruz**
Machining	1hr	2hrs
Assembly	2,5hrs	2hrs

The production manager has provided you with the following budget data for the month of April:

	Machining	**Assembly**
Capacity	700hrs	1 000hrs
Fixed costs (R)	R14 000	R20 000

	Mac R	**Ruz R**
Selling price (unit)	300	200
Materials	160	60
Salesmen's commission	60	40

Under the regulation controls in terms of the Environmental Act, the maximum permitted output of either product in April is 400 units. The Managing Director of Bee (Pty) Ltd, an accountant by profession, has been trying various permutations in order to develop an optimal production plan that will maximise profit for April. He has approached you for help with the calculation of shadow prices as he has confidence in CTA students, whom he calls "a smart bunch". He has provided you with the following **correct** linear programming model:

M + 2R	≤	700	(machining capacity)
2,5M + 2R	≤	1 000	(assembly capacity)
M	≤	400	(maximum output of Mac constraint)
R	≤	400	(maximum output of Ruz constraint)
M	≥	0	
R	≥	0	

Note:

M = number of units of Mac produced and sold.
R = number of units of Ruz produced and sold.

PART A

REQUIRED		**Marks**
(a)	Write a **memorandum** to Mr Johnson in which you **critically review** the calculation performed by Junior Mercury. Advise Mr Johnson about the **appropriateness** of decision(s) taken by his nephew in calculating the price (providing calculations and reasons) and the **accuracy** of the price. Your detailed contract price should be an **addendum** to the report.	(25)
(b)	Mr Johnson intends to set a mark-up of between 15% and 20% on the total cost of the contract. He has asked you to suggest an appropriate basis to help him decide on a mark-up percentage and to explain the limitations of cost-plus pricing.	(6)
Ignore taxation.		
TOTAL		**31**

PART B

REQUIRED		**Marks**
(c)	Calculate the value to Bee (Pty) Ltd of an independent **marginal** increase in the available capacity for **each** of machining and assembly, assuming that the capacity of the other department in each case is not altered and the output maxima continue to apply.	(6)
(d)	State the principal assumptions underlying the linear programming calculation.	(3)
TOTAL		**9**

(ICAEW adapted)

QUESTION 20

KNYSNA SPECIALIST SUPPLIERS **40 marks**

(Source: Test 3 2012 MAC4862)

Knysna Specialist Suppliers (KSS) trades as a supplier of fresh produce, specialised meat products and non-perishable foodstuffs to the restaurant and catering industry, on land and sea. Mr Te Water, the owner of KSS, is an experienced ocean-going yachtsman and yacht builder who settled in Knysna seven years ago and started up KSS.

KSS is operated on a segmental basis with an operations manager responsible for each segment, whilst Mr Te Water and a small team are responsible for co-coordinating functions at the central office. The central office is an independent 'segment'. Transfers between segments, only when unavoidable, take place on a variable cost basis. Mr Te Water bases the annual salary increases on the margin obtained by each segment in the most recent financial year and the segmental (central office excluded) staff bonuses on the residual income generated. For this purpose a weighted average cost of capital of 20% per annum is used. Bonuses amount to 25% of residual income. A residual loss in one year is carried over to the next year in determining the bonus due. An extract from the management accounts of KSS for the financial year ended 30 April 2012 shows the following information for the central office and the three operational segments:

	Note	Central		Eastern Cape	Shipping	Southern Cape
Turnover (R'000)	1	284,0		3 276,0	2 340,0	5 200,0
Variable production costs	2	45,8	(R'000)	40%	45%	50%
Marketing and distribution	3	36,0	(R'000)	10%	8%	12%
Direct fixed costs (R'000)		330,0		1 024,4	624,0	938,60
Indirect fixed costs (R'000)	4	514,5				
Interest paid (R'000)	5	210,0				
Net working capital turnover	6			3 times	2 times	4 times
Direct non-current assets (R'000)		390,0		708,0	380,0	1 450,0
Non-current assets (R'000)	7	1 972,0				
Residual loss – prior year (R'000)				-	-	100,8

Notes

1. The turnover of the central office relates to consultancy work performed by Mr Te Water for a listed hotel group. The sales of the segments are largely contract based.

2. The operations manager of each segment is responsible for the acquisition and distribution of products in their designated area.

3. Sales persons' costs form part of the marketing and distribution costs which are all largely sales related. Sales/contract prices are set in consultation with the responsible operations manager.

The marketing and distribution cost incurred by the central office relates to KSS's social responsibility programme and is considered to be a variable cost.

4. Indirect fixed costs include Mr Te Water's drawings of R360 000, which his auditor considers to be only 45% of a market related salary.

5. Interest paid relates to a loan with a fixed interest rate of 12% p.a.. Mr Te Water considers the interest paid as a non-allocatable cost. The loan was used to acquire a refrigeration facility used by all the segments on an ad hoc basis.

6. The net working capital turnover ratios are based on each segment's turnover.

7. These non-current assets include the refrigeration facility referred to in note 5.

8. The company tax rate is currently 28% p.a.. All expenses incurred by KSS may be considered tax deductible. KSS paid tax in respect of the prior year as there were no tax allowances claimable. A provisional tax payment of R150 000 was made on 30 April 2012 in respect of the 2012 financial year.

George Dairy Bottlers

Mr Te Water inherited this business from an uncle, but it has been struggling for the past 2 years due to intense market competition from leading dairy brands. The business produces flavoured milk and its results for the year ending 30 April 2012 were as follows:

	R
Sales	910 000
Cost of sales	812 800
Gross profit	97 200
Premises lease and operating costs	96 400
Administrative salaries	63 000
Loss for the year	(62 200)

Mr Te Water spends approximately 10% of his time on this business, but no cost is accounted or paid for. In evaluating the situation, Mr Te Water listed the following:

- A major market player is preparing to open a dairy plant in the area.
- Cost of sales consists of:

	R
Variable input costs	600 000
Other variable costs	25 500
Fixed staff costs	152 000
Depreciation	35 300
	812 800

- A number of production staff with an annual salary cost of R39 060 can reasonably be accommodated in KSS. The remainder of the production staff as well as the administrative staff will have to be legally retrenched with a payment equal to 2 weeks wages for every year's service. The particular employees have on average three (3) year's service. Mr Te Water will in the event of retrenchment add a further two (2) weeks wages for each of these employees.

- Depreciation relates to the bottling plant and a delivery vehicle. The bottling plant will probably fetch R40 000 net on auction, creating a book profit of R13 500. The delivery vehicle can be transferred to KSS.

- The contracts with milk suppliers can be terminated without further cost as all suppliers have been offered contracts by the major player.

- The lease contract of the premises still has 18 months to run. The lessor is prepared to accept 25% of the next 12 months rental of R90 000 for early termination, as prime industrial space is currently at a premium in George.

- Other closure costs will amount to R3 700.

REQUIRED		**Marks**
(a)	Determine the segmental performance (central office excluded) on 30 April 2012 based on their: (i) Margin (ii) Residual income You may ignore the 2012 bonuses payable for this calculation.	 (5½) (7½)
(b)	Calculate KSS's breakeven point for the year ended 30 April 2012. You may ignore the 2012 bonuses payable for this calculation.	(6)
(c)	Briefly discuss the impact of the current transfer price policy on segmental operations.	(3)
(d)	Briefly list the risks that KSS should consider, given their business environment.	(6)
(e)	Calculate KSS's tax liability for the year ended 30 April 2012.	(5)
(f)	Determine whether Mr Te Water should continue or discontinue the business of George Dairy Bottling. List the factors you consider relevant to your calculations and decision.	(7)
TOTAL		**40**

QUESTION 21

ROYAL JAMS (PTY) LTD **40 marks**

(Source: Test 1 2011 TOE408W (MAC4862)

Royal Jams (Pty) Ltd manufactures two kinds of fruit jam, namely apricot jam and orange marmalade which are sold in 500g glass jars.

The fruit is washed and the apricot stones and orange pips and central pith are removed. The fruit is soaked for a few hours and then boiled with sugar and commercial pectin (gelling agent). The jam is then poured into sterilised jars, cooled and sealed. Finally the labelling and packaging is completed.

Royal Jams uses a vacuum boiler to boil the jam, resulting in the reduction of the jam's boiling temperature to between 65-80°C. The lower boiling temperature results in the water being driven off using less energy than conventional boiling methods.

Extract from the budgeted statement of comprehensive income for year ended 28 February 2011:

	Apricot	**Marmalade**	**Total**
	R	**R**	**R**
Turnover	7 480 000	12 852 500	20 332 500
Less: Cost of sales	?	?	?
Opening inventory	(210 000)	(200 000)	(410 000)
Materials	(3 541 500)	(3 024 000)	(6 565 500)
Direct labour	(450 000)	(960 000)	(1 410 000)
Fixed manufacturing overheads	?	?	(8 460 000)
Closing inventory	?	?	?
Gross profit	?	?	?
Less: Selling & distribution costs			(1 053 325)
Less: Other administrative costs			(550 000)
Profit before interest and tax			?

The budgeted quantities for the year were as follows:

	Apricot	**Marmalade**	**Total**
	kg	**kg**	**kg**
Opening inventory	15 000	10 000	25 000
Produced	450 000	480 000	930 000
Sold	(440 000)	(485 000)	(925 000)
Closing inventory	25 000	5 000	30 000

The actual selling price per jar was the same as the budget. The actual quantity and cost price of opening inventory was the same as the budget. There were no raw materials or work-in-progress included in opening or closing inventory for both budget and actual.

The actual production and sales quantities for the year were as follows:

	Apricot Kg	Marmalade Kg	Total Kg
Produced	445 000	487 000	932 000
Sold	(442 000)	(486 000)	(928 000)

The actual material and direct labour costs per kilogram were the same as the budget.

Fixed manufacturing overheads are allocated to inventory using machine hours. The budgeted machine hours were 67 500 hours for apricot jam and 144 000 hours for marmalade while actual hours were 66 750 hours for apricot jam and 146 100 hours for marmalade. There was no difference between the budgeted and actual hours per kilogram.

The actual fixed manufacturing overheads for the year amounted to R8 560 000. The higher than budget expense was driven by more maintenance on the vacuum boiler than expected due to the boiler being close to its scheduled five yearly overhaul.

Variable selling and distribution costs are 1% of turnover (both budget and actual). Actual fixed selling and distribution costs were R820 000 for the year. Other administrative costs were 5% higher than budget.

Mr Seville, the accountant at Royal Jams, is investigating alternative ways of allocating the fixed manufacturing overheads. He has analysed the budgeted fixed manufacturing overheads and found the following:

Activity area	Cost Driver	Total cost R	Apricot jam	Marmalade
Soaking of fruit	Soaking hours	1 845 000	3,75 hours / batch	12,5 hours / batch
Boiling	Boiling hours	5 220 000	0,06 hours / 100kg	0,25 hours / 100kg
Bottling & packaging	Packaging hours	1 395 000	2,0 hours / batch	2,0 hours / batch
Total		8 460 000		

Apricot jam and marmalade are manufactured in batches of 2 500kg each.

REQUIRED		Marks
(a)	Prepare the **actual** statement of comprehensive income for the year ended 28 February 2011 up to the profit before interest and tax line. Use machine hours as the basis to allocate fixed manufacturing overheads to inventory.	(19)
(b)	Calculate the total **actual** costs attributable to each type of jam using activity based costing (ABC) and the available information.	(7)
(c)	Calculate the **budgeted** break-even jars (units) for the year ended 28 February 2011.	(8)
(d)	During March 2011 the manufacturing plant was not in operation for three weeks due to the overhaul of the vacuum boiler which takes place once every five years. This will result in the production of only 870 000kg (apricot jam and marmalade) for the year ended 29 February 2012. The auditors have approached Mr Seville with concerns that the fixed manufacturing overhead rate for the year ended 28 February 2011 was therefore calculated incorrectly.	
	(i) Discuss what impact the reduced production in the coming year and the costs of the overhaul will have on the fixed manufacturing overhead rate.	(4)
	(ii) Discuss what impact this will have on the fixed manufacturing overheads (Rand) and rate in the flexible budget for the year ended 29 February 2012.	(2)
TOTAL		**40**

QUESTION 22 **40 marks**

SA HEATING COMPANY (PTY) LTD

(Source: Test 1 2011 DIPAC26 (MAC4861))

Please note that the standard of the DIPAC26 questions is not always comparable to MAC4861 standards but that these questions can be used for revision purposes for MAC4861 as they test basic principles and concepts that are necessary for MAC4861/2.

SA Heating Company (Pty) Ltd produces small heaters (SH) as well as large commercial heating systems (CHS) in their factory in Spartan. The management accountant, Mr Khan, has extracted the following numbers in respect of the financial year ended 28 February 2014:

Budgeted	**SH**	**CHS**
Production capacity – units	120 000	180
Sales – units	120 000	180
Sales price (Rand per unit)	240	100 000
Materials (Rand per unit)	95	42 000
Labour (Rand per unit)	57	10 500
Variable overhead (Rand per unit)	19	7 000
Fixed overhead (Rand per unit)	19	10 500
Sales commission (% of sales price)	5	7

Production labour is considered a fixed cost. The two production lines are supported by the administrative and maintenance functions, with the following fixed cost information:

	SH	**CHS**	**Maintenance**	**Admin**
Annual cost (R)			3 000 000	4 000 000
Administration (allocation %)	50	40	10	
Maintenance (allocation %)	49	49		2

Maintenance is currently not costed as a production cost. A detailed analysis of the administrative and maintenance functions on an activity base revealed the following:

	% of cost*	SH	CHS
Maintenance			
Scheduled maintenance visits	40	6	4
Call-out visits	30	15	5
Rework hours ☼	30	1 700	100
Administration			
Orders processed	10	150	150
Invoicing and debtors accounts (Hours)	30	1 350	450
HR transactions	40	1 800	200
General transactions	20	400	100

* For purposes of the activity analysis inter-departmental activities and transactions between the maintenance and administrative functions were ignored.

☼ Defective items are repaired by the maintenance team and returned to the production lines – 'reworked'.

Actual

- All costs, sales prices and units approximated the budgeted information, except for the sales quantities, which were 115 000 SH units and 170 CHS units respectively for the year ended 28 February 2014.
- Opening inventory on 1 March 2013 consisted of 6 000 SH units at R180 each and zero CHS units.
- The company incurred financial costs of R1 590 000, of which R1 050 000 was considered to be fixed.

REQUIRED		**Marks**
(a)	Calculate SA Heating Company's budgeted breakeven point in Rand for the year ended 28 February 2014.	(9)
(b)	Allocate the maintenance and administrative costs to the SH and CHS production lines, using the step method (repeated distribution method) of allocation. Show the cost per unit manufactured.	(5)
(c)	Allocate the maintenance and administrative costs to the SH and CHS production lines, using an activity based costing approach. Show the cost per unit manufactured.	(14)
(d)	Calculate the value of closing inventory on 28 February 2014, based on the existing costing practice.	(3)
(e)	Discuss the validity or not of the current maintenance costing policy. Explain the impact of this policy on SA Heating Company's business with supporting calculations or reference thereto.	(6)
(f)	What is the impact of the lower **actual** sales achieved for the year ended 28 February 2014 on SA Heating Company's breakeven point and profitability.	(3)
TOTAL		**40**

QUESTION 23

MALEMA SPORTS LTD **40 Marks**

(Source: Test 2 2011 TOE408W (MAC4862))

Malema Sports Ltd (MSL) manufactures mainly cricket bats and hockey sticks made of wood. This business has helplessly watched sales decline due to the economic recession that gripped the country. In a bid to keep afloat, sales staff was significantly reduced and management has been searching for ways to improve this situation. Turnover has dropped markedly and is currently estimated at R5,5 million for the year. Further, the company currently operates at 45% of total production capacity and fixed costs are projected at R3,0 million for the current year. These fixed costs comprise management and administration staff salaries, rent, depreciation of plant, machinery and equipment and other fixed overheads. Management has forecast poor profits for the current year as the contribution margin hovers around 38%. The company has also already spent R860 000 on a new hockey stick (refer HOCKEY STICK section), which is not included in the fixed costs projection. Though the company is financially sound with adequate cash and cash equivalents, shareholders are gravely concerned with the return on equity which has been decreasing in recent years. Total assets amount to R6,250 million. Malema Sports Ltd has a weighted average cost of capital of 19%.

SPECIAL ORDER

The South African Cricket Board has requested the company to manufacture and supply 10 000 miniature cricket bats for its schools programme. Following the resounding success of the 2010 Soccer World Cup, the Cricket Board aims to continue building on the sport legacy left by the spectacle in the country and also encourage national support for the Proteas in the 2011 Cricket World Cup in India. It is however not anticipated that there will be a need for further supplies of these bats in future. Management is very keen to obtain this order and has therefore enlisted your assistance in preparing a possible low quote.

The company has manufactured a batch of 10 bats to specification and the production manager has painstakingly prepared the following production sheet; as production will take places in batches of 10 bats each:

Raw materials for the batch

Straight pine 0,125 cubic meters
Varnish 1,00 litre

Direct labour

The first batch was completed in 7 hours. The production manager believes that their experience can help them derive an accumulative average learning curve rate of 90% and thereby improve labour time by 4,55 hours per batch.

Total variable overheads for the batch: R13,00

MSL is confident that the manufacturing costs for the remaining bats will be in line with these details.

Additional information:

- The company has 100 cubic metres of the straight pine and has no other use for this wood in the current production range. The original cost was R902 per cubic metre, replacement cost is R946 per cubic metre and the wood can be resold for R824 per cubic metre. The varnish is a standard item used in several of their products and the inventory on hand of 800 litres was purchased for R16 per litre.

- The company has allowed their labour levels to drop to a minimum and will therefore need to employ temporary staff to work on the special order. Due to high unemployment levels in the country, labour should be easily obtainable at an anticipated rate of R8 per hour. The production supervisor will have to work overtime in order to supervise the production of the bats. It is anticipated that the supervisor will have to work a total of 25 hours overtime at a rate of R44 per hour. In addition to overtime, one third of his salary for the month of the manufacture will be charged to the special order. His basic salary is R5 000 per month and fringe benefits are estimated at 15% of the basic salary.

- The bookkeeper has requested your guidance in setting the fixed overhead recovery rate which is ridiculously high due to low production volumes. Fixed overhead costs include the depreciation of plant and machinery which will be available to the project due to the spare production capacity. The depreciation charge for the complete project amounts to R8 750.

- Management has set the selling price at a markup of 20% on cost.

HOCKEY STICKS

The wooden hockey sticks currently manufactured compete in the lower end of the market. All available market feedback indicates that this market is in decline as sport is not a compulsory activity and hockey is not a mainline sport. MSL has therefore embarked on a research project to develop a hockey stick made of composite materials including carbon fibres, fibreglass and polyvinylchloride foam. Research costs to date amount to R860 000 as previously stated. The project has temporarily been put on hold as one of the directors, Mr Wiseman Floyida, has asked whether MSL should not rather negotiate with an international manufacturer to be their African agent.
MSL's marketing manager has forecast the following annual sales for a new composite hockey stick, especially if sold at a competitive price.

Sales volume	Probability
5 000	0,15
8 000	0,40
12 000	0,35
17 000	0,10

Whether the new hockey stick is manufactured or imported and distributed, the machinery and equipment used in the current manufacturing process will not be useable for the new hockey stick.

REQUIRED		Marks
(a)	**Calculate** the selling price that should be used for this special order quote, setting out your workings. **Briefly comment** on the choice of relevant costs.	(14)
(b)	What is the **minimum price** that the company could charge for this project and why?	(4)
(c)	**Discuss** briefly the implications of supplying a product at relevant cost in the long term.	(3)
(d)	**Calculate** whether Malema Sports Ltd will achieve a residual income for this year or not. The **special order** should be **ignored** for this purpose.	(4)
(e)	In measuring Malema Sports Ltd's financial performance, **recommend and motivate** a course of action regarding the expenditure on the new hockey stick.	(4)
(f)	**Calculate** the standard deviation of the new composite hockey sticks' expected sales and indicate clearly whether this is a high or low risk project.	(7)
(g)	Briefly **list** the issues to be considered in the discontinuance of the existing wooden hockey stick.	(4)

QUESTION 24

HARARI LTD AND SHELENI LTD **40 marks**

(Source: Test 2 2011 DIPAC26 (MAC4861))

Please note that the standard of the DIPAC26 questions is not always comparable to MAC4861 standards but that these questions can be used for revision purposes for MAC4861 as they test basic principles and concepts that are necessary for MAC4861/2.

THIS PAPER CONSIST OF TWO INDEPENDENT PARTS

PART A **[20]**

Harari Ltd uses a standard absorption costing system to control the manufacturing costs of its single product. The following standards have been set:

		R per Unit
Direct material	2 kgs at R6 per kg.	12
Direct labour	1 hour at R7 per hour	7
Fixed overheads	1 hour at R9 per hour	9
Total production cost		28

The fixed overhead standard cost per unit is based upon a normal budgeted monthly production of 4 000 units. Actual results for the most recent month were:

Production	4 300 units.
Direct material	Cost R56 000 for 9 000 kgs.
Direct labour	Cost R32 800 for 4 600 hours paid. Only 4 000 hours were worked.
Fixed overheads	R35 000.

No direct material inventory is held. All products produced are currently sold at R40,00 per unit with sales commission of 5% payable on the sales price.

Gwheri Ltd is a sister company of Harari Ltd in the Zimbo Group Ltd. Gwheri Ltd wishes to purchase 800 adapted units a month from Harari Ltd for a period of 8 months. Ghweri Ltd will use these adapted units in a product that they are contracted to deliver to a new mining venture. Harari Ltd's factory manager has indicated that their maximum production capacity will be 4 400 units per month of which 400 will be manufactured by working overtime.

Employers have indicated that they are willing to work the overtime hours at a premium of only 30% on the current direct labour cost, considering the existing high unemployment environment. The adapted product to be delivered by Harari Ltd will require additional material costing R1,40 per unit.

PART B **[20]**

Sheleni Ltd produces three chemicals X, Y and Z. The selling price and cost per litre for each of these products are budgeted as follows:

	X R/litre	**Y R/litre**	**Z R/litre**
Selling price	100	120	120
Direct materials	20	16	21
Direct labour (R12 per hour)	18	24	27
Other direct expenses	–	3	–
Variable overhead	12	16	18
Fixed overhead	6	8	9

Notes

1. The fixed overhead is absorbed on the basis of labour hours, based on a budget of 440 hours per month.

2. Maximum demand for each product for month 4 is as follows:

   X = 150 litres
   Y = 40 litres
   Z = 60 litres

3. Included in the maximum demand totals is an unavoidable commitment to a major customer to supply 15 litres per month of each of the three products.

4. During month 4 there is a shortage of labour hours that will restrict production. The total number of labour hours available is 375 hours.

5. Sheleni is able to produce and sell fractions of a litre.

PART A – HARARI LTD **(20 Marks)**

REQUIRED		Marks
(a)	**Calculate the following variances:** (i) Direct material price (ii) Direct material usage (iii) Direct labour rate (iv) Labour idle time (v) Direct labour efficiency (vi) Fixed overhead expenditure (vii) Fixed overhead volume	(7)
(b)	**Explain the meaning and suggest one potential cause of each of the following variances:** (i) Direct labour rate; (ii) Labour idle time;	(4)
(c)	**Calculate** the transfer price per unit that Harari Ltd should charge Ghweri Ltd to ensure that they are not worse off on a monthly basis.	(9)

PART B – SHELENI LTD **(20 Marks)**

REQUIRED		
(a)	Determine the production mix that will maximise profit in month 4 and calculate the resulting profit.	(12)
(b)	After completing the production plan you are informed that new environmental controls on pollution are to be introduced from the beginning of month 4. These controls relate to the production of product Z only, and will incur an additional fixed cost of R1 000 per month for each month that Z is manufactured. In addition you also learnt that an overseas supplier will supply as many litres of chemical Z as Sheleni Ltd needs at a cost of R100 per litre. Importation of Z will not incur the additional fixed pollution control costs. **Using the above information calculate how much chemical Z (if any) should be purchased by Sheleni Ltd from the overseas supplier in month 4.**	(4)
(c)	Briefly explain **FOUR** factors, other than costs or selling price, which should be taken into consideration when deciding whether to subcontract the manufacture of chemical Z.	(4)

QUESTION 25

LEKKER RUSKS (PTY) LTD **40 marks**

(Source: Test 1 2010 TOE408W (MAC4862) adapted)

Lekker Rusks (Pty) Ltd is a processor and distributor of the old South African favourite, rusks. The owner Mr Biscuit, uses his grandmother's secret family recipe to manufacture and package rusks for resale.

Lekker Rusks currently offers two types of rusks, buttermilk and muesli, both which are processed and sold in 1kg boxes.

Budgeted Data for the 28 February 2010 year end are as follows:

		Buttermilk	**Muesli**	**Company**
Units sold (boxes)		1 500 000	700 000	
Per unit (Rands)		**R**	**R**	**R**
Sales price		38,50	55,00	
Direct materials		12,00	15,50	
Costs in Total (Rands)				
Manufacturing overheads	Fixed			45 000 000
Manufacturing labour	Fixed			25 000 000
Admin salaries	Fixed			777 000
Sales and distribution cost	Fixed and variable			8 280 000

Mr Terry, the management accountant, provided the following information in respect of the actual figures for the year ended 28 February 2010:

Sales	R/unit	39,00	57,00	
Materials	R/unit	12,70	15,00	
Manufacturing overheads	R			44 000 000
Manufacturing labour	R			24 000 000
Admin salaries	R			777 000
Sales and distribution costs	R			8 000 000
Inventory – 1 March 2009 (R)		2 000 000	1 485 000	
Inventory – 28 February 2010		?	?	

Additional information:

		Buttermilk	**Muesli**
1.	Quantity schedule – actual (boxes)		
	Sales	1 400 000	600 000
	Opening inventory	100 000	55 000
	Closing inventory	80 000	30 000

No inventory of raw materials or work in process are held.
All baked rusks are packaged and taken into inventory.

Assume that there are no expired goods on hand. i.e. all inventory have been sold before expiry date and thus nothing should be written off.

Assume that the budgeted closing inventory = actual closing inventory (per above quantity schedule).

2. The inventory on 1 March 2009 includes fixed manufacturing overhead and fixed manufacturing labour.

3. Manufacturing labour and fixed overheads are based on normal capacity of 1 000 000 hours per year.

   For the year ending 28 February 2010 the following information is applicable:

	Buttermilk	**Muesli**
Budgeted labour hours	450 000	550 000
Actual labour hours	460 000	555 000

4. Admin salaries and sales and distribution costs are divided on a 50/50 basis between the two products.

   R3 000 000 of the budgeted sales and distribution costs are fixed and the actual fixed costs equalled the budgeted fixed costs.

5. There is no difference between budgeted and actual depreciation.

6. Mr Terry, believes that the traditional costing system may be providing misleading information. He has developed an activity based analysis of the 2010 manufacturing overhead costs which is shown in the following table:

Activity	Cost Driver	Cost R'000
Materials handling	Set-ups	21 300
Quality control	Batches	2 100
Baking	Baking hours	10 500
Packaging time	Packaging hours	4 600
Not identifiable	-	5 500
		44 000

The **actual** data regarding the 2010 production of rusks are presented below:

	Buttermilk	Muesli
Batch size	10 000 boxes	10 000 boxes
Set-ups	3 per batch	4 per batch
Baking time	1 hour / 100 boxes	1 hour / 100 boxes
Packaging time	0,1 hr / 100 boxes	0,1 hour / 100 boxes

REQUIRED		Marks
(a)	Prepare an income statement for the year ended 28 February 2010 on the absorption costing basis. Show as much detail as possible and round all calculations off to 3 decimals.	(20)
(b)	Using an activity based approach, calculate the **actual** manufacturing overhead cost of buttermilk and muesli rusks for the year ended 28 February 2010.	(12)
(c)	Prepare a memo to Mr Terry to briefly explain the impact of the ABC approach on the allocation of the two products' manufacturing overhead cost.	(4)
(d)	Calculate Lekker Rusks (Pty) Ltd's Rand budgeted break-even point based on expected sales of 1 350 000 boxes buttermilk and 450 000 boxes muesli.	(4)

Ignore tax for all calculations above.

QUESTION 26

ANTIHISTAMINES; CYCLE ADDICTS; ABC **40 marks**

(Source: Test 1 2010 DIPAC26 (MAC4861))



Please note that the standard of the DIPAC26 questions is not always comparable to MAC4861 standards but that these questions can be used for revision purposes for MAC4861 as they test basic principles and concepts that are necessary for MAC4861/2.

PART A **20 marks**

Antihistamines (Pty) Ltd manufactures medication for allergies which is then sold to doctors and pharmacies.

Antihistamines (Pty) Ltd only specialises in the production of one type of tablet which is sold per kg.

The management accountant has extracted a portion of the trial balance for the year ended 31 December 2009:

		Note	Dr R'000	Cr R'000
Sales		1		142 500,0
Delivery costs		2	500,0	
Depreciation	Factory and equipment		7 500,0	
	Administrative		250,0	
Interest paid		3	320,0	
Materials purchased		4/5	60 000,0	
Overhead costs	Fixed		4 794,2	
	Variable		2 380,4	
Salaries	Administrative	5	4 090,5	
	Sales		2 166,0	
Packaging costs			500,0	
Travelling			7 000,0	
Wages manufacturing		6	15 995,0	
Inventory – 1 January 2009			21 450,0	

1. Sales for the year took place at an average net price of R9,50 per kilogram.
2. Delivery costs vary with sales.
3. Interest paid relates to interest on a long term loan.
4. Materials were purchased at an average net cost of R4,00 per kilogram.
   On 31 December 2009 there was 250 000 kilograms of raw materials in inventory i.e. closing inventory.

5. Antihistamines (Pty) Ltd determined the cost of 1 kg of tablets (1 January 2009 to 31 December 2009) to be:

	R
Material (1,25kg input @ R4/kg)	5,00
Depreciation-manufacturing	0,40
Overheads – fixed	0,40
Overheads – variable	0,20
Packaging	0,15
Wages	0,50
Value of finished goods	6,65

The cost per kilogram is based on production of 12 000 000 kilograms of tablets.

6. Administrative salaries are fixed. Sales salaries are variable i.e. they vary according to sales.
7. Manufacturing wages are considered to be of a fixed nature.
8. Inventory 1 January 2009: Finished goods valued at R6,50 per kg
   There were no raw materials opening inventory on this date.

REQUIRED	**Marks**
Calculate Antihistamines (Pty) Ltd's profit for the year ended 31 December 2009 using an absorption costing basis. **Note: <u>Do not calculate</u> the under/over recovery of fixed overheads** Closing inventory is valued at standard cost. Ignore taxation for this section.	(20)



From the information and notes identify which costs are variable and which are fixed.
From the Required note that Absorption costing is used, therefore allocate all fixed *and* variable *manufacturing costs* to products. Note that the standard cost of finished goods is given.

PART B **12 marks**

Select the correct answer for each question below. Show **ALL** calculations clearly.

Cycle Addicts (Pty) Ltd manufactures mountain bikes for health fanatics. The company is now busy with its budget for the Super mountain bike line for 2011 and has the following information available:

Total cost if 5 000 units are produced	R35 000 000
Total cost if 7 000 units are produced	R45 000 000

The company expects to produce and sell 6 000 units of Super mountain bike in 2011.
The selling price is R10 000 per mountain bike.
The company cannot produce more than 7 000 units per annum.

1. Calculate the break-even point for the Super mountain bike. The following quantity should be sold for the company to achieve its break-even point:

   (a) 7 000 units
   (b) 8 000 units
   (c) 2 000 units
   (d) 6 000 units
   (e) none of the above (3)

2. What profit will the company realise at its expected sales level of 6 000 units?

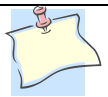
   (a) R10 000 000
   (b) R20 000 000
   (c) R30 000 000
   (d) R60 000 000
   (e) none of the above (3)

3. What units must the company sell in order to earn a profit of R30 000 000?

   (a) 6 000 units
   (b) 2 000 units
   (c) 8 000 units
   (d) 7 000 units
   (e) It is not currently possible for the company to earn a profit of R30 000 000 (3)

4. What is the company's margin of safety?

   (a) 66,67%
   (b) 28,57%
   (c) 200%
   (d) 0%
   (e) none of the above (3)



Total cost given = variable + fixed costs.
Note: Two Rand values (total cost) and two production levels given. Therefore use high-low.

PART C **8 marks**

Select the correct answer for each question below. Show ALL calculations clearly**.**

ABC (Pty) Ltd, manufactures two products, A and B. The production of these products gives rise to a fixed manufacturing overhead cost of R 2 000 000 per annum, which has traditionally been allocated to the products based on labour hours required in the production process. The financial director wants to change the accounting system to an ABC system, and has asked for your assistance.

The following information has been provided:

	Number of units	Labour hours	Machine hours	Number inspections	Number set-ups
A	20 000	1 600 000	20 000	800	200
B	60 000	6 400 000	80 000	1 200	300
	80 000	8 000 000	100 000	2 000	500

Overhead costs	**R**
Maintenance of machinery	480 000
Quality control	920 000
Set-up costs	600 000
Total	2 000 000

5. The **total** overhead cost that would be allocated to Product B on a traditional costing system using labour hours as an allocation would be:

   (a) R1 600 000
   (b) R26,67
   (c) R1 500 000
   (d) R25,00
   (e) None of the above (4)

6. The overhead cost per unit allocated to Product B, based on activity based costing system should be:

   (a) R26,67
   (b) R25,00
   (c) R33,33
   (d) R21,60
   (e) None of the above (4)

QUESTION 27

ZIMPOPO LTD **40 marks**

(Source: Test 2 2010 TOE408W (MAC4862))

Mr Knowledge Chinamaso is the management accountant of Zimpopo Ltd, a company manufacturing consumer products. Mr Chinamaso is preparing his management meeting packs and has prioritised the following items:

A. POLOKWANE DIVISION

The divisional accountant has reported a profit of R55 600 for April 20X0, made up as follows:

	R	
Budgeted profit	42 000	
Sales volume variance	8 400	A
Standard profit on actual sales	33 600	
Selling price variance	2 400	A
	31 200	
Manufacturing variances	24 400	F
Material price	10 000	F
Material usage	1 500	A
Labour rate	2 000	A
Labour efficiency	1 500	A
Variable overhead expenditure	2 000	A
Variable overhead efficiency	3 500	F
Fixed overhead expenditure	12 650	F
Fixed overhead (volume) efficiency	5 250	F
Manufacturing profit	55 600	

The budget for April 20X0 contains the following data:

Sales volume	15 000	Units
Production volume	15 000	Units
Materials purchased	7 500	Kg
Materials used	7 500	Kg
Labour hours	11 250	Hours

	R
Sales revenue	210 000
Material costs	45 000
Labour costs	45 000
Variable overhead costs	24 000
Fixed overhead costs	54 000

Additional information:

- There were 1 000 completed product units in inventory on 1 April, but no materials or work-in-process.
- Inventory of raw materials and completed product units are valued at standard cost.
- During April 15 500 product units were produced.
- The actual sales revenue for April was R165 600.
- 10 000 kg of materials were purchased in April.

REQUIRED		**Marks**
(a)	**REVISION -** To evaluate Polokwane's performance, Mr Chinamaso requires the following **information**:	
(i)	the actual sales volume	(3)
(ii)	the actual quantity of materials used for production in April	(3)
(iii)	the actual cost of materials in April	(2)
(iv)	the actual labour hours worked in April	(3)
(v)	the value of inventory on 30 April 20X0	(4)

B. Musina Division

The Musina Division manufactures three products. At the end of April 20X0 the divisional accountant forwarded the following information for its profitability review:

Product	**L**	**I**	**M**
Production capacity (units)	100 000	120 000	100 000
Sales (units)	100 000	120 000	80 000
	R'000	**R'000**	**R'000**
Sales revenue	1 500	1 440	880
Costs	1 550	1 400	760
Materials	500	480	240
Labour	400	320	160
Overheads	650	600	360
Profit/(Loss)	(50)	40	120

Additional information:

- All units produced are sold, except for product line M where the production volume is limited to the sales volume.
- 40% of the labour cost for each product is fixed in nature.
- Fixed administration overheads of R1 200 000 in total have been apportioned to each product on the basis of units sold and are included in the overhead costs above. All other overhead costs are variable in nature.
- The Musina division used assets of R675 000 in their division and applies Zimpopo Ltd's cost of capital of 15% for evaluations.
- The divisional accountant has recommended that production of product L be ceased and that the spare capacity be utilised to manufacture an additional 100 000 units of product M.

Mr Knowledge Chinamaso has investigated further and found that:

- Ceasing production of product L would eliminate the fixed labour cost of this product as well as 20% of the fixed administration overhead apportioned to it.
- Increasing production of product M by 100 000 units would cause the fixed labour cost of product M to double, the variable labour cost to rise by 20% and unit sales price to decrease by R1,50 per unit in order to move the additional units.

REQUIRED		**Marks**
(b)	Assess whether production of product L should be ceased by considering: (i) Contribution (ii) Residual income (RI)	 (8) (10)
(c)	The Gweru division has put in an offer to buy 40 000 units of product M at R7, 00 per unit should production of product L be changed to product M. Calculate the effect of this transfer price on the Musina division.	 (4)
(d)	List the five primary types of transfer prices that companies can use.	(3)

QUESTION 28

KISUZU LTD; BW LTD **40 Marks**

(Source: Test 2 2010 DIPAC26 (MAC4861))

Please note that the standard of the DIPAC26 questions is not always comparable to MAC4861 standards but that these questions can be used for revision purposes for MAC4861 as they test basic principles and concepts that are necessary for MAC4861/2.

PART A **32 marks**

Kisuzu Ltd prepared the following budget for the production and sales of 100 000 branded shirts, showing the following unit information:

	R
Sales price (per shirt)	120
Manufacturing cost:	
Materials – 2m^2 @ R18/ m^2	36
Labour – 1 hour @ R14 / hour	14
Variable production overhead – 0,25 machine hours @ R16 / hour	4
Fixed production overhead – 0,5 machine hours @ R24 / hour	12
Variable selling costs (per shirt)	6
Fixed administrative costs (per shirt)	18

Actual results for the year ended 30 April 2014 showed the following:

1.
   - Opening inventory — Nil
   - Shirts produced — 110 000
   - Shirts sold — 84 000

   60 000 shirts were sold at the budgeted price and the remainder at R115 per shirt. The sales persons earned the budgeted rate (percentage) on the sales.

2. Costs incurred for the year were:

   - Materials: 230 000 m^2 costing R4 255 000. There were no materials in inventory on 30 April 2014.

   - Labour: 105 000 hours were worked at a total labour cost of R1 550 000. This includes long-serving bonuses of R80 000 paid to qualifying factory workers. Factory labour is considered a variable cost.

   - Variable production overhead of R410 000.

   - Fixed production overhead of R1 191 500.

   - Fixed administrative costs of R1 901 000.

3. On 29 April 2014 Kisuzu Ltd received an enquiry from Balls Ltd as to whether Kisuzu Ltd would be prepared to sell 15 000 shirts to Balls Ltd at a special price of R90 per shirt. These shirts (now in inventory) will require additional embroidering and new packaging at a cost of R16 per shirt. No sales commission will be payable on this 'order'. Balls Ltd will pay for the shirts on delivery thereof. The request was not processed as a firm order for the 30 April 2014 year end.

4. Kisuzu Ltd's effective weighted average cost of capital (WACC) was 16%. An average R15 million of assets was employed in the business.

REQUIRED		**Marks**
(a)	Calculate the following for the year ended 30 April 2014:	
	(i) Material price variance	(1)
	(ii) Material usage variance	(1)
	(iii) Labour rate variance	(2)
	(iv) Labour efficiency variance	(1)
	(v) Variable production overhead expenditure variance	(1)
	(vi) Fixed production overhead absorbed, indicating the amount over or under absorbed.	(2)
	(vii) Actual commission paid on sales	(3)
(b)	Indicate, with the necessary supporting calculations and motivation whether or not Kisuzu Ltd should accept Balls Ltd's enquiry as a special order. [Calculations 5 marks and motivation 4 marks]	(9)
(c)	Calculate the impact of Balls Ltd's order on Kisuzu Ltd's **actual** residual income for the year ended 30 April 2014. (You may assume that the request turned into an accepted order and was processed for the year ending 30 April 2014). [You may round your calculations to 1 decimal iro R'000]	(12)

PART B **8 marks**

Four types of material will be needed for a special project being considered by BW Limited:

Material	Quantity (units) needed for contract	Quantity already in inventory	Purchase price of units already in inventory (R per unit)	Current purchase price (R per unit)	Current resale price (R per unit)
A	200	400	40,00	44,00	38,00
B	1 400	600	35,00	33,00	25,00
C	400	800	20,00	21,00	10,00
D	300	400	25,00	0,00	10,00

Material A is in regular use. Neither materials B nor C is currently used. B has no foreseeable use in the business, but C could be used on other jobs in place of material currently costing R16 per unit. D is not available on the market any more.

PART B

REQUIRED		**Marks**
(d)	Calculate the (relevant) material cost for the special project and briefly give your supporting motivation.	(8)

QUESTION 29 **56 Marks**

(Source: SAICA ITC June 2013, Paper 4, Question 2)

EFICAZ (PTY) LTD ('EFICAZ')

Eficaz (Pty) Ltd ('Eficaz') is an engineering company that manufactures and supplies consumable products for use in the mining and industrial fluid transfer industries. The key raw material used in its manufacturing processes is steel and Eficaz uses a range of machining technologies to make end products.

The current chief executive officer of Eficaz, Mr Rob Oliveira, acquired 100% of the shares in issue in 2004 from the founding shareholders. The business has been transformed since then by expanding the product range and focusing on production efficiencies. Mr Oliveira would like to grow the business further in anticipation of the sale of the company in a few years' time.

Eficaz's target gross profit margin is 30% and the company has achieved or exceeded this over the past three financial years.

Eficaz has unutilised manufacturing capacity. For the year ending 31 December 2013, it is estimated that Eficaz will have 18 400 machine hours and 25 600 direct labour hours available for the manufacture of additional products. A total of 132 000 direct labour hours (excluding any overtime) is available for the 2013 financial year.

Potential opportunities

The company has been approached by two potential customers to manufacture specific products for them on an ongoing basis. Details of the estimated manufacturing costs and proposed selling prices of these products are summarised in the table below:

Potential customer	**Notes**	**XYZ Mining**	**ABC Industrial**
Product code		**M134**	**IN223**
Proposed selling price per unit	1	R120,00	R80,00
Raw material costs per unit manufactured		R40,00	R15,00
Estimated production time per unit manufactured			
Machine hours		0,50	0,20
Direct labour hours	2	0,60	0,50
Number of units required annually by potential customer	3	35 000	60 000
Estimated variable manufacturing overheads per unit		R7,50	R8,50
Fixed manufacturing overheads per unit	4	R6,50	R2,60

Notes

1 XYZ Mining has indicated that its existing supplier charges R120 per unit. XYZ Mining is dissatisfied with this supplier as the order lead times are far too long (currently 30 days from order to delivery whereas XYZ Mining requires products to be delivered within 15 days of order). The M134 product is currently manufactured by Eficaz and sold to a variety of mining customers.
Product IN223 would replace one of the existing products currently purchased and used by ABC Industrial. The existing product has some safety deficiencies and ABC Industrial has identified Eficaz as the appropriate partner to manufacture the new product (IN223), given its engineering expertise and manufacturing excellence. ABC Industrial has patented the IN223 product design and will not make this product available to anyone else in South Africa. It is estimated that Eficaz will need to make a once-off investment of R308 180 in plant and machinery modifications to enable it to manufacture product IN223. ABC Industrial has offered to pay R50 000 of the plant and machinery modification costs. Eficaz plans to expense its share of the modification costs, as the company is uncertain as to how to allocate these costs to future units of IN223 manufactured.

2 The average direct labour cost for both the IN223 and M134 products is R60 per hour. Employees directly involved in manufacturing have indicated that they are prepared to work overtime, if necessary, provided they are paid 1,5 times their normal hourly rate.

3 XYZ Mining has committed to purchase 35 000 units of M134 in 2013. However, it cannot accurately predict volumes in future years. If commodity prices decline, XYZ Mining may have to scale back on mining operations in the short term until prices recover.

ABC Industrial is prepared to enter into a three-year contract with Eficaz for 60 000 units per annum to provide Eficaz with some assurance of recovering the initial investment in the supply arrangement. ABC Industrial estimates that annual order volumes of IN223 will increase by 10% per annum from 2014 for the foreseeable future.

4 Annual fixed manufacturing overheads are budgeted to be R1 300 000 in the 2013 financial year. These overheads are allocated to all products based on estimated machine hours to manufacture individual products, multiplied by an overhead recovery rate. The overhead recovery rate is determined as fixed manufacturing overheads divided by the total available machine hours. There are 100 000 machine hours available for the manufacture of products in 2013. Planned machine hours, excluding the potential M134 and IN223 opportunities, are 81 600 hours for the 2013 financial year.

Mr Oliveira is uncertain which opportunity Eficaz should pursue – should Eficaz manufacture M134 for XYZ Mining or IN223 for ABC Industrial? Eficaz has insufficient manufacturing capacity to pursue both opportunities and will for the next two years not have the financial resources to invest in further plant and machinery to expand capacity. It does, however, have sufficient resources to invest in modifying the existing plant and machinery to pursue the IN223 opportunity.

Selling prices

Eficaz has been grappling with the best way to determine selling prices for its products. Historically, it has estimated product manufacturing costs and added a standard mark up on costs to arrive at the selling price. Mr Oliveira is becoming increasingly concerned about this approach as it tends to result in the company favouring products which have higher manufacturing costs. He has heard that a more appropriate way of pricing products is to determine a standard recovery rate per machine hour and add this recovery to manufacturing costs to arrive at a selling price.
This alternate price determination strategy is illustrated below for an existing product (not M134 or IN223):

	Per unit R
Raw material costs	60,00
Direct labour costs	30,00
Variable manufacturing overheads	10,00
Fixed manufacturing overheads	6,50
Total manufacturing cost per unit	106,50
Machine hour recovery (0,5 hours x R80,00 per hour)	40,00
Selling price	146,50

REQUIRED		**Marks**	
		Sub-total	**Sub-total**
(a)	Calculate and evaluate which of the M134 and IN223 products would be the most profitable for Eficaz to manufacture in the 2013 financial year. Justify any decisions made to exclude specific manufacturing costs.	(15)	(15)
(b)	Identify and discuss the factors to be considered by Eficaz in evaluating whether to pursue the opportunity to supply XYZ Mining or ABC Industrial. *Communication skills – logical argument; clarity of expression*	(15) (2)	(17)
(c)	Calculate the selling prices Eficaz should charge for the M134 and IN223 products if it used a machine hour recovery rate methodology and the recovery rate is R80 per hour. Include fixed manufacturing overheads allocated to products in your calculations.	(10)	(10)
(d)	Discuss the potential merits and pitfalls of adopting a selling price strategy of adding a machine hour recovery rate to manufacturing costs per unit. *Communication skills – logical argument*	(8) (1)	(9)
(e)	Estimate the allocation of fixed manufacturing overheads to the M124 and IN223 products during the 2013 financial year based on direct labour hours as opposed to machine hours.	(5)	(5)
Total			56

QUESTION 30

AERO AFRICA LTD **100 marks**

(Source: SAICA ITC January 2013, Paper 2, Part A)

Aero Africa Ltd ('AA') is an airline that listed on the Johannesburg Securities Exchange in June 2012. AA is a decentralised company with the following divisions:

- Aero Africa;
- Aero Africa Mobilechefs (catering division);
- Aero Africa Technical (the technical division which provides aircraft maintenance services); and
- Aero Africa Travel Centres (Aero Africa's own branded travel agencies).

United Auditors Inc., a firm of Registered Auditors, has been the external auditor of AA for the past ten years. United Auditors Inc., based in Johannesburg, is currently comprised of three audit partners and 30 trainees. Kingston Jacobs was the audit engagement partner for AA prior to his resignation from United Auditors Inc. in February 2012.

Governance arrangements

Prior to its listing, all the shares in AA were owned by Mr Marc Brooney, a highly successful South African businessman. Mr Brooney introduced a policy ten years ago whereby directors on AA's Board were appointed for a three-year period only, in order to promote transformation. Since then the only person who has been reappointed was Ms Luhle Jacobs, who was the information technology director. To date, Ms Jacobs (wife of Mr Kingston Jacobs) has been a director for ten and a half consecutive years.

Prior to its listing, Mr Brooney consistently had to provide financial support to AA to enable it to continue as a going concern. Therefore to eliminate the need for on-going financial support of the airline, Mr Brooney proposed that –

- AA be listed at R50 per share;
- he would own 51% of the shares;
- all directors would receive share options on listing and annually thereafter;
- Mr Jacobs would be appointed as the chief executive officer (CEO); and
- Ms Jacobs would be appointed as the chief operating officer (COO).

Investors did not, however, accept Mr Brooney's proposed listing price and shareholding. Rather, they believed that a market related price was R20 per share, and that Mr Brooney should own no more than 40% of the shares. Mr Brooney agreed to these conditions but added a condition of his own, namely that his shareholding would permit him to appoint the majority of the Board members. This was accepted and the Memorandum of Incorporation of AA was amended to reflect this.

Once listed and with the approval of Mr Brooney, Kingston and Luhle Jacobs nominated and appointed the following persons to the Board of Directors for a three-year period:

- Non-executive directors

    - Mr Ryan Abbott (22), a qualified aircraft technician;
    - Mr Nathan Barclay (70), a retired marketing manager who is now a director of various companies; and
    - Mr Tyler Callan (24), a travel agent.

- Executive director

    - Ms Alexia Viljoen (41) CA (SA), as the chief financial officer (CFO).

There are no other directors. An audit committee and remuneration committee were established once the company had listed. In terms of a Board resolution the internal audit services have been outsourced to United Auditors Inc.

Within a few months of the listing, AA stated at a press conference that Ms Viljoen had 'resigned with immediate effect to pursue personal interests'. In her letter of resignation to the Board of Directors of AA, Ms Viljoen stated the following:

> The high director turnover rate at AA is a consequence of the three-year contract appointments made while it was a private company. This has resulted in the following practices, which are continuing even though the company is now a public company and listed:
>
> - Upon their appointment, directors negotiate very large payments that would be payable to them if they were not reappointed after their three-year term.
> - The rigging of tender processes is common practice – tenders are awarded to companies or individuals who secretly pay incentives to the executive directors.
> - There is no focus on employee recruitment, remuneration or incentives, which has resulted in poorly skilled employees and a generally unhappy employee base.
>
> When the CEO signed the engagement letter for the reappointment of United Auditors Inc. as the company's registered auditor despite my reservations, and without the knowledge of the shareholders, I had no other option but to resign.

Shortly thereafter the public shareholders of AA began raising concerns about the company's commitment to compliance with the Companies Act, 2008 and its application of *The Revised Code of and Report on Governance Principles for South Africa* (King III).

AA and its flight operations

The core business of the airline is to move people and goods by air. AA operates in three distinct markets:

- Domestic: AA has the most extensive domestic schedule of all airlines operating in the South African domestic market;
- Regional: AA is one of the leading carriers and serves 20 destinations across the African continent apart from its South African destinations; and
- International: AA connects South Africa to all continents of the globe via ten direct routes.

AA is segmented on a geographic basis into South Africa, the rest of Africa, Asia, Australia, Europe, the Middle East, North America and South America. The CEO and COO of AA are currently reviewing the performance of each of these segments and want a detailed performance review of the South American segment, because this is one of the very few segments which is performing well.

South American segment

AA has landing rights for a daily flight from South Africa to the international airport at São Paolo, Brazil. As the agreement entails operating a single landing slot, the airline has one return flight per day on the Johannesburg to São Paolo ('JNB-GRU') route. This route has proved to be very popular among travellers as AA is the only African airline linking South America to South Africa. AA currently has no other landing slots in South America.

AA started using two new Airbus A330-200 aircraft for this route at the beginning of the financial year ending on 31 December 2012. These aircraft represent the latest technology on offer from Airbus in Europe, with one of their most important features being their fuel efficiency. The aircraft are financed by means of an operating lease entered into between AA and a European based aviation leasing company. The aircraft are configured with 36 business class seats and 186 economy class seats.

The financial results for the JNB-GRU route for the year ended 31 December 2012 are summarised below:

	Notes	Budget R	Actual R
Passenger sales revenue	1	842 678 400	909 381 200
Cargo services revenue	2	19 600 000	19 800 000
Total revenue		862 278 400	929 181 200
Direct operating expenses		(218 305 824)	(257 941 320)
Accommodation and refreshment costs	3	(14 625 360)	(15 331 680)
Catering	4	(21 491 520)	(23 354 240)
Jet fuel costs	5	(163 522 944)	(200 163 600)
Navigation, landing and parking fees	6	(18 666 000)	(19 091 800)
Indirect operating expenses		(474 018 000)	(490 931 000)
Aircraft lease costs	7	(42 140 000)	(42 570 000)
Electronic data costs	8	(16 278 000)	(16 456 000)
Other allocated overheads	9	(415 600 000)	(431 905 000)
Operating profit for the period		169 954 576	180 308 880

Notes

1. The budget was based on the following key assumptions:

Number of one-way flights	732
Passenger numbers (average per one-way flight)	
Business class	24
Economy class	148
Average one-way fare per passenger	
Business class	R27 000
Economy class	R3 400

An evaluation of the actual results revealed the following:

- Two flights out of Johannesburg had to be cancelled – one for technical reasons and the other due to low passenger numbers. The passengers booked on these flights as well as those booked on the return flights from São Paolo, which had to be cancelled as well, were accommodated on AA flights on other days.

- A total number of 19 656 business class passengers had been carried on the route at an average one-way fare of R28 250, while 110 656 economy class passengers had been carried on the route at an average one-way fare of R3 200 per passenger.

It is AA's policy to price a return ticket at double the one-way fare on this particular route.

2. AA had budgeted to transport a total of 14 000 tonnes of cargo on the JNB-GRU route for the year. Because AA has standing contracts with their customers in terms of which customers are charged for the volume of cargo transported, it considers this revenue to be fixed in nature. The actual quantity of cargo transported was 5% more than the budgeted quantity.

3. This represents the cost of the subsistence allowance paid to each crew member. A fixed number of cabin crew works on the Airbus A330-200 to São Paolo and the crew has a three-night stop-over in São Paolo. The budgeted and actual allowance was US $150 per crew member per day. The average budgeted exchange rate was US $1: R7,40 while the average actual exchange rate was US $1 : R7,80.

4. Each passenger is served two meals on each flight and it is airline policy not to carry any extra meals on a flight. The budgeted cost per business class meal was R180, while the actual cost per business class meal was R200. The budgeted and actual cost per economy class meal was R70.

5. The standard flying time to and from São Paolo is ten hours. AA budgeted for the aircraft to consume 4 800 litres of jet fuel per hour at a budgeted price per tonne of R4 654. The actual quantity of jet fuel used on the route was 33 360 600 litres and the actual average flying time was 9 hours and 45 minutes. Fuel is consumed evenly throughout the flight.

6. These costs are incurred per flight. The budget was based on an average cost per flight of R25 500.

7. This represents the annual operating lease costs of the two aircraft deployed to this segment. The operating lease payments amount to €4 300 000 for the year. The budget was based on an average exchange rate of €1 : R9,80.

8. This is the allocated portion of the information technology costs of AA. A review of the allocation indicates that the correct allocation rates have been used in respect of the budgeted and actual allocations.

9. This represents the portion of the overheads AA incurred at corporate level and apportioned to the different business segments and business units.

Performance assessment of the South American segment

AA makes use of variance analysis to evaluate the performance of its segments. The AA group does not prepare flexed budgets but instead compares actual results to original budgets.

At a recent meeting of the Board of Directors, the COO of AA was asked how critical business class passengers were to the profitability of the JNB-GRU route. The COO agreed to investigate the issue and report back at the next meeting.

Jet fuel

The CEO is particularly pleased with AA's revenue performance in the 2012 financial year given that the actual results exceeded the budgeted results in tough economic times. However, he is concerned about jet fuel costs which were much higher than budgeted. AA did not hedge against jet fuel price increases and exchange rate fluctuations in the 2012 financial year. The cost of jet fuel has increased dramatically in recent months and the Board of Directors of AA has requested its executive management to investigate ways in which jet fuel prices could be hedged in future. Jet fuel prices are dependent on the prevailing Brent crude oil prices in US dollar.

ONLY READ AFTER QUESTION INFORMATION (all questions)

PART A – REQUIRED		**Sub-total**	**Total**
(a)	Calculate the following variances for the year ended 31 December 2012: • Passenger sales price variance; • Passenger sales margin mix variance; • Passenger sales margin quantity variance; • Jet fuel price variance; and • Jet fuel consumption variances. For the jet fuel consumption variances, provide as much detail as possible regarding the differences between budgeted and actual amounts.	(20)	(20)
(b)	Calculate the number of business class and economy class passengers that AA needed to transport in the 2012 financial year in order to break even. For the purpose of your calculations, assume the same ratio of business class passengers to economy class passengers as the actual ratio for the year.	(10)	(10)
(c)	Analyse and discuss the importance of business class passengers to the profitability of the JNB-GRU route. *Communication skills – clarity of expression*	(8) *(1)*	(9)
(d)	Explain how AA could hedge against jet fuel price increases.	(6)	(6)
(e)	Identify and describe four major risks faced by AA with respect to the JNB-GRU route. *Communication skills – logical argument; clarity of expression*	(8) *(2)*	(10)
Total for Part A			**55**

PART B – REQUIRED			
		Sub-total	**Total**
(f)	Discuss, with reasons, any concerns you have about the current (i.e. post-listing) corporate governance arrangements of AA. *Communication skills – clarity of expression; logical argument*	(43) (2)	(45)
Total for Part B			(45)
TOTAL			**100**

QUESTION 31

EXPERIENCE AFRICA HOLDINGS LTD **100 marks**

(Source: SAICA QE1 2011 Paper 3 Part B)

Experience Africa Holdings Ltd ('EA Holdings') is the holding company of a group of eco-tourism companies which provides world-class wildlife and outdoor experiences to discerning travellers. The group is committed to sustainable conservation and the empowerment of local communities. The group owns and operates safari lodges in South Africa, Namibia and Botswana.

EA Holdings has three wholly-owned subsidiaries, which operate and manage the group's safari lodges in their respective countries, namely —

- Experience Africa SA (Pty) Ltd ('EA SA'),
- Wildlife Botswana (Pty) Ltd ('Wildlife Botswana'), and
- Wildlife Namibia (Pty) Ltd ('Wildlife Namibia').

EA Holdings was founded 15 years ago by Dr Joe Player, an environmentalist and entrepreneur. His objective was to provide tourists with the opportunity to experience Africa's unique wildlife whilst promoting sustainable conservation of wilderness areas. EA Holdings listed on the alternative exchange of the Johannesburg Securities Exchange (JSE) in 2008. Immediately prior to listing, Dr Player retired as chief executive officer of the company and became the non-executive chairman of EA Holdings.

Shareholding and board of directors

The current shareholders of EA Holdings are as follows:

Player Family Trust	37%
Employee Staff Trust	5%
Unit trusts	8%
Public investment corporation	12%
General public	38%

The current directors of EA Holdings are as follows:

Dr Joe Player	Non-executive chairman
Ms Jane Player-Carter	Chief executive officer
Mr Sonny Ndlovu,CA(SA)	Financial director
Mr Thabiso Mleng	Operations director
Mr Jan Johnsen	Independent non-executive director
Mr Jim Jackson	Independent non-executive director
Ms Jayne Naidoo	Independent non-executive director

All executive directors have employment contracts (dealing with salaries, benefits and performance incentive payments). Non-executive directors receive directors' fees plus fees for attending meetings of the board of directors and its sub-committees. The Remuneration Committee is responsible for setting and monitoring the remuneration policies of the group and approving incentive schemes. The members of the Remuneration Committee are Mr Johnsen (chairman), Dr Player and Ms Naidoo. EA Holdings also has an Audit Committee consisting of Mr Jackson (chairman) and Ms Naidoo. A Risk Committee has also been established.

Auditors

Dlamini and Partners Inc. has been the auditors of EA Holdings and EA SA for the past nine years. The engagement partner for the past six years, Bob Dlamini, is an experienced auditor of ecotourism businesses.

Wildlife Botswana and Wildlife Namibia are audited by local audit firms in their respective countries. Independence-auditors Inc. was appointed as the auditor of Wildlife Botswana during October 2010.

The financial year end of the group is 31 October.

Information regarding EA SA's operations

EA SA offers 'all inclusive' travel packages to the upper end of the tourist market. Customer service is of paramount importance and the group strives to provide each guest with a unique and memorable experience. EA SA owns numerous safari lodges in the Welgevonden Private Game Reserve ('Welgevonden') in the Limpopo Province of South Africa. The company also owns a fleet of small aircraft to transport guests from OR Tambo International Airport in Gauteng to the game reserve and back again.

EA SA at present only offers six-night packages to guests and the standard all inclusive package includes the following:

- A return flight from OR Tambo International Airport to Welgevonden in one of EA SA's aircraft;
- Six nights' accommodation at one of the safari lodges, including all meals and beverages;
- Daily game drives; and
- A hot-air balloon excursion.

The EA SA Welgevonden safari lodges can together accommodate a maximum of 150 six-night packages per month. For the past three years EA SA's lodges have accommodated an average of 105 packages per month.

EA SA bases its six-night package price on cost plus 20%, which at present amounts to R18 384 excluding VAT. The current estimated cost of the package is as follows:

Cost of the all inclusive six-night package

	Notes	R
Accommodation	1	8 000
Food and beverages	2	3 000
Cleaning and other services	3	500
Game drives	4	900
Aircraft and fight costs	5	1 500
Hot-air balloon excursion	6	1 200
Electricity	7	220
Total cost		15 320

Notes

1. Accommodation represents the fixed costs of operating the lodges, excluding those costs mentioned below, and includes lodge maintenance costs, administration costs and municipal rates. These expenses are allocated to the six-night package based on normal occupancy levels of 70% for the lodges throughout the year.

2. Food and beverage costs represent the average cost of food and beverages consumed by each guest during a six-night stay.

3. Cleaning and other service expenses are the cost of employing staff to clean the lodges and service the rooms. Cleaning staff are able to clean and service the lodges if they operate at the normal 70% occupancy level. If the lodges are more than 70% full, cleaning staff work overtime at R150 per hour. Cleaning staff spend an extra 30 minutes per day cleaning and servicing the lodges for each additional guest above the 70% occupancy level.

4. The cost of daily game drives comprises the salaries of game rangers, maintenance costs of vehicles and the cost of complimentary hampers provided for each guest. The complimentary hampers cost R400 each and include a blanket for the game drives, a game viewing booklet and mini binoculars. Each guest receives one of these hampers during the course of their six-night stay at the lodge (they do not receive a hamper each time they go on a game drive). Game rangers are permanent employees of EA SA.

5. The total cost of collecting guests from OR Tambo International Airport, transferring them to the lodge, and returning them to OR Tambo International Airport at the end of their stay amounts to R157 500 per month. This cost is allocated to each package based on the number of guests per month at the 70% occupancy level. EA SA has sufficient aircraft capacity to cope with a 100% occupancy level.

   Some guests travel to EA SA's safari lodges by road and at their own expense. These guests do not receive a reduction in the total package price. Mr Ndlovu, the financial director, believes that this is the correct treatment. In his opinion, 'the cost of running the fleet is mainly fixed and is incurred regardless of whether clients use the service or not'. Thus the cost of the air transport service is allocated to all packages sold. Market research has indicated that many guests find safari lodge packages which do not include air transport services in their prices, to be more affordable than the EA SA package.

6. The hot-air balloon excursions offered to guests are outsourced to Eagle Eye Balloon Tours ('Eagle Eye') which operates from Welgevonden. Eagle Eye charges EA SA a fee of R1 200 per guest per hot-air balloon excursion. This fee is applicable for up to 105 guests in any particular month. If the monthly number of guests is higher, Eagle Eye has to rent extra hot-air balloons and charges an incremental fee of R800 for each guest over 105 guests (i.e. R800 plus the original R1 200) to cover the increased costs. Eagle Eye has indicated that it could increase its capacity and reduce the fee to R1 000 per guest if EA SA could guarantee at least 140 guests per month.

7. Electricity costs vary according to occupancy levels. The total electricity cost per month at an occupancy level of 70% is R23 100. Total electricity costs would amount to R31 200 per month if the safari lodges operated at full capacity.

A new business venture for the South African operation

The directors of EA SA were satisfied with the company's results for the year ended 31 October 2010. The number of guests increased as a result of the FIFA 2010 Soccer World Cup and it is hoped that many of these tourists will return in the future. At the National Tourism Board Conference entitled Tourism 2010 and beyond, Mr Sonny Ndlovu learnt that tour bus operators which offered bus tours around Gauteng to foreign guests during the World Cup had been highly successful. Sonny subsequently did some market research into a venture to utilise the monthly spare capacity at the EA SA safari lodges in South Africa.

Sonny prepared the following business proposal which has been approved by the directors of EA SA:

- EA SA will enter into an agreement with Interstate Bus Tours (Pty) Ltd ('Interstate'), a tour bus company wholly owned by Petunia Ndlovu, Sonny's wife. (Interstate is audited by Eric Eager, a sole practitioner.) Interstate will sell an all-inclusive package for R17 000, consisting of three days and nights at the EA SA Welgevonden safari lodges and four days and nights touring famous landmarks in and around Gauteng. The portion of the package that relates to the bus tour will include all meals, accommodation and sightseeing expenses. The three-night stay at the Welgevonden safari lodges will entitle guests to all the privileges of the standard six-night package, including the hot-air balloon excursion, daily game drives, food and beverages and all other lodge services. The existing game rangers and vehicles will be able to cope with the increased demand for daily game drives as most guests do not use this service every day. Over and above these normal privileges, EA SA needs to provide entertainment on one evening per package. This would entail a cost of R2 000, eight times per month. This package price does not include flights to and from OR Tambo International Airport because Interstate will take responsibility for transporting guests.

- Market research indicates that there would be a demand for 120 such packages per month. Sonny was particularly thrilled with this proposal as it would solve the current problem of unutilised capacity at the Welgevonden safari lodges. The demand from Interstate is equivalent to 60 normal all inclusive six-night packages sold by EA SA.

- Interstate will take responsibility for the booking, invoicing and general administration of the all-inclusive packages.

- Per the agreement, Interstate will pay EA SA R4 000 for each guest for the three-night stay at EA SA's Welgevonden safari lodges. Interstate has requested that EA SA reserve a minimum of 120 three-night packages at the Welgevonden safari lodges for their all-inclusive packages.

- Interstate will charge EA SA a management fee of 5% of the three-night stay revenue for the booking, invoicing and administrative functions they perform.

The terms and conditions of the venture were unanimously approved at a directors' meeting on 15 September 2010 at which all the directors were present, for implementation in early 2011. The board of directors furthermore decided that the pricing of the standard six-night package should remain at R18 384 per guest despite EA SA's cost-plus pricing policy.

Interstate cost structure

In anticipation of the new business venture, Petunia is considering two options for delivering the four-day bus tour section of the package. The first option is for Interstate to deliver all the services themselves. The second option is to outsource the service to another tour bus operator, which will take responsibility for all aspects except booking, invoicing and general administration.
Petunia has prepared the following monthly revenue and cost structure sheet:

	Notes	Option 1 R	Option 2 R
Revenue(120 x R17 000)		2 040 000	2 040 000
EA SA three-night package (120 x R4 000)		(480 000)	(480 000)
5% management fee payable by EA SA		24 000	24 000
Accommodation, meals and sightseeing expenses	1	(420 000)	-
Bus costs	2	(220 000)	-
General administration costs	3	(60 000)	(60 000)
Outsourced cost (120 x R7 500)	4	-	(900 000)

Notes

1. Accommodation, meals and sightseeing expenses relate to the costs incurred during the four day and four night bus tours.

2. Bus costs include depreciation, the salary of the bus driver, insurance and diesel fuel costs.

3. General administration costs consist mainly of the recently appointed manager's salary. The manager will be responsible for the booking, invoicing and general administration of the bus tour packages.

4. Interstate would be able to outsource the four-day bus tour experience to Gauteng Tours at a cost of R7 500 per guest. This price includes the bus service, accommodation and meals as well as sightseeing expenses.

Extract from the minutes of a meeting of the board of directors of EA Holdings

MINUTES OF THE BOARD MEETING
held on 15 December 2010

Present	**Other attendees**	**Apologies**
Directors		
Dr Joe Player Ms Jane Player-Carter Mr Sonny Ndlovu Mr Thabiso Mleng Mr Jan Johnsen Mr Jim Jackson	Mr Jimmy Naicker (company secretary)	Ms Jayne Naidoo

Approval of the minutes of the previous meeting — 15 September 2010

The minutes of the meeting held on 15 September 2010 were approved subject to a few editorial corrections.

Election of board chairman

Dr Player expressed the desire to step down as chairman because he wants to spend more time with his wife and grandchildren. Dr Player will remain on the board as a non-executive director. He proposed that, in line with the company track record of success due to tight family control, his daughter, Jane Player-Carter, succeed him as chairman, while also remaining chief executive officer of the company.
After a short discussion by the board it resolved to appoint Ms Player-Carter as executive chairman. It was further resolved that no 'big fuss' would be made regarding the appointment, and that except for a JSE SENS announcement, no further publicity will be given to this matter.

Audit Committee feedback

Mr Jackson reported back on the recent Audit Committee meeting and raised a specific area of disagreement with the external auditors, namely that the external auditors are uncomfortable with the R4 000 package fee to be paid by Interstate. They are of the opinion that it is below cost to EA SA and that the transaction is not at arm's length.

Additional key business risks

Mr Ndlovu reported that the Risk Committee (which he chairs) has identified the following two additional key business risks facing the group, of which the board should be aware:

- **Regulation and compliance**: The risk that uncertainty surrounding new laws, regulations and compliance therewith is stalling business decision making and planning; and
- **Cost cutting**: Consumer price inflation and pressure from low-cost competitors can have a negative effect on the market share of the company.

PART A		
REQUIRED		**Marks**
(a)	Discuss the following in terms of the Companies Act, 2008 (Act 71 of 2008): • Whether the consolidated annual financial statements of EA Holdings are required to be audited; and • Any matters to be considered with regard to the re-appointment of the current auditors of the company.	(3)
(b)	Discuss any ethical, corporate governance and statutory issues arising from EA SA entering into a new business venture with Interstate as was presented to and approved by the board of directors of EA Holdings.	(19)
(c)	Discuss whether the board of directors of EA Holdings complies with the King Code of Governance Principles for South Africa, 2009 (King Ill), regarding the appointment of Ms Player-Carter as chairman. Also discuss how Diamini and Partners Inc. should respond to the appointment.	(7)
(d)	Discuss the fraud risks and their potential impact on the audit report arising From the new business venture between Interstate and EA SA.	(7)
Presentation marks: Arrangement and layout, clarity of explanation, logical argument and language usage.		(2)
TOTAL MARKS		**38**

PART B		
REQUIRED		**Marks**
(e)	Calculate the impact that entering into the agreement with Interstate will have on EA SA's monthly operating profits.	(18)
(f)	Identify and outline the key factors that EA SA should have considered prior to entering into the new business venture with Interstate.	(18)
(g)	Critically discuss the pricing strategies and policies adopted by EA SA in pricing its standard six-night all inclusive package at the Welgevonden safari lodges.	(10)
(h)	Calculate the breakeven point, both in terms of revenue amount and number of guests, and the degree of operating leverage for each of the two cost structure options for delivering the four-day bus tour experience by Interstate. Based on the results of your calculations recommend, with reasons, which option is best for Interstate.	(13)
Presentation marks: Arrangement and layout, clarity of explanation, logical argument and language usage.		(3)
TOTAL MARKS		**62**

QUESTION 32 **65 marks**

EXCEL LEISURE LTD ('EXCEL') **40 marks**

(Source: SAICA QE I 2010, Paper 1, Question 3 adapted)

(ONLY PART B IS APPLICABLE)

Amounts in the question exclude VAT, except where indicated.

Excel Leisure Ltd ('Excel') operates four golf driving ranges, two in Johannesburg and one each in Cape Town and Durban. The business strategy has been to acquire land in suitable locations and develop golf driving ranges. Excel then develops family-orientated restaurants on the driving range sites to diversify income streams. Excel has traded well over the past couple of years and is intent on listing on the Alternative Exchange of the Johannesburg Securities Exchange within the next 18 months. The sole shareholder and Chief Executive Officer of Excel, Mr Adamant, is keen to list his business to enable him to sell some shares and to facilitate a share incentive scheme for company employees. The company's financial year end is 31 December.

Golf driving range division

Driving ranges require significant space and the average size of the properties owned is 75 hectares (or 750 000 square metres). Excel owns all properties from which it operates, and generally targets sites in outlying urban areas. It is hoped that the properties owned will become more valuable in time as residential development occurs around the driving ranges.

Revenue has historically been generated from two sources at the driving ranges. First, the public pays for using golf balls (charged per bucket of 50 golf balls). Second, Excel sells beverages and snacks at the golf shops.

Excel introduced golf driving range memberships with effect from the 2009 financial year. The key benefits of becoming a member are unlimited usage of the golf driving range (as opposed to paying per bucket of golf balls hit on the range) and a 10% discount on all food and beverage purchases from the restaurant situated on the site. New members are charged a once-off entrance fee of R750. Entrance fees are not refundable in the event of individuals ceasing to become members.

The annual membership (membership) period is from 1 January to 31 December each year. Membership fees in 2009 were R1 200 per member (regardless of whether members join in January or later in the respective year). Membership fees are payable upfront by new members. Members renewing their memberships are required to pay fees on or before 31 January every year. The introduction of membership proved popular with the public and 5 000 memberships in total were sold at the four driving ranges during 2009.

Restaurant division

The restaurants situated at the golf driving ranges cater for golfers and non-golfers. The décor and facilities are family orientated, and the restaurants serve a wide variety of meals. Restaurants are open seven days a week for breakfast, lunch and dinner.

Excel introduced a customer loyalty programme for non-members of the golf driving ranges in January 2009. Customers electing to join the loyalty programme are issued with loyalty cards (cards with magnetic strips containing individuals' details). Customers are encouraged to present loyalty cards to waitrons when paying for purchases at restaurants and these cards are then swiped at the point-of-sale devices.

Loyalty points are captured electronically, recording amounts spent and updating the balance of loyalty points. The loyalty reward is one point for each R10 spent at restaurants which can be redeemed for food and beverages at the restaurants (in full or partial payment of bills).

Loyalty points can also be redeemed for buckets of golf balls to be hit on the driving ranges.

Finance and administration division

The finance and administration division is located at the Durban driving range and is responsible for all the accounting, finance, human resources and administration functions of the driving ranges and restaurants. Divisional statements of comprehensive income are prepared monthly. Statements of financial position are prepared only at the company level and not per division. The trial balance of Excel for the year ended 31 December 2009 is set out below:

EXCEL LEISURE LTD
TRIAL BALANCE AT 31 DECEMBER 2009

	Notes	Debit R000	Credit R'000
Driving range division			
Entrance fees received	1		3 750
Annual membership fee income	1		6 000
Usage of golf balls revenue	2		7 200
Sales of beverages and snacks	2		6 300
Cost of sales: Beverages and snacks		4 200	
Salaries and wages	3	3 240	
Maintenance of driving ranges		2 160	
Replacement of driving range golf balls		800	
Other expenses	3	480	
Depreciation: Driving range equipment	4	400	
Depreciation: Driving range buildings	4	40	
Driving range divisional profit			11 930
Restaurant division			
Sale of food and beverages	5		43 200
Cost of food and beverages sold		19 640	
Salaries and wages	3	5 950	
Depreciation: Restaurant equipment	4	1 120	
Depreciation: Restaurant buildings	4	360	
Linen cleaning and replacement		1 080	
Other expenses	3	720	
Marketing expenses	6	430	
Restaurant divisional profit			13 900

EXCEL LEISURE LTD
TRIAL BALANCE AT 31 DECEMBER 2009
(continued)

	Notes	Debit R000	Credit R'000
Salaries and wages	3	1 800	
Bank charges	7	1 100	
Depreciation: Finance and administration office Equipment	4	125	
Electricity	8	820	
Water	9	780	
Advertising	10	560	
Other expenses	3	450	
Finance and administration divisional loss		4 035	
Share capital			1 000
Retained earnings – 1 January 2009			47 000
Trade and other payables			3 800
Land, at cost	11	30 000	
Restaurant buildings, at cost	4	18 000	
Accumulated depreciation: Restaurant buildings			1 440
Driving range buildings, at cost	4	2 000	
Accumulated depreciation: Driving range buildings			160
Driving range equipment, at cost	4	4 000	
Accumulated depreciation: Driving range equipment			1 600
Restaurant equipment, at cost		8 960	
Accumulated depreciation: Restaurant equipment			4 480
Finance and administration office equipment, at cost		650	
Accumulated depreciation: Finance and administration office equipment			350
Inventories	12	1 800	
Other current assets		640	
Cash and cash equivalents		15 575	
		85 660	85 660

Notes

1. Entrance and membership fees received are recognised as revenue upon receipt. Members pay entrance fees and membership fees directly into Excel's bank account through Internet banking transfers.

2. Revenue generated by the driving ranges in respect of golf ball usage and sales of beverages and snacks is on a cash basis only.

3. Salaries, wages and other expenses are fixed in nature.

4. Excel records property, plant and equipment at cost and does not revalue assets subsequent to initial recognition. Land is not depreciated and other items of property, plant and equipment are depreciated on a straight-line basis over their expected useful lives. The carrying amounts of property, plant and equipment are equivalent to their tax bases.

5. Customers can pay by credit card or cash at the restaurants. In 2009, 60% of revenue at the restaurants was on a cash basis. Members claimed a total of R480 000 discount at the restaurants during 2009. This discount does not include any effect of the loyalty programme (see note 6). Restaurant revenue has been recorded net of these discounts claimed by members.

1. Marketing expenses represent the direct costs of food and beverages provided, in exchange for the redemption of loyalty points at the restaurants. Excel has not recognized any other entries in respect of loyalty points.
7. Excel pays deposit fees of 1,5% on cash banked (all cash is banked and no cash is used to pay expenses). Banks deduct 2% from all credit card payment transactions in the restaurants and settle amounts owing to Excel net of these transaction fees.

8. The restaurants are responsible for 80% of Excel's electricity consumption and the balance is attributable to the driving ranges. Local authorities charge Excel on the basis of electricity consumption and there is no fixed cost element in monthly billings.

9. The driving ranges are responsible for 90% of Excel's water consumption and the balance is consumed by the restaurant division. Water suppliers charge Excel a variable rate per kilolitre of water used.

10. Excel advertises on local radio stations and through outdoor advertising to promote the driving range facilities and its restaurants.

11. The fair value of land owned by Excel at 31 December 2009 was recently determined as R50 million by an independent property expert.

12. R1 500 000 of inventories are attributable to the restaurant division and R300 000 is attributable to the driving range division.

13. Excel's liability for normal taxation in the 2009 financial year is to be calculated and accrued for following the external audit.

Allocation of finance and administration division expenses

The finance and administration division does not allocate expenses directly attributable to the division's activities to the driving range and restaurant divisions.

PART B (ONLY THIS SECTION IS APPLICABLE)

(c)	Identify and describe the potential advantages and disadvantages of selling the usage of the golf driving ranges through annual memberships, as introduced in 2009, from the perspective of Excel Leisure Ltd.	(10)
(d)	Determine the contribution margins and net profits of the driving range and restaurant divisions for the year ended 31 December 2009 using the variable costing approach. Ignore interest and taxation.	(16)
(e)	Calculate the return on total assets ratio for the driving range division and the restaurant division for the year ended 31 December 2009.	(6)
(f)	Identify and discuss the arguments for and against allocating the overheads of the finance and administration division to the driving range and restaurant divisions.	(6)
Presentation marks: Arrangement and layout, clarity of explanation, logical argument and language usage.		(2)
Total		**40**



SECTION A: COSTING – SUGGESTED SOLUTIONS

QUESTION 1

GAUTENG RAIN GUTTERS (PTY) LTD **SUGGESTED SOLUTION**

(Source: Exam 2013 MAC4862)

R/W – Right/Wrong mark ***C – Consequential mark***

Part (a) Contribution per installation and contribution per combined metre

	Domestic	**Commercial**	**Construction**	**Total**
Avg metres of gutter per installation	60m	145m	975m	
Avg metres of downpipe per installation	35m	95m	585m	
Combined metres of gutter & downpipe	95m	240m	1 560m	
No. of installations	520	240	63	823

	Domestic	**Commercial**	**Construction**		
	R	**R**	**R**		
Turnover per installation (95 x R125; 240 x R95; 1 560 x R85)	**11 875**	**22 800**	**132 600**		1½ r/w (½ each)
Direct material per installation	(3 500)	(10 650)	(61 230)		
- Gutter (60 x R35; 145 x R42; 975 x R37)	**(2 100)**	**(6 090)**	**(36 075)**		1½ r/w (½ each)
- Downpipes (35 x R40; 95 x R48; 585 x R43)	**(1 400)**	**(4 560)**	**(25 155)**		1½ r/w (½ each)
Contribution to overheads per installation	8 375	12 150	71 370		
Contribution per combined metre (8 375/95; 12 150/240; 71 370/1 560)	**88,16**	**50,63**	**45,75**		1½ c (½ each)
Alternative: Dom: R125 – (60/95 x R35) – (35/95 x R40) = **R88,16** Com: R95 – (145/240 x R42) – (95/240 x R48) = **R50,63** Con: R85 – (975/1560 x R37) – (585/1560 x R43) = **R45,75**					

Students often incorrectly adjusted for wastage. The contribution per combined metre was often forgotten or calculated without taking the ratio of domestic to industrial usage into account. Students also got confused between the calculations per metre, per installation and in total and mixed them.

Part (b) Operating profit margin percentages – ABC

Activity	Domestic	Commercial	Construction	Total	
Ordering (orders)	3 x 520 = **1 560**	5 x 240 = **1 200**	8 x 63 = **504**	3 264	(1½) r/w
Travelling (km)	70 x 520 = **36 400**	185 x 240 = **44 400**	290 x 63 = **18 270**	99 070	(1½) r/w
Rolling gutters (gutter metres)	60 x 520 = **31 200**	145 x 240 = **34 800**	975 x 63 = **61 425**	127 425	(1½) r/w
Installing (combined metres)	95 x 520 = **49 400**	240 x 240 = **57 600**	1 560 x 63 = **98 280**	205 280	(1½) r/w
Other (working days)	1 x 520 = 520	2,5 x 240 = 600	5 x 63 = 315	1 435	(1½) r/w

	Domestic	Commercial	Construction	Total	
	R	**R**	**R**	**R**	
Contribution to overheads – as above	4 355 000	2 916 000	4 496 310	11 767 310	
Overheads allocated based on activity drivers	(2 738 020)	(3 128 093)	(3 633 887)	(9 500 000)	
Ordering (orders)	(195 956)	(150 735)	(63 309)	(410 000)	(1½) c
Travelling (km)	(789 947)	(963 561)	(396 492)	(2 150 000)	(1½) c
Rolling gutters (gutter metres)	(523 979)	(584 438)	(1 031 583)	(2 140 000)	(1½) c
Installing (combined metres)	(1 010 717)	(1 178 488)	(2 010 795)	(4 200 000)	(1½) c
Other (working days)	(217 422)	(250 871)	(131 707)	(600 000)	(1½) c
Operating profit / (loss)	1 616 980	(212 093)	862 423	2 267 310	
Operating profit %	**26,19%**	**-3,88%**	**10,32%**	**11,34%**	2 c
					(½ each)

A concerning number of students calculated the driver quantities incorrectly by merely adding the given quantities together e.g. 1 + 2,5 + 5 = 8,5 days. This is not ABC. It is fundamentally incorrect. The question required the operating profit percentages not only for the three markets but also for the company as a whole. Students often ignored this requirement. Some students also calculated the operating profit % by dividing the operating profit the contribution instead of the turnover. The alternative calculation was often done by merely adding the R value per installation together and presenting this as the company value – also on a % basis. This is an illogical format. Answering all the required items and using correct formulas are important.

Part (c) Challenges

Obtaining *management buy-in* for adopting an ABC methodology.	(1)
Choosing *appropriate cost drivers* for each activity.	(1)
Developing a *reliable data base* on the chosen cost drivers. This could be *expensive*. *Staff skills* and commitment in recording accurate data could be a challenge.	(1) (1) (1)
Deciding *how to handle costs that may be common across several activities,* e.g. insurance, especially when they represent a significant portion of total costs.	(1)
Choice of the *time period* to calculate cost rates per cost driver.	(1)
Any other valid point.	(1)
	Total 8
	Max 5

Some students only focussed on the possible difficulties in recording accurate data for each type of cost driver instead of implementation and operational challenges.

Part (d) New manufacturing investment

Metres of downpipe per year (domestic and industrial combined)	**Probability**	**Expected metres – Year 1**	**Year 2 2% growth**	**Year 3**	**Year 4**
50 000	50%	25 000			
60 000	30%	18 000			
70 000	20%	14 000			
		57 000	**58 140**	59 303	60 489
		(1) r/w	(½) r/w		

	Domestic	**Industrial**	
	R	**R**	
Selling price per metre – External	39,00	50,00	
Variable cost per metre (R39 x 100/145; R50 x 100/145)	**(26,90)**	**(34,48)**	(2) 1 each r/w
Contribution per metre – External	12,10	15,52	

Purchase price / Saving per metre – GRG	**38,00**	**49,00**	(1) for using R38 & R49 for saving
Variable cost per metre (R39 x 100/145; R50 x 100/145)	(26,90)	(34,48)	
Contribution per metre – GRG	11,10	14,52	

Demand – Volume	**Years**				
	1	**2**	**3**	**4**	
	metres	**Metres**	**Metres**	**metres**	
External market	57 000	58 140	**55 000**	**55 000**	(1) ½ each r/w. Limited due to max capacity
- Domestic (70%)	**39 900**	**40 698**	**38 500**	38 500	(1½) ½ each Y1, Y2, Y3 r/w
- Industrial (30%)	**17 100**	**17 442**	**16 500**	16 500	(1½) ½ each Y1, Y2, Y3 r/w
GRG internal use – Given	90 000	90 000	95 000	95 000	
- Domestic (70%)	**63 000**	**63 000**	**66 500**	66 500	(1½) ½ each Y1, Y2, Y3 r/w
- Industrial (30%)	**27 000**	**27 000**	**28 500**	28 500	(1½) ½ each Y1, Y2, Y3 r/w
Total	147 000	148 140	150 000	150 000	Note: Y3 & Y4 limited to max
- Domestic (70%)	102 900	103 698	105 000	105 000	Capacity
- Industrial (30%)	44 100	44 442	45 000	45 000	



Internal sales (with no machine) will be acquired at the supplier's or savings price, which is then relevant as any buyer will be able to acquire the volumes at this price. The new machine has a limited capacity which comes into play in year 3 onwards.

Investment evaluation	**Years**						
	0	**1**	**2**	**3**	**4**	**5**	
	R	**R**	**R**	**R**	**R**	**R**	
Sales – External							
- Domestic (Volume x R39)		**1 556 100**	**1 587 222**	1 501 500	1 501 500		(1) ½ each Y1, Y2 c
- Industrial (Volume x R50)		**855 000**	872 100	**825 000**	825 000		(1) ½ each Y1, Y3 c
Savings – GRG							
- Domestic (Volume x R38)		**2 394 000**	**2 394 000**	2 527 000	2 527 000		(1) ½ each Y1, Y2 c
- Industrial (Volume x R49)		**1 323 000**	1 323 000	1 396 500	**1 396 500**		(1) ½ each Y1, Y4 c
Variable costs							
- Domestic (Total vol x R26,90)		**(2 767 655)**	**(2 789 119)**	(2 824 138)	(2 824 138)		(1) ½ each Y1, Y2 c
- Industrial (Total vol x R34,48)		**(1 520 690)**	(1 532 483)	**(1 551 724)**	(1 551 724)		(1) ½ each Y1, Y3 c
Fixed costs – excl depr, rent (R1 350 – ((R3m-500k)/4) – (5% x 1 350k))		**(657 500)**	(657 500)	(657 500)	**(657 500)**		(1) ½ each Y1, Y4 r/w
Rent (5%x1 350k) + 3% growth p.a		**(67 500)**	**(69 525)**	(71 611)	**(73 759)**		(1½) ½ each Y1,2,4r/w
Cashflows		1 114 755	1 227 696	1 145 027	1 142 879		
Market research (R10k) – **sunk**							(½) r/w
Investment	**(3 000 000)**						(½) Must be Y0. r/w
Sale at end of life					**500 000**		(½) Must be Y4. r/w
Working capital	**(250 000)**				**250 000**		(1) ½ each Y0, Y4 r/w
Tax – 50% of current year		11 934	(73 877)	(76 304)	(146 003)		
Tax – 50% of previous year			11 934	(73 877)	(76 304)	(146 003)	(1) for 50% lag c
Net cash in-/outflow	(3 250 000)	1 126 689	1 065 753	994 846	1 670 572	(146 003)	
Discount factor @ 15%	1,0000	0,8696	0,7561	0,6575	0,5718	0,4972	(1) for 15% r/w
NPV p.a.	(3 250 000)	979 730	805 862	654 127	955 155	(72 589)	
Total NPV	72 285						(½) c
Positive NPV, therefore **accept** investment							(½) c

Note 1: Tax calc							
Cashflows above		1 114 755	1 227 696	1 145 027	1 142 879		
Wear-and-tear (40:20:20:20)		**(1 200 000)**	**(600 000)**	**(600 000)**	**(600 000)**		(1) r/w
Tax recoupment on sale					**500 000**		(½) Must be Y4 r/w
		(85 245)	527 696	545 027	1 042 879		
Tax @ **28%**		(23 869)	147 755	152 608	292 006		(1) for 28% r/w

Alternative – on contribution basis	**Years**						
	0	**1**	**2**	**3**	**4**	**5**	
	R	**R**	**R**	**R**	**R**	**R**	
Sales – External							
- Domestic (Volume x R12,10)		**482 928**	**492 586**	**465 983**	465 983		(1½) ½ each Y1, Y2, Y3 c
- industrial (Volume x R15,52)		**265 345**	**270 652**	**256 034**	256 034		(1½) ½ each Y1, Y2, Y3 c
Savings – GRG							
- Domestic (Volume x R11,10)		**699 517**	**699 517**	**738 379**	738 379		(1½) ½ each Y1, Y2, Y3 c
- industrial (Volume x R14,52)		**391 966**	**391 966**	413 741	**413 741**		(1½) ½ each Y1, Y2, Y4 c
Fixed costs – excl depr, rent (R1 350 – ((R3m-500k)/4) – (5% x 1 350k))		**(657 500)**	(657 500)	(657 500)	**(657 500)**		(1) ½ each Y1, Y4 r/w
Rent (5%x1 350k) + 3% growth p.a		**(67 500)**	**(69 525)**	(71 611)	**(73 759)**		(1½) ½ each Y1,2,4 r/w
Cashflows		1 114 755	1 227 696	1 145 027	1 142 879		

One of the most common errors made was that students did not take the savings into account in the appraisal or used the wrong rates when they did do so. Capital investment appraisals utilise the relevant cost approach. The savings that GRG obtains due to not having to purchase the downpipes anymore are a relevant benefit. Students also often forgot to include the variable cost of the internal transfers in the evaluation when not following the contribution approach.
It was concerning to note the number of students who made basic mathematical errors. The most common was that the total volume sold was multiplied by the sum of the two sales prices (eg. R39+R50=R89). Another error was that the total volume sold was multiplied by the average of the two sales prices (R44,0) (implying 50:50 sales of domestic and industrial) although the sales were 70% domestic and 30% industrial. A weighted average price (70%:30%) was acceptable.
Many students did not separate their tax calculations from the main appraisal and did everything as one calculation. This resulted in non-tax items such as working capital being included in the tax calculation and on the other hand that non-cash flow items such as wear-and-tear were deducted in the appraisal instead of only including the net tax benefit of the wear-and-tear allowance.

Total 28
Max 26

Part (e) – Hurdle rate

• The discount rate used to calculate the net present value (NPV) in a discounted cash flow (DCF) analysis to assess an investment opportunity is based on WACC and is often **adjusted for high/low risks or uncertainty** i.e. a higher/lower investment hurdle rate is applied.	(1)
• The projected cash flows in a DCF analysis are usually subject to a degree of uncertainty. A high hurdle rate could in certain circumstances **encourage bias in the projections.**	(1)
• Where a risk adjustment takes place as an **adjustment to the discount rate or to expected cash flows, or combination of both approaches**, • it is important to **avoid double counting, miscounting or understating risk**. • The **danger** of building up "additive models" for a variety of risk factors is **over discounting for risk**. (Based on IFAC (2012). Exposure draft: Project and Investment Appraisal for Sustainable Value Creation, Principle C3)	(1) (1) (1)
	Total 5
	Max 3

A fair number of students could identify that the hurdle rate was applicable in the case of risk or uncertainty but very few considered the dangers of using a hurdle rate.

Part (f) – Qualitative factors

TO: *Gauteng Rain Gutters (Pty) Ltd*
FROM: *Consultant*
DATE: *25 September 2013*
RE: *Qualitative factors to consider before investing in new machinery*

This memorandum is drafted in response to your request dated xx/xx/2013. It sets out – in brief – the qualitative factors that GRG should consider before investing in the new manufacturing machinery, apart from the financial aspects already considered.

• Will the **quality** of the downpipes be the same or better than that of the current supplier? If the quality is worse, then GRG's installation division will be negatively impacted.	(1)
• Will the **delivery time** of the new division be the same or better than that of the current supplier?	(1)
• Will **skilled staff** be available to operate the new machinery?	(1)
• Will **suitable managers** be available to manage the new division or will the current management have sufficient time and expertise to manage the new division without neglecting the installation division?	(1)

• Will **government policy and plans** impact on the growth in demand for downpipes (e.g. provision of schools, hospitals, low-cost housing)?	(1)
• Will the new machinery be **reliable** and not break-down causing delivery interruptions and delays?	(1)
• Will GRG have the **ability / capacity to maintain** the new machinery?	(1)
• How easily accessible would **replacement parts** be?	(1)
• What will happen **after the four years** useful life of the machinery? Will the company replace it? If not, will they be able to source downpipes again from their current supplier at the same price?	(1)
• Will the **relationship with the current supplier** be negatively impacted if downpipes are no longer purchased from them and GRG is in competition with them in the provision of downpipes (e.g. possible volume discounts on other purchases from the supplier)?	(1)
• Are GRG's current **premises** suitable for the new division or will new premises have to be obtained? Would existing premises need to be modified?	(1)
• What **environmental** impacts will the new division have e.g. pollution and disposal of waste?	(1)
• What **social** impacts will the new division have e.g. job creation, labour relations with current employees?	(1)
• Any other valid point (max 1). Note that it should be a qualitative factor, **not quantitative / financial**.	(1)
	Total 14
	Max 6

• *Communication skills – layout and structure*	(1)
• *Communication skills – appropriate style*	(1)



A large number of students included *quantitative* factors in their discussions instead of only qualitative factors e.g. source of financing for the investment or impact of price changes. These factors earned no marks. Most students presented their answers in the format of a memorandum although some did not include all the components of the memorandum, especially concluding remarks and closing. A very important matter is that students are not allowed to include their names (or personal information that could identify them) in their exam scripts at all. It is a very serious contravention of exam regulations to do so. Another matter is that students should not include "CA(SA)" even if a fictitious name is used. Use of the CA(SA) designation is reserved for qualified Chartered Accountants only.

Part (g) – Break-even and margin of safety (Year 1)

	Year 1	
	R	
Total sales & savings (External + GRG) (1 556 100 + 855 000 + 2 394 000 + 1 323 000)	6 128 100	
Total variable costs (2 767 655 + 1 520 690)	(4 288 345)	
Total contribution	1 839 755	(1) c
Contribution margin % (1 839 755 / 6 128 100)	30,02%	(1) c

Break-even Rand value = $\frac{\text{Total fixed costs}}{\text{Contribution \%}}$

= $\frac{\text{R1 350 000}}{\text{30,02\%}}$ (1) r/w for fixed costs

= R4 496 759 (1) c for correct formula application

Margin of safety = $\frac{\text{Current sales – Break-even sales}}{\text{Current sales}}$

= $\frac{\text{R6 128 100 – R4 496 759}}{\text{R6 128 100}}$

= 26,62% (1) c for correct formula application

The break-even Rand value was required, however many students tried to calculate it by first calculating the break-even in units. This indicates a lack of knowledge of the appropriate formula to use and resulted in wasted time. Many students omitted the savings and related variable cost from their calculations. Although many students could correctly calculate the margin of safety percentage there were also numerous students who could not apply the formula correctly.

Part (h) Transfer pricing

First year

Minimum price = incremental cost + opportunity cost

Opportunity cost:

As calculated in (e):

	Year 1 Metres
External market	57 000
- Domestic (70%)	39 900
- Industrial (30%)	17 100
GRG internal use – Given	90 000
- Domestic (70%)	63 000
- Industrial (30%)	27 000
Total	147 000
- Domestic (70%)	102 900
- Industrial (30%)	44 100

The total demand of 147 000 is within the maximum capacity of 150 000. It will therefore not be necessary to sacrifice any external sales to meet internal usage and consequently there is **zero opportunity cost in year 1**. (1) r/w

Incremental cost = variable cost + any increase in fixed cost

There are **no increases in fixed costs**. (½) r/w
Incremental cost is therefore only variable cost (as calculated in (e)):
Variable cost per domestic metre = R39 x 100/145 = **R26,90** (½) c
Variable cost per industrial metre = R50 x 100/145 = **R34,48** (½) c

Maximum transfer price in 1st year = Market price
Domestic = **R38** per metre (½) r/w
Industrial = **R49** per metre (½) r/w

Total: 3½
Max: 3

Fourth year

Opportunity cost:

External demand (Y4) = 57 000 x $(1,02)^3$
= 60 489 metres (½)

Total demand (Y4) = 60 489 + 95 000 [GRG] (½)
= 155 489

Maximum capacity = 150 000 metres

Therefore 5 489m will be sacrificed. (½)

Opportunity cost = contribution lost
GRG will sacrifice the external sales with the lowest contribution:

	Domestic R	Industrial R	
Selling price per metre – External	39,00	50,00	
Variable cost per metre (R39 x 100/145; R50 x 100/145)	(26,90)	(34,48)	
Contribution per metre – External	12,10	15,52	(1) c

Domestic has the lowest contribution.

Opportunity cost = 5 489m x R12,10 (1) c
= R66 417

Incremental cost = (66 500 x R26,90) + (28 500 x R34,48) + 0 [fixed cost] (2) c
= R1 788 850 + R982 680
= R2 771 530

Minimum transfer price in 4th year = R2 771 530 + R66 417
= R2 837 947

Domestic = R26,90 + ((R66 417 x 70%)/66 500) (1) c
= R27,60 per metre

Industrial = R34,48 + ((R66 417 x 30%)/28 500) (1) c
= R35,18 per metre

Maximum transfer price in 4th year = Market price
Domestic = R38 per metre (½) c
Industrial = R49 per metre (½) c

Total: 8½
Max: 7



Although most students identified the variable cost per metre as the minimum transfer price very few considered opportunity cost or incremental fixed costs in the first year. The question about the transfer price, in the **fourth year** was poorly answered. Many students only gave a vague discussion with no calculations. Those students that did do a calculation often did not identify that the opportunity cost was the **contribution lost** on the external sales sacrificed and that this should be allocated across all the units transferred when calculating the price per metre. Opportunity cost refers to contribution lost, not sales price.

Other factors that GRG should consider

- Are the prices to be charged by GRG (R39 & R50) market related prices? (It might be possible to ask a higher price.) (1)
- Is the market for downpipes a perfectly competitive market? (1)
- Are there no incremental fixed costs? Will the fixed costs be the same even at much lower levels of production (e.g. if only the external demand was manufactured)? (1)
- Is it not possible to increase production capacity in year 4 by means of overtime? This will result in increased labour costs and maintenance costs. (1)
- Will the managers of the two division's performance bonus be based on the results of the divisions? (1)
- The final transfer price between the two divisions will be between the minimum and maximum prices and should be negotiated to allow the manufacturing division some profit. This should lead to goal congruence and profit maximisation for the company as a whole. (1)
- Transfers at the minimum variable cost could de-motivate the manufacturing division since they will not make any profit on the transfers. (1)
- Any other valid point (max 1). (1)

Total **8**
Max **4**

- *Communication skills – logical argument* (1)

Part (i) ROI, residual income and other measures

ROI and residual income as bases for performance measurement:

- If the ROI (return on investment) and RI (residual income) calculation is to measure the performance of the divisional manager, he/she might be de-motivated if uncontrollable costs were included or conversely could be unfairly advantaged if uncontrollable income was included. (1)
- ROI and RI are both short-term financial measures. This could motivate decisions that are beneficial in the short-term but less beneficial in the long-term. (1)
- They ignore qualitative issues. (1)
- They ignore the time value of money. (1)
- ROI is widely used and most managers understand it. (1)
- ROI is expressed in percentage terms which makes it comparable between different sizes of projects, divisions and companies. (1)

- RI is expressed in absolute (monetary) terms which make a comparison between different sizes of divisions more difficult. (1)
- RI is more flexible than ROI since different cost of capital charges can be used for investments that have different levels of risk. (1)
- The use of ROI to evaluate managers might not encourage goal congruence in the company e.g. ROI of proposed project may be more than WACC but less than current ROI of division which could lead to incorrect rejection because it will lower the average ROI of the division. Also vice versa. (1)
- Projects with a positive NPV might be rejected. (1)
- ROI and RI can vary depending on which basis assets are valued (e.g. opening or closing or average balances). (1)
- ROI and RI can vary depending on the accounting treatment of assets e.g. impairment of assets will lower the value of assets which could lead to higher ROI and RI in future. (1)
- ROI and RI are not suited to service based industries because of low capital investment. (1)
- ROI does not take into account the risks of a division or project (unless it is compared to a risk adjusted WACC for the particular division). (1)
- RI encourages managers to act in the best interests of the company as well as their own best interests. (1)
- Any other valid point. (Max 1) (1)

Total 16
Max 4

List and motivate other (non-ratio) measures

- **Trends** in the following measures could be monitored to try to **improve** them: (1)
  - wastage percentages (1)
  - time taken per installation (1)
  - customer waiting time after request and before installation time (1)
  - number of customers (1)
  - market share (1)
  - staff turnover rate (staff retention) (1)
  - revenue per employee (1)
  - training costs per employee (1)
  - number of injuries on duty (1)
  - average number of days sick leave per employee (1)
  - number of insurance claims for incidents (breakages at customer premises, injuries, vehicle accidents) (1)
  - number of fines for traffic offences whilst travelling to and from customer premises (1)
  - amounts spent with BEE suppliers (also as a percentage of total procurement spend) (1)
  - number of complaints (1)
  - cost of repairing gutters and downpipes after complaints of poor quality installations (also as a percentage of turnover) (1)

- Customer satisfaction questionnaires could be used to highlight problems as well as good service. (1)
- Annual employee satisfaction surveys could indicate if employees are becoming more or less satisfied. (1)
- Any other valid point. (Max 1) (1)

Total 19
Max 4

- *Communication skills – logical argument* (1)



Despite the numberous marks available, students did not do well in this question. Many students only gave a general definition or explanation of ROI and RI but did not discuss whether they were good (or bad) bases for performance measurement **and why** (i.e. advantages and disadvantages). Simply stating that they are good (or bad) measures does not suffice. The section about other measures was also poorly answered with many students listing financial measures (e.g. EVA, IRR) or not accurately defining what the measure should be, i.e. merely stating "quality", social responsibility" or "brand" does not say what the **measure** is.

Part (j) Risks – two proposals

Proposal 1 – Angolan contract

- Currency risk: Rand could strengthen against the US$ (1)
- Risk due to inflation in Angola (1)
- Credit risk of the Angolan government – they could default or take long to pay causing cashflow problems for GRG (1)
- Social or political problems in Angola could hinder movement of employees around the country and safety of employees (Political / country risk) (1)
- Reliance on Angolan infrastructure which is likely to be poor. (1)
- Potential barriers to free capital flow in and out of Angola (exchange controls) (1)
- Legal risk: risk relating to the Angolan legal and regulatory environment (lack of knowledge thereof in particular) (1)
- Risk of contravention of legislation and corporate governance and negative impact on GRG's reputation if bribes have to be paid in Angola (1)
- The existing workforce is already working at full capacity. Sending local employees to Angola would result in a loss of business in SA. (1)
- If employees are hired in Angola there is a risk that they might not be competent in using GRG's machines and methods. There could also be a communication / language problem (cultural risk). (1) (1)

- Higher wages might have to be paid to SA employees in Angola to compensate for being away from home. (1)
- Any other valid point. (Max 1) (1)

Total 13
Max 3

Proposal 2 – Dept of Defence

- Credit risk of the SA government – they could take long to pay causing cashflow problems for GRG (1)
- The existing workforce is already working at full capacity. Using existing employees on the project would result in a loss of existing business. (1)
- If new employees are hired in the northern provinces there is a risk that they might not be competent in using GRG's machines and methods. (Training requirements). (1)
- Higher wages might have to be paid to existing employees on the project to compensate for being away from home. (1)
- Poor infrastructure in certain of the northern provinces. (1)
- Any other valid point. (Max 1) (1)

Total 6
Max 2

- *Communication skills – appropriate style* (1)



Students generally did well in this section although some only listed generic risks (e.g. political risk, legal risk or operational risk) without discussing **how or why** this was applicable to GRG and the two proposals.

QUESTION 2

SKY (PTY) LTD **SUGGESTED SOLUTION**

(Source: Exam 2012 MAC4862)

R/W – Right/Wrong mark ***C – Consequential mark***

Part (a)

Product:	**Alpha**	**Nebula**	**Terra**	**Total**	
	R	**R**	**R**	**R**	
Sales (Given)	1 300 000	2 275 000	2 120 000	5 695 000	
Direct materials (Given)	(450 000)	(910 000)	(940 000)	(2 300 000)	
Direct labour (Given)	(400 000)	(1 040 000)	(640 000)	(2 080 000)	
Overhead:					
Machine related	(80 000)	(78 000)	(32 000)	(190 000)	(2) (½ each) R/W
Batch related	(50 000)	(65 000)	(40 000)	(155 000)	(2) (½ each) R/W
Gross contribution	320 000	182 000	468 000	970 000	
Product specific OH	(330 000)	(75 000)	(185 000)	(590 000)	(2) (½ each) R/W
Net contribution	(10 000)	107 000	283 000	380 000	
General overhead				(15 000)	(1) R/W
Operating profit				365 000	
Finance cost				(97 900)	(1) R/W
Net profit				267 100	(1) R/W

Overheads:
Machine hour related = R190 000 / 47 500 machine hours = R4 per machine hour
Batch related = R155 000 / 310 batches = R500 per batch.

Very few students produced the layout required incorporating ABC and product specific overheads. It was clear that this section of the work was not studied at all. Product specific OH is treated as a direct cost, therefore net contribution. It is common that the newly prepared statement will change the result of the given loss-making product.

Part (b) **Max 8**

• The original profit statement shows that two of the antenna products are profitable while one is making losses. This statement is unsuitable for decision making given that the statement incorrectly indicates Nebula as being loss making.	(1)
• Although it shows the revenues and costs attributed to each product it makes arbitrary assumptions with regard to the company's overhead costs and uses these assumptions to allocate the overhead costs to each product using a blanket recovery rate.	(1)

• The original profit statement does not consider the causes of the overhead costs and the extent to which they are avoidable if the company were to decide on particular courses of action.	(1)
• For example, some of the overhead costs are product specific and would be avoided if the product were to be discontinued. This is not clear from the statement that has been presented.	(1)
• Alternative: the revised statement does not allocate facility related overhead costs (i.e. R15m) to the different products – hence the revised statement indicates the avoidable overheads only.	(1)
	Max 4

 The comments refer directly to the statements given and prepared. Regurgitation of theory garnered no marks. This also applies to part (c). Some of the comments are valued without doing the calculations.

Part (c)

• The revised profit statement shows that **Product Alpha** has a negative net contribution and therefore from a financial perspective it should be **discontinued** as this would **increase the company's profits by R10 000** given that only avoidable fixed costs have been taken into account in calculating the R10 000.	(1) (1)
• This is **in contrast to the original profit statement** which showed that products Alpha and Terra were profitable and product Nebula was loss making.	(1)
• Products **Nebula and Terra** should be **continued** because they both have a positive net contribution.	(1)
• The discontinuance of product Alpha will **release resources** that were previously used by that product – this would be particularly relevant if limiting factors exist. If there is sufficient demand for products Nebula or Terra then Sky may be able to **increase its output** of these other products or new products and increase its profits by even more than R10 000.	(1) (1)
	Max 4

Part (d)

The specific fixed cost of product Alpha is **R330 000**. (1) R/W
(**Total** FC = R330 000)

Contribution per unit = R320 000 / 10 000 units (1) C
(must divide by 10 000)
= R32 per unit

$$\text{Break-even point in } \textbf{units} = \frac{\text{Fixed costs}}{\text{Contribution per unit}}$$
= R330 000 / R32 (1) C
= 10 312,500 units
= 10 313 units
= 104 batches (Must round up) (1) C

ALTERNATIVE: (Batches)

The specific fixed cost of product Alpha is **R330 000**. (1) R/W
(**Total** FC = R330 000)
No. of batches = 10 000 units / 100 units per batch = 100 batches
Contribution per batch = R320 000 / 100 batches
= R3 200 per batch (1) C
Break-even point in **batches** = R330 000 / R3 200 (1) C
= 103,125 batches
= 104 batches (Must round up) (1) C

ALTERNATIVE:

The specific fixed cost of product Alpha is **R330 000**. (1) R/W
(**Total** FC = R330 000)
The gross contribution to sales ratio of Alpha is R320 000/R1 300 000 = **0,24615 (or 24,615%)** (1) C

Therefore, the breakeven sales value is R330 000 / 0,24615 = **R1 340 625**; (½) C
and the breakeven sales volume is R1 340 625 / R130 = 10 312,500 units = **10 313 units** (rounded up).
(½) C
However, since production must be in batches of 100 units then to break even **104 batches** would have to be produced and sold. (1) C


The breakeven 'required' referred to a product level breakeven,implying that the usual (normal) contribution has to be adjusted for direct fixed costs.The product specific FOH is deemed the fixed cost for calculation purposes.This 'required' was another indicator of the layout required for (a).

Part (e)

JIT purchasing involves having an arrangement with a small number of key suppliers where the supplier is able to provide raw materials or components **on demand or with a very short lead time**.	(1)
This means that the company can hold zero or very little inventory thus **reducing the costs involved with holding inventory** including **storage costs, insurance costs and obsolescence costs**.	(2)
The **costs involved with ordering inventory** may however **increase**.	(1)
The use of a small number of suppliers should also **reduce administrative costs** for the company and result in **greater quantity discounts**.	(1)

The successful operation of a JIT purchasing system involves the company working together with their suppliers to ensure that they can **rely on receiving supplies at the right time and at the required quality level**. This should result in a **reduction in quality control costs** for the company. Quality standards should improve resulting in **lower wastage** in the production process	(1) (1) (1)
JIT aims to **reduce lead times** which should improve the company's ability to meet customer delivery dates and to respond to changes in market needs, thereby improving competitiveness.	(1)
Cash flow should improve / **Less cash** is tied up in inventory / Any other point related to improved cash flow.	(1)
	Max 4

 Note the focus on costs in the above.

Part (f)

Residual income measurement – divisions

	Antenna R'000	**Converter R'000**	**Processor R'000**		
Operating profit	365	6 160	12 450	(1) R/W	For all 3
Add back: Allocated FOH cost	15	1 694	4 235	(2) R/W	1 – con-verter 1 - other
[Converter: R4,9m – R3,206m = R1,694m]					
Controllable profit	380	7 854	16 685		
WACC charge (12%)	(106,8)	(1 455)	(3 943,2)	(3) R/W	(1 each)
Divisional assets (Given)	890	12 125	32 860		
Residual income (RI)	273,2	6 399	12 741,8	(1½) R/W	

 Many students used the incorrect base profit eg. NPBT and/or did not add back the allocated FOH which is not controllable. Performance mostly focussed on profit margins, ignoring the use of assets altogether.

One other measure:				
RI as % of Turnover (RI/TO)	4,80%	12,70%	10,19%	(1½) C
OR				OR
Return on assets (RI/Div. assets)	30,70%	52,78%	38,78%	(1½) C
OR				OR
Return on assets or investments (Controllable profit / Div. Assets)	42,70%	64,78%	50,78%	(1½) C

- The converter division **performs best** in terms of the other measure while the processor division has the highest RI. (1) C
  OR: any other acceptable, relevant conclusion.

*RI & another measure – **Max 7***

Other information

- Capacity utilisation of all divisions (1)
- Market share % for their products (1)
- Growth experienced (1)
- Return of malfunctioning products/Quality measures (1)
- Divisional bad debt, discounts etc. (1)
- Cash generated by division – debtors days o/s (1)
- Age of asset base / When last assets were replaced (1)

*Other information – **Max 2***

Max **9**

 ROA and RI/Assets use the numbers already there, thus no additional calculations. The other information is almost generic where multiple products or divisions are considered.

Part (g)

The internal sales volume is 70 000 converters. The converter division could have sold a further 42 000 converters to the external market if it had extra capacity or reduced its internal sales (i.e. sales foregone). Therefore 42 000 converters were sold to the processor division at market price and the balance was sold at variable cost.

An analysis of the sales is therefore as follows:

	Alternative: Internal	**Internal**		**External**	**Total**	
	@ weighted avg.	**@ cost**	**@ MV**	**@ MV**		
No. of converters	**70 000** *@R332,50*	**28 000** *@R250*	42 000 *@R387,50*	70 000 *@R387,50*	140 000	(1) for 28 000 R/W (or alt. 70 000)
		R'000	R'000	R'000	R'000	
Sales value	**23 275**	**7 000**	**16 275**	**27 125**	**50 400**	(2) (½ each) R/W
Variable cost (R250 each)	17 500	7 000	10 500	17 500	35 000	

Variable cost per converter = R35 000' / 140' = R250 per converter
Current market price per converter = R387,50 (given)

 Students struggled with producing a logical layout and this lead to poorly structured and illogical comments in section (h).

Part (h)

Transfers @ market value: Transfers are done at incremental (variable) cost plus opportunity cost (i.e. lost contribution) which is fair.	(1)
Transfers @ cost: The converter division has sold converters to the processor division without deriving any financial benefit. If the processor division had bought them at market value the cost to the processor division would have been **R10,85m** (28 000 units x R387,50 market price) which is **R3,85m** (R10,85m – R7m) greater than the current transfer price. Alternative: (R387,50 – R250) x 28 000 = **R3,85m** profit	(1) (1)
While it **may not be appropriate** for the processor division to **pay the full market price** it does seem **unfair that all of the profit** from the use of these components **accrues to the processor division** and therefore a transfer price that accrues some reward to the converter division for the supply of the components would be **fairer to both divisions**.	(1) (1) (1)
Any transfer price **above variable cost** would **reduce** the profits of the processor division and **increase** those of the converter division by the same amount. For example if the difference between variable cost and market price were shared equally then the change in profit of each division would be R1,925m (R3,85m ÷ 2).	(1) (1)
External demand decrease/increase: If the **external demand** for the components were to **decrease**, then **more** of the components supplied to the processor division would be **transferred at variable cost** thus **lowering** the profits of the converter division, but i**ncreasing** the profits of the processor division. If the **external demand** were to **increase** then the opposite effect would occur until all of the internal transfers were being made at the external selling price.	(1) (1)
	Max 6

Part (i)(i)

Revised sales and variable cost analysis – including increase in capacity and reduction in variable cost

	Internal		**External**	**Total**	
	@ cost	**@ MV**	**@ MV**		
Previous units	28 000	42 000	70 000	140 000	
Increase / Decrease	20 000	(20 000)	20 000	20 000	
New units	**48 000** *@R200*	**22 000** *@R387,50*	**90 000** *@R387,50*	**160 000**	(2) (½ each) R/W
	R'000	R'000	R'000	R'000	
Sales value	**9 600**	**8 525**	**34 875**	**53 000**	(2) (½ each) R/W
Variable cost (R200 each)	9 600	4 400	18 000	32 000	

New variable cost per unit = R250 – (R250 x 20%) = R250 – R50 = R200 per unit
Current market price per converter = R387,50 (given)

The inability to start off with a logical structure that showed movements/changes clearly, again lead to the downfall of many. Students read 'transfer pricing' and applied their (theoretical) knowledge without thinking it through. It should again be noted that part (i) was from the Converter divisions' perspective and part (ii) from the businesses' perspective – all savings to be considered. A further complicating factor was the investment cost (of R30,8m). This implied that a NPV approach was required, which few considered, despite all the components, (period CoC, NPV) being stated in the question.

Internal sales at cost:

Volume has increased but variable cost per unit has decreased and consequently the transfer sales price per unit has also decreased. No contribution is derived from these internal sales therefore **no benefit** to the converter division. (1)

Sales at market value (internal and external)

Volume has remained the same (sales @ MV): (1)
Before = 42 000 + 70 000 = 112 000 units
After = 22 000 + 90 000 = 112 000 units

Sales price per unit has remained the same.
Therefore no benefit to converter division in terms of increased sales value for sales @ MV: (1)

Before = R16 275' + R27 125' = R43 400'
After = R8 525' + R34 875' = R43 400'

Internal sales however increase by R2 600 000 annually (R9,6m – R7m) which together with the saving in variable costs amounting to R3 000 000 (R35m – R32m) results in a total **annual savings** of **R5 600 000.** (1)

OR

Reduction in variable cost of units sold at market value (internal and external) is however a benefit to the converter division:

Saving = 112 000 x R50 = **R5,6m per annum** (1)

Using the 12% annuity factor for 7 years this saving has a present value of:
R5,6m x 4,564 = **R25,558m** (OR n =7, i = 12, PMT = 5,6m, COMP PV) (1) C
(If NPV calculated)

Since the capital cost of the equipment is R30,8m with no residual value the investment is **not financially viable** from the converter division's perspective. (NPV = (R30,8m) + R25,558m
= (R5,242m) (negative)) (1)

Max 8

Alternative

The investment has two effects: the increase in the converter division's capacity to 160 000 units and the 20% reduction in its variable cost. From the converter division's perspective the benefit of these effects is **diluted** due to the internal sales and the transfer pricing policy.	
Increase in capacity If the capacity of the converter division is increased to 160 000 then it will increase its external sales, but in doing so will **reduce the volume of external sales foregone** by selling the components to the processor division. Capacity increases by 160 000 – 140 000 = 20 000 units (to be sold externally) External sales foregone reduces: 42 000 – 20 000 = 22 000 units Therefore new internal sales @ MV = 22 000 units And new internal sales @ cost = 28 000 + 20 000 = 48 000 units Therefore the effect of the additional capacity would be to **transfer an additional 20 000 units (volume) at cost**. Thus there is **no financial benefit** to the converter division in terms of internal sales. The total value of internal sales will decrease. External sales will increase by 22 000 x R387,50 = R8,525m per annum. However, total sales at market value will remain the same = 22 000 (internal) + 90 000 (external) = 112 000 units.	(1) (½) (½) (1) (1)

Reduction in variable cost The converter division sells 50% of its present capacity internally, and 28/70 of this is transferred to the processor division at variable cost, therefore any cost savings arising in respect of this portion will be **passed on to the processor division** due to the transfer pricing policy. The cost saving that will accrue to the converter division will therefore be limited to items sold at market value. This amounts to: Items sold at market value = 50% + (42/70 x 50%) = 50% + 30% = 80% (existing sales) (OR ((42 000 + 70 000) / 140 000) = 80%) Variable cost of items sold at market value = 80% x R35m = **R28m per annum** 20% cost saving thereon = **R5,6m per annum**	(1) (1) (1) (1)
Using the 12% annuity factor for 7 years this saving has a present value of: R5,6m x 4,564 = **R25,558m** (OR n =7, i = 12, PMT = 5,6m, COMP PV)	(1)
Since the capital cost of the equipment is R30,8m with no residual value the investment is **not financially viable** from the converter division's perspective.	(1)
	Max 8

Part (i)(ii)

However, if the investment were to be evaluated from the position of the whole organisation then consideration would be given to the benefits that accrue to the processor division as a consequence of the transfer pricing policy. These benefits can be identified as the difference between the original and revised values of internal sales. The original value was R23,275m (R7m + R16,275m - see above).	
The revised transfer value will be: 48,000 converters @ revised cost of R200 (or 48k x R250=R12m–20%) R **9,600m** 22,000 converters @ market value R387,50 R **8,525m** Total R18,125m	(1) R/W (1) R/W
A saving to the processor division of **R5,150m** per annum (i.e. R23,275m – R18,125m).	(1) R/W
If this were added to the converter division saving of R5,6m, the total is **R10,750m** per annum which would have a present value of **R49,063m** (i.e. R10,750m x 4,564) (or n =7, i = 12, PMT = 10,75m, COMP PV) which clearly makes the investment of R30,8m worthwhile.	(1) (1)
OR	
The incremental saving will be R5,150 x 4,564 = **R23,504m** (or n =7, i = 12, PMT = 5,15m, COMP PV) which together with the variable cost saving of R25,558m totals **R49,062m** which will make the project feasible.	(1) (1)
	Max 4

Part (j)

Impact of new equipment on product cost

• Current cost (depreciation) R3 206 000 ÷ 140 000 (per unit)	R22,90	(1) R/W
• New equipment R30,8m ÷ 7 ÷ 160 000 (per unit)	R27,50	(1) R/W
Increased overhead cost	4,60	
• Variable cost (saving) 20% x R250 or R250 – R50 = R200	50,00	(1) R/W or R200
Net decrease in product cost	45,40	(1) C
		4

Students were lost in this rather basic costing exercise where capacity, fixed and variable cost all changed. Compare the costs that will change and as normal in a relevancy exercise, ignore allocated costs.

Part (k)

Strategy for installing new equipment

- New equipment should be installed incorporating the festive season closure. (Do not order immediately and do not install as soon as possible.) (1)
- This will lead to loss of production of 3 – 4 weeks – preferably in the new calendar year. (1)
- Based on the current depreciation per unit (FOH per (j)), the equipment is good for a further (R2 061 000 ÷ 22,9) 90 000 units. (1)
  OR R3 206 000/12 = R267 167 per month, therefore R2 061 000/R267 167 = 7,7 months
- Current (2012) monthly production is (140 000 ÷ 12) 11 667 units (weekly 2 858). (1)
- The lost production (approximately 11 667) should be manufactured in the period September to December before closure by way of (1)

  - longer shifts (overtime) (1)
  - weekend manufacture (1)

- Excess units manufactured in the designated period will not be a problem as

  - demand is stated as a minimum of 112 000 units externally (1)
  - it may provide buffer inventory allowing workers to get fully conversant with new equipment. (1)

- Must include in plan an opportunity to train staff to use the new machine. (1)

Max 5

Students exhibited a complete lack of understanding of the industrial environment in SA as well as reality, with most strategic comments simply not being implementable. One will attempt to minimise the impact on your production. For example: Eskom attempts to schedule most of their maintenance work in the summer months when usage is lower.

Part (l)

	Note	R	
Marketing manager – meeting	1	Nil	
Material A (110 x R22,50)	2	2 475	(1)
Material B (40 x R19,00) OR (30xR19)+(10xR19)= 570 + 190	3	760	(1)
Components (500 x R8,00)	4	4 000	(1)
Direct labour	5	4 720	
(160 x R27,00)		4 320	(1)
(10 x R40,00)		400	(1)
Machine hours (25 x R15,00)	6	375	(1)
Fixed overhead	7	Nil	
Total relevant cost		**12 330**	

Notes:

1	The **marketing manager** has already had this meeting with the potential client, therefore the relevant cost is Nil firstly because it is a **past cost**, and secondly because even if it were future the manager is paid an annual salary and therefore there is **no incremental cost** to the processor division.	(1)
2	**Material A** is in **regular use** by the processor division and consequently its relevant value is its **replacement cost**. The historical cost is not relevant because it is a past cost and the resale value is not relevant since the processor division is not going to sell it since the material is in regular use and therefore must be replaced.	(1)
3	**Material B** is to be purchased for the contract therefore its **purchase cost is relevant**. Although only 30 litres are required for the work the minimum order quantity is 40 litres and as the processor division has no other use for this material and there is no indication that the unused 10 litres can be sold, the **full cost of purchasing the 40 litres is the relevant cost**.	(1) (1)
4	The **components** are to be purchased at a cost of R8,00 each. This is a relevant cost because it is future expenditure that will be incurred as a result of the work being undertaken.	(1)
5	**Direct labour:** Since **75 hours** of spare capacity are available which have a **zero relevant cost**, the relevant cost relates only to the other 160 hours (235 – 75). The division has two choices: either use its existing employees and pay them overtime at R32,00 per hour which is a total cost of **R5 120** (160 x R32,00); or engage the temporary staff which incurs their cost of **R4 320** (160 x R27,00) plus a supervision cost of **R400** (10 x R40,00) which equals **R4 720**. The relevant cost is the cheaper of these alternatives which is to use the temporary employees.	(1) (1)
6	**Machine hours:** The machine is currently being leased and it has spare capacity so it will either stand idle or be used on this work. The **lease cost will be incurred regardless (irrelevant)** so the only relevant cost is the **incremental running cost of R15,00 per hour**.	(1) (1)
7	**Fixed overhead costs** are incurred whether the work goes ahead or not so it is **not a relevant cost**.	(1)

Max 14

Students exhibited a dismal level of knowledge iro relevancy. This was exacerbated by unit costs being mixed up with total costs. Explanations often took the format of 'this is relevant' or 'not relevant'. The basic considerations of change as a result of the decision and cash flow were ignored. Remember this layout as there are always the 'soft' marks to consider.

Part (m)

Two constraint possibilities were 'highlighted': machine production as less hours available due to repairs and the fire at the supplier of material G resulting in a specific quantity of G being available.

Calculation of constraints

Machine hours	**Hours**	Alternative	
Required:			
- Green: 7 300 kg x 0,20 hrs/kg	1 460	1 460	(½) R/W
- Grow: 5 000 kg x 0,15 hrs/kg	750	750	(½) R/W
- Nutri: 3 000 kg x 0,30 hrs/kg (or 2 800kg x 0,30hrs)	900	840	(½) R/W
Total (original budgeted machine hours)	3 110		
Needed		3 050	
Available (3 110 hrs – 180 hrs)	2 930	2 930	(1) R/W
Shortage = Constraint	(180)	(120)	

Material G	**Kg**	Alternative	
Required:			
- Green: 7 300 kg x 0,25 kg	1 825	1 825	(½) R/W
- Grow: 5 000 kg x 0,50 kg	2 500	2 500	(½) R/W
- Nutri: 3 000 kg x 0,40 kg (or 2 800kg x 0,40kg)	1 200	1 120	(½) R/W
Total	5 525	5 445	
Available	5 600	5 600	
Surplus, therefore NO constraint for Material G	75	155	(1) R/W

Variable cost per product

[Remember this short-cut]:	**Green R**	**Grow R**	**Nutri R**	
Total cost (given)	113 150	85 000	59 400	
Less: Fixed overhead (given)	(29 200)	(15 000)	(18 000)	
Variable cost	83 950	70 000	41 400	(1½) R/W
No. of kilograms	7 300	5 000	3 000	
Variable cost per kg	11,50	14,00	13,80	(1½) R/W
Machine hours per kg	0,20	0,15	0,30	
Variable cost per machine hour	57,50	93,33	46,00	(1½) R/W
Ranking based on **lowest** variable cost	2	3	1	(1½) C

Constraints are often driven by contribution per limiting factor. In this case, the lowest **cost** per limiting factor (machine hours) was used. The obvious may not be the answer as no prices were given in the question. Students worked with total cost or assumed that the 5 600kg of material G was a limiting factor.

Production plan

	Machine hours	
Available	2 930	(½) R/W
Committed production	(1 325)	
Green 4 000 kg x 0,20 machine hrs	(800)	(½) R/W
Grow 500 kg x 0,15 machine hrs	(75)	(½) R/W
Nutri 1 500 kg x 0,30 machine hrs	(450)	(½) R/W
Available	1 605	
1st ranked: Nutri: 3 000 kg needed but limited to 2 800 kg less committed kg of 1 500. Therefore additional = 2 800 – 1 500 = **1 300 kg** x 0,30 machine hrs	(390)	(2) C
Available	1 215	
2nd ranked: Green: 7 300 kg needed (not limited) less committed kg of 4 000. Therefore additional = 7 300 – 4 000 = 3 300 kg x 0,20 machine hrs	(660)	(1) C
Available	555	
3rd ranked: Grow: 5 000 kg needed (not limited) less committed kg of 500. Therefore additional = 5 000 – 500 = 4 500 kg x 0,15 machine hrs = 675 hrs. Only 555 available. 555 hrs / 0,15 = **3 700 kg**	(555)	(2) C
	–	

The production plan should always first take care of the **committed production**, then the remainder.

Summary	**Committed Kg**	**Additional Kg**	**Total Kg**	
Green	4 000	3 300	**7 300**	(½) C for 7 300
Grow	500	3 700	**4 200**	(½) C for 4 200
Nutri	1 500	1 300	**2 800**	(½) C for 2 800
Total	6 000	8 300	**14 300**	

Max 15

Part (n)

Normal tax implication for the Foundation:

- As the foundation is a registered Public Benefit Organisation (PBO) **all** amounts received or accrued to the company (otherwise than from business activities) will be **exempt** from normal tax in terms of Section 10(CN). (1)

Normal tax implications for the Company:

- The inventory donated would have been deducted in terms of **Section 11(a)** when it was **purchased** OR when included in **opening stock** (Section 22(2)). (2)

- When the inventory is donated there must be **recoupment** in terms of Section 22(8). As the institution to which it is donated qualifies for a Section 18A deduction (i.e. a Public Benefit Organisation) Section 22(8)(c) applies and the amount to be included must be the same as the amount previously deducted. Therefore the **cost price** of the stock donated must be included as a recoupment and not the market value of the stock. (2)

- The inventory donated qualifies as a **deduction in terms of section 18A** (donated to a Public Benefit Organisation). The value deductible will be the amount included in terms of section 22(8) (i.e. the **cost price**). The deduction will be **limited to 10%** of the taxpayer's taxable income. (2)

Max 5


Airy fairy comments were produced. Students must be able to exhibit their ability to apply their all-in knowledge at this letter.

Final comment: the required order followed the structure of the question information provided as is inevitably the case. Use this to your advantage when prioritising and attempting sections.

Presentation (1)
e.g. use of tables
Language (1)
Logic (1)


The average mark for this paper was 40%.

QUESTION 3

CORLEONIE'S (PTY) LTD **SUGGESTED SOLUTION**

(Source: Exam 2012 MAC4862)

Part (a):

Draft the budgeted Statement of Comprehensive Income, for both lasagnas according to the direct/variable costing system.

Some students did not know what 'direct/variable' meant and others implemented the principles incorrectly. The layout requested assisted in doing sections (d) and (e), although the majority of students ignored this. Some produced an actual instead of the required **budgeted** statement.

	Budget units		Actual units		
	Julio	**Carrie**	**Julio**	**Carrie**	
Opening inventory	8 000	3 000	8 000	3 000	
Manufactured	60 000	15 000	59 000	16 000	
Sales	(65 000)	(17 000)	(65 000)	(17 000)	
Closing inventory	**3 000**	**1 000**	2 000	2 000	(2) 1 each R/W*

***** Part of the 4 marks allocated to closing inventory.

Basic knowledge often incorrectly applied: OI + Manufactured – CI = Sales

CORLEONIE'S (PTY) LTD STATEMENT OF COMPREHENSIVE INCOME FOR THE YEAR ENDED 30 SEPTEMBER 2012

	Julio	**Carrie**	**Julio R**	**Carrie R**	**Not required Total R**	
Sales	R50	R45	3 250 000	765 000	4 015 000	(1) R/W
Variable cost			(733 353)	(180 483)	(913 836)	
Opening inventory (1)	R11	R11	88 000	33 000	121 000	(2) R/W
Manufacturing cost			679 200	158 100	837 300	
Flour, butter, dough(2)	R3,08	R2,80	184 800	42 000	226 800	(1) R/W
Plastic	R0,24	R0,24	14 400	3 600	18 000	(1) R/W
Mince (3)	R4,00		240 000		240 000	(1) R/W
Vegetables (4)		R2,00		30 000	30 000	(1) R/W
Direct labour (11)	R1,50	R3,00	90 000	45 00	135 000	(1) R/W
Overheads (5)	R2,50	R2,50	150 000	37 500	187 500	(1) R/W
Closing inventory Based on average(6 & 7)			(33 847)	(10 617)	(44 464)	(4) R/W
			2 516 647	584 517	3 101 164	

Variable selling/admin(8)	(600 000)	(100 000)	(700 000)	(1) R/W
Contribution	1 916 647	484 517	2 401 164	
Fixed cost	1 860 000	390 000	2 250 000	
Manufacturing (9)	960 000	240 000	1 200 000	(2) R/W
Selling and admin(10)	900 000	150 000	1 050 000	(2) R/W
Net profit	56 647	94 517	151 164	

1. R216 000/8 000 = R27 – R16 = R11 x 8000 = R88 000 and R81 000/3 0000 = R27 – R16 = R11 x 3 000 = R33 000
2. R14 x 0,22kg x 60 000 = R184 800 and R14 x 0,20kg x R15 000 = R42 000
3. R283 200 – R43 200= R240 000
4. R2 x 15 000 = R30 000
5. R2,50 x 60 000= R150 000 and R2,50 x 15 000 = R37 500
6. R88 000 + R679 200 = R767 200 / (8 000+60 000) = **R11,28 x 3 000** = R33 847
7. R33 000 + R158 100 = R191 100 / (3 000+15 000) = **R10,617 x 1 000** = R10 617 (**1 Mark each)**
8. R1 750 000 x 40% = R700 000 x 6/7 = R600 000 and R700 000 x 1/7 = R100 000 (must not include in production cost) **(1 mark for logic)**
9. R1 200 000 x 60 000/75 000 = R960 000 and R1 200 000 x 15 000/75 000 = R240 000
10. R1 750 000 x 60% = R1 050 000 x 6/7 = R900 000 and R1 050 000 x 1/7 = R150 000
11. R15 x 6/60 x 60 000 = R90 000 and R15 x 12/60 x 15 000 = R45 000

Max 15

Not all costs were equal (2, 3, 4, 11) as some students assumed.

Part (b): Indicate why you would prefer a variable/direct costing system to an absorption costing system.

- A variable costing system is more ***useful*** for management **short-term decision-making**. (1)
- As *profits* are not ***distorted*** by the **fixed cost** allocated to the product cost. (1)
- The effect that a ***change in volume*** will have on the profit of the company or division is therefore much easier to isolate and explain. (1)
- The use of a variable costing system allows for better *divisional/product* **performance measurement**. (1)
- It **highlights those *cost that can vary*** in the short term, and which should therefore be manageable. (1)
- Profits are not influenced by *changes in* opening and closing ***inventory.***
- Fixed costs are accounted for in the correct period (1)

Max 5

Part (c): Determine the variances of: Dough Premix, Mince and Selling and Administration



Quite a number of different approaches were possible. Students are reminded that they should apply the approach that they understand best and feel comfortable with.

Approach 1

	Julio			Carrie			
	Original Budget	**Actual**	**Variance**	**Original Budget**	**Actual**	**Variance**	
Premix	184 800	191 160	(6 360)	42 000	38 880	3 120	(2)
Mince	240 000	283 200	(43 200)				(1)

	Total		
S & A	**Budget**	**Actual**	**Variable**
Fixed	1 050 000	1 020 000	30 000 (1)
Variable	700 000	735 000	(35 000) (1)

Approach 2

Material (Dough Mix) Price variance

The price variance per kg	= R14,00 – R13,50 = R0,50 F	(1)
This implies (R230 040 ÷ R13,50)	= 17 040 kg was bought	(1)
Price variance (0,5 x 17 040kg)	= R8 520 F	(1)

Material (Dough Mix) Quantity variance

Standards allowed:		
59 000 x 0,22kg + 16 000 x 0,2	= 16 180 kg	(1)
Usage variance (17 040kg – 16 180kg)	= 860 kg	(1)
Usage variance (860kg x R14)	= R12 040 U	(1)

Compared to Original budget (**840kg** x R14) = R11 760 (**16 200kg – 17 040kg**)
where : (R226 800 ÷ R14 = 16 200kg)

OR per PRODUCT: **(Julio + Carrie)**

Price variance: R7 080F + R1 440F = R8 520F (1½ per product)
Where: (R191 160 ÷ R13,50) x R0,50 = R7 080F **and** (R38 880 ÷ R13,50) x R0,50 = R1 440F

Quantity variance : R16 520A + R4 480F = R12 040A (1½ per product)
Where: (14 160kg – 12 980kg) x R14 = R16 520A **and** (2880kg – 3200kg) x R14 = R4 480F

Flexed vs Original budget: R13 440A + R1 680F = R11 760A (1½ per product)
Where: (14 160 x R14) – R184 800 = R13 440A **and** (2880kg x R14) – R42 000 = R1 680F

Alternative

Even though a price reduction was negotiated (**good relationship with supplier) (must recognise it was because of Alberto)** and only 59 000 were manufactured, Julio still has an unfavourable variance. This would indicate an **inefficiency variance**. R191 160/59 000 = R3,24 / R13,50 = 0,24 kg. This is 20 grams more than budgeted. Alberto must investigate **why so much more dough mix was used on the Julio (any reason, can be spoilage or wrong standards)**. For the Carrie R38 880/16 000 = R2,43/R13,50 = 0,18kg. This is 20g less than budgeted. (2)

Mince

The price of R20 per kg, remained constant. (**or R0 variance)** (1)
This indicates a serious ***inefficient usage* of the mince** or Alberto has the standard completely wrong (spoilage or wrong standards only). R283 200/59 000 = R4,80 / R20 = 0,24kg against the budgeted R240 000/60 000 = R4 / R20 = 0,20kg. This 40grams discrepancy should be a serious concern for Alberto. (2)
Usage variance **R43 200** (1)

Selling and Admin

The variable variance was unfavourable by (R35k/R700k) 5% which is **not unreasonable** but **should be investigated further** (must talk about selling expenses), this is not linked. A conscious decision was probably made to cut back on the fixed cost (director's fees?) to ensure no major overall variance. (2)

Max 16

Part (d): Impact of operating at full capacity

Full capacity does **not** imply that the fixed costs will increase. It implies that the existing fixed costs will be spread over more units, ie each unit will absorb/carry less of the cost.

		Julio	**Carrie**	
Budget (Units)		60 000	15 000	
Capacity utilisation		80%	40%	
Full capacity (Units)		**75 000**	**37 500**	(1)R/W
Time taken per unit (hours) (6/60min and 12/60min)		0,1	0,2	(1)R/W
Total time (hours)		7 500	7 500	(1)R/W
Rate per hour	= R1 200 000 ÷ 15 000 h			
	= R80 per hour			(1)R/W
Or for	Julio: 0,1hr x R80 = R 8			(½)R/W
	Carrie: 0,2 x R80 = R16			(½)R/W

Note that the two products take different times, this was mostly ignored.

1. The current equal rate of **R16 should be changed** to reflect machine usage. (2)R/W

2. On the current allocation basis the effective rate is (R1 200 000 ÷ 75 000 units) R16 but Julio carries 80% (60 000/75 000) of the cost and Carrie 20% (15 000/75 000), (1)C
   Which **distorts** both the costing and the pricing. (1)

3. Incorrect decisions may be taken by the business regarding the products. (1)

If production quantities are used **max 2 points.** 6
Max

Part (e) Budgeted Breakeven – point in Rand for Corleonie's



The breakeven in **Rand** is required for the business. Most students ignored this and calculated the product break-evens. Likewise, the information from section (a) was ignored. A total incorrect approach and numbers will not yield a pass mark.

Total contribution (R1 916 647 + R484 517)	R2 401 164	(1)	R/W
Total sales – given	R4 015 000		
Contribution %	59,80%	(1)	C
Total fixed cost (R1 860 000 + R390 000)	2 250 000	(1)	R/W

$$\therefore \text{ BEP} = \frac{R2\ 250\ 000}{0{,}5980}$$

= R 3 762 238 or R3 762 542 (1) C

4

Part (f): Calculate the actual fixed overhead manufacturing cost per unit for Carrie and Julio, based on the current overhead allocation method and the fixed manufacturing cost per unit, should Alberto decide to implement ABC.



Actual fixed overhead **manufacturing** cost to be used. Students lost marks by not calculating the current rate or ABC unit rate.

Current allocation

Actual fixed	R1 170 000		
Lasagnas manufactured (59 000 + 16 000)	75 000	(1)	R/W
Rate per unit	R15,60	(1)	R/W

Lasagnas	Set-up cost		Ordering		Handling		Inspection	
Julio	(2 t)	24	(½)	12	(59t x 2)	**118 000**	(8 856 ÷ 9)	**984**
Carrie	(1 t)	12	(½)	12	(16t x 3)	**48 000**	(2 400 ÷ 6)	**400**
		36		24		166 000		1 384
	(1) R/W		(1) R/W		(2) R/W		(2) R/W	

Note that Handling and Inspection are driven by the activities and **not** proportionally (eg 2/5 or 9/15). This was a common error which indicates lack of interpretation and ultimately understanding of ABC.

Activity area	Total	Julio	Carrie	
Set-up costs	120 000	80 000	40 000	(1)R/W
Ordering of materials	59 700	29 850	29 850	(1)R/W
Handling materials	770 300	547 563	222 737	(1)R/W
Quality control	220 000	156 416	63 854	(1)R/W
	1 170 000	813 829	356 441	
Units		**59 000**	**16 000**	
Rate per unit		R13,79	R22,28	(2)C
				Max 12

Part (g): How might **Corleonie's Ltd** use the ABC information to make better decisions?

- **Pricing decisions**: The Julio's price can be dropped with almost R2,00, which can possibly lead to bigger volumes. The Carrie's price should be increased. (2)
- **Product design** decisions: ABC provides a roadmap as to how to reduce the costs of individual products. (1)
- **Process improvements**: Each activity is now highlighted and can be questioned. (1)
- **Product profitability** analysis becomes more informative. (1)
- **More accurate and detailed** than traditional method (1)

Note: If General comments only 1 mark; students should focus on making comments relevant to the business scenario. Also relevant to sections h, i and j.

Max 4

Part (h): Explain how ABC will affect the **break-even point** and viability of **Carrie**

- The **fixed cost** of R1 170 000 remains **the same**. (1)
  The allocation **to the two different products however**, changed. The fixed cost on the traditional method allocated is R920 400 (Julio – (R1 170 000 x 59 000/75 000)) and R249 600 (Carrie – (R1 170 000 x 16 000/75 000)) and with the ABC R813 829 (Julio) and R356 171 (Carrie). (1)
- The break-even point for Julio will ***decrease*** (lower fixed cost) and the break-even point for *Carrie* ***will increase****,* higher fixed cost. (2)
- The question here is, if the new allocation is so accurate, can the fixed cost be **classified as product specific**. If not, the break-even will remain the same. If it can, the break-even will no longer be weighed but **calculated separately**. (2)

For Carrie:		**R**	
Contribution to fixed cost		484 517	
Fixed cost per traditional method		390 000	
Surplus (as per (a))		**94 517**	(1)C
Contribution to fixed cost		484 517	
Fixed cost per ABC method	(390 000 – **240 000** + 356 171)	506 171	(2)R/W
Surplus (Shortfall)		**(21 654)**	(1)C

- On the face of it Carrie makes a loss and should be terminated. However, if Julio has to carry all or the bulk of it, **Corleonie's will not be better off**. (1)
- Other factors to consider:
    - Can Carrie's **price be increased?**. It is R5 lower than Julio, why is this price used? (2)
    - Will more **marketing improve sales** of Carrie? (1)
    - Can current **capacity be swopped** between the two products in the interim – more Julio's. (1)

Max 8

Part (i): Discuss the reasons why Gaby should be sceptical about the idea of Luca getting involved with Corleonie's?

- It is always risky because of **arguments to involve family in business**, more so when a family recipe is used. (1)
- At this point in time, Corleonie's might be ***too small*** to involve a third person. (1)
- Alberto and Gaby **worked very hard** on this company and had a battle with their business. (1)
- ***Control* can become a problem**, how many shares will Luca receive indications that it is a company)? (1)

- What **contribution** can Luca make to the business?
    - **Experience** of this industry (1)
    - **Financial (anything about contributing towards business)** (1)
    - **Marketin**g / sales experience (1)

- What type of **personality** does Luca have? Combative work in a team, overbearing? (1)
- She (Gaby) may fear being **sidelined.** (1)
- Will have to **share profits** (1)

1 each, max 6

Part (j): What are the benefits of franchising and possible disadvantages?

From Corleonie's perspective, as they are considering it – not general.

Advantages:

- **Capital upfront** fee will strengthen the business. (1)
- Products may be **distributed** nationally. (1)
- *Continuing revenue streams:* If successful Alberto can benefit from **royalties.** (1)
- ***Brand development**:* Franchisee contributions to advertising can expand the Corleonie's brand. (1)
- ***Economies of scale***: If properly managed Alberto can leverage suppliers or products can be centrally manufactured. **Pricing** (2)

Disadvantages:

- If separate manufacturing premises, **quality control** becomes a major issue. (1)
- ***Legal regulation***: Can be a lot of red tape for Alberto and will have to register with FASA. (1)
- ***Size***: Corleonie's might be too small to advertise a franchise and may not attract bank support. (2)
- **Training**: Franchisor has to invest in (expensive) training. (1)

  Any other valid point – max of (1)

  Max 6

Part (k): Calculate the minimum selling price per Romeo for this order?

	Requirements	On hand	Needed/Relevant
Flour, butter and dough (1)	1 650 kg	300kg	1 350 kg
Mince (2)	750 kg	150 kg	750 kg
Vegetables (2)	750 kg	200 kg	750 kg
Plastic holders	7 500 holders	6 000 holders	7 500 new holders

Mince and vegetables are in regular use and if used, will have to be replaced.

Calculations:

1) 7 500 x 0,22kg = 1 650kg
2) 7 500 x 0,1 = 750kg

Variable cost	**Reason**	**R**	
Flour, butter and dough on hand (300kg x R1) Saving	Close to disposal date	(300)	(1)
Flour, butter and dough purchase (1 350kg x R14)		18 900	(1)
Mince on hand (150kg x 0)			
Mince purchase (750kg x **R25 (½ if only price)**	Regular use	18 750	(1)
Vegetables			
Vegetables purchased (750kg x **R1,10**) **½ if only price**	Regular use	825	(1)
Plastic holders on hand	Cannot be used (N1)	0	(1)
Plastic holders purchase (10 000 x 0,30)	Order quantity	3 000	(1)
Fixed cost – Salary allocated		0	
Labour (R20 x 12/60 x 7 500)	Additional time	30 000	(1)
Training	Not relevant	0	(1)
Fixed manufacturing cost	Not relevant	0	(1)
Total relevant cost		71 175	
Units		7 500	
Min price per unit (R71 175 / **7 500**): See notes 4 & 6		R9,49	(1)C

Students could not apply the principle of relevant costing. There was also often a mix of per unit and total R information. One or the other should be applied.

Notes: These may be relevant issues:

1. The existing holders are **pre-labelled** and cannot be used for this project. (1)
2. Actual **time materially exceeded the budget** time (1)
3. Mince and vegetables in regular use, **new (excess) quantities** will be used in normal production. (1)
4. **Possible cost increases**: Mince 40g x 7 500 x 25 and labour 25% of R30 000 = R15 000 or R2 per unit. (1)
5. There is adequate **spare capacity.** (1)
6. Should Alberto not consider **some profit mark-up**? (1)

Max 10

Part (I): Issues re-corporate sales [corporate implies a business or organisation]

- What will **the additional cost** be get products tested by corporate, as well as **time spent** to get introductions / follow through the process. (2)
- What **prices** and/or discounts will **corporate require**. (1)
- What will cost of **shelf space be**? (1)
- Impact on working capital (1)
- What **terms do Corleanie's** get from their suppliers; encounter **cash flow problems**? (2)
- Current production **capacity** may have to be **expanded**. Is **finance** available? (2)

Max 5

The average mark for this paper was 36%.

QUESTION 4

ANANI GROUP **SUGGESTED SOLUTION**

(Source: Supplementary exam 2012 MAC4862 adapted)

(a) **Bedrooms department – ABC profit statement**

Fixed general overheads = 70% x R17 500
= **12 250** (1) r/w

Only fixed overheads are subject to ABC.

Activity drivers:

Items sold	= 1 000 + 1 500 + 4 000	= 6 500
Purchase orders	= 1 000 + 900 + 2 500	= 4 400
Floor area	= 16 400 + 10 000 +14 400	= 40 800
Consultations	= 798 + 200 + 250	= 1 248

	Cost R'000	Driver Quantity	Recovery rate/driver R	Bedroom Qty	Cost R'000	
Sales staff	6 644	**4 400**	**1 510,0**	900	**1 359**	(1½) r/w
Consultation	2 496	**1 248**	**2 000,0**	200	**400**	(1½) r/w
Warehouse	3 055	**6 500**	**470,0**	1 500	**705**	(1½) r/w
Admin	3 050	**5 648**	**540,0**	1 100	**594**	(1½) r/w
General	12 250	**40 800**	**300,2**	10 000	**3 002**	(1½) r/w
					6 060	(½) r/w

Be aware that a short-cut approach (i) will not always work correctly and (ii) if details of calculations are not shown, marks may be lost.

ABC Profit Statement for 3 months

	R'000	
Sales	11 250	
Cost of sales	(3 750)	
Gross profit / contribution	7 500	
Variable general overheads (R17 500 x 30% x R3 750 / R27 650)	(712)	(1) r/w
Contribution	6 788	(1) c
ABC fixed costs	(6 060)	(1) c
Operating profit	728	
		12

(b) **Evaluation of Bedroom closure**

Breakeven: **ABC and Traditional**

Contribution % $\frac{\text{R6 788 000}}{\text{R11 250 000}}$

= 60,34% (1) c

OR: Contribution per unit: $\frac{\text{R6 787 975}}{\text{1 500 units}}$ $\frac{\text{R6788 000}}{\text{1 500 units}}$

R4 525,32 per unit **or** =R4525,33 per unit (2)

Based on 'traditional' cost allocation

$\frac{\text{8 250 000 – 712 000}}{\text{0,6034}}$ OR $\frac{\text{R7 538 000}}{\text{R4 525.33}}$ (1) c

= R12 495 000 = 1 666 units (1) r/w

Based on ABC (1) c

$\frac{\text{R6 060 000}}{\text{0,6034}}$ OR $\frac{\text{R6 060 000}}{\text{R4 525,33}}$

= R10 043 000 = 1 340 units (1) r/w

Inventory turnover (R'000):	**BFN**	**BR**	**LR**	
	$\frac{\text{6 300}}{\text{1 475}}$	$\frac{\text{3 750}}{\text{950}}$	$\frac{\text{17 600}}{\text{2 550}}$	
=	4,27 times	3,95 times	6,90 times	½ each r/w Max (1 ½)
Gross profit =	$\frac{\text{(21 000 - 6 300)}}{\text{21 000}}$	$\frac{\text{(11 250 – 3 750)}}{\text{11 250}}$	$\frac{\text{(44 000 – 17 600)}}{\text{441 000}}$	
=	70%	66,7%	60%	½ each r/w Max (1 ½)
Turnover (in R'000) per m^2	$\frac{\text{21 000}}{\text{16 400}}$	$\frac{\text{11 250}}{\text{10 000}}$	$\frac{\text{44 000}}{\text{14 400}}$	
=	1,28	1,125	3,06	½ each r/w Max (1 ½)

For calculating a minimum of any 3 indicators (½) r/w
Calc max 10

Although the question specified break-even it is a good starting point for any closure evaluation. Profitability and inventory turn are also pretty standard and a hint regarding the latter was given. Question info should also be linked to practical industry measures: for a shop: turnover per size or number of staff; cash vs credit sales etc.

Comment:

- The *method of allocation* influences Bedroom department's (BRD) profitability, as can be seen from both the profit statement and BEP using ABC. At this stage, in terms of an ABC approach, BRD is generating a profit. (2)
- BRD *exceeded its target inventory* turn comfortably – better than Breakfast Nook. (1)
- The *Margin* of this department (GP/Contribution) is again the *second best*. (1)
- The impact of follow-up orders should be considered ie customer ordering first from one department and then another. (1)
- The use of space lags that of BFN marginally (1)
- This department should not be closed (1)

Other factors to consider other than above:
One would need to do a relevant costing exercise in deciding finally on the closure of the division. (1)
Based on an ABC approach, which would be the more realistic of the two, a positive margin of safety exists which indicates that more units are being sold than are required to break even – this would also explain the profitability of BRD at this stage. (1)

Discussion max 6

(c) Strategic issues

Customers:

- It appears that transactions are on a *cash basis*. Will the *provision of credit* improve turnover? This will however need to be financed. (2)
- Are *guarantees / warrantees* given and if so what is the *cost* thereof or the cost of replacement products. (2)
- Are customers kept on a *database* and provided with *promotional* information. (2)
- What is the extent of *repeat customers*? Is *customer satisfaction* measured? (1)
- What is the extent of *multiple department support*? (1)
- Which products are bought most – Scandinavian (imported) or local (own manufactured)? (1)

Consultants:

- Customer satisfaction relies on the availability of consultants at the lifestyle stores, do they have sufficient consultants. (1)
- What is the ratio of successful consultations to total consultations?

- Do consultations *add value* in respect of an increased sales value per order or repeat orders. This would require an analysis of sales per consultant to be prepared. (2)
- Are the consultants knowledgeable across the three departments or specialists? (1)

Max 9

The above all relate to practical issues which relate to the improvement of sales, profitability and cash.

(d) Structure change – remuneration

Cost saving	= (R1 359 + R400) x 50%		
	= R879,5	(1)	r/w
4% Commission	= R11 250 x 0,04		
	= R450	(1)	r/w
Additional sales	= (R879,5 – R450) / ,04		
	= R10 737	(1)	r/w
Total sales	= R11 250 + R10 737 = R21 987,5 (**OR** R879,5 / 0,04)	(1)	r/w

At this level, staff will be in the same position, but the department will be better off as the GP/Contribution will exceed the commission.
At levels **above** R21 987,5 the department and staff will be better off. (1)

Max 4

The set-off between the cost saving and commission often confused students.

(e) Impact of current production level – Lindela factory

Normal capacity given	27 000 hours		
Budgeted time per unit	0,6 hours (36min)		
Normal full capacity (27 000/0,6)	45 000 units	(1)	r/w
FOH – budgeted (x R60)	R2 700 000	(1)	r/w
Under recovery 40% x R2 700 000	R1 080 000	(1)	r/w
Additional cost per unit R1 080 000 ÷ 45 000 x 60% (OR new rate = R60 + R40 = R100)	R40 per unit	(1)	r/w

- Based on the current cost structure Lindela should recover the additional R40 per unit, hence push product cost up. (1)
- If they don't, their margin is not as presented. (1)
- Current standard margin (1 160 – 725) / 725 = 60% or 38% margin on selling price. (1)

- If they wish to keep the margin intact, their price of R1 160 should be reviewed and increased. (1)
- The budgeted profit for the year based on external sales only at full capacity is 45 000 x (R1 160 – R725) = R19,575m. (1)
- Based on external sales (if all the current capacity was to be used for external sales), at the current capacity it will only be R10,665m [(27 000 x R435) – R1,080]. (1)
- The incorrect price is passed on to group departments/divisions. (1)

Max 8

 Capacity, the related overhead rates and their impact on total cost and eventual pricing were problematic. Students should ensure that they are conversant with the different aspects. Consistent and material under-recovery of fixed overheads must be reflected in a revised cost of the product.

(f) Strategic impact of holding 8 weeks inventory

- Inventory on current production level: 27 000 x 0,15 m^3 x 8/50 = 648 m^3 (½) r/w
  Value 648m^3 x R3 500 = R2 268 000 (½) r/w
  At full capacity (÷ 0,6) = R3 780 000 (1) r/w

- The Mukwa is not readily available; therefore *JIT* cannot be applied.
- EOQ can be used to determine the order size and with the estimated lead time available, a re-order point determined. (1)
- The factory can thus *run up to 8 weeks* or if capacity is ramped up a maximum of *4,8 weeks* (648 / (45 000 / 50 x 0,15)) without any interruptions in production taking place. (2)

- A *storage facility* for the above quantity is required, with *safeguards* and especially fire insurance. (2)

- What safeguards for *continuous supply* are in place eg contracted suppliers. (1)

- Strategically, the Lindela factory should consider whether *substitute materials* exist which could be used in the manufacturing of the product.
- The *opportunity cost of holding* the inventory needs to be considered. (1)
- Any other valid comment – Max 1 (1)

Max 8

 Many students treated the wood inventory as if it was a consumable item eg it would go bad or become obsolete. JIT model was often proposed, but by the very nature of this commodity, these cannot be utilised as indicated in note 1.

(g) Production level after proposals

	Group %	External %	Export %	Total %	
Current: 60% capacity	80	20	0	-	
Normal capacity (0,8 x 0,6; 0,2 x 0,6)	**48**	**12**	**0**	60	(1½) r/w
Design change (0,48 x1,25; 0,12 x1,67)	**60**	**20**	**10**	90	(1½) r/w
Price decrease (factor 1,10)	**66**	**22**	10	98	(1) r/w

Students did not follow a step-by-step logical approach and produced incorrect quantities when starting with 80% and 20% as a base.

Alternative	**Lifestyle units**	**External units**	**Export units**	**Total Units**	
Current sales	**21 600**	**5 400**	-	**27 000**	(1½) r/w
Design change	5 400	3 600	4 500	13 500	
	27 000	**9 000**	**4 500**	40 500	(1½) r/w
Price decrease (10% increase)	**2 700**	**900**		3 600	(1) r/w
	29 700	9 900	4 500	44 100	

Within the normal capacity of 45 000 units
44 100 / 45 000 = 98%

Prices after the proposal:

Group: R1 160 – (56 – 26)	=	R1 130,0		(1) r/w
Less 5%		56,5		(1) r/w
		1 073,5		
External: R1 160 less 5%		R1 102,0		(1) r/w
Export: R1 160 less 10%		R1 044,0		(1) r/w
@ rate of R10,8 = €1		€96,7		(1) r/w
			Max	8

Basic relevancy and mathematical rules were not applied, which caused pricing to be problematic.

(h)	**Mark-up % on units transferred**	**R**	
	FOH need not change as capacity per (g) is 90% plus		
	Standard cost given	725,00	(1) r/w
	Additional material	45,00	(1) r/w
	Packaging saved	(30,00)	(1) r/w
	Total cost	740,00	
	Transfer sales price per (g)	1 073,50	(1) c
	Mark-up (R)	333,50	(1) c
	Mark-up (%) (333,5 / 740)	45,1 %	(1) r/w
			6


Mark-up often calculated incorrectly on the sales price.

(i) Naldnak factory – Options A and B

IDC		**1**	**2**	**3**	**4**	**5**	
Open bal		27 000 000	26 730 000	26 441 100	26 131 977	**25 801 215**	(1)rw
Interest	7%	**1 890 000**	1 871 100	1 850 877	1 829 238	**1 806 085**	(2)rw
Pmt	40%/5	**(2 160 000)**	(2 160 000)	(2 160 000)	(2 160 000)	**(2 160 000)**	(1)rw
Closing bal		26 730 000	26 441 100	26 131 977	25 801 215	**25 447 300**	(1)rw
Net cash		25 650 000					

Alternative:
n = **5**; i = **7**; PV = **27 000 000**; pmt = **(2 160 000)** (4) r/w
Comp FV = **25 447 300 (bullet payment)** (1) r/w

Euro loan		**1**	**2**	**3**	**4**	**5**	
Open bal (€)	10,8	**2 500 000**	2 537 500	2 575 563	2 614 196	**2 653 409**	(2)rw
Interest (€)	1,5%	**37 500**	38 063	38 633	39 213	**39 801**	(2)rw
Closing bal (€)		2 537 500	2 575 563	2 614 196	2 653 409	**2 693 210**	(1)rw
Net cash		25 920 000					

Alternative:
n = **5**; i = **1,5**; PV = **2 500 000** (3) r/w
Comp FV = **2 693 210 (Bullet payment)** (2) r/w


The three items requested were often not supplied or clearly indicated. Many students followed the alternative route, but then got the components mixed up with incorrect results. The applicable interest rate proved especially problematic. Calculation steps must be clearly shown and students should familiarise themselves with the functions of their calculators to enable them to determine any component.

Cash available for the project (Loan amount less cost)

Option A:
Cost = R27 000 000 x 5% = R1 350 000
Cash = R27 000 000 – R1 350 000 = **R25 650 000** (1) r/w

Option B:
Cost = €100 000 x R10,80 = R1 080 000
Cash = R27 000 000 – 1 080 000 = **R25 920 000** (1) r/w

Forward €/R exchange rate for 5 year pmt

Exchange rate required: R25 447 300 ÷ €2 693 210 (1) c
R9,45 (therefore R9,45 = €1) (1) c

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(j) Issues regarding options A and B

(i) *Option A*

- Option A generates interim cash flows over the 5-year period (from DTI). (1)
- The grant payment would be subject to conditions - how onerous can these become? (1)
- The interest rate is fixed but low (below prime), a change in interest rates will not impact on their planning. Decreasing interest rates would create interest rate risk for the company. (1)
- Planning (above) re final settlement can be done and monitored throughout the term. Will there be adequate cash available for final bullet payment? (1)
- There is no currency impact. (1)

Option B

- Interest rate below that of option A but is in Euro terms – in terms of interest parity principles any benefit of lower interest rates in one country would be eliminated through foreign exchange differences. (1)
- Most currency hedges, hedging the repayment in year 5, will only be available one year before it is due. (1)
- Will there be adequate cash available for payment at end of 5 years? (1)
- Can their product be exported to Europe? (1)
- What measurable benefit can be derived from special status? (1)

Max 7

(ii) *Joint venture*

- Is the JV administered by way of *contract* in terms of:
  - shareholding – would they need to take up shares in the JV entity? (1)
  - assets contributed – what assets and what is the value thereof, which Anani would need to contribute to the JV? (1)
  - security provided for loans/facilities – what is the extent of the security to be provided on behalf of the JV? (1)
  - additional funding required (R8m) – where will the funding be sourced from and at what cost? (1)
  - existing and new staff – will additional people be able to be recruited? (1)
  - termination – when will the JV terminate? (1)
  - *Any possible synergies and savings* (1)

Max 5



Students struggled with the joint venture concept; this is a prominent part of the business environment.

Logic (1)
Layout (1)
Legibility (1)



The average mark for this paper was 36%.

QUESTION 5

ANANI GROUP **SUGGESTED SOLUTION**

(Source: Supplementary exam 2012 MAC4861)

(a) **Bedrooms department – ABC profit statement**

Fixed general overheads = 70% x 17 500
= 12 250 (1) r/w

Only fixed overheads are subject to ABC.

Activity drivers:

Items sold	= 1 000 + 1 500 + 4 000	= 6 500
Purchase orders	= 1 000 + 900 + 2 500	= 4 400
Floor area	= 16 400 + 10 000 + 14 400	= 40 800
Consultations	= 798 + 200 + 250	= 1 248

	Cost R'000	**Driver Quantity**	**Recovery rate/ driver**	**BRD**	**Cost R'000**		
Sales staff	6 644	4 400	1 510,0	900	1 359	(1½)	r/w
Consultation	2 496	1 248	2 000,0	200	400	(1½)	r/w
Warehouse	3 055	6 500	470,0	1 500	705	(1½)	r/w
Admin	3 050	5 648	540,0	1 100	594	(1½)	r/w
General	12 250	40 800	300,2	10 000	3 002	(1½)	r/w
					6 060	(½)	r/w

Be aware that a short-cut approach (i) will not always work correctly and (ii) if details of calculations are not shown, marks may be lost.

Profit Statement for 3 months

	R'000		
Sales	11 250		
Cost of sales	(3 750)		
Gross profit / contribution	7 500		
Variable general costs (17 500 x 30% x 3 750 / 27 650)	(712)	(1)	r/w
Contribution	6 788	(1)	C
ABC fixed costs	(6 060)	(1)	C
Operating profit	728		
		12	

(b) Discuss the advantages and disadvantages that may arise for Anani from introducing activity-based costing in its stores.

Advantages

- In modern systems, overhead functions include a lot of non-factory-floor activities such as product design, quality control, production planning and customer services. ***ABC is concerned with all overhead costs and so it takes management accounting beyond its 'traditional' factory floor boundaries,*** helping Anani to understand more clearly the origins of its costs. (1)

- Once the necessary information has been obtained, ABC is similar to traditional absorption costing. ***This simplicity is part of its appeal, although some apportionment of certain overhead costs may need to be done on an arbitrary basis***. (1)

- The ***complexity of the business has increased, with wider product ranges, shorter product life cycles and more complex processes***. ABC ***recognises this complexity with its multiple cost drivers***, and so provides an adaptable tool for Anani to use by highlighting the activities that generate costs. (1) (1)

- In a more ***competitive environment, companies must be able to assess product profitability realistically***. ABC can facilitate a good understanding for companies such as Anani of what drives overhead costs. (1) (1)

- This ***enables better decision making, for example by providing more information on whether or not to close the Bedrooms department***, or how to price certain products. (1)

Disadvantages

- It should also be noted that the ***introduction and maintenance of an ABC system might be time consuming to set up.*** (1)

- The ***information needed by the new system may not currently be available***. (1)

- Also the ***costs of introducing and maintaining an activity-based costing system may exceed the benefits*** of such a costing system. (1)

Max 10

(c) Evaluation of Bedroom closure

Breakeven:	**ABC and Traditional**
Contribution %	R6 788 / R11 250
	= 60,34% (1) C

OR: Contribution per unit:	R6 787 975 / 1 500 units	Or	R6 788 000 / 1 500 units
	= R4 525.32 per unit		**=R4 525.33 per unit**

Based on 'traditional' cost allocation

$\frac{8\,250\,000 - 712\,000}{0{,}6034}$ OR $\frac{R7\,538\,000}{R4\,525.33}$ (1) c

= R12 495 000 = 1 666 units (1) r/w

Based on ABC (1) c

$\frac{R6\,060\,000}{0{,}6034}$ OR $\frac{R6\,060\,000}{R4\,525{,}33}$

= R10 043 000 = 1 340 units (1) r/w

		BFN	**BR**	**LR**	
Inventory turnover:		$\frac{6\,300}{1\,475}$	$\frac{3\,750}{950}$	$\frac{17\,600}{2\,550}$	
	=	4,27	3,95	6,90	½ each r/w Max (1 ½)
Gross profit					
	=	$\frac{(21\,000 - 6\,300)}{21\,000}$	$\frac{(11\,250 - 3\,750)}{11\,250}$	$\frac{(44\,000 - 17\,600)}{441\,000}$	
	=	70%	66,7%	60%	½ each r/w Max (1 ½)
Turnover (in R'000) per m^2		$\frac{21\,000}{16\,400}$	$\frac{11\,250}{10\,000}$	$\frac{44\,000}{14\,400}$	
	=	1,28	1,125	3,06	½ each r/w Max (1 ½)

For calculating a minimum of any 3 indicators (½) r/w
Calc max 10

Although the question specified break-even it is a good starting point for any closure evaluation. Profitability and inventory turn are also pretty standard and a hint regarding the latter was given. Question info should also be linked to practical industry measures: for a shop: turnover per size or number of staff; cash vs credit sales etc.

Comment:

- The *method of allocation* influences BRD's profitability, as can be seen from both the profit statement and BEP using ABC. At this stage, in terms of an ABC approach, BRD is generating a profit. (2)
- BRD *exceeded its target inventory* turn comfortably – better than Breakfast Nook. (1)
- The *Margin* of this department (GP/Contribution) is again the *second best*. (1)

- The impact of follow-up orders should be considered ie customer ordering first from one department and then another. (1)
- The use of space lags that of BFN marginally (1)
- This department should not be closed (1)

Other factors to consider other than above:
One would need to do a relevant costing exercise in deciding finally on the closure of the division. (1)
Based on an ABC approach, which would be the more realistic of the two, a positive margin of safety exists which indicates that more units are being sold than are required to break even – this would also explain the profitability of BRD at this stage. (1)

Discussion max 6

(d) Variance calculation

Sales revenue:		**R**		
Internal (transfer price)	1 160 – (56 – 26) x 2 000	2 260 000		(2)
External	1 160 x 300	348 000		(1)
		2 608 000		
Variances:				
Material A				
Price	= 370 x (3 600 – 3 500)	37 000	U	(1)
Quantity	= (2 300 x 0,15 – 370) x 3 500	87 500	U	(1)
Material C				
Price	= 2 280 x (21,50 – 21,60)	228	F	(1)
Quantity	= (2 300 x 1 – 2 280) x 21,60	432	F	(1)
Packaging				
External	300 x 56 – 17 000	200	U	(1)
Internal	2 000 x 26 – 51 000	1 000	F	(1)
Fixed overhead				
Expenditure variance	274 000 – 270 000	4 000	U	(1)
Volume variance	270 000 – 2 300 x 60	132 000	U	(1)
The above is underrecovered				(1)
Budgeted fixed overhead	= (27 000 ÷ 0.60) x 60	R2 700 000		(1)
For 5 weeks (5/50)		R270 000		(1)
Maximum				**13**

Note that (i) a 5-week period is specified and (ii) that the output for this period is 2 300 units. This output drives the 'quantity' variances. Based on the FOH, the production budgeted for this period would be 45 000 ÷ 50 x 5 = 4 500 vs only 2 300 actually manufactured, clearly under budget. The initial FOH calc is driven by the normal capacity.

(e) Variance investigation

- The significant variances are in respect of *material A* and fixed *overhead volume*. (2)
- The reason for the higher price should be determined and whether the higher price will continue in future, so that the standard cost and price can be adjusted. (1) (1)
- Quantity is negative: is this a *workers efficiency (skill)* issue or due to the *wood quality.* (2)
- Corrective steps are required for the aforementioned. (1)
- The FOH expenditure variance is a function of spending vs budget. (1)
  It will only be corrected more control on expenditure. (1)
- Any other valid point max 1 (1)

Max **7**

Students covered all the variances in their solutions. The focus should be on material variances.

(f) Impact of holding 8 weeks inventory

- Inventory on current production level: 27 000 x 0,15 m^3 x 8/50 = **648 m^3** (½) r/w
  Value 648 x 3 500 = **R2 268 000** (½) r/w
  At full capacity (÷ 0,6) = **R3 780 000** (1) r/w

- The **Mukwa is not readily available**; therefore ***JIT* model cannot be applied.**
- EOQ can be used to determine the order size and with the estimated lead time available, a re-order point determined. (1)
- The **factory can thus *run up to 8 weeks*** or if capacity is ramped up or a maximum of ***4,8 weeks*** (648 / (45 000 / 50 x 0,15)) without any interruptions in production taking place. (2)
- A ***storage facility* for the above quantity is required**, with ***safeguards* and especially fire insurance.** (2)
- What **safeguards for *continuous supply*** are in place eg contracted suppliers. (1)
- Strategically, the Lindela factory should **consider whether *substitute materials* exist** which could be used in the manufacturing of the product. (1)
- The ***opportunity cost of holding* the inventory** needs to be considered. (1)
- Any other valid comment – Max 1 (1)

Max 6

Many students treated the wood inventory as if it was a consumable item eg it would go bad or become obsolete. JIT model was often proposed, but by the very nature of this commodity, these cannot be utilised as indicated in note 1.

(g) Production level after proposals

	Group %	External %	Export %	Total %	
Current: 60% capacity	80	20	0	-	
Normal capacity (0,8 x 0,6; 0,2 x 0,6)	**48**	**12**	**0**	60	(1½) r/w
Design change (0,48 x1,25; 0,12 x1,67)	**60**	**20**	**10**	90	(1½) r/w
Price decrease (factor 1,10)	**66**	**22**	**10**	98	(1) r/w

Students did not follow a step-by-step logical approach and produced absurd quantities when starting with 80% and 20% as a base.

Alternative	**Lifestyle units**	**External units**	**Export units**	**Total units**	
Current sales	**21 600**	**5 400**	-	**27 000**	(1½) r/w
Design change	5 400	3 600	4 500	13 500	
	27 000	**9 000**	**4 500**	40 500	(1½) r/w
Price decrease (10% increase)	**2 700**	**900**		3 600	(1) r/w
	29 700	9 900	4 500	44 100	

Within the normal capacity of 45 000 units
44 100 / 45 000 = 98%

Prices after the proposal:

Group: R1 160 – (56 – 26)	=	**R1 130,0**	(1) r/w
Less 5%		**56,5**	(1) r/w
		1 073,5	
External: R1 160 less 5%		**R1 102,0**	(1) r/w
Export: R1 160 less 10%		**R1 044,0**	(1) r/w
@ rate of R10,8 = €1		**€96,7**	(1) r/w
		Max	**8**

Basic relevancy and mathematical rules were not applied, which caused pricing to be problematic.

(h) Mark up % on units transferred

FOH need not change as capacity per (g) 90% plus

Standard cost given	725,00	(1)
Additional material	45,00	(1)
Packaging saved	(30,00)	(1)
Total cost	**740,00**	
Sales priced per (g)	1 073,50	(1)
Mark-up(R)	333,50	(1)
Mark-up % (R333,50/R740,00)	45,1%	(1)
		6

Mark-up often calculated incorrectly on the sales price.

(i) Bullet payment, etc

IDC	**1**	**2**	**3**	**4**	**5**	
Option A	27 000 000	26 730 000	26 441 100	26 131 977	**25 801 215**	(1)
Interest (@7% p.a)	1 890 000	1 871 100	1 850 877	1 829 238	**1 806 085**	(2)
Pmt	(2 160 000)	(2 160 000)	(2 160 000)	(2 160 000)	**(2 160 000)**	(1)
Closing bal	26 730 000	26 441 100	26 131 977	25 801 215	**25 447 300**	(1)
Cost	1 350 000					
Nett cash	25 650 000					(1)
Option B						
Euro loan (R10.8 = €1)	2 500 000	2 537 500	2 575 63	2 614 196	2 653 409	(2)
Interest (@1.5% p.a)	37 500	38 063	38 633	39 213	39 801	(2)
Closing bal	2 537 500	2 575 563	2 614 196	2 653 409	2 693 210	(1)
Cost	1 080 000					
Net cash	25 920 000			Exchange rate required	9,45	(3)

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Alternative for Option A**:**	
n = **5**; i = **7**; PV = **27 000 000**; pmt = **(2 160 000)**	(4) r/w
Comp FV = **25 447 300**	(1) r/w

Alternative for Option B**:**	
n = **5**; i = **1.5**; PV = **2 500 000**; pmt = **(0)**	(4) r/w
Comp FV = **2 693 210**	(1) r/w

(ii) *Joint venture*

Students struggled with the joint venture concept; this is a prominent part of the business environment.

- Is the JV administered by way of *contract* in terms of:
  - **Shareholding** – would they **need to take up shares in the JV entity?** (1)
  - **Assets contributed** – **what assets and what is the value** thereof, which Anani would need to contribute to the JV? (1)
  - **Security provided for loans/facilities** – what is the extent of the security to be provided on behalf of the JV? (1)
  - **Additional funding required (R8m)** – where will the funding be sourced from and at what cost? (1)
  - **Existing and new staff** – will additional people be able to be recruited? (1)
  - **Termination** – when will the JV terminate? (1)
  - ***Any other valid point – Max (1)*** (1)

Max 5

Logic	(1)
Layout	(1)
Legibility	(1)

QUESTION 6

FILMSTARS (PTY) LTD **SUGGESTED SOLUTION**

(Source: Exam 2011 TOE408W (MAC4862))

(a) Assess the financial feasibility of producing the DVD's in **real terms**, computing both the net present value (NPV) and the internal rate of return (IRR). For this assessment, **assume no delay in respect of the tax cash flows exists**.

Assumption: All cash flows will increase at the general rate of inflation resulting in current prices being equal to real cash flows. Real cash flows must be discounted at a real discount rate while a nominal required rate of return is given.

Use this formula

1 + nominal rate	= (1 + real rate) x (1 + anticipated inflation)	
1 + 0,14	= (1 + real rate) x (1 + 0,0556 rate)	
∴ Real rate	= **8%**	(1) R/W

Contribution per DVD sold
= Selling price – Material – Labour – Variable overhead – Distribution cost
= R40 – R16 – R4 – R3 – R4 = **R13**

(2) for R13 R/W
or
(1) for R10 R/W



Note: Allocated fixed overheads do not represent incremental cash flows and are therefore not relevant to the decision.

	R		
∴ **165 000** units x R13 =	2 145 000		(1) R/W for 165 000
Taxation 28%	**(600 600)**		(1) C
Annuity for five years @ **8%**	1 544 400	x 3,993	(1) R/W for 8% & 5y
[n = **5**; i = 8; PMT = 1; COMP PV= 3,993]			
or [n = 5; i = 8; PMT = 1 544 400; COMP PV]			
Present value - Annuity	6 166 789		
Redesign and set-up after tax	**(4 000 000)**		(1) R/W
NPV	**2 166 789**		(1) C

Using a financial calculator –

IRR ∴ CFi_0	= R- 4 000 000	
CFi_{1-5}	= R 1 544 400	
∴ IRR	= **26,9%**	(2) if correct Or (1) if Incorrect

As both the **NPV is positive** and the **IRR exceeds the real cost of capital of 8%**, the production of DVD's is financially feasible. (2) 1 for general recommendation

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Many students forgot to convert the nominal rate to a real rate or did the conversion incorrectly despite the formula being given in the **REQUIRED** part. Many students spent unnecessary time doing long capital budgeting calculations for each of the five years when all that was needed was the calculation of the present value of an **annuity**. The contribution was also frequently calculated incorrectly due to the exclusion of distribution costs or the inclusion of allocated fixed overheads. The identification of relevant and irrelevant up-front costs was also a problem. Students also often spent unnecessary time calculating the annual revenues and cost including inflation in order to discount at the nominal rate despite the question asking for the assessment in **real** terms.
Students often forgot to calculate the IRR or did not know how to do it. Note the IRR should be compared to the nominal return of 14% instead of 8%.

(b)(i) Determine the Price (P) at which the DVD project's NPV becomes zero.

A few alternatives are given for each what if calculation as students had different approaches that made sense to them. Use the one that you understand best and are comfortable with. The **principle** remains the same: by how much can the NPV change before it becomes zero. The time value of money and the five year period remain considerations. As what ifs are real life events, it should be considered a priority aspect of all topics where it can be applied.

NPV can drop by:	= R2 166 789 ÷ 3,993	
[or PV = 2 166 789; n=5; i=8; Comp PMT]	= **R542 646,90** p.a.	(1) C
Per unit (R542 647,90 ÷ **165 000** units)	= R3,289 for 165 000 (after tax cash flow)	(1) R/W
Before tax (**÷ 0,72**)	= R4,57	(1) R/W
Sales price (R40 – R4,57)	= **R35,43** per unit	(1) C
		4

OR

		(C)	(C)	(R/W)	(R/W)
NPV (R0)	= [**165'** (P – **R27**)(1 – **0,28**) (3,993)] – R4m				
	= [(165'P – R4 455') 2,8750] – R4m				
	= 474 375P – R12 808 125 – R4 000 000				
474 375 P	= R16 808 125				
P	= **R35,43** per unit				(4)

OR

After tax cashflow needed p.a.		
[n = 5; i = 8%; PV = (4m); Comp PMT]	= R1 001 825,82	
Before tax cashflow (contribution)		
(R1 001 825,82 ÷ 0,72)	= R1 391 425	
Costs (R27 x 165 000)	= R4 455 000	
Sales (R1 391 425 + R4 455 000)	= R5 846 425	
Sales price (R5 846 425 ÷ 165 000)	= **R35,43** per unit	(4)

(ii) Determine the Volume (V) at which the DVD project's NPV becomes zero.

NPV can drop by	= **R542 646,90**	(1) C
Before tax (**÷ 0,72**)	= R753 676,20	(1) R/W
Number of DVD's (**÷R13**)	= 57 975	(1) C
Volume (165 000 – 57 975)	= 107 025 DVD's	(1) C
		4

OR

(C) (C) (R/W) (R/W)

NPV (R0) = [V (**R40** – **R27**) (1 – **0,28**)(3,993)] – **R4m**

= [V13 (2,8750)] – R4m

= V37,373 – R4m

37,373V = R4 000 000

V = 107 023 DVD's (4)

OR

After tax cashflow needed p.a.		
[n = 5; i = 8%; PV = (4m); Comp PMT]	= R1 001 825,82	
Before tax cashflow (contribution)		
(R1 001 825,82 ÷ 0,72)	= R1 391 425	
Volume (R1 391 425 ÷ R13)	= 107 033 DVD's	(4)

Discuss your results and suggest appropriate management action.

Discussion and recommendations:

Sensitivity (Price) = R4,57 / R40

= 11,43% (1) C

The price can drop by R4,57 (11,43%) before the NPV becomes negative. (1) C

Sensitivity (Volume) $= \frac{57\,975}{165\,000}$

= 35,1% (1) C

The volume can drop by 57 975 DVD's (35,1%) before the NPV becomes negative. (1) C

The results suggest that the project's NPV is more sensitive to price variations than to changes in volume. Pricing policy should be reviewed to ensure that prices do not decline by more than 11%. A focus on advertising will ensure that demand is kept at the proposed price. Advertising will lower contribution, assuming that these costs are variable. (C) ito% (2) R/W

Max 4

(c) Calculate the NPV of the proposed investment in the movie production company.

Expected value of annual sales

Sales (R)	Probability	Total (R)
8 000 000	0,2	1 600 000
10 000 000	0,4	4 000 000
14 000 000	0,3	4 200 000
20 000 000	0,1	2 000 000
		11 800 000

(2)R/W

	R'000							
Year	**0**	**1**	**2**	**3**	**4**	**5**	**6**	
Purchase of company	(800)							(1) R/W
Sales 5%		11 800	**12 390**	13 010	13 660	**14 343**		(2) R/W y2,5
Legal/professional		(40)	(40)	(40)	(40)	(40)		(½) R/W
Lease rentals		(24)	(24)	(24)	(24)	(24)		(1) R/W
Studio hire (R360' x 3)		**(1 080)**	(1 080)	**(1 350)**	(1 485)	**(1 634)**		(3) R/W y1,3,5
Camera hire (R80' x 3)		(240)	(240)	(240)	(240)	(240)		(1) R/W
Technical staff (R1 040' x 3)		**(3 120)**	(3 338)	(3 572)	(3 822)	**(4 090)**		(2) R/W y1,5
Screenplay (R100' x 3)		**(300)**	(336)	(376)	(421)	**(472)**		(2) R/W y1,5
Actors salaries (R1 400' x 3)		(4 200)	(4 494)	(4 809)	(5 145)	**(5 505)**		(1) R/W y5
Costumes/wardrobe (R120' x 3)		(360)	(360)	(360)	(360)	(360)		(1) R/W
Set design (R300' x 3)		(900)	(900)	(900)	(900)	(900)		(½) R/W
Non-production staff wages		(120)	(128)	(137)	(147)	(157)		(1) R/W
Lost income (office accommodation)		(40)	(40)	(40)	(40)	(40)		(1) R/W
Cash flow before tax	(800)	1 376	1 410	1 162	1 036	881	-	
Taxation at **28%**	-	-	**(385)**	(395)	(325)	(290)	**(247)**	(3) 1 for 28%, R/W 1 for flow in year 2 C + 1 for tax in Y6 R/W
Net cash flow	(800)	1 376	1 025	767	711	591	(247)	
Discounted at **14%**	1	0,877	0,769	0,675	0,592	0,519	0,456	(1) for 14%
Present value of cash flow	(800)	1 207	788	518	421	307	(113)	
∴ NPV	R2 329						Max	(1) C 20



The **reading** and identification of information were critical. Some costs were driven by the number of movies made (three) and when not applied correctly cost students marks. Students often counter with 'I have done all the rest correctly', which we consider irrelevant. If this exercise was prepared on incorrectly read information in real life, you would have been fired or severely reprimanded.

(d) Distinguish between specific (diversifiable) and market (non-diversifiable) risk with specific reference to Filmstars (Pty) Ltd's activities

Specific (diversifiable) risk –

- is specific to an individual company, for example impact on cash flows resulting from a new competitor or a technology change, which makes one of the company's products obsolete,
- such as more popular films / better (improved) decoder.

(1) Example

(1) reason must be appl

Market (non-diversifiable) risk –

- arising from macroeconomic factors that affect the returns of all companies, for example in overall consumer demand due to spending ability
- for instance the impact which a recession would have on consumer spending.

(1) Example

(1) reason

4

Theory not studied and not applied to the given scenario.

(e) Making use of the risk measure – coefficient of variation – comment on the risk associated with the company's anticipated sales of produced films for the current year.

R	**EV per (c)**	**R – EV**	**$(R - EV)^2$**	**xP**	**σ**
Rm	**Rm**	**R'000**	**Rm**		**Rm**
8	11.8	-3 800	14 440 000	0,2	2 888 000
10	11.8	-1 800	3 240 000	0,4	1 296 000
14	11.8	2 200	4 840 000	0,3	1 452 000
20	11.8	8 200	67 240 000	0,1	6 724 000
					12 360 000
	(1) EV	(2)	(1)		(2)

R/W: If no's incorrect max 3

$\sigma = \sqrt{12\ 360\ 000}$

= R3,5157m

(1) C for calc.$\sqrt{}$

$CoV = \frac{\sigma}{EV}$

$= \frac{R3,5157m}{R11,800m}$

(1) C

= 0,298 OR 29,8%

(1) R/W

In the calculation of the standard deviation, instead of deducting the expected value (R11 800 000 per c) from the possible sales values (column 1), students often deducted the values obtained during the expected value calculation eg. R16 000 000 per c from the possible sales of R8m. Several students did not attempt the calculation at all, which could be an indication that they did not study the topic, although it was tested during the year with a resultant poor outcome.

Comment:

- Risk is indicated by the **relative dispersion** in the probability distribution (coefficient of variation).
- The risk associated with the produced film sales is low as evidenced by the relatively low coefficient of variation and relatively low standard deviation. (2)

Max 10

(f) Briefly explain how the tax treatment of capital purchases can affect an investment decision.

Capital allowances are usually granted for capital purchases that *accelerate* the write-off of these types of assets (eg. 40% : 20% : 20% : 20%). As they tend to reduce the tax payable during the initial periods, a *positive cash* inflow is also *initially accelerated*. Given the time value of money, *this contributes to a positive investment in capital* and *reduces the capital cost of the project*. Increases NPV. (2)

Students did not link the capital allowances granted for tax purposes to the cash flow impact in the capital budgeting process and its positive effect on the NPV.

(g) Calculate the budgeted breakeven point in **Rand** for Filmstars (Pty) Ltd's decoder section.

$$\text{BEP} = \frac{\text{Total fixed cost}}{\text{Contribution \%}} = \frac{34\,500\text{hrs} \times \text{R}38{,}50/\text{hr}}{,35} = \frac{\text{R}1\,328\,250}{,35}$$

(1) R/W for top
(1) R/W for bottom

= R3 795 000

(h) **Address the following in respect of the order received by** Filmstars (Pty) Ltd for 6 000 decoders to be supplied to Hot TV:

(i) Calculate the **minimum price** at which the decoders could be supplied

	R	
Materials		
(R123,65 x 6 000 units)	**741 900**	(1) R/W
Variable overhead		
(R24,60 x 2,63hr x 6 000 units)	**388 188**	(1) R/W
W1		
Incremental fixed cost	**28 500**	(1) R/W
Total	1 158 588	
Lowest SP per decoder		
(÷ 6 000)	193,13	(1) C

W1 *Cumulative average learning time per unit*

4 batches will be required (1) R/W

Col. 1 Batch	Col. 2 Average Time (Prev. hr x 85%)	Col. 1 x Col. 2 Cumulative	
1	5 463,0	5 463	(1) C for process
2	4 643,5	9 287	
4	3 947,0	15 788	

15 788 hrs ÷ 6 000 units = **2,63** hours per unit (1) R/W

OR

$Y = a X^{b}$

$Y = 5\,463 \text{ hrs} \times 4^{\frac{\log 0.85}{\log 2}}$

$Y = 3\,947{,}02 \text{ hrs}$ (1) process

Total time for the first four batches = 4 x 3 947,02 hrs = 15 788 hrs
Time per unit = 15 788 hrs ÷ 6 000 units = **2,63** hours per unit (1) R/W

(ii) Calculate the price(s) at which the decoders could be supplied on the assumption that the decoder division does not want to make a loss during the ensuing year

(a) Assume further orders from Hot TV

Current free capacity 34 500 x 70%	= 24 150 hrs	(1) R/W
At 2,63 machine hours per unit this equates to (÷ 2,63) (or 8 050 when 3 hours p.u. used)	= 9 182 units	(1) C

The total price will have to recover the variable costs, the loss and the incremental fixed costs. (2)

W2 *Budgeted profit / (loss)*

	R	
Contribution (R3 627 000 x 35%) x 30%	**380 835**	(1) R/W
Fixed costs (34 500 hrs x R38,50)	**(1 328 250)**	(1) R/W
(Loss)	(947 415)	
Materials (R123,65 x 9 182 units)	**1 135 354**	(1) C
Variable overhead (R24,60 x 24 150 hrs)	**594 090**	(1) C
	1 729 444	
Add: Incremental fixed costs	**28 500**	(1) R/W
Loss for the year	**947 415**	(1) C
	2 705 359	
Price per unit (÷ 9 182)	**R294,64**	(1) C

In this case Hot TV can sell the decoders at **R305** which gives them a profit of **R10,36** per unit.

Total direct profit = R10,36 x 9 182 units			(1)	(2) R/W (1) C

(b) Assume the order remains at 6 000

	R		
Total cost as before	1 158 785	(1)	C
Loss for the year	947 415	(1)	C
	2 106 200		
Cost per unit (÷ 6 000)	R351,03	(1)	C
In this case Hot TV can sell the decoders at **R400** and make a profit of **R48,97** per decoder.		(2)	R/W
Total direct profit = R48,97 x 6 000	R293 820	(1)	C
	Max	15	15

Make a final recommendation to management regarding a possible course of future action.

We recommend a purchase of 6 000 decoders at a unit cost of R351,03 each. (1) C

Based on all the information presented to us, Hot TV will make a profit of

- R293 820 on 6 000 units bought and sold
- R95 126 on 9 182 units bought and sold.

If Hot TV is convinced that they can recover the **differential profit** *of R198 694* through **additional subscriptions** and that their **growth** is the more important factor, you should recommend the higher production figure of **9 182** decoders. (3)

(1)

Max 4

This part was a challenge to many students. Students often did not identify which costs were relevant or incremental and equally often confused the fixed production overheads (R38,50 per machine hour) with the variable production overheads (R24,60 per machine hour). Numerous students could not do the learning curve calculations correctly. Students often did not identify that the division was currently trading at a loss, which implied the loss had to be recovered first (equivalent to a fixed cost then and standard basic principle).

(i) Briefly explain whether the subsidization process used by Hot TV is reasonable and sustainable from a business perspective.

- The numbers currently involved are *low*, for a *new entrant*, so it should be sustainable for some time. (2)
- It is a policy actively and successfully pursued by the major player in the market. (1)
- It is based on the premise that it will generate repeat and continuous income from the subscriber – *recurring income*. (1)
- Subscribers may more easily *convert* to more expensive bouquets and thereby increase income. (1)
- The sales adage that it is cheaper to hold onto an existing customer than to obtain new customers is the fundamental base. (1)
- Possible additional income from guarantees and insurance. (1)
- The process appears reasonable and sustainable from a business perspective. (1)

Max 5

Too gain market share – This comment is too generalised and received **no** mark. At this level students must direct their comments at the given scenario.
Several students did not understand what was meant by the subsidisation process or confused Filmstars and Hot TV.

(j)(i) Calculate the Fixed overhead over or under allocated during September 2011 for/by the wiring set division.

Production of 1 320 wire sets during September 2011 falls in the range [1 100 – 1 500] and the high–low method should be applied to this range:

$$\therefore \text{ VC} = \frac{\text{R37 950 – R36 115}}{\text{1 500 – 1 100 sets}}$$

$$= \frac{\text{R1 835}}{\text{400 sets}}$$

= **R4,59** per set (1) R/W

∴ FC = R36 115 – (1 100 sets x R4,59)
= **R31 066** (based on rounded off figures) (1) R/W

Many students did not read the given information carefully and consequently applied the high-low method incorrectly when calculating the fixed and variable overheads. They used the highest and lowest values in the whole table instead of the applicable range. Many students used the actual fixed overheads instead of the budgeted fixed overheads when calculating the over or under allocation.

Average quantity for range of [1 100 – 1 500] = 1 300 sets

∴ FOH rate	= R31 066 ÷ 1 300 sets	
	= **R23,897** per set	(1) R/W
FOH allocated (x 1 320)	= R31 544,04	(1) R/W
Budget	R31 066,00	
Over – allocated	R478,04	(2) 1 for over 1 amount
		6

Budget 1 300 vs actual of 1 320, therefore good production and additional recovery. Recovery of overhead and over/under recovery are regular examination features.

(ii) Calculate the normal loss incurred during September 2011 for/by the wiring set division.

Given: Loss of 10% of completed products is considered normal ∴ 1 320 x 10% = 132 sets

Production cost:	**R**	
Material and labour		
[(R19 800 + R17 754)/1 320] x 132	3 755,40	(1) R/W
VOH (132 x R4,59)	605,88	(1) R/W
FOH (132 x R23,897)	3 154,40	(1) R/W
Recovery (132 x R10)	(1 320,00)	(1) R/W
Net loss for September	6 195,68	
		4

In the calculation of the normal loss students sometimes forgot to include the fixed overheads as well as the recovery of R10 per rejected set.

The average mark for this paper was 38%.

QUESTION 7

SMILE (PTY) LTD **SUGGESTED SOLUTION**

(Source: Exam 2011 DIPAC26 (MAC4861))

Notes will focus on areas of concern or where consistent errors occurred.

R/W – Right/Wrong mark ***C – Consequential mark***

Part (a)(i): How many litres of direct material were purchased (and used) in total to produce the actual quantity of toothpaste manufactured during the 2011 year?

Actual	**Mint**	**Citrus**	**OR**	**Total**	
	Tubes	Tubes		Tubes	
Sales	135 000	57 000		192 000	
Opening inventory -	(7 000)	(6 000)		(13 000)	
Closing inventory +	6 000	8 000		14 000	
Produced	134 000	59 000		193 000	(1) R/W
Equivalent litres produced (÷ 10)	13 400	5 900		19 300	(1) R/W
Litres purchased (Produced ÷ 95 x 100)	14 105	6 211		20 316	(1) C

① Students make mistakes when calculating the missing component and this is considered a **fundamental** error. The normal layout for the 4 components: OI + Production/Purchases – CI = COS units ⇨ sales units.

② The conversion to equivalent litres not done as the 100 ml either misread or not linked to the required 'litres of material'.

③ Where the conversion was correctly executed (ie ÷ 95 x 100), the consequential mark is awarded. Students often ignored this normal loss.

Part (a)(ii): What is the actual direct material cost per tube of toothpaste, per product line?

	Mint	**Citrus**	**Total**	
Litres purchased (Produced ÷ 95 x 100)	14 105	6 211	20 316	
Cost (R) (Litres x R42,75)	603 000	265 500	868 500	
Cost per tube (Cost ÷ tubes produced) (R) [868 500 ÷ 193 000]	4,50	4,50	4,50	(1) R/W

Alternative
R42,75 ÷ 95 x 100 = R45,00 per litre produced
R45,00 per litre ÷ 10 tubes per litre = R4,50 per tube (1) R/W

Part (b)(i): Calculate the fixed overhead recovery rate for the company using machine hours as the allocation base.

Budgeted fixed overhead rate

$$\frac{\text{R 600 000}}{\text{(1 750 hrs + 960 hrs)}}$$

$$= \frac{\text{R 600 000}}{\text{2 710 hrs}}$$

= R221,40 / hour (1) R/W

Fixed overhead (recovery) rates are based on the budgeted value(s) and quantities.

Part (b)(ii): Calculate the amount of overhead that was budgeted to be allocated to each product, per tube.

Budget	**Mint Tubes**	**Citrus Tubes**	**Total Tubes**	
Sales	120 000	62 000	182 000	
Opening inventory -	(7 000)	(6 000)	(13 000)	
Closing inventory +	12 000	4 000	16 000	
Produced	125 000	60 000	185 000	(1) R/W

Some students ignored the 'budgeted' instruction. This implies the consequential marks at the end of the section cannot be awarded.

Budgeted machine hours per tube

Mint

$$\frac{\text{1 750 hours}}{\text{125 000 tubes}} = \text{0,014 hr/tube}$$

Citrus

$$\frac{\text{960 hours}}{\text{60 000 tubes}} = \text{0,016 hr/tube}$$

Budgeted overhead cost per tube

Mint: R221,40/hour x 0,014 hr/tube	R3,10		(1) C
Citrus: R221,40/hour x 0,016 hr/tube		R3,54	(1) C

Part (b)(iii): Calculate the over- or under-recovery of fixed manufacturing overheads for the company. State whether the calculated amount represents an over- or under-recovery of fixed overheads.

Fixed overheads	**R**	
Mint: R3,10 x 134 000 actual tubes produced	415 400	
Citrus: R3,54 x 59 000 actual tubes produced	208 860	
Total actual fixed overheads allocated to production	624 260	(1) C
Total budgeted fixed overheads	600 000	
Over recovery	24 260	(1) R/W

Note: ½ mark for **R24 260** and ½ mark for **Over** recovery



Recoveries are determined by measuring against the budgeted amount. Students are uncertain when it is over (quantity more than budget: 134 000 + 59 000 vs 125 000 + 60 000) or under.

Part (b)(iv): Calculate the fixed manufacturing overhead expenditure variance. State whether the variance is favourable or adverse.

Fixed overheads	**R**	
Total actual fixed overheads expense	620 000	
Total budgeted fixed overheads	600 000	
Adverse expenditure variance	20 000	(1) R/W

Note: ½ mark for **R20 000** and ½ mark for **Adverse**

Part (c)(i): Calculate the actual total fixed selling and distribution cost per product line for the 2011 financial year.

High-low method	**Mint**		**Citrus**		
	Tubes sold	**R**	**Tubes sold**	**R**	
Current year	135 000	135 900	57 000	111 510	
Previous year	132 000	135 360	55 800	111 234	
Difference	3 000	540	1 200	276	(2) R/W
Variable cost per tube sold:					
Mint: R540 / 3 000 tubes		0,18			(½) R/W
Citrus: R276 / 1 200 tubes				0,23	(½) R/W
Current year total cost (given)		135 900		111 510	
Current year variable cost:					
Mint: 135 000 tubes x R0,18		(24 300)			(½) C
Citrus: 57 000 tubes x R0,23				(13 110)	(½) C
Fixed cost		111 600		98 400	(1) C

Part (c)(ii): Calculate the actual contribution per tube per product for the 2011 financial year, as well as the actual contribution percentage per tube per product.

Note:

Step 1: Calculate **total budgeted cost** per line item by multiplying **budgeted production** (per part (b)(ii)) by **budgeted cost per tube** (given).
Step 2: **Deduct /add variances** (given) from/to **total budgeted cost** per line item (per step 1) to calculate **total actual cost** per line item.
Step 3: Divide **total actual cost** per line item (per step 2) by **actual production** (per part (a)(i) if calculated per product line, alternatively, calculate here) to get **actual cost per tube** per line item.
Step 4: Calculate **contribution** per tube per product line.

Step 1: Total budget:	**Mint**		**Citrus**		
	Calculation	**R**	**Calculation**	**R**	
Antibacterial agent	N/A	**–**	60 000 x R0,50	**30 000**	(½) R/W
Packaging – tubes	125 000 x R0,40	**50 000**	60 000 x R0,40	**24 000**	(1) R/W
Packaging – boxes	125 000 x R0,15	**18 750**	60 000 x R0,15	**9 000**	(1) R/W
Direct labour	125 000 x R0,60	**75 000**	60 000 x R0,80	**48 000**	(1) R/W
Variable overhead	125 000 x R2,00	**250 000**	60 000 x R2,20	**132 000**	(1) R/W
	R3,15	**393 750**	R4,05	**243 000**	

② The descriptive 'direct' for material and labour (and the 'costs per tube' (N5) indicated their variable nature.

② Students do not understand the implication of a variance: eg Antibacterial agent favourable variance (F) of 2 270 will decrease the budgeted cost. The variances thus allow one to move from budget to actual.

③ In step 3 the actual production quantities are calculated – see earlier comment.

④ Contribution is determined by deducting *all* the variable costs from the given actual sales prices.
⑤ Note the alternative calculations; this is often possible in a cost exercise: the proverbial more ways than one to kill a cat. Always follow the route that you are comfortable with.
⑥ Some students attempted step 4 on the basis of standard costs and variances which distorted the costs per tube and limited the number of marks awarded.

Step 2: Total	**Budget**		**Variance (given)**		**Actual**		
	Mint R	**Citrus R**	**Mint R**	**Citrus R**	**Mint R**	**Citrus R**	
Antibacterial agent	–	30 000	–	2 270 F	–	27 730	(½) C – Dec cost
Packaging – tubes	50 000	24 000	8 960 A	1 960 A	58 960	25 960	(1) C – Inc cost
Packaging – boxes	18 750	9 000	2 670 F	1 920 F	16 080	7 080	(1) C – Dec cost
Direct labour	75 000	48 000	25 500 A	5 100 A	100 500	53 100	(1) C – Inc cost
Variable overhead	250 000	132 000	42 120 A	750 A	292 120	132 750	(1) C – Inc cost
	393 750	243 000	73 910 A	3 620 A	467 660	246 620	

Step 3:

Actual production per product line as calculated in part (a)(i) or alternatively calculate here:

Mint: 135 000 – 7 000 + 6 000 = 134 000 tubes (1) R/W
Citrus: 57 000 – 6 000 + 8 000 = 59 000 tubes (1) R/W

Actual R per tube:	**Mint**		**Citrus**		
	Calculation	**R***	**Calculation**	**R***	
Antibacterial agent	N/A	**–**	27 730 / **59 000**	**0,47**	(½) C – Divided
Packaging – tubes	58 960 / **134 000**	**0,44**	25 960 / **59 000**	**0,44**	(1) C by actual
Packaging – boxes	16 080 / **134 000**	**0,12**	7 080 / **59 000**	**0,12**	(1) C quantities
Direct labour	100 500 / **134 000**	**0,75**	53 100 / **59 000**	**0,90**	(1) C
Variable overhead	292 120 / **134 000**	**2,18**	132 750 / **59 000**	**2,25**	(1) C

* Note: May be given per line, 1½ mark each: for example ((125 000 x 0,4) + 8 960))/134 000 = 0,44

Step 4: Contribution	**Mint R**	**Citrus R**	
Selling price – given	16,20	20,00	(½) R/W
Variable costs:	(8,17)	(8,91)	
Direct material – per part (a)(ii)	(4,50)	(4,50)	
Antibacterial agent	–	(0,47)	See above 1½ mark per item R/W
Packaging – tubes	(0,44)	(0,44)	
Packaging – boxes	(0,12)	(0,12)	
Direct labour	(0,75)	(0,90)	
Variable overhead	(2,18)	(2,25)	
Variable selling and distribution cost – per part (c)(i)	(0,18)	(0,23)	
Contribution per tube	8,03	11,09	(2) C
Contribution percentage (R8,03 / R16,20; R11,09 / R20,00)	**49,6%**	**55,5%**	(2) C

Alternative 1:

Actual = Standard cost (budget) ± Variance ± Flex adjustment (refer to alternative 2)

If the calculation was (incorrectly) only done as follows: Standard cost (budget) ± Variance, then a maximum of 9 marks plus 3 marks if calculations were shown, can be earned. The (incorrect) calculations and answers would be as follows:

		Mint R	Citrus R	
Antibacterial agent:	Citrus: R0,50 – (R2 270 / 59 000)	–	0,46	(1)
Packaging – tubes:	Mint: R0,40 + (R8 960 / 134 000)	0,47		(1)
	Citrus: R0,40 + (R1 960 / 59 000)		0,43	(1)
Packaging – boxes: Mint:	R0,15 - (R2 670 / 134 000)	0,13		(1)
	Citrus: R0,15 - (R1 920 / 59 000)		0,12	(1)
Direct labour:	Mint: R0,60 + (R25 500 /134 000)	0,79		(1)
	Citrus: R0,80 + (R5 100 / 59 000)		0,89	(1)
Variable overhead: Mint:	R2,00 + (R42 120 / 134 000)	2,31		(1)
	Citrus: R2,20 + (R750 / 59 000)		2,21	(1)
				(3) if calc's shown

Alternative 2:	Mint R	Citrus R	
Selling price	16,20	20,00	(1) R/W
Budgeted costs			
(4,00 + 0,4 + 0,15 + 0,6 + 2,00)	7,15		(2) R/W
(4,00 + 0,5 + 0,4 + 0,15 + 0,8 + 2,20)		8,05	(2) R/W
	9,05	11,95	(2) R/W
Materials variance	(0,50)	(0,50)	(2) R/W
Average variance per tube (excl mat) – W1	(0,55)	(0,06)	[4] see below*
Variable S + D per c (i)	(0,18)	(0,23)	
Contribution per tube	7,82	11,16	
Flex adjustment – W2	0,21	(0,07)	(2) C
Actual contribution per tube	8,03	11,09	(2) C
W1			
Total variances			
(8 960 – 2 670 + 25 500 + 42 120)	73 910		*(1) R/W
(-2 270 + 1 960 – 1 920 + 5 100 + 750)		3 620	(1) R/W
Production per (a)(i) – Actual	134 000	59 000	(2) R/W
W2 – Flex adjustment			
Variable cost (excluding material) – Budget	3,15	4,05	
(x 125 000 ÷ 134 000)	2,94		
(x 60 000 ÷ 59 000)		4,12	
	0,21	0,07	

Alternative 3:	Mint R	Citrus R	
Antibacterial agent, packaging, labour, VOH:			
Total budgeted cost (125 000 x R3,15) (60 000 x R4,05)	393 750	243 000	(4½) R/W
Total variances – adverse	73 910	3 620	
Total actual cost	467 660	246 620	(4½) R/W
Per unit (467 660 / 134 000) (246 620 / 59 000)	(3,49)	(4,18)	(4½) R/W
Direct material – per part (a)(ii)	(4,50)	(4,50)	
Variable selling and distribution cost – per part (c)(i)	(0,18)	(0,23)	
	(8,17)	(8,91)	
Selling price – given	16,20	20,00	(½) R/W
Contribution per tube	8,03	11,0 9	(2) C

Max 18

Part (c)(iii): Calculate the actual number of tubes of each product line that Smile had to sell for the 2011 financial year in order to break even.

Note to markers: If break-even was calculated individually for each product: Max 4 marks of which 2 marks are for fixed costs.



There was no indication in the question of how some of the fixed costs should be split between the products. The correct approach is to determine the weighted contribution based on the sales mix and not do an arbitrary split.

	R	
Fixed manufacturing overheads – given	620 000	(½) R/W
Fixed selling & distr. costs: R111 600 + R98 400 – per part (c)(i)	210 000	(1) C
Administrative costs – given	420 000	(½) R/W
Total fixed costs	1 250 000	

Weighted contribution based on sales mix:

(1) Weighting Correct

(1) C – contrib. R

= ((135 000 / 192 000) x R8,03) + ((57 000 / 192 000) x R11,09)
= (70,313% x R8,03) + (29,687% x R11,09)
= R5,65 + R3,29
= R8,94 per tube

Alternative:

(1) Qts Correct

(1) C - Contrib R

= ((135 000 x R8,03) + (57 000 x R11,09)) / 192 000
= (R1 084 050 + R632 130) / 192 000
= R1 716 180 / 192 000
= R8,94 per tube

Break-even number of tubes:

= Fixed costs / Weighted contribution per tube
= R1 250 000 / R8,94 per tube (1) C – if Answer Given
= 139 821,03 tubes
= 139 822 tubes *(round-up)*

Mint: 135 000 / 192 000 x 139 822 (1) C
= 70,313% x 139 822
= 98 313 tubes

Citrus: 57 000 / 192 000 x 139 822 (1) C
= 29,687% x 139 822
= 41 510 tubes

Part (c)(iv): Calculate the total percentage margin of safety relating to the actual results for the 2011 year and comment on your answer. Also give advice to management on how the results of the ratio may be improved. (Calculation = 2 marks; Comments / advice = 5 marks)

(192 000 – 139 822) / 192 000 = 27,176% (2) 1 for above → C 1 for correct below



The MoS is based on the actual sales. Note that a single comment cannot earn 5 marks. The last detailed bullet illustrates one's grasp of the principle.

Comments and advice:

- Sales revenue (Rand) can decline by 27,2% before the company will start making losses. (Assumption that accounting profit = tax profit). (1)
- The fixed and variable selling and distribution costs have not increased from the previous year. This could indicate a higher than inflation increase in the coming year which is a cause for concern since it would increase the break-even point, thus reducing the margin of safety. (2)
- Sales of citrus are *5 000 tubes below budget* (57 000 actual v. 62 000 budget) which could be a cause for concern. Efforts should be made to improve the sales of citrus *as higher contribution %*. (2)
- The margin of safety can be improved by increasing actual sales and / or decreasing break-even sales. (1)
  - Actual sales can be increased by *increasing sales volume* and / or *increasing sales price per unit* (but this could potentially reduce volumes). (2)
  - Break-even sales can be decreased by *reducing or containing fixed costs* and / or *improving the contribution per unit*. The latter can be achieved by reducing or containing costs and / or increasing sales price per unit (2)

Max (5)

Part (d)(i): Calculate the minimum selling price per tube of Strawberry toothpaste that Smile will have to ask to break even on the once-off order.

	Strawberry	
	R	
Direct materials (basic ingredients and flavouring)	2,00	
Packaging – tubes	0,25	
Packaging – boxes	0,12	
Direct labour	0,55	
Variable overhead	1,80	
Variable cost per tube	4,72	(1½) R/W
	R	
Rental of machine	75 000	(½) R/W
Manufacturing overheads	35 000	(½) R/W
Delivery costs	7 000	(½) R/W
Total fixed cost for the order	117 000	

Number of tubes ordered	30 000	
Fixed cost per tube (R117 000 / 30 000 tubes)	R3,90	(1) R/W
Minimum selling price to break-even	**R**	
Variable cost per tube	4,72	
Fixed cost per tube	3,90	
	8,62	(1) C

Alternative:
R117 000 + (30 000 x 4,72) = R258 600 / 30 000 = R8,62 (2) C

5

The term 'break-even' in the required indicates that the incremental fixed costs should also be considered in terms of the 'selling price **per tube'.**

Part (d)(ii): Discuss the conditions that must generally apply for such a special pricing order to be acceptable.

- Is *sufficient capacity* available for all the resources needed to complete the order? If some *resources are fully utilised*, the selling price must cover the *opportunity costs* of the scarce resources. (1) (1)
- (Or: Are there any *fixed contracts* for the other two products that have to be complied with which may make it impossible to accept the once-off contract?) (1)
- The selling price of the special order *will not affect future selling prices*. Customers will not demand the lower selling price on normal orders. (1)
- The order will utilise unused capacity for *only a short period* and capacity will be released for use on more profitable opportunities. (1)

Max (3)

Part (d)(iii): Discuss the qualitative factors to be considered.

- A selling price higher than the minimum price will provide a contribution to fixed costs which would not otherwise have been received. (1)
- Gain entry to a new market. Possibility of more/repeat orders in future. (2)
- The children using the toothpaste could later become users of the adult toothpaste. (1)
- If the parents of the children using the toothpaste perceive the toothpaste to be of high quality, they could start using the adult toothpaste. (1)
- How accurately are the costs of the new product calculated? (1)
- Will the materials and new packaging be available on time and be of the right quality? (1)
- Will there be sufficient skilled staff? (1)
- Reliability of other party to the contract: will payment for the 30 000 tubes definitely be received? (1)

- Goodwill lost if sales of the other two products are sacrificed? (1)
- Any restraint placed by counter-party on future development of a similar product in company's own name? (1)
- Safety of product, especially for children: fluoride could be harmful in large doses. (1)
- Any other valid point (max 1) (1)

Max (4)

Part (e)(i): Calculate the 2012 budgeted fixed manufacturing overhead cost per tube for both Mint and Citrus using Activity Based Costing (ABC). Round total costs to the nearest Rand.

Activity	**Mint**	**Citrus**	**Total**	
No. of batches:			89 batches	
M: 145 000 tubes / 2 500 tubes per batch	58 batches			(½) R/W
C: 62 000 tubes / 2 000 tubes per batch		31 batches		(½) R/W
Store costs:			387 orders	
M: 58 batches x 4 orders/batch	232 orders			(½) R/W
C: 31 batches x 5 orders/batch		155 orders		(½) R/W
Set-up costs:			120 set-ups	
M: 58 batches x 1 set-up/batch	58 set-ups			(½) R/W
C: 31 batches x 2 set-ups/batch		62 set-ups		(½) R/W
Power:			4 295 kJ	
M: 58 batches x 50 kJ/batch	2 900 kJ			(½) R/W
C: 31 batches x 45 kJ/batch		1 395 kJ		(½) R/W
Depreciation:			3 000 hrs	
M: 1 950 machine hours (given)	1 950 hrs			(½) R/W
C: 1 050 machine hours (given)		1 050 hrs		(½) R/W
Store costs:			180 000	
M: 232 orders / 387 orders x R180 000	107 907			(½) C
C: 155 orders / 387 orders x R180 000		72 093		(½) C
Set-up costs:			120 000	
M: 58 set-ups / 120 set-ups x R120 000	58 000			(½) C
C: 62 set-ups / 120 set-ups x R120 000		62 000		(½) C
Power:			105 000	
M: 2 900 kJ / 4 295 kJ x R105 000	70 896			(½) C
C: 1 395 kJ / 4 295 kJ x R105 000		34 104		(½) C

Activity	**Mint R**	**Citrus R**	**Total R**	
Depreciation:			255 000	
M: 1 950 hrs / 3 000 hrs x R255 000	165 750			(½) R/W
C: 1 050 hrs / 3 000 hrs x R255 000		89 250		(½) R/W
Total	402 553	257 447	660 000	



Take note that in some scenarios a proportional allocation of activity costs will yield an incorrect answer.

Per tube:

M: R402 553 / 145 000 tubes	R2,78		(½) C
C: R257 447 / 62 000 tubes		R4,15	(½) C

Part (e)(ii): Assume the 2012 budgeted fixed manufacturing overhead cost using traditional absorption costing is R2,96 and R3,73 per tube of Mint and Citrus respectively. Compare the traditional and ABC 2012 budgeted fixed manufacturing overhead costs per tube and briefly comment on the impact of ABC on the cost per tube which could have pricing and profit consequences.

2012 budgeted fixed manufacturing overhead cost per tube	**Mint R**	**Citrus R**	
Traditional (based on machine hour rate)	2,96	3,73	
ABC	2,78	4,15	
Difference	0,18	(0,42)	(1) C

- Citrus uses more machine hours per tube to produce than mint and therefore the traditional approach allocates more overhead costs to each tube of citrus. (1)
- The ABC approach allocates overheads more accurately by allocating the overheads to the product that used more of the overheads as identified by the cost drivers. (1)
- Citrus also uses more resources to produce than mint as measured in terms of cost drivers (ABC, e.g. volume of Citrus 30% of total products, but store use 40%, etc.) resulting in more overhead costs being allocated to each tube of citrus. (2)
- Mint is allocated less overheads using ABC compared to the traditional approach while citrus is allocated more overheads using ABC. (1)
- As per traditional ABC theory, the lower-volume product is undercosted whilst the higher-volume product is overcosted. (1)
- The higher-volume product is in fact SUBSIDISING the lower-volume product. (1)
- Using ABC mint is actually more profitable than suggested by the traditional approach while citrus is less profitable. (2)
- If the selling prices have been determined using the traditional approach, Smile could consider reducing the price of mint slightly and increasing the price of citrus. (2)

Max (5)



A general discussion of the affect of ABC on fixed overhead costs will not earn more than one mark (at best). The discussion *must* focus on the specific scenario given.

Part (f): A foreign investor is interested in taking up a majority share in the company. The investor's initial investigation, however, indicated that the current return on capital achieved by Smile is not sufficient to make an investment worthwhile. The investor has asked management to consider improving the profitability of the company in 2012 by either increasing the selling price of the products by 10% or reducing the size of the tubes by 10%. The financial director has asked for you to discuss the possible consequences of both of these proposals. Note that calculations are NOT required.

Increase selling price by 10%:

- If demand is inelastic, the sales Rand value will increase and profits will increase accordingly. (1)
- Demand could decrease if the market is sensitive to price changes. This is quite likely since there are a number of close substitutes in the toothpaste market. The decline in demand could be so great that a lower overall profit is achieved after the price increase. (2)
- It may be difficult to recover market share lost in this manner. (1)
- The price increase will be higher than the inflation rate. This could damage the image of the company. (1)
- Any other valid point (max 1) (1)

Max (4)

Decrease size of tubes by 10%:

- Costs will decrease, although probably not with a full 10% and profits will increase accordingly (cost savings will depend on whether tube, box and etc, also reduce in price / cost). (1)
- Customers could perceive the size decrease as a price increase and if demand is inelastic, demand could decrease, resulting in lower profits. (1)
- The reduction in size could be perceived by customers as being "sneaky" which could damage the image of the company. (1)
- Clients may perceive competitors' products as higher value as higher quantity for similar/same price. (1)
- Production machinery may have to be modified to accommodate the new size e.g. packaging machine may have to be modified for smaller tubes and boxes. This could result in additional costs. (1)
- Any other valid point (max 1) (1)

Max (4)

Part (g): Define incremental and zero-based budgeting and summarise the main differences between the two budgeting techniques.

- Incremental budgets are based on existing operations and past budgets. They are then adjusted for expected changes during the new budget period (e.g. changes in product mix, volumes and prices). (2)
- Incremental budgets assume that past activities are still appropriate. They are not questioned or examined again. Subsequently past inefficiencies and waste are accepted and continue in the future. (2)
- Zero-based budgeting does not perpetuate past inefficiencies. It seeks to reduce non-value adding activities through justification of each activity and cost on a cost/benefit basis. (2)
- Zero-based budgeting is mainly used for discretionary costs and support activities and works from the assumption that existing expenditure should be justified from scratch every year. (2)
- Zero-based budgeting is more costly and time-consuming to use than incremental budgeting. (1)
- Incremental budgeting done annually and/or ZBB usually every 3 to 5 years (rolling basis). (1)

Max (5)

Part (h): Calculate the expected sales (in Rand) for the 2012 sales budget for both the Mint and Citrus product lines. Assume that the company will incur the additional marketing cost and that Mint and Citrus will be the only two products sold in the 2012 year.

Current market share: 192 000 tubes = 6%
Total market = 192 000 tubes / 6%
= 3 200 000 tubes (1) R/W

Growth in total market during 2012 year:
Total market = 3 200 000 tubes x 1,01 (1) C
= 3 232 000 tubes

Smile's 6,5% market share in 2012 year:
New market share = 3 232 000 x 6,5% (1) C
= 210 080 tubes

New sales mix:
Mint: 210 080 x 71% = 149 157 tubes (½) C
Citrus: 210 080 x 29% = 60 923 tubes (½) C

Price increase:
Mint: R16,20 x 1,065 = R17,25 per tube (1) R/W
Citrus: R20,00 x 1,06 = R21,20 per tube (1) R/W

2012 Budget:	**R**	
Mint: 149 157 tubes x R17,25	2 572 958	(1) C
Citrus: 60 923 tubes x R21,20	1 291 568	(1) C
	3 864 526	



The conversion of market share to total market proved problematic. This was required to calculate the new split for the products.

Part (i): The managing director has heard of the residual income approach for purposes of evaluating performance. Briefly explain what the residual income approach is and name one advantage of the residual income approach.

- Where residual income is used for evaluating the *performance of divisional managers*, it is defined as controllable contribution less a cost of capital charge on the investment controllable by the divisional manager. (2)

- Where residual income is used for evaluating the *economic performance of the division*, it can be defined as divisional contribution less a cost of capital charge on the total investment in assets employed by the division. (2)

- Residual income is the amount with which shareholders' wealth is increased by generating a return on assets used that is higher than the cost of the financing of those assets. (1)

- Residual income encourages managers to act in the best interests of the company (/make decisions in line with NPV approach, which maximises shareholder wealth). (1)

- Residual income is flexible because different cost of capital percentage rates can be applied to investments that have different levels of risk. (1)

Max (3)

QUESTION 8

EWORX LTD **SUGGESTED SOLUTION**

(Source: Supplementary exam 2011 TOE408W (MAC4862))

a. ROI / ROE / RI

			2011 R'000	2010 R'000	
• **ROI =**	EBIT / Net total assets	=	28 994 / 98 826	27 618 / 92 494	
		=	29.3%	29.9%	(2)
• **ROE =**	NPAT / Equity	=	20 223 / 87 441		
		=	23.1%		(1)
• **RI =**	Operating profit - (total assets x WACC)	=	28 994 (1) – (98 826 (1) x 15%)		(2)
		=	14 170,1 ('000)		
					5

EBIT should be used in the calculation of Return on Investment (ROI) and Residual Income (RI), whereas NPAT is used for ROE.

b. ROI as measurement tool

- Activities such as manufacturing, data monitoring and Fastbook.com eg (1) rely on **equipment / use of hardware** to generate income and the use of **ROI seems fair**. (3)
- Software development and hardware installation eg (1) are **less dependent** on the use of equipment and their returns may **not be reasonable using** ROI – overstated returns. (3)

Max 5

Some students did not differentiate between the use of ROI in asset intensive divisions vs those that are less asset intensive. Required: '… stated business activities …'.

c. EVA

- EVA in basic application should consider future benefits (revenue) derived from current costs / expenditure and then adjust the Income Statement (SOCI) and Balance Sheet (SFP) for these costs. (2)
- EVA$^{(TM)}$ = Conventional divisional profit ± accounting adjustments – cost of capital charge on divisional assets. (2)

In terms of the above:

- Development of software packages may / will generate future revenue for **more** than one year through **royalties or sales** and therefore comply with the basic definition. The **research and development costs** will be **amortised** over the

period of the benefits being obtained. **EVA can be applied.** (3)

- Manufacturing of electronic components will provide a future benefit in that the cost of **inventory will be deferred**, but no other benefit will be derived. **EVA will not apply**. (2)
- Data monitoring services may refer to the development of a monitoring platform which **will generate future benefits**, thus **EVA will apply OR** the **costs related to the monitoring process (no future benefits**) where **EVA will not apply**. (2)

Max 9

Many students struggled to explain Economic Value Added (EVA) and consequently did not give appropriate discussions regarding the implementation thereof for the various required activities.

d. Break-even for hardware installation division for 2011

				R'000	
Revenue	(R9 764 – R732)			9 032	(1)
Variable costs				(1 457)	
Contribution				7 575	(1)
Contribution %				83,87%	(1)
Fixed costs		=	8 313 + 420 + 920 + 1 260		
		=		10 913	(2)
Less Fixed revenue				(732)	(1)
				10 181	
BEP	Fixed costs	=		10 181	
	Contribution %			,8387	
		=	(R'000)	12 139	(1)
					7

Basic principle: *fixed revenue* should be set off against fixed costs and be excluded from total revenue. In most organisations revenue will have some link to variable costs. As a consequence the alternative solution below was worth fewer marks.

Alternative (less correct answer):

				R'000	
Revenue				9 764	
Variable costs				(1 457)	
Contribution				8 307	(1)
Contribution %				85,08%	(1)
Fixed costs as above	8 313 + 420 + 920 + 1 260	=		10 913	(2)
Break-even	Fixed costs	=		10 913	
	Contribution %			,8508	
		=	(R'000)	12 827	(1)
					5

e. Closure of hardware installation division

Adjust profits for the target return:

		2011	2010	
Net loss – given		(2 606)	(629)	(1)
Target profit	(2011: R3 600 @ 15%); (2010: R1 230 – R720)	540	510	(1)
Adjusted loss		(2 066)	(119)	
Loss seems to increase from 2010 to 2011.				(1)

Therefore look at remnants after closure or non-avoidables:

		2012 R'000	
Income (2012 fixed monitoring fees)		(548)	(1)
25% of fixed labour (25% x R8 313')	(2 078) carried by group.	2 078	(1)
Rental	Carried by group.	420	(1)
Other FC, carried by group direct & indirect	(OR 920 x 60% = 552)	920	(1)C
Cost recovery	Other will now carry.	720	(1)
Net cost remaining if closed	Now carried by HI	3 590	
Once-off cost – retrenchment		1 198	(1)
∴ Do not close HI			(1)
		Max	**7**

Alternative:

		2011 R'000	
Lost revenue	R9 764 – R732	9 032	(1)
Labour	75% x R8 313	(6 235)	(1)
Variable cost		(1 457)	(1)
Income lost while fixed costs remain		1 340	(1)
Once off retrenchment costs of R1 198k		1 198	(1)
which will decrease contribution of other divisions?			(1)
Do not close hardware installation division.			(1)
		Max	7

Other issues:

- Workers required for maintenance contracts. (1)
- Penalties / outsourcing costs when cancelled (legal risk). (1)
- Can remaining 15% fulfil above? (1)
- Loss of expertise resulting from the retrenchment of staff. (1)
- Labour procedures to be followed for retrenchment. Staff morale. (1)
- Tax implications – can tax loss be carried over? (1)
- Impact of closing this division on revenue generation of other divisions e.g. software and data monitoring. (1)

Max 3
10

Relevancy items related to closure and continuance were often mixed up. The required qualitative issues were often ignored or too general in nature.

f. Bonus payable – telecommunication division – year one – market-based approach

		R	
Sales price		156,00	
Margin allowed (30% x R156,00)		(46,80)	(1)
Processing costs		(25,00)	(1)
Selling price / transfer price		84,20	
Component product line		**R'000**	
Revenue	60 000 + 60 000 = 120 000 units x R84,20 (N1)	10 104	(1)C
Variable costs	120 000 units x R51	(6 120)	(1)
Relevant fixed cost		(2 700)	(1)
Profit for year 1		1 284	

Bonus: 4% of (R1 284k – (5% x R10 104k)) = R31 152 (1)C

Profit line profits exceed 5% of revenue thus bonus will be payable in respect of the above. (1)
N1: Capacity = 450 000 mach. hrs / 2,5 hrs per unit = 180 000 units. No capacity constraint exists. (OR 120 000 units x 2,5 hrs = 300 000 hrs < 450 000 hrs) (1)

Max **7**

'READING': It seems that students did not always read carefully how to calculate the prices for the partner and the internal transfer in part (g). Students sometimes ignored the internal transfers when calculating the profit.

g. Profit year 2 – cost-based approach

Transfer price		**R**	
Variable costs		51,00	
Mark-up (70% x R51)		35,70	(1)
Transfer price – internal		86,70	(1)

Revenue		**R'000**	
Partner sales	(80 000 units x R84,20)	6 736	(1)C
Transfer	(60 000 units x R86,70)	5 202	(1)
External / export	(180k max – 80k – 60k = 40 000 units x ($12 x R7,20))	3 456	(2)
		15 394	
Variable costs	(180 000 units x R51)	(9 180)	(1)
Fixed Cost		(2 700)	(1)
Profit for year 2		3 514	
			8



Most students realised that there was a capacity constraint in year 2, but many students did not read the given information carefully and therefore reduced the partner sales or internal transfers instead of the exports. The application of mark-up (basic principle) remains problematic.

h. Impact on Eworx Ltd of funding the new component product line from internal sources

- Internal funding implies using own cash created by Eworx / equivalent to equity. (1)
- Is the additional fixed cost an **annual cash** cost or the **depreciation** on equipment bought? (2)
- In the case of the latter, what is the **capital outlay** and the **period for** which it will be operational? (2)
- Cash used for the above implies Eworx Ltd. will earn less interest on bank funds. (1)
- Own funding – equity – implies the WACC will increase as equity is more expensive than debt. (1)
- Improved gearing. (1)
- Reduced / strained liquidity. Less cash available for other products / operations. (1)
- Using own cash (no new loan) results in lower expenses (don't have to pay interest on loan) and don't have obligation to repay capital and interest. (1)
- Short term funds should not be used for long term project (3 years) (1)
- Any other valid point (Maximum 1) (1)

Max 8



Some students do not understand that funding from **internal sources** excludes obtaining loans and issuing debentures (debt).

i. Risks that Eworx Ltd should consider in terms of the new component product line

- Capacity utilisation – what is **period** as per above (equipment or cost)? (1)
- Capacity utilisation – partner contract for 3 years (1 – 3) and export contract for 3 years (2 – 4). (2)
- Full capacity for years 2 and 3 but what about the other years. (2)
- Can internal use be increased somehow, otherwise potential under-utilisation. (1)
- To what extent will costs actually increase – impact of industry inflation. (1)
- Export prices – what will happen to the **exchange rate** and / or will Eworx Ltd **hedge itself**? (2)
- Change in technology – stock obsolete? (1)
- Skilled staff available for expansion? (1)
- Possible manipulation of revenue and profit by management to get bonus. (1)
- Is strategic partner creditworthy? (1)
- Reputation of strategic partner and quality of their product (1)
- Any other valid point. Maximum of 1 (1)

Max 10

Many students did not note that the component was an existing component (on a new **product line**) used internally and assumed that it was a new component or that it was also sold externally to other third parties. The risks given were therefore not always appropriate.

j. Client service division.

(i) The profitability of each line of business, based on the contribution per processing hour.

		Holidays R'000	Tickets R'000	Flights R'000	Total R'000	
Revenue		7 685,00	3 770,00	3 045,00	14 500,00	(2) ½ per column
Less: Variable costs		(4 103,79)	(1 896,31)	(1 827,00)	(7 827,10)	
Direct costs		5 870,00	2 670,00	2 260,00	10 800,00	(2) ½ per column
Less: Advertising (given)		(1 766,21)	(773,69)	(433,00)	(2 972,90)	(2) ½ per column
Contribution	**A**	3 581,21	1 873,69	1 218,00	6 672,90	(2)
Contribution as % of revenue	(%)	46,60	49,70	40,00	46,02	(1)C% cal
Central costs (R'000) (given)		1 157,00	845,00	767,00	-	(1)r/w
Processing hours (000) (i.e. central costs ÷ R65/hr)	**B**	17,80	13,00	11,80	42,60	(1)r/w
Contribution/hr (Rand)	(**A/B**)	201,19	144,13	103,22	-	(1)C **12**

Part (i) driven by the contribution per processing hour. Some students simply ignored 'each line of business' and worked on a total basis.

(ii) The total revenue which Fastbook.com must generate to meet the target of a 10% increase in profit in the 2012 financial year.

		R'000	
Fixed costs:			
Advertising	(1 766,21 + 773,69 + 433)	2 972,90	
Increase	(160 + 55 + 35)	250,00	(1)
Central costs	(1 157 + 845 + 767)	2 769,00	(0.5)
Increase		215,00	(0.5)
Upgrade of system	(819k ÷ 3)	273,00	(0.5)
Forecast fixed costs		6 479,90	(1)

		R'000	
Profit:			
Current	(658 + 255 + 18)	931,00	(0.5)
Increase (10%)		93,10	(0.5)
Target profit	(723,8 + 280,5 + 19,8)	1 024,10	(0.5)
Required contribution	(1 024,10 + 6 479,90) = 46,02% of revenue	7 504,00	(1)C
Thus Revenue (R'000)	= 7 504 x 100 / 46,02	16 305,95	(1)C
			7



Most students forgot to include the cost of the upgrade in the fixed costs. As the **total revenue** was required, it was not necessary to show the line calculations, but the **total** contribution % (from (i)) had to be used.

(iii) Whether the upgraded system will provide sufficient processing hours to generate the revenue required to achieve the profit target as identified in j(ii) above.

	Holidays R'000	Tickets R'000	Flights R'000	Total R'000	
Current revenue	7 685,00	3 770,00	3 045,00	14 500,00	
% of total	53	26	21		(2)
Required revenue – per (ii)	8 642,16	4 239,55	3 424,25	16 305,95	(2)
Contribution as % of revenue – per (i)	46,60	49,70	40,00		(1.5)
Contribution (R'000)	4 027,25	2 107,06	1 369,70		(1.5)
Contribution/hr (Rand) – per (i)	R201,19	R144,13	R103,22		(1.5)C
Hours required	20 017	14 619	13 270	47 906	(1)C

Analysis of Hours:			
Hours available before upgrade	42 600		
13% increase	5 538		(0.5)
Hours available after upgrade	48 138		(1)
Hours required	47 906		
Surplus hours	232		(1)C
			12

Alternative using processing hours:

	Holidays R'000	Tickets R'000	Flights R'000	Total R'000	
Current revenue	7 685,00	3 770,00	3 045,00	14 500,00	
% of total	53	26	21		(2)
Required revenue	8 642,16	4 239,55	3 424,25	16 305,95	(2)
Processing hours (000)	17,8	13	11,8	42,6	(1.5)
Current revenue per processing hour	R431,74	R290	R258,05		(1.5)
Required revenue ÷ current revenue per processing hour (hours)	20 017	14 619	13 270	47 906	(2.5)C
Analysis of Hours:					
Hours available before upgrade				42 600	
13% increase				5 538	(0.5)
Hours available after upgrade				48 138	(1)
Hours required				47 906	
Surplus hours				232	(1)C
					12

Alternative:

Hours available before upgrade	42 600	
13% increase	5 538	(0.5)
Hours available after upgrade	48 138	(1)

Weighted current contribution/hr = R6 672,9 / 42,6hrs = R156,64/hr

48 138 hours x R156,64/hr = (R'000) 7 540,34 new contribution

New contribution – Required contribution = 7 540,34 – 7 504 = (R'000) 36,34 surplus contribution
Therefore surplus hours = (R'000) 36,34 / R156,64/hr = 232 surplus hours

Alternative:

Hours available before upgrade	42 600	
13% increase	5 538	(0.5)
Hours available after upgrade	48 138	(1)

Weighted current revenue/hr = R14 500 / 42,6hrs = R340,38/hr

48 138 hours x R340,38/hr = (R'000) 16 385,21 new revenue

New revenue – Required revenue = 16 385,21 – 16 305,95 = (R'000) 79,26 surplus revenue
Therefore surplus hours = (R'000) 79,26 / R340,38/hr = 233 surplus hours

Alternative:

Required revenue	16 305,95
Contribution as % of revenue	46,02%
Contribution (R'000)	7 504,00

Increase in contribution = (7 504 – 6 672,90) / 6 672,90 = 12,45%
Therefore hours required = 42 600 x 1,1245 = 47 906

Analysis of Hours:		
Hours available before upgrade	42 600	
13% increase	5 538	(0.5)
Hours available after upgrade	48 138	(1)
Hours required	47 906	
Surplus hours	232	(1)C

Alternative:

Increase in revenue = (16 306 – 14 500) / 14 500 = 12,45%
Therefore hours required = 42 600 x 1,1245 = 47 906

Analysis of Hours:		
Hours available before upgrade	42 600	
13% increase	5 538	(0.5)
Hours available after upgrade	48 138	(1)
Hours required	47 906	
Surplus hours	232	(1)C



A number of alternatives are presented. Students should note that it is best to stick to the logic/approach that they understand best and are comfortable with. The process (to be used) can be described as follows:

- Apply the current turnover split to the required turnover (53, 26, 21).
- Use the contribution %'s from section (i) to determine the contribution value.
- Use the existing contribution/hour (section (i)), which should not change, to calculate the number of hours required per activity.
- Compare the total hours required to the total hours available after the upgrade.

The average mark for this paper was 43%.

QUESTION 9

MATOJENI LTD **SUGGESTED SOLUTION**

(Source: Supplementary exam 2011 DIPAC26 (MAC4861))



Students often state: 'I have done most of the calculations correctly, why don't I pass the section/question. When we consider the mark allocation, we also consider what the impact of a particular mistake will be in/on a business. Will it lead to an incorrect decision, loss and eventual failure in real life? Where this is the case, a pass mark **cannot** (and will not) be awarded for part correctness. Regardless of current positions, 90% plus of CTA students, end up in commerce and industry, where (their) decisions play a critical role in the continuance and sustainability of the business. Fundamental issues should thus be correctly addressed and handled. As most students do not at this stage have hands-on business experience, it is even more imperative that text books and other referred to reading matter be adequately consulted.

[35]

PART A

(a) Using the variances and other relevant information above comment on the performance of the following managers: **(10)**



Students were not always sure what the responsibilities of the different managers were and consequently used incorrect variances in their comments. It is a clear indication that prescribed text books and other relevant reading matter are not consulted.

(i) The purchasing manager

- The purchasing manager has clearly bought a cheaper product, saving R48 000. The cause of this is not specified and it could be due to good buying or negotiation, reductions in quality or changes in overall market conditions. (1)
- We are told the market for buying seeds is stable, so there is more likely to be an internal reason for the problem. The material usage variance is significantly adverse, indicating that much more waste than normal has occurred in month 1. (1)
- This suggests that the quality of the seed bought was poor and as a result a R52 000 excess loss has occurred. It is possible that the waste was caused by the labour force working poorly or too quickly and this has to be considered. (1)
- The sales price achieved is also well down on standard with the sales price variance showing an R85 000 loss of revenue and (therefore) profit. (1)
- We are told that the market for sales of brown rice is stable and so it is reasonable to presume that the fall in sales price achieved is as a result of internal quality issues rather than general price falls. This may have also led to a fall in the volume of sales, another R21 000 of adverse variance. (1)
- In conclusion the purchasing manager appears mainly responsible for a loss of R110 000 (R85 000 + R52 000 + R21 000 – R48 000) taking the four variances above together. (1)

(ii) The production director

- The production director has increased wage rates and this has cost an extra R15 000 in month 1. However one could argue that this wage increase has had a motivational effect on the labour force. (1)

- The labour efficiency variance is R18 000 favourable; and so it is possible that a wage rise has encouraged the labour force to work harder. Academic evidence suggests that this effect might only be temporary as workers get used to the new level of wages. (1)

- The plant has recorded an adverse idle time variance of R12 000. Idle time can have many causes, including, material shortages, machine breakdowns, unclear work instructions and poor bill of materials/assembly drawings. However, we are told the machines are running well and the buyer has bought enough rice seeds. This could point to a probably once – off incidence of unclear work instructions or poor bill of materials/assembly drawings as a probable cause. (1)

- In conclusion the increase in the wage rate did cost more money despite a marginal improvement in morale and productivity. The total of the three variances above is R9 000 Adverse (R18 000 – R12 000 – R15 000). (1)

(iii) The maintenance manager

- The maintenance manager has decided to delay the annual maintenance of the machines and this has saved R8 000. This will increase profits in the short term but could have disastrous consequences later. (1)

- In this case only time will tell. If the machines breakdown before the next maintenance then lost production and sales could result. (1)

- The maintenance manager has only delayed the expenditure and not prevented it altogether. A saving of R8 000 as suggested by the variance has not been made. It is also possible that the adverse variable overhead expenditure variance has been at least partly caused by poor machine maintenance. (1)

- The variance calculated is not the saving made as it represents a timing difference only. The calculation also ignores the risks involved. (1)

Maximum **10**

(b) Calculate the variances for month 2 in as much detail as the information above allows **(10)**


The status of a variance (adverse or favourable) should be clearly indicated. Where this is not done or incorrectly done, the deduction is that the student does not understand the topic or underlying principle. Note that the sales volume variance is expressed in terms of contribution and that the FOH variance in this case only referred to the expenditure variance.

Sales price

Formulae	(SP – AP) AQ	
Variance [AP: R1 800 000 / 8 000 tonnes = R225 per tonne]	(R240 – R225)8 000 t	
	R120 000 Adverse	(1)

Sales volume

Formulae	(BQ – AQ) Std Contribution/Unit	
Variance	(8 400 t – 8 000 t) R56	
	R22 400 Adverse	(1)

Direct material price

Formulae	(SP – AP) AQ	
Variance [AP: R660 000 / 12 000 tonnes = R55 per tonne]	(R60 – R55)12 000 t	
	R60 000 Favourable	(1)

Direct material usage

Formulae	(SQ – AQ) SP	
Variance [SQ: 8 000 tonnes rice x 1,4 = 11 200 tonnes seed]	(11 200 t – 12 000 t)R60	
	R48 000 Adverse	(1)

Direct labour rate

Formulae	(SR – AR) AH	
Variance [AR: R303 360 / 15 800 hrs = R19,20 per hour]	(R20 – R19,20)15 800 hrs	
	R12 640 Favourable	(1)

Labour idle time

Formulae	(AHP – AHW) SR	
Variance	(15 800 hrs – 15 000 hrs) R20	
	R16 000 Adverse	(1)

Direct labour efficiency		
Formulae	(SHA – AHW) SR	
Variance [SHA: 8 000 tonnes x 2 hrs = 16 000 hrs]	(16 000 hrs – 15 000 hrs) R20	
	R20 000 Favourable	(1)

Variable overhead expenditure		
Formulae	(SR – AR) AHW	
Variance [AR: R480 000 / 15 000 hrs = R32 per hr]	(R30 – R32)15 000 hrs	
	R30 000 Adverse	(1)

Variable overhead efficiency		
Formulae	(SHA – AHW) SR	
Variance [SHA: 8 000 tones x 2 hrs = 16 000 hrs]	(16 000 hrs – 15 000 hrs) R30	
	R30 000 Favourable	(1)

Fixed overhead expenditure		
Formulae	Budget – Actual	
Variance	R210 000 – R200 000	
	R10 000 Favourable	(1)

Marginal costing treats the fixed cost as a period cost, which does not impact on the cost of sales and contribution. CoS is directly linked to the sales quantity (8 400 units in the calculations below).

(c) Reconcile the budget profit to the actual profit using marginal costing principles **(15)**

Calculate Budget Profit for Period 2

	R	**R**

As the variances were calculated in part (b), it would be good examination practice to simply add them all up and include one line item 'Variances per (b) …' Note that as we are working with marginal costs, the fixed expenditure variance is shown separately.

PART B **[30]**

(a) A good transfer pricing system should avoid sub-optimal decision making by divisional mangers if overall objectives of the organisations are to be attained. List five objectives of a transfer pricing system. **(5)**

The transfer price should:

- Provide an adequate reward to the supplying division to compensate for the resources used; (1)
- Provide the receiving division with access to resources at a reasonable cost; (1)
- Allow divisional performance to be assessed on a basis which is commercial; (1)
- Motivate divisional managers to achieve corporate goals; (1)
- Maximise overall company profits. (1)

(b) Assess the argument that a transfer pricing system will provide no benefits to Telematics, and discuss the conflicts which the company must resolve if a transfer pricing system is introduced. **(5)**

- Although it is correct to say that the transfer price will represent income to one division and a cost to the other, and that the income and cost will cancel one another out, it is not correct to say that this will have no effect on profit (1)

- An effective transfer pricing system will encourage divisional managers to take appropriate decisions which will lead to overall company profit being maximised. (1)

- On the other hand an ineffective transfer pricing system will lead to divisional managers taking decisions which will maximize the profit for their own division, but will not maximise overall company profit. (1)

- This can arise because the basic premise of divisionalisation is that each division is autonomous. This basic premise will encourage managers to seek the best result for their division, irrespective of the effect on the rest of the company. (1)

- This means that an effective transfer pricing system is likely to be designed (and may even be imposed) by head office. As a result, the concept of divisional autonomy is somewhat undermined. There is therefore a tension between the concept of divisional autonomy and head office control. (1)

- The extent to which managers will be satisfied with a transfer pricing system, and therefore managerial behaviour, is likely to be influenced by two factors. One of these will be the extent to which divisional managers perceive that their decisions are constrained by head office involvement (more involvement = less autonomy = less satisfaction). The second will be the extent to which managers feel that the transfer price provides adequate reward for the effort and resources which have been utilised. (1)

- Consequently, the transfer pricing system will have a significant influence on the motivation of divisional managers and therefore divisional and corporate performance. The greater the extent to which the transfer pricing system can achieve a balance between maximising overall company profits and maintaining divisional autonomy, the more successful it will be. (1)

Maximum **5**

(c) Calculate the bonus which would be payable to the manager of the component division in the first year of divisionalisation in each of the proposed transfer pricing systems. **(15)**

Calculate Target Based Transfer price Per Unit

	R	R	
Selling price of final product		78,00	(½)
Less:			
Margin (25% x R78)	19,50		(1)
Processing cost	12,50	(32,00)	(1)
Selling price/Transfer price		46,00	(1)

Calculate Bonus Based On Target Transfer price

Capacity 450 000 hours ÷ 2,5 = 180 000 units (1)

		R'000	R'000	
Revenue	180 000 units at R46,00		8 280	(½)
Less Costs:				
Variable cost	180 000 units at R26,80	4 824		(½)
Fixed costs		2 700	(7 524)	(½)
Profit			756	(½)
Target profit (R8 280 000 × 5%)			(414)	(½)
Excess			342	(½)
Bonus	4% of R342 000		13,680	(1)

Calculate Cost-based Transfer price Per Unit

Per Unit	R	
Variable cost	26,80	(0.5)
Mark up (70% x R26,80)	18,76	(0.5)
Transfer price	45,56	(1)

Calculate Bonus Based On Cost-based Transfer price

		R'000	R'000	
Revenue – External sales	60 000 units at R46,00	2 760,0		(½)
Revenue – Transfers (180k-60k)	120 000 units at R45,56	5 467,2	8 227,20	(½)
Less Costs:				
Variable cost	180 000 units at R26,80	4 824		(½)
Fixed costs		2 700	(7 524,00)	(½)
Profit			703,20	(½)
Target profit (R8 227 200 × 5%)			(411,36)	(½)
Excess			291,84	(½)
Bonus	4% of R291 840		11,674	(1)

Bonus calculations were often done without calculating the transfer price, partly due to students not understanding the target price concept. In calculating the target price, the given external market price (R78) was not used and/or the profit margin was added and not deducted. Reading time should be used to identify/highlight core issues in the given information. Note that in the target system all units are sold at the same price, whereas in the cost based system there is price differentiation.

(d) Recommend the basis for calculating a transfer price which is likely to be appropriate for use by Telematics. **(5)**

- The fact that an external customer is the catalyst for establishing divisions and therefore transfers pricing means that the transfer price should be based on market price. The emergence of the partner creates a market where it appears none previously existed. As the market price will be based on negotiations with an external partner, it will provide a degree of objectivity. This will mean that the transfer price will be perceived as equitable. (1)

- In market-based systems, the transfer price may be set below the external sales price. This is to reflect the fact that certain costs (e.g. advertising, packaging, distribution) are not incurred on internal transfers. (1)

- Telematics has been approached by the partner, so it can be argued that most of the possible savings considered above in respect of market-based transfer prices do not apply. For example, as there has not been any external market for the component in the past, there will not be any advertising costs associated with the component. (1)

- If there will be any packaging and distribution costs to be incurred in connection with sales to the partner, it should be possible to ensure that the price negotiated is calculated to take account of such costs. In that way, it will be possible to arrive at a final price for the component itself. This should be the transfer price, as it will be an objective, market-based price, and will exclude those elements which typically need to be "stripped out" from the market price to arrive at the transfer price. (1) (1)

PART C **[35]**

(a) (i) – (iii): Please refer to Eworx Ltd solution (Supplementary exam 2011 TOE408W), part (j)(i) – (j)(iii)

(b) Briefly discuss the limitations of your analysis in (a). **(4)**

- **Sales mix -** The assumption that the sales mix will be unchanged may not prove to be justified. Each line of business generates widely differing contribution per processing hour. This means that any shift in the sales activity from holidays to flights will have a detrimental effect on profitability. (2)

- **Cost behaviour** - The analysis is based on only two classifications of cost – fixed and variable. This may be a simplistic analysis. If costs have been incorrectly classified, the analysis will be inaccurate. (1)

- **Cost estimates** - The accuracy of the analysis is highly dependent on the accuracy of the directors' estimates of increases in the fixed costs. (1)

QUESTION 10

[100]

JUG AND BREW GROUP LTD - SUGGESTED SOLUTION

(Source: Exam 2010 TOE408W (MAC4862))

Overall comment

In general, it appeared as if many candidates failed to properly interpret the required-section. This is evident from the answers presented, which often did not address that which was required, or represented a haphazard collection of thoughts.

In this regard, candidates could improve by implementing the following approach:

- Read the required section several times;
- Analyse the required section in detail to determine exactly what is required; and
- Plan the answer before attempting, by using a 'mind-map' or similar technique.
- Present an answer in a clear, well organised manner.

In general, where discussion was required, candidates often offered illogical answers or presented a mere 'laundry list' of points. Normally, where discussion is required, marks are only allocated to logical deductions that were applied to the scenario in question.

(a) Using direct labour hours as the base for assigning manufacturing overhead cost to products, undertake the following:

(i) Calculate the predetermined rate that will be used during the year. **[1]**

Predetermined rate **=** $\frac{\text{R22 000 000}}{\text{50 000 Hours}}$ **= R440,00 per Hr** **(1)r/w**

The overhead is given as R22m and should not include admin costs or depreciation. The hours are direct labour hours and **not** activity hours.

(ii) Calculate the unit product cost of one kilogram of the **Ethiopian and Zimbabwean roast.** **[2]**

	Ethiopian R	**Zimbabwean R**	
Direct materials (given)	14,50	12,90	
Direct labour (given)	0,48	0,48	
Manufacturing overhead (0,01 Hrs x R440)	4,40	4,40	
Packaging	3,02	2,72	
Total unit product cost	**22,40**	**20,50**	**(1)r/w**
	(1) r/w		

The overhead must be assigned in sync with the labour standard of 0,01 hours per bag. Using the rate of R440 per (i) for 1 kg gives a product cost of R458,00 per kg of Ethiopian coffee, **clearly** not realistic. Sales and distribution cost is **not** a product cost.

(b) Using activity based costing as a basis for assigning manufacturing overhead cost to products, undertake the following:

(i) Calculate the total amount of manufacturing overhead cost assigned to the **Ethiopian and Zimbabwean** for the year.

Workings:

Cost driver rates

The R22m overhead relates to the annual activity of **all** the products (p 8)

Cost Pool	Measure	Annual Activity	Budgeted Annual Cost R	Rate R	
Purchasing	Purchase orders	5 000 orders	5 600 000	1 120/order	(1)r/w
Material Handling	Number of set-ups	2 500 setups	1 930 000	772/setup	(1)r/w
Quality Control	Number of batches	1 250 batches	900 000	720/batch	(1)r/w
Roasting	Roasting hours	237 500 hours	10 450 000	44/hour	(1)r/w
Blending	Blending hours	80 000 hours	1 920 000	24,00/hour	(1)r/w
Packaging	Packaging hours	75 000 hours	1 200 000	16,00/hour	(1)r/w
Total cost			**22 000 000**		

The annual activities and total costs **cannot** be allocated to the Ethiopian and Zimbabwean products as they are related to **all** the products. This is a fundamental mistake and cannot be marked through.

Budgeted production (E)	80 000 kg (Z)	30 000 kg
Purchase order size	20 000 kg	5 000 kg
∴ Orders	4	6
Batch size	5 000 kg	3 000 kg
∴ Batches	16	10

Cost driver outputs per product:

Number of purchase orders:

Ethiopian: 80 000 kg ÷ 20 000 kg per order = 4 orders
Zimbabwean: 30 000 kg ÷ 5 000 kg per order = 6 orders

Number of batches:

Ethiopian: 80 000 Kg ÷ 5 000 kg per batch = 16 batches
Zimbabwean: 30 000 kg ÷ 3 000 kg per order = 10 batches

Number of setups:

Ethiopian: 16 batches × 2 setups per batch = 32 setups
Zimbabwean: 10 batches × 1 setup per batch = 10 setups

Roasting hours:

Ethiopian: 1,5 hours × (80 000 kg ÷ 100 kg) = 1 200 hours
Zimbabwean: 1,5 hours × (30 000 kg ÷ 100 kg) = 450 hours

Blending hours:

Ethiopian: 0,5 hours × (80 000 kg ÷ 100 kg) = 400 hours
Zimbabwean: 0,4 hours × (30 000 kg ÷ 100 kg) = 120 hours

Packaging hours:

Ethiopian: 0,3 hours × (80 000 kg ÷ 100 kg) = 240 hours
Zimbabwean: 0,3 hours × (30 000 kg ÷ 100 kg) = 90 hours

Other calculations:

Summary Cost driver outputs per product:

Activity	Driver	Ethiopian	Zimbabwean	
Purchasing	Purchase orders	4 orders	6 orders	**(1)r/w**
Material handling	Number of set-ups	32 set-ups	10 set-ups	**(1)r/w**
Quality control	Number of batches	16 batches	10 batches	**(1)r/w**
Roasting	Roasting hours	1 200 hrs	450 hrs	**(1)r/w**
Blending	Blending hours	400 hrs	120 hrs	**(1)r/w**
Packaging	Packaging hours	240 hrs	90 hrs	**(1)r/w**

Costs assignment to products:

Activity	Rate	Drivers Ethiopian	Drivers Zimbabwean	Costs Ethiopian	Costs Zimbab-wean	
	R			**R**	**R**	
Purchasing	1 120/order	4 orders	6 orders	4 480	6 720	(1)r/w
Material handling	772/setup	32 set-ups	10 set-ups	24 704	7 720	(1)r/w
Quality control	720/batch	16 batches	10 batches	11 520	7 200	(1)r/w
Roasting	44/hour	1 200 hrs	450 hrs	52 800	19 800	(1)r/w
Blending	24,00/hour	400 hrs	120 hrs	9 600	2 880	(1)r/w
Packaging	16,00/hour	240 hrs	90 hrs	3 840	1 440	(1)r/w
Total costs				**106 944**	**45 760**	(1)r/w

Students often obtained total costs without showing any supporting calculations whatsoever; where these totals were incorrect it limited the marks severely.

(ii) Using the data calculated under **b (i)** above, compute the overhead cost per Kg of the **Ethiopian and Zimbabwean roast.** (Work to 2 decimal places) **[2]**

		Ethiopian	Zimbabwean	
		R	**R**	
Total overhead cost assigned (above)	(A)	106 944	45 760	
Number of kgs manufactured	(B)	80 000	30 000	(1)r/w
Cost per kg	(A) ÷ (B)	1,34	1,53	(1) c

Cost per kilogram based on production and not the net movement.

(c)**(iii)** Prepare abc absorption costing statements of comprehensive income **[2]**

	Ethiopian	Zimbabwean
	R	**R**
Direct materials (given)	14,50	12,90
Direct labor (given)	0,48	0,48
Manufacturing overhead (as above)	1,34	1,53
Packaging	3,02	2,72
Total unit product cost	**19,34**	**17,63**
	(1) c	(1) c

When compared to a(ii), the change in cost per kilogram is clear and in this case as a result of the ABC approach.

Statement of comprehensive income

	R	R		
Sales (80 000 x 35,00)(40 000 x 33,50)	2 800 000	1 340 000	(2)	r/w
Opening inventory (5 000 x 17,40)(20 000 x 15,87)	87 000	317 400	(2)	c
Production (80 000 x 19,34)(30 000 x 17,63)	1 547 200	528 900	(2)	c
Closing inventory (5 000 x 19,34)(10 000 x 17,63)	(96 700)	(176 300)	(2)	c
Cost of sales	1 537 500	670 000		
Gross profit	1 262 500	670 000		
Sales and distribution (2,8m x 3%)(1,34m x 3%)	84 000	40 200		
	(1)	(1)		r/w
Net profit per product	1 178 500	629 800		
Net profit %	42,1%	47,0%	(2)	c

- No ABC information regarding opening inventory was given; either existing or the ABC cost may be used, with the existing cost more logical.
- Production and closing inventory unit cost must be the same.
- More informative to calculate the sales and distribution cost per product.
- Company costs should not be shown as a product income statement is required.

Although the Ethiopian coffee creates more profit (Rand) than the Zimbabwean coffee, this is due to the quantity sold (double). A percentage profit or profit per kilo indicates that the Zimbabwean coffee is more profitable.

Look at the mark allocation:
- 2 marks for opening inventory if volume is correct
- 2 marks for closing inventory and production if volume is correct and the correct unit cost calculated earlier is carried through.

(c) Explain the accounting treatment

This is **not** a change in accounting policy with its associated mumbo jumbo, simply a restatement of cost as absorption costing is still applied.

- Absorption costing is still applied. (1)
- The method of calculation (abc) causes a decrease in value. (1) c
- The adjustment should be written off to cost of sales. (1)

3

(d) Briefly discuss the ethical issues and additional required information

- The ethical issues concern the marketing manager and not an auditor or a chartered accountant, thus issues relevant to the majority of staff members.
- Students went to town on the other information, **incorrectly** focussing on the special order and the issues relating thereto, such as price, capacity, capability etc.

Ethical issues arising

- Fully paid rugby holiday represents an *expensive gift*. This may be construed as a "*bribe*" to influence purchasing decision. (2)
- Can J & B rely on *feedback* from Mr Singh re UK firm? (1)
- Should J & B enter into a *business relationship* with UK firm given their undue hospitality towards Mr Singh? (1)
- Even if trip did not influence procurement decision, public disclosure of issue may harm J & B's *reputation*. (1)
- Can J & B use information in Mr Singh's *personal email*? Because it was sent from work/on UK's server, presumption could be that this information *is that of the firm*? (2)

Further information required

- Is the marketing manager able *to influence or make the decision* to sell the products at a lower price? (1)
- What is the *value* of the holiday? (1)
- What is J & B's *policy* re acceptance of gifts? (1)
- Will the trip occur *prior to making decision* to sell to UK? (1)
- Have *any other* J & B employees/directors accepted gifts from suppliers in the past? (1)
- Did marketing manager *apply for leave* for the week? (1)

Comments

- *Non-disclosure* sends a strong message that he was uncomfortable disclosing his trip to superiors. (1)
- Mr Singh *fit to continue* in his current position? Impact of his actions on staff morale? (1)

Max 9

(e) Calculate the cash flow impact of changing the debtors' days and inventory turn

- Students do **not** understand the term 'inventory turn' and to a lesser extent debtors days. Debtors relate to credit sales and inventory to purchases.

A decrease in debtors days ⇒ lower debt, more cash.
An increase in inventory turn ⇒ lower inventory, more cash.

Current value Debtors	$\frac{47}{365}$ x 87 430	=	11 258,1	(1)r/w
Proposed value	$\frac{40}{365}$ x 87 430	=	9 581,4	(1)r/w
Net improvement			1 676,7	
Inventory				
Current value	$\frac{34{,}100}{12}$	=	2 841,7	(1)r/w
Proposed value	$\frac{34\ 100}{15}$	=	2 273,3	(1)r/w
Net improvement			568,4	

The total improvement in cash flow will thus be **R2 245,1** plus the **interest** earned / saved thereon. (2)r/w

(f) Explain the impact of a transfer price on Jug and Brew

The transfer price is evaluated on the basis of the contribution of the product(s), taking into account relevancy concepts. Fixed costs should **only** be considered on an incremental cash basis. The existing position must be compared with the proposed position, which very few students did. Students did not cover the discussion section, mostly performance and business issues, well.

Pricing	**Ethiopian**	**Zimbabwean**	
Normal absorption			
Prices requested	20,20	18,30	(1)r/w
Variable mfct (22,4 – 4,40)(20,5 – 4,40)	(18,00)	(16,10)	(1)r/w
Sales and distribution	-	-	(1)r/w
Contribution	2,20	2,20	
Kg to be ordered	3 000	2 000	
Value (R)	6 600	4 400	(1)r/w

Alternative

Activity based costing		
Prices requested	20,20	18,30
Variable mfct (19,34 – 1,34)(17,63 – 1,53)	18,00	16,10
	2,20	2,20

The current contribution is (((35 – 18) x 3 000) + (33,50 – 16,10) x 2 000) = R85 800 (1)r/w
OR The lost contribution therefore amounts to R74 800 (1)r/w

- The transfer price order will make a *contribution to fixed costs*, albeit lower. (1)
- The *performance measures* (ROA/ROI/ROE) *will drop* very slightly on a total basis given the small quantities involved in the transfer. (1)
- The danger in allowing coffee shop a transfer price is that the *other group shops* will insist on being *treated the same*. (2)
- This will lead to a more material decrease in contribution / profit for the manufacturing, lower returns and future arguments on these issues - overall. (1)
- Paying market prices, the group coffee shops *must compete* with other coffee shops. (1)
- on service, ambience etc. This (competition) will be *distorted* with market prices do not apply. (1)
- If the demand for coffee increases and the company could alternatively sell the quantity being transferred, in the *external market*, they definitely would not consider accepting the transfer price being requested then. (1)
- The request should *not be accepted*, based on the qualitative issues. (1)

Max 8

(g) Calculate a price for the UK special order



Note the batch approach stated in the solution. Again, relevancy of costs have to be considered. Students by and by ignored the specific ABC costs. Very few students were able to calculate the existing mark-up, an area of particular concern.

As batch information is given, it follows that on ABC relevancy approach is required.

	Alternative (per kg)	**R**	
Beans (50 000 x 1,80 x 7,80)	14,04	702 000	(1) r/w
Labour (50 000 x 0,01 x 1,5 x 48)	0,72	36 000	(1) r/w
Purchasing (1 order)	0,02	1 120	(½) r/w
Materials handling (2 set-ups) – 2 batches	0,03	1 544	(½) r/w
Quality control (2 batches x 720)	0,03	1 440	(1) r/w
Roasting, blending, packaging	-	-	(1) r/w
[Not relevant, fixed machine cost]			
Packaging bags (50 000 x 1,70)	1,70	85 000	(1) r/w
	16,54	827 104	
Mark-up ((33,50 – 20,50)/20,50) x 100/1	63,4%	63,4%	(1) c
Value	10,49	524 384	(1) c
Price to be quoted	27,03	1 351 487	
Per kg (÷ 50 000)		R27,03	

Max 6

(h) Evaluate Mr Ramphele's new product [19]



Different approaches and layouts were possible. The following common mistakes were found:

- Refurbishment cost ignored
- Sales growth ignored or alternatively continued beyond periods stipulated
- Depreciation not excluded from cash costs
- Financing cost shown as part of the evaluation
- Guarantee cost ignored or not taken into year 6.
- Upgrade cost ignored and/or not considered in tax calculation
- Wear and tear calculated for five (?) years
- Detailed calculations not shown
- Recommendation not made

CAPITAL BUDGETING

Cash contribution	**Basic**	**Advanced**	
Sales price	1 200	3 500	(1) r/w
Manufacturing cost	900	2 275	(1) r/w
Depreciation	(220)	(535)	(1) r/w
Cash cost	680	1 740	
Distribution and marketing	48	210	(1) r/w
	472	1 550	

Years	**0**	**1**	**2**	**3**	**4**	**5**	**6**
Basic		3 000	3 600	4 320	4 320	4 320	
Advanced		1 000	1 100	1 210	1 331	1 331	
R'000							
Cash contribution							
Basic		1 416,0	1 699,2	2 039,0	2 039,0	2 039,0	
Advanced		1 550,0	1 705,0	1 875,5	2 063,1	2 063,1	

Alternative Basic

Sales	3 600	4 320	5 184	5 184	5 184
	(2 040)	(2 448)	(2 937,6)	(2 937,6)	(2 937,6)
Manufacturing cost	(2 700)	(3 240)	(3 888)	(3 888)	(3 888)
Deprecation	660	792	950,4	950,4	950,4
Distribution and marketing	(144)	(172,8)	(207,4)	(207,4)	(207,4)
	1 416,0	1 699,2	2 039	2 039	2 039

Advanced

Sales	3 500	3 850	4 235	4 658,5	4 658,5
	(1 740)	(1 914)	(2 105,4)	(2 315,9)	(2 315,9)
Manufacturing cost	(2 275)	(2 502,5)	(2 752,9)	(3 028,0)	(3 028,0)
Depreciation	535	588,5	647,5	712,1	712,1
Distribution and marketing	(210)	(231)	(254,1)	(279,5)	(279,5)
	1 550	1 705	1 875,5	2 063,1	2 063,1

R'000							
Premises	(600)						
Equipment	(7 500)					300	
Materials	(1 100)					0	
Guarantee cost							
Basic			(128,0)	(128,0)	(128,0)	(128,0)	(128,0)
Advanced			(75,0)	(75,0)	(75,0)	(75,0)	(75,0)
Upgrade				(1 000)			
Taxation							
Tax calc @ **28%**	(168,0)	9,5	(476,3)	(339,2)	(671,7)	(1 175,5)	56,8
Net cash in		2 966	3 201,2	3 711,5	3 899,1	3 899,1	(203,0)
Premises	(600)						
Wear and tear		(3 000)	(1 500)	(1 500)	(1 500)		
Upgrade				(1 000)			
Equipment rec						300	
Taxable	(600)	(34)	1 701,2	1 211,5	2 399,1	4 199,1	(203,0)
Tax benefit / (cost)	168,0	9,5	(476,3)	(339,2)	(671,7)	(1 175,7)	56,8
Net CF	(9 032,0)	2 975,5	2 724,9	2 372,3	3 227,4	3 023,6	(146,2)
DF @ 18%	1,0	,848	,718	,609	,516	,437	,370
NPV	(9 032,0)	2 523,2	1 956,5	1 444,7	1 665,3	1 321,3	(54,1)

= (175,1) (1) for negative value

∴ Do not proceed based on current information. (1) r/w

Max 19

An alternative layout

Year	**0**	**1**	**2**	**3**	**4**	**5**	**6**	
Total sales		7 100 000	8 170 000	9 419 000	9 842 500	9 842 500		
Sales								
Basic		3 000	3 600	4 320	4 320	4 320		(1)
Price		1 200	1 200	1 200	1 200	1 200		(1)
Total		3 600 000	4 320 000	5 184 000	5 184 000	5 184 000		
Advanced		1 000	1 100	1 210	**1 331**	1 331		(1)
Price		3 500	3 500	3 500	3 500	3 500		(1)
Total		3 500 000	3 850 000	4 235 000	4 658 500	4 658 500		
Manufacturing costs		(3 780 000)	(4 362 000)	(5 043 000)	(5 253 540)	(5 253 540)		
Basic		(2 700 000)	(3 240 000)	(3 888 000)	(3 888 000)	(3 888 000)		(1)
Less: Depreciation		660 000	792 000	950 400	950 400	950 400		(½)
Advanced		(2 275 000)	(2 502 500)	(2 752 750)	(3 028 025)	(3 028 025)		(1)
Less: Depreciation		535 000	588 500	647 350	712 085	712 085		(½)
Units returned			(203 000)	(203 000)	(203 000)	(203 000)	(203 000)	
Basic			(128 000)	(128 000)	(128 000)	(128 000)	(128 000)	(½)
Advanced			(75 000)	(75 000)	(75 000)	(75 000)	(75 000)	(½)
Distribution and marketing		(354 000)	(403 800)	(461 460)	(486 870)	(486 870)		
Basic		(144 000)	(172 800)	(207 360)	(207 360)	(207 360)		(½)
Advanced		(210 000)	(231 000)	(254 100)	(279 510)	(279 510)		(½)
Net profit in cash		2 966 000	3 201 200	3 711 540	3 899 090	3 899 090	(203 000)	
Tax payable	168 000	9 520	(476 336)	(339 231)	(671 745)	(1 175 745)	56 840	(1)
One times								
Premises	(600 000)							(½)
Equipment	(7 500 000)					300 000		(½)
Refurbishment				(1 000 000)				(½)
Raw material	(1 100 000)							(1)
Net cash	(9 032 000)	2 975 520	2 724 864	2 372 309	3 227 345	3 023 345	(146 160)	
PV Factor @ 18%	1	0,8475	0,7182	0,6086	0,5158	0,4371	0,3704	(1)
	(9 032 000)	2 521 753	1 956 997	1 443 787	1 664 664	1 321 504	(54 138)	

Tax calculation

Net profit cash		2 966 000	3 201 200	3 711 540	3 899 090	3 899 090	(203 000)	(1)
Wear and tear		(3 000 000)	(1 500 000)	(1 500 000)	(1 500 000)			(1)
Premises	(600 000)							(½)
Refurbishment				(1 000 000)				(½)
Recoupment						300 000		(½)
Taxable income	(600 000)	(34 000)	1 701 200	1 211 540	2 399 090	4 199 090	(203 000)	
Tax @ 28%	(168 000)	(9 520)	476 336	339 231	671 745	1 175 745	(56 840)	(1)

Present value (177 432)

Conclusion: The Net Present Value is negative. Do not undertake this project. (1)

Max 19

(i) Impact of Mr Adam's observations regarding futures and forex



The cost increase was generally ignored, as was the hedging required.

Futures

Cost of beans will increase by 30%: (1)

(30% x 34,1m) 10,23m (1)r/w

Counter by

- Holding inventory of beans (1)
- Fix prices now by contract (1)

Forex

Cost of beans will increase (1)

(34,1 x 8,50 ÷ 7,80) – 34,1 3,1m (1)r/w

OR (44,3 x 8,50 ÷ 7,80) – 44,3 4,0m (1)r/w

Counter by

- Natural hedge ⇨ exports (1)
- Fixed $ prices in purchase contracts (1)

Max 6

(j) Impact of franchising operation on Jug and Bean Group Ltd



Very poorly answered. Students do not seem to know how franchises operate. The initial **sale** of shops was ignored.

- The existing 10 shops (franchises) must be *valued as going concerns* and should determine whether they will *realise cash* in excess of *set-up costs of R1,5m*. (3)
- New franchises will contribute as on *annual basis* through the *management service fees* and operations costs, which may exceed existing profits generated by J&B. (2)
- The *advertising fee will be neutral* as it has to be spent on advertising. (1)
- The *initial fees* will yield a small cash income. (1)
- *Profit and cash* from own shops will be *lost* in future. How profitable are the coffee shops at present? (1)
- The cash generated by the franchise operations can be used for *expansion* or investment purposes or alternatively to lower the company's gearing – the *liquidity* therefore will be benefit the company. (1) (1)
- *Higher success* of coffee shops – managers have personal interest, therefore work harder. (1)

Max 6

QUESTION 11

(Source: Exam 2010 DIPAC26 (MAC4861))

SWEET ROAST LTD

[34]

PART A

(a) Using direct labour hours as the base for assigning manufacturing overhead cost to products, undertake the following:

(i) Calculate the predetermined rate that will be used during the year. **[1]**

Calculation errors mostly a problem. Students should review/assess their answers for reasonableness before proceeding to the next section. For example; Using R44 in the unit cost required under b(ii), would clearly not make sense when compared to the given cost.

Predetermined rate $\frac{\text{R2 200 000}}{\text{50 000 Hours}}$ **= R44,00 per Hr** (1)r/w

(ii) Calculate the unit product cost of one kilogram of the **Ethiopian and Malawi roast.** **[3]**

	Ethiopian		**Malawi**	
	R		**R**	
Direct materials (given)	4,50		2,90	
Direct labor (given)	0,24		0,24	
Manufacturing overhead (0,02 Hrs x R44)	**0,88**		**0,88**	(1)r/w
Total unit product cost	5,62	**(1) *r/w***	4,02	(1)r/w

(b) Using activity based costing as a basis for assigning manufacturing overhead cost to products, undertake the following:

(i) Calculate the total amount of manufacturing overhead cost assigned to the **Ethiopian and Malawi roast** for the year. **[19]**

Workings:

Cost driver rates

Note: Calculations below may be combined with the cost driver outputs calculation to determine the cost assignment in a single calculation and may still be correct. Marks should be allocated for correct calculations.

Cost Pool	Measure	Yearly Activity	Budgeted Annual Cost R	Rate R	
Purchasing	Purchase Orders	2 000 Orders	560 000	280/Order	(1)r/w
Material Handling	Number of set-ups	1 000 setups	193 000	193/setup	(1)r/w
Quality Control	Number of batches	500 Batches	90 000	180/batch	(1)r/w
Roasting	Roasting Hours	95 000 Hours	1 045 000	11/Hour	(1)r/w
Blending	Blending Hours	32 000 Hours	192 000	6/Hour	(1)r/w
Packaging	Packaging Hours	24 000 Hours	120 000	5/Hour	(1)r/w
Total cost			**2 200 000**		

Alternative

		Total cost	Allocation to the 2 products	
Purchasing	(12/2 000 x 560 000)	560 000	3 360	
Material handling	(48/1 000 x 193 000)	193 000	9 264	
Quality control	(24/500 x 90 000)	90 000	4 320	
Roasting	(1 932/95 000 x 1 045 000)	1 045 000	21 252	
Blending	(1 932/32 000 x 192 000)	192 000	11 592	
Packaging	(1 932/24 000 x 120 000)	120 000	9 660	
		2 200 000	59 448	(4)

Cost driver outputs per product:

This step is required to do the assignment of costs to the product. A lot of students did the assignment based on the yearly activity given in the question, a fundamental mistake which **cannot** be marked through. Such a faulty decision in a business will let the business fail.

Concentrate on utilising reading time effectively.

Number of purchase orders:

Ethiopian: 80 000 Kgs ÷ 20 000 Kgs per order = 4 orders
Malawi: 4 000 Kgs ÷ 500 Kgs per order = 8 orders

Number of batches:

Ethiopian: 80 000 Kgs ÷ 5 000 Kgs per batch = 16 batches
Malawi: 4 000 Kgs ÷ 500 Kgs per order = 8 batches

Number of setups:

Ethiopian: 16 batches × 2 setups per batch = 32 setups
Malawi: 8 batches × 2 setups per batch = 16 setups

Roasting, Blending and Packaging hours:

Ethiopian: 2,3 hours × (80 000 Kg ÷ 100 Kg) = 1 840 hours
Malawi: 2,3 hours × (4 000 Kg ÷ 100 Kg) = 92 hours

Summary Cost driver outputs per product:

Activity	Driver	Ethiopian		Malawi	
Purchasing	Purchase Orders	4 Orders	(½)r/w	8 Orders	(½)r/w
Material Handling	Number of set-ups	32 Set-ups	(½)r/w	16 Set-ups	(½)r/w
Quality Control	Number of batches	16 Batches	(½)r/w	8 Batches	(½)r/w
Roasting, Blending and Packaging	Roasting Hours	1 840 Hrs	(½)r/w	92 Hrs	(½)r/w

Costs assignment to products:

Note: Mark through where students used the incorrect amounts, correctly in the below cost assignment calculation.

Activity	Rate	Mark	Drivers Ethiopian	Drivers Malawi	Costs Ethiopian	Mark	Costs Malawi	Mark
	R				**R**		**R**	
Purchasing	280/Order		4 Orders	8 Orders	1 120	(1)	2 240	(1)
Material Handling	193/setup		32 Set-ups	16 Set-ups	6 176	(1)	3 088	(1)
Quality Control	180/batch		16 Batches	8 Batches	2 880	(1)	1 440	(1)
Roasting, Blending and Packaging	22/Hour	(1) r/w	1 840 Hrs	92 Hrs	40 480	(1)	2 024	(1)
Total Costs					**50 656**	(½)r/w	**8 792**	(½)r/w

Max 19

(ii) Using the data calculated under **b (i)** above, compute the overhead cost per Kg of the **Ethiopian and Malawi roast.** (Work to 2 decimal places) **[3]**

Refer earlier comment re reasonableness.

	Ethiopian R	Malawi R	Mark
Total overhead cost assigned (above) (A)	**50 656**	**8 792**	(1)r/w
Number of Kgs manufactured (B)	**80 000**	**4 000**	
Cost per Kg (A) ÷ (B)	**0,63**	**2,20**	(2) c

(iii) Calculate the unit product cost of one kilogram of the **Ethiopian and Malawi roast.** [3]

	Ethiopian	Malawi	Mark
	R	**R**	
Direct materials (given)	4,50	2,90	
Direct labor (given)	0,24	0,24	
Manufacturing overhead (as above)	**0,63**	**2,20**	(1) c
Total unit product cost	5,37	5,34	(2) r/w

(c) In bullet point form, prepare short notes for the marketing manager on your findings under (a) and (b) above and clearly indicate the implications to the company of using direct labour as the base for assigning manufacturing overheads to products. [5]

General comments often made, ignoring the question environment and/or the calculations referred to.

- Analysis of the company's data shows that several activities other than direct labour drive the company's manufacturing overhead costs. (1)
- These activities include purchase orders issued, number of setups for material processing, and number of batches processed. (1)
- The company's present costing system, which relies on direct labour time as the sole basis for assigning overhead cost to products, significantly undercosts low-volume products, such as the Malawi coffee, and significantly overcosts high-volume products, such as our Ethiopian coffee. (1)c (1)c

Note: Mark through based on calculations in a (ii) and b (iii) when reasonable

- An implication of the traditional costing analysis is that our low-volume products may not be covering the costs of the manufacturing resources they use. (1)
- Under our present (traditional) costing and pricing system, our high-volume products, such as our Ethiopian coffee, may be subsidizing our low-volume products. Some adjustments in prices may be required. (1)
- In markets where product prices are highly inelastic overcosting or undercosting products may distort reported profits per product line. (1)

Max (5)

PART B **[32]**

(a) Based on normal capacity, calculate the unit product cost under both absorption and marginal costing. [3]



Calculations often based on the production quantity of 260 000 units and not the specified 'normal capacity'. Normal capacity also clearly linked to the budgeted costs in the question detail.

	Absorption	**Marginal**	
	R	**R**	
Direct materials and labour (26 400 000/240 000)	110,00	110,00	(1) r/w
Variable production overhead (4 800 000/240 000)	20,00	20,00	(1) r/w
Fixed production overhead (14 400 000/240 000)	60,00	-	(1) r/w
Total	190,00	130,00	

Alternative calculation:

	Absorption	**Marginal**	
	R	**R**	
Direct materials, labour and Variable production overhead ((26 400 000+4 800 000)/240 000)		130,00	(2) r/w
Direct materials, labour, Variable production overhead and Fixed production overhead ((26 400 000+4 800 000+14 400 000)/240 000)	190,00		(2) r/w

Note: 1 mark allocated for the cost and 1 mark allocated for the 240 000

(b) Calculate the fixed production overhead absorbed during the period, and the extent of any under/over absorption. **[2]**



Students still battle to link the terms absorbed and production. If, like in this case, **more** units are produced (260 000 u) than the budget level (240 000 u), **over** recovery takes place.

	R	
Fixed production overhead incurred	14 400 000	(1) r/w
Fixed production overhead absorbed (260 000 x R60)	15 600 000	(1) r/w
Over-recovery (*1Mark for Over- recovery*)	1 200 000	

(c) Calculate the operating profit for the period using absorption costing. Max **(9)**

	R	
Sales (230 000 x R300)	69 000 000	(1) r/w
Less Cost of sales *(Alternative 1- COS mark allocation)*	(42 500 000)	
Opening inventory (40 000 x R190)	7 600 000	(2)*
(1) r/w (1) c		
Direct materials and labour (260 000 x R110)	28 600 000	(1) r/w
Variable production overhead (260 000 x R20)	5 200 000	
Fixed production overhead (260 000 x R60)	15 600 000	
Closing inventory (70 000 x R190)	(13 300 000)	(2)*
(1) r/w (1) c		
Over-recovery of fixed production overhead	(1 200 000)	(1) c
(1 mark for Over-recovery)		
Gross Margin	26 500 000	(1) r/w
Less Period Costs:		
Variable selling and administration overhead (230 000 x R10)	(2 300 000)	(1) r/w
Fixed selling and administration overhead	(9 600 000)	(1) r/w
Operating profit	14 600 000	

Max (9)

Note *: 1 mark for the correct cost and 1 mark for the correct volume

Alternative 2- COS mark allocation:

	R	
Less Cost of sales	(42 500 000)	
Opening inventory (40 000 x R190)	7 600 000	(2)* r/w
(1)r/w (1) c		
Direct materials, direct labour, variable and fixed production overhead (260 000 x R190)	49 400 000	(1) r/w
Closing inventory (70 000 x R190)	(13 300 000)	(2)* r/w
(1)r/w (1) c		
Over-recovery of fixed production overhead	(1 200 000)	(1) c
(1 mark for Over-recovery)		
		(6)

Note *: 1 mark for the correct cost and 1 mark for the correct volume

Alternative 3- COS mark allocation:

	R	
Less Cost of sales	(42 500 000)	
Direct material, direct labour, variable and fixed production overhead cost	43 700 000	(1) r/w
[40 000 + 260 000 – 70 000= 230 000]		
(1) r/w (1) r/w		
[230 000 x R190] *(2 mark for R 190)*		
Over-recovery of fixed production overhead	(1 200 000)	
(1 mark for Over-recovery)		(1) *c*
		(6)

(d) Calculate the operating profit for the period using marginal costing Max (9)



Compare the unit cost of both opening and closing inventory between the two methods.

	R	
Sales (230 000 x R300)	69 000 000	(1) r/w
Less Production cost of sales *(Alternative 1- COS mark allocation)*	(29 900 000)	
Opening inventory (40 000 x R130)	5 200 000	
(1)r/w (1) c		
Direct materials and labour (260 000 x R110)	28 600 000	(1) r/w
Variable production overhead (260 000 x R20)	5 200 000	
Closing inventory (70 000 x R130)	(9 100 000)	
(1)r/w (1) c	39 100 000	
Less Variable selling and administration overhead (230 000 x R10)	(2 300 000)	(1) r/w
Contribution *(1 mark for contribution only if variable selling and administrative cost was included above)*	36 800 000	(1) r/w
Less Fixed Costs:		
Fixed selling and administration overhead	(9 600 000)	(1)
Fixed production overhead	(14 400 000)	(1)
Operating profit	12 800 000	

Max (9)

Alternative 2- COS mark allocation:

	R	
Less Production cost of sales	(29 900 000)	
Opening inventory (40 000 x R130)	5 200 000	
(1)r/w (1) c		
Direct materials, direct labour, variable production overhead (260 000 x R130)	33 800 000	(1) r/w
Closing inventory (70 000 x R130)	(9 100 000)	
(1)r/w (1) c		

(3)

Alternative 3- COS mark allocation:

	R
Less Production cost of sales	(29 900 000)
Direct material, direct labour, variable production overhead [40 000 + 260 000 – 70 000= 230 000] (1) r/w (1) r/w (1) r/w [230 000 x R130] (1) r/w (1) c	29 900 000

(e) Reconcile the profit figures which you calculated under (c) and (d) above. **[2]**



Often left out. The difference should be the change in inventory (from 40 000 to 70 000) multiplied with the fixed production unit cost. As more of this fixed production cost is transferred into the next period, the absorption profit will be higher than the marginal costing profit.

Absorption costing profit	14 600 000	(½) c
Fixed overhead included in inventory increase (30 000 x R60) (1)	(1 800 000)	
Marginal costing profit	12 800 000	(½) c

Note:
(½) Mark allocated for 30 000 and (½) for R 60 and not the R 1 800 000 movement

(f) State the situation in which the profit figures calculated under both absorption costing and marginal costing would be the same. **[1]**



This situation concept not readily grasped by students.

- The profit figure will be the same with both systems whenever production equals sales and therefore opening inventory equals closing inventory. (1)

(g) List the arguments (advantages) put forward for the use of absorption and marginal costing systems respectively. **[6]**

Arguments for the use of absorption costing:

- Absorption costing does not understate the importance of fixed costs. (1)
- Absorption costing avoids the possibility of fictitious losses being reported. (1)
- Internal profit measurement should be consistent with external reporting and absorption costing achieves this objective (SA GAAP) (1)
- There will be relevant accounting packages available. (1)
- If there is no real inventory in the business this benefit will not be fully utilised. (1)

Max 3

Arguments for the use of marginal costing:

- Easier to understand Cost-Volume-Profit relationships. (1)
- The Marginal costing technique is appropriate for decision making as it highlights those costs (and revenues) which will change as a result of the decision under review being put into effect. (1)
- As fixed costs are mostly overheads and under marginal costing these are all treated as period costs and charged into the income statement therefore marginal costing avoids arbitrary allocation of overheads to units of output. (1)

- Reporting profit on a marginal costing basis will be more closely relates to changes in sales volume and are less affected by changes in inventory levels. (1)
- An understanding of the behaviour of costs and the implications of contribution is vital for accountants and managers as the use of marginal costing for decision making is universal. (1)
- Costs will be clearly identified to improve management thereof over the short term (1)
- Exercise of control (via budgets) will be improved (1)
- Prevents capitalisation of overheads in unsaleable inventory (1)

Max 3

PART C **[14]**

Signs may be inverse thus mark accordingly.

(a) Prepare a statement that compares the net benefits of the two contracts. **[14]**

	Mangaung	**Bela Bela**	
	R'000	**R'000**	
Material A in inventory (R32 400 x 0.90)	29 160		(1) r/w
Material B in inventory (R37 200 x 2)		74 400	(1) r/w
Material A on order (R45 600 x 0.90)	41 040		(1) r/w
Material A not yet ordered	90 000		(1)
Material C not yet ordered		106 800	(1)
Direct labour	129 000	165 000	(1)
Plant on site: Depreciation	Nil	Nil	(1)
Rental income: Mangaung *(Alternative 1)*	(9 000)	Nil	(1)
Rental income: Opportunity cost for Bela Bela (Alternative 2) [Note that it should not be included under both contract options]	*Nil*	*9 000*	(1)

Site foremen	Nil	Nil	(1)
Site foremen (Alternative 3)	*51 000*	*51 000*	(1)
Temporary accommodation for foremen	10 200	8 400	(1)
Interest	Nil	Nil	(1)
Central overheads	Nil	Nil	(1)
Penalty payment		42 000	(1)
Total relevant costs *(Alternative 1)*	(290 400)	(396 600)	
Total relevant costs (Alternative 2)	*(299 400)*	*(405 600)*	
Total relevant costs (Alternative 3)	*(341 400)*	*(447 600)*	
Contract price	432 000	528 000	(1)
Net benefit *(Alternative 1)*	**141 600**	**131 400**	(1)c*

Relevancy concerns cash, change and (alternative) opportunity. Consider material A: (1) no resale vs alternative use or R0 vs R32 400 x 90% →select the highest value (2) Change – alternative use (3) Cash – replaces the physical buying of the alternative. All components that are relevant will comply with these basic issues. Items not relevant should be shown as having a nil value iro the decision. In this particular question no motivation was requested, doing this wasted time.

Note*: Mark allocated if calculation was performed irrespective if a net benefit or a net loss was calculated

Conclusion: Based on the above calculation the Manguang project should be undertaken as it generates the highest net benefit. (1)*c*

(14)

QUESTION 12

OOSCAR LTD **SUGGESTED SOLUTION**

(Source: Supplementary exam 2010 TOE408W (MAC4862))

(a) **Value of debenture**

The interest receivable and the market rate were often used on an after-tax basis, yielding a lower value which does not accurately represent the actual cash flows involved. Some candidates incorrectly interchanged the rates.

Interest 11,5%
Discount rate 9,25%

∴ Value	= (115 x 2,52) + (1 050 x 0,767)	1 each or	(2)
	= R1 095,15		

(b) **WACC using target ratio**

- Gordons formula incorrectly applied.
- Debenture interest not used after tax.
- Target ratio (2:1) swapped – indicates poor reading and attention to detail.

K_e	= $\frac{105 \times 1,08}{3\,500} + 8\%$		
	= 11,24%		
K_d	= 9,25% x 0,8	(2)	r/w
	= 7,4%	(1)	r/w
WACC	= (0,667 x 7,4%) + (0,333 x 11,24%)	(1)	r/w
	= 8,68%	(1) Target Ratio	
		5	

(c) **Statement of comprehensive income – Variable costing basis**

Income statement generally done well, except for the following:

- Fixed costs not excluded from inventory.
- Skilled labour (fixed cost) not adjusted.
- Operating costs (variable) included in manufacturing costs.
- Contribution not shown.
- Overhead under recovered and expenditure variance not treated as fixed costs (ignored) or incorrectly handled – added back the wrong way.

	R'000	
Revenue	310 600	
Cost of sales	110 000	
Opening inventory	30 000	(½)
Production:		
Materials and components	75 490	(½)
Labour unskilled	10 750	(½)
Overhead variable	7 250	(½)
Packaging	6 510	(½)
Closing inventory	(20 000)	(½)
Contribution – manufacturing	200 600	
Operating costs	24 100	(½)
Contribution	176 500	(1)
Less:	149 000	
Manufacturing fixed costs (note 1)	69 600	(2)
Administrative costs	40 300	(½)
Finance costs	39 100	(½)
Profit before tax	27 500	(½)
		8

Note 1 Manufacturing fixed costs (see above)

Labour skilled	21 000	(½)
Overhead fixed	27 300	(½)
Overhead under recovered	20 700	(½)
Expenditure variance	600	(½)
	69 600	

Notes were mostly left out or not focussed on the accounting process and the treatment of inventory. This is a popular QE area.

Notes to management

1. A change from absorption to variable cost accounting during the year will necessitate a write-off of the fixed costs included in inventory at the point of change. (1)

2. Inventory for financial accounting (GAAP) purposes should be valued on the absorption costing method, including fixed manufacturing costs. (1)

   It should also include an adjustment (proportionate) in respect of the under recovery of the fixed costs as the factory only runs at 70% of capacity. (1)

3. The change in profit can be attributed to the fact that inventories levels have dropped from the levels that were maintained at the start of the financial year (closing inventory levels are less) resulting in a higher profit for the variable/marginal costing basis. (1)

Max 10

(d) **Manufacturing division's break-even point**



The variable costing income statement (part c) should have enabled students to easily calculate the contribution percentage and apply the BE formula. This often did not happen. The sequence of the 'required' sections often assist students and should be read attentively before starting to answer the question.

Contribution % per (c) 176 500 / 310 600 = 56,83% (1)(c)

$$\text{BEP} = \frac{69\,600 + 40\,300 + 39\,100}{0{,}5683} \quad (1)\text{r/w}$$

= R262 185,46 (1)(c)

3

(e) **Impact of new credit policy – debtors**



- Very unstructured approaches.
- Existing debtors amount not calculated, a comparative/change number not possible.
- Sales not split between cash and credit.
- Even where cash sales were calculated, the new debtors figures were often still calculated on the total sales – lack of understanding and/or lack of concentration evident.
- Bad debt ignored as part of the debtors amount – percentage (of debtors) then not 100.



The layout shown in the solution follows a step by step approach which can be easily applied to all similar situations.

Current debtors amount	=	142 000 x 71/360	
	=	28 005,6	(1)
New policy:			
Cash sales (142 000 x 0,1)	=	14 200,0	(1)
Credit sales (142 000 x 0,9)	=	127 800,0	(1)

	%	R'000	Days	Debtors	
Discount takers	15	19 170	12	639,0	(1)r/w
30 days	65	83 070	30	6 922,5	(1)r/w
60 days	17	21 726	60	3 621,0	(1)r/w
Bad debt	3	3 834	360	3 834,0	(1)r/w
	100	127 800		15 016,5	(1)(c)
	(1)				

Decrease in debtors book (additional cash)

	=	28 005,6 – 15 016,5	
	=	12 989,1	(1)r/w
			10

(f) Impact of new credit policy – costs

- Very poorly answered, mostly due to unstructured approaches.
- The solution required was for 10 months, but an annual approach was also acceptable. Students however often mixed up the 2 eg showing admin costs incurred on an annual basis but other calculations or 10 months, not acceptable.
- Cash saving not shown.
- Interest amounts incorrectly calculated.
- Change in bad debt not shown.

Cash saving per (e):		14 200,0 + 12 989,1 – (7% x 14 200)		
	=	26 195,1		(1)(c)
OR				
Cash available		26 195,1		
		On an annual basis	10 months	
Interest earned	=	26 195,1 @ 7,5%		
	=	1 964,6	1 637,2	(1)(c)
Interest on overdue accruals (12/10)				
21 726 x 12% x 30/360		260,7		
			(217,3)	(1)r/w
Costs incurred:				
Debtors clerk		210,0	175,0	(1)r/w
IT		120,0	100,0	(1)r/w
Cash discount (12/10)		1 192,8		(1)r/w
14 200 x 7%			994,0	
Settlement discount (12/10)		690,1		(1)r/w
19 170 x 3%			575,1	
Bad debt (12/10) see below		2 044,8	1 704,0	(1)r/w

Bad debts calculated as follows:

3% * R142m * 90% (for 10m) = R3 834 000	= difference of R1 704 000	(1)r/w
1,5% * R142m (for 10m) = R2 130 000		(1)r/w
From a cost perspective, the change is not beneficial.		(1)
		11

(g) **New contract assessment**



- Both a cost per product or total basis could have been used.
- The split between the products (70/30) not calculated or shown.
- Material requirement often done on a total basis for the 3 year period. This made relevancy rules irrelevant. As FIFO is applied, inventory held/required should be assessed on an **annual** basis. Where two uses are possible (eg. Plastitex can be sold or used in an alternative product), the **highest** cost is used for this *decision* assessment (opportunity approach). Any purchase after the initial inventory is used, must show the (latest) acquisition price clearly.
- Skilled labour is moved to the contract line for 2 years and therefore the actual cost (R50) is relevant. Students often used a rate of R5 (R50 – R45) Question: who then pays the R45 which they actually earn as part of the new R50 rate? To the rate should be added the opportunity cost (R10 per hour contribution) lost because of their move. This disappears after 2 years as new staff are now trained and can take over. This change must be shown clearly.

NB: Note that more layouts than those presented are possible.

Calculations: Quantity (70:30) **Contribution per set type**	**Individual Quantity** **P**	 **L**	**Plastitex R'000**	**Leather R'000**	
	2 100	900			
Set of frames @ R4 000	4 000	4 000	8 400	3 600	(1)r/w
Material:					
Required – 10m^2 x 2 100 units					
= 21 000 m^2 plastitex					(1)r/w
20 000 m^2 x R100 per m^2			2 000		(1)r/w
1 000 m^2 x R110 per m^2	1 004,76		110		(1)r/w
Leather					
Required – 9 m^2 x 900 units					
= 8 100					(1)r/w
8 100 m^2 x R140 m^2		1 260		1 134	(1)r/w
Labour: Skilled					
2 hours x 2 100 units x R50	100		210		(1)r/w
4 hours x 900 units R50		200		180	(1)r/w
Opportunity cost					
2 hours x 2 100 units x R10	20		42		(1)r/w
4 hours x 900 units x R10		40		36	(1)r/w
Labour: unskilled					
3 hours x R15 x 2 100 / 900	45	45	94,5	40,5	(1)r/w
Variable overheads:					
2 hours x R20 / 4 hours x R20	40	80			
2 100 units / 900 units			84	72	(1)r/w
	5 209,76	5 625	10 940,5	5 062,50	
Sales: R9 000 x 2 100			18 900		
R12 000 x 900	9 000,00	12 000		10 800	(1)
Annual – year 1	3 790,24	6 375	7 959,5	5 737,5	
Total				13 697,0	(1)

Year 2:			
Plastitex:	[9 000 – (4 000 + (10 x110) + (2 x 60) + (3x15) + (2 x 20)] x 2 310	= 8 535,5	(2)
Leather	(6 375 x 989) + (6 375 + (9 x (140 – 180)]	= 6 310,9	(1)
		14 846,4	
Year 3:			
Plastitex	[9 000 – (4 000 + (10 x 110) + (2 x 50) + (3x15) (2 x 20)] x 2 520	= 9 361,8	(2)
Leather	[12 000 – (4 000 + (9 x 180) + (4 x 50) + (3x15) + (20 x 4)] x 1 080	= 6 539,40	(3)
		15 901,20	
			22

The project evaluation gave rise to the following:

- The discount rate was 11%. An adjusted discount rate implied all forecast numbers had to be adjusted as well.
- Although the special equipment has a useful life of 4 years, the contract is for only 3 years. An asset recovery is required after the three years.
- Wear and tear was sometimes taken over 5 years (ie 120%!!)
- The cleaning cost of R500 000 was relevant but often ignored.
- Depreciation and/or the allocated cost were not extracted out of the incremental fixed costs – the *cash cost* was R1,5 m.
- The given tax rate of 20% was changed to 28%.

Project evaluation

Year	**Cashflow**	**R'000**	**PV factor @ 11%**	**Present value**		
0	Equipment	(20 000)	1,0	(20 000)	(1)	
0	Rigging and cleaning	(500)	1,0	(500)	(1)	
1	Contribution 13 697,0 x 0,8	10 957,6	0,901	9 872,8	(1)	(c)
2	Contribution 14 846,4 x 0,8	11 877,1	0,812	9 644,1	(1)	(c)
3	Contribution 15 901,2 x 0,8	12 720,9	0,731	9 299,0	(1)	(c)
1-3	Cash fixed cost 1,5m x 0,8	(1 200,0)	2,444	(2 932,8)	(1)	
1	Wear and tear tax benefit 20m x 40% x 0,2	1 600,0	0,901	1 441,6	(1)	
2-3	Wear and tear tax benefit 20m x 20% x 0,2	800,0	1,543	1 234,4	(1)	
3	Recovery 20m x 25% OR 5 m	5 000,0	0,731	3 655,0	(1)	
1	Rig and cleaning 500 x 0,2	100,0	0,901	90,1	(1)	
3	Recovery 1 m x 0,2	(200,00)	0,731	(146,2)	(1)	
	NPV			11 658,0		
	[Tax rate = (1)]		(1)			

Guideline:

Another layout with detailed line calculations follows on the next page. Students should always follow/use a format/layout that they understand the best and are comfortable with. This is applicable to the majority of management accounting topics.

Alternative layout

Year	**R'000**	**0**	**1**	**2**	**3**		
Equipment		(20 000)				(1)	
Rig and cleaning		(500)				(1)	
Contribution			13 697,0	14 846,4	15 901,2	(1)	(c)
Fixed cost			(1 500,0)	(1 500,0)	(1 500,0)	(1)	
Recovery (5%)					5 000,0	(1)	
Tax			(739,4)	(1 869,3)	(2 280,2)	(1)	(c)
NCF			12 197,0	13 346,4	14 400,7	(1)	(c)
Wear and tear			(8 000,0)	(4 000,0)	(4 000,0)	(2)	
Rig and cleaning			(500)				
Recovery					1 000,0	(1)	
Taxable			3 697,0	9 346,4	11 401,2		
Tax @ 20%			739,4	1 869,3	2 280,2	(1)	(c)
NCF		(20 500,0)	11 457,6	11 477,1	17 121,0		
DF @ 11%		1,0	0,901	0,812	0,731	(1)	
NPV		(20 500,0)	10 323,3	9 319,4	12 515,5		
		= 11 658,2					
∴ Proceed with the contract						(1)	

Calculation

1. Depreciation R20m x 25% = R 5 m
   After 3 years R15 m
   Wear and tear: 3 years R16 m
   R 1 m

2. Cash fixed cost:
   Total 7 700
   Less: Depreciation 5 000
   Rental 1 200
   'Cash' 1 500

Max 35

CAPITAL BUGDET

YEARS				1	2	3
Quantity				3 000	3 300	3 600
Material				2 100	2 310	2 520
	Own			2 000		
	Buy out			100	2 310	2 520
Leather				900	990	1 080
	Own			900	988.8888889	
	Buy out				1,111111111	1 080
Labour						
Material	Skilled	Required		2	2	2
		Price		60	60	50
	Unskilled	Required		3	3	3
		Price		15	15	15
Leather	Skilled	Required		4	4	4
		Price		60	60	50
	Unskilled	Required		3	3	3
		Price		15	15	15
Variable overheads				20	20	20
Sales price		Mat		9 000		
		Leather		12 000		
Sales		(2 100 x 9 000)		18 900 000	20 790 000	22 680 000
Material						
Set of frames		(2 100 x 4 000)		8 400 000	9 240 000	10 080 000
Material @ 10 maximum	Own	(2 000 x 10 x 100)		2 000 000	-	-
	Buy out	(100 x 10 x 110)		110 000	2 541 000	2 772 000
Labour	Skilled	(2 100 x 2 x 60)		252 000	277 200	252 000
	Unskilled	(2 100 x 3 x 15)		94 500	103 950	113 400
Variable overheads		(2 100 x 20)		84 000	92 400	100 800
Profit on material				7 959 500	8 535 450	9 361 800
Leather		(900 x 12 000)		10 800 000	11 880 000	12 960 000
Leather						
Set of frames		(900 x 4 000)		3 600 000	3 960 000	4 320 000
Leather @ 9 maximum	Own	(900 x 9 x 140)		1 134 000	1 246 000	-
	Buy out			-	1 800	1 749 600
Labour	Skilled	(900 x 4 x 60)		216 000	237 600	216 000
	Unskilled	(900 x 3 x 15)		40 500	44 550	48 600
Variable overheads		(900 x 4 x 20)		72 000	79 200	86 400
Profit on Leather				5 737 500	6 310 850	6 539 400
				13 697 000	14 846 300	15 901 200
Profit on material				7 959 500	8 535 450	9 361 800
Profit on Leather				5 737 500	6 310 850	6 539 400
Equipment			(20 000 000)			
Set up			(500 000)			
Fixed costs				(1 500 000)	(1 500 000)	(1 500 000)
Profit before tax				12 197 000	13 346 300	14 401 200
Recoupment						5 000 000
Tax pay				(839 400)	(1 869 260)	(2 280 240)
Tax pay				100 000		
Profit after tax/cf			(20 500 000)	11 457 600	11 477 040	17 120 960
PV Factor			1	0,901	0,812	0,731
			(20 500 000)	10 323 298	9 319 356	12 515 422
			11 658 076			

YEARS		1	2	3
Tax Calc				
Profit before tax	**(500 000)**	**12 197 000**	**13 346 300**	**14 401 200**
Wear and tear		**(8 000 000)**	**(4 000 000)**	**(4 000 000)**
Recoupment				**1 000 000**
Taxable Inc	**(500 000)**	**4 197 000**	**9 346 300**	**11 401 200**
Tax payable @ 20%	**(100 000)**	**839 400**	**1 869 260**	**2 280 240**

(h) Standard cost variances

The efficiency variances were either left out or done incorrectly. Keep in mind that a standard for the actual output is determined by applying flexible budgeting principles.

Skilled labour rate variance
7 500 x (55 – 50) = R37 500 U (1)
Skilled labour efficiency variance
(7 500 – (2 100 x 2 + 900 x 4)) x 50 = R15 000 F (1)
Unskilled labour rate variance
9 600 x (17 – 15) = R19 200 U (1)
Unskilled labour efficiency variance
(9 600 – 3 000 x 3) x 15 = R 9 000 U (1)
4

(i) Risk considerations

Many students approached the risk discussion from an incorrect angle, that Ooscar is the exporter. Ooscar is a supplier to the exporter.

- Labour unrest curtailing production and delivery of products – refer NUM strike. (2)
- Dependent on vehicle manufacturers, manufacturing division turnover double that of retail. Vehicle sales dependent on economy. (2)
- Retail divisions sales (after market) dependent on free cash available to consumers, quite depressed at the moment. (2)
- Target structure of capital may be indicative of high debt which is risky from an inventor perspective. Can decrease market accessibility. (2)
- Interest paid very high and liquidity poor. May be left with too low cash resources/buffer. (2)
- Debtor's days outstanding beyond target. May lead to higher bad debt in time and detrimental to cash flows. (2)
- Cost plus 15% transfer price between divisions may lead to acrimony between divisions as (target?) returns not met. (2)
- Underutilised capacity may imply fixed costs will become difficult to be met. (2)
- The technology being used could pose a technology risk. (2)
- If the debentures are not replaced in 3 years with an alternative source of funding, then the company may be exposed to liquidity risk. (2)
- In economic cycles such as the one being experienced at present, the company would be exposed to a higher degree of credit risk in respect of its accounts receivable. (2)
- Any other valid point, 2 marks. (2)

Max 10

QUESTION 13 – SUGGESTED SOLUTION

(*Source: Supplementary exam (2010) DIPAC26 (MAC4861)*

DREAMY CREAMY

(a) Budgeted breakeven point



The budgeted breakeven point in units was required. Notes 1 and 7 disclose that budget levels equal actual levels. The unit indicator implies that cost information given per unit should be used in the calculations. Students often get confused when using the formula approach (given as an alternative) and should refrain from using it.

Variable costs	**Strawberry Surprise**	**Grape Treat**	
Yoghurt	1,60	2,00	(½)
Apple juice	0,18	0,05	(½)
Grape juice	0,16	0,30	(½)
Strawberry juice	0,35	0,06	(½)
Flavourants / colourant	0,50	0,50	(½)
Bottles	0,45	0,45	(½)
Labelling	0,75	0,75	(½)
	3,99	4,11	
		Max	3
Fixed costs			
Manufacturing overheads given	400 000	216 000	(1)
H/O costs			
500 000 x 0,20	100 000		(½)
270 000 x 0,20		54 000	(½)
	500 000	270 000	
Break even points	= (7,90 – 3,99)	(7,40 – 4,11)	(2)
(1 mark for rounding up)	= 127 878	82 067	(2)
			9

Alternative

Break even Strawberry Surprise

Px = SP – VC – FC
0 = 7,90x – 3,99x – 400 000 – 100 000
0 = 3,91x – 500 000
x = 127 877,2 round up 127 878 (2)

Break even Grape Treat

Px = SP – VC – FC
0 = 7,40x – 4,11x – 216 000 – 54 000
0 = 3,29x – 270 000
x = 82 066,87, round up therefore = 82 067 (2)

(b) Statement of Comprehensive Income for the year ended 31 December 2010



Lost marks were mostly due to:

- Sloppy reading, not applying the cost changes per the notes.
- Simple calculation errors.
- Erratic recording of numbers eg: apple juice R90 000, grape juice R80 (instead of R80 000).
- Actual prices or cost per unit as requested not disclosed.

The actual statement of comprehensive income which was required, inferred that the absorption method had to be used.

	Strawberry Surprise R	**Grape Treat R**	
Sales			
Strawberry Surprise: 7,90 x 1,06 = 8,37			(1)
Total sales = 8,37 x 480 000 = 4 019 520	4 017 600		(½)
Not rounded alternative (4 019 520)			
Grape Treat: 7,40 x 1,06 = 7,84			(1)
Total sales = 7,84 x 270 000 = 2 117 880		2 116 800	(½)
Not rounded alternative (2 117 880)			
Opening inventory	0	0	(1)
Yoghurt			
Strawberry Surprise: 1,60 x 0,9 = 1,44			(1)
Total = 1,44 x 500 000 = 720 000	720 000		(½)
Grape Treat: 2 x 0,9 = 1,80			(1)
Total = 1,80 x 270 000 = 486 000		486 000	(½)
	345 000	110 700	
Apple juice			
Strawberry Surprise: 0,18 x 500 000	90 000		(½)
Grape Treat: 0,05 x 270 000		13 500	(½)
Grape Juice			
Strawberry Surprise: 0,16 x 500 000	80 000		(½)
Grape Treat: 0,30 x 270 000		81 000	(½)
Strawberry Juice			
Strawberry Surprise: 0,35 x 500 000	175 000		(½)
Grape Treat: 0,06 x 270 000		16 200	(½)
Flavour / colourant			
Strawberry Surprise: (0,5 + 0,2) x 500 000	350 000		(½)
Grape Treat: (0,5 + 0,2) x 270 000		189 000	(½)

MAC4861
MAC4862

	Strawberry Surprise R	Grape Treat R	
Bottles			(1)
0,45 x 1,05 = 0,47			
Strawberry Surprise: 0,47 x 500 000	235 000		(½)
Alternative not rounded (236 250)			
Grape Treat: 0,47 x 270 000		126 900	(½)
Not rounded alternative (127 575)			
Labels	376 000	203 500	
Strawberry Surprise: 0,75 x 500 000	375 000		(½)
Grape Treat: 0,75 x 270 000		202 500	(½)
Delivery of labels	1 000	1 000	(1)
Manufacturing overhead	400 000	216 000	(1)
Closing inventory			
Strawberry Surprise: 4,852 (rounded 4,85) x 20 000	(97 000)	-	(3)
Alternative (97 040)			
Per unit			
(1,44 + 0,47 + 0,18 + 0,16 + 0,35 + 0,7 + 0,75 + 0,8 + 0,002) x 20 000			
OR RATHER: Total cost R2 426 000/500 000 x 20 000			
Cost of sales	2 329 000	1 332 100	
Gross profit	1 688 600	784 700	
Head office costs			
Strawberry Surprise: 0,2 x 500 000	100 000		(½)
Grape Treat: 0,2 x 270 000		54 000	(½)
Net profit	1 588 600	730 700	
		Max	20

Alternative layout:

Sales (7,9 x 1,06 x 480 T/ 7,4 x 1,06 x 270T)	4 017 600	2 116 800	(3)
Cost of sales	2 329 000	1 332 100	
Production:			
Opening inventory	0	0	(1)
Yoghurt (1,6 x 0,9 x 500T / 2 x 0,9 x 270T)	720 000	486 000	(3)
Other variable (2,61 x 500T/2,33 x 270T)	1 305 000	629 100	(6)
[0,45 x 1,05 = 0,47]			(1)
Delivery of labels	1 000	1 000	(1)
Manufacturing overhead	400 000	216 000	(1)
Closing inventory	(97 000)	0	(3)
Gross profit	1 688 600	784 700	
HO costs	100 000	54 000	(1)
Net profit	1 588 600	730 700	

(c) Evaluate special order proposal



Students lost marks by

- Not using the specified (relevant) costs
- Taking the allocated costs as relevant
- Calculation errors
- Missing the specified period (reading!!)
- The motivation focuses on the benefits and constraints caused by the new order, whereas students tended to be very general in their discussion eg profit will increase.

It is good practice to stick with the information as given, in this case unit information. By applying the quantity to each item, a lot more time is required for all the calculations and there is more chance for error.

	R	
Selling price	5,00	(½)
Yoghurt	1,44	
Apple Juice	0,18	
Grape Juice	0,16	(1)
Strawberry Juice	0,35	
Bottling	0,50	(1)
Labels	0,60	(1)
Flavour / colourants (Actual 0,7 – 0,2 saving)	0,50	(1)
Overheads	0,30	(½)
Head office costs	-	(½)
Profit per unit	0,97	(½)
Total profit		
x 2 500 units x 7 months	R16 975,00	(1)

- Dreamy Creamy may, based on financial figures only, decide to sell to Baby Bear (Pty) Ltd, as they will make an additional profit of R2 425 per month. (1)
- However, the company should consider whether such a small amount (compared to the company's normal profit the project will only add 0.73% more profit) is worth its while and the changes to the production process (for which no cost was budgeted). (2)
- Exposure to a new client in Gauteng – possibility of more permanent orders (1)
- Additional contribution will contribute towards fixed costs (1)
- Toddlers may 'convert' to normal users (1)
- The validity of the projections should be considered: the costs were estimated by a production-assistant. (1)
- Consider whether the new shape bottles and labels will be available on short notice: the project starts 'tomorrow': will the new materials be sourced in time for delivery. (1)
- Consider the impact of doing additional work on the company's machinery: does it lead to additional wear-and-tear? Will machinery require maintenance more quickly? (1)
- Quality is a huge issue with children: consider the reputational risk (Nestlé was very hard hit by the infant deaths caused by its substitute milk product) – is it worthwhile taking this risk, and should liability insurance (if something should go wrong) be considered? (2)
- Any other valid point (maximum 1) (1)

Max 10

(d) Milk process evaluation



The project evaluation showed the following:

- Students often converted each line item to an annual cost, wasting valuable time.
- Supporting calculations often not shown.
- Capital cost in the incorrect periods.
- Not incorporating the vehicle cost as a capital cost.
- Incorrectly showing the financing cost as a relevant cash flow.
- Capacities used incorrectly, impacting on sales and costs.
- Not picking up the dates correctly in the projection.
- This is a stand alone project ('…no other activities…') and therefore the tax loss should be carried forward from year to year.
- Recommendations should always be made and be in line with the calculated value.

Calculations

1.	**Income – per day**	**R**	
	Packed 250 000 x 4,10	1 025 000	(½)
	Bulk 50 000 x 3,50	175 000	(½)
		1 200 000	
	Annual (x 360)	432 000 000	(1)
	Half (x 180)	216 000 000	(1)
2.	**Costs – per day**		
	Milk supply *(NOTE to markers: this can (and likely will) be shown separately, and should be marked as well! – would be R248 400' for full years and R124 200' for half year)*	2,30	(1)
	Bag	0,60	(½)
	Labour	0,15	(½)
	Fixed overhead	0,25	(½)
	Collection and storage	0,05	(½)
	Delivery cost	0,10	(½)
	Management	0,05	(½)
		3,50	
	Packed (x 250 000)	875 000	(1)
	Bulk (2,90 x 50 000)	145 000	(1)
		1 020 000	
	Annual (x 360)	367 200 000	(1)
	Half year (x 180)	183 600 000	(1)

Alternative:

For retailers (packed): 3,5 x 250 000 x 360	315 000 000
Half year	157 500 000
For F & V (bulk): 2,9 x 50 000 x 360	52 200 000
Half year	26 100 000

(315 000 + 52 200 = 367 200)

Alternative 2: (if milk supply is shown separately)

(Milk supply would be R248 400 000 per year and R124 200 000 per half year). Then other variable cost:

For retailers (packed): 1,2 x 250 000 x 360	108 000 000
Half year	54 000 000
For F & V (bulk): 0,6 x 50 000 x 360	10 800 000
Half year	5 400 000

(248 200 + 108 000 + 10 800 = 367 200)

3. **Wear and tear - Plant and equipment**

Capex	R90m
40% (year 1)	36m
30% (year 2)	27m
30% (year 3)	27m

(2)

4. **Wear and tear – Trucks**

Capex	R1,2m
25% (year 1) x 0,5	150 000
25% (year 2)	300 000
25% (year 3)	300 000
25% (year 4)	300 000
25% (year 5) x 0,5	150 000

(2)

Project evaluation (R'000)

Year	0	1	2	3	4	5	6	(1)
Capex P & E	(65 000)	(25 000)						(2)
Capex Trucks		(1 200)						(1)
Inflow		32 400	64 800	64 800	64 800	64 800	32 400	(1)
Tax		-	(9 450)	(10 500)	(18 060)	(18 102)	(9 072)	(1)
Tax calculation								
Tax loss			(3 750)					(1)
Inflow		32 400	64 800	64 800	64 800	64 800	32 400	(1)
W & T- P & E		(36 000)	(27 000)	(27 000)				(1)
W & T Trucks		(150)	(300)	(300)	(300)	(150)		
Taxable income		(3 750)	33 750	37 500	64 500	64 650	32 400	(1)
Tax @ 28%		-	9 450	10 500	18 060	18 102	9 072	(1)
NCF	(65 000)	6 200	55 350	54 300	46 740	46 698	23 328	
DF @ 18%	1,0	0,847	0,718	0,609	0,516	0,437	0,370	(1)
NPV	(65 000)	5 251,4	39 741,3	33 068,7	24 117,8	20 407,0	8 631,4	
=	66 217,6							
Per calculator =	66 215,9							
∴ Proceed with the facility								(1)
							Max	25

(e) Mr Farmer's income differential



The question specifically required the differential income per day, often ignored by students. The management fee payable by the project is an income for Mr Farmer, often missed.

Current net income

800 x 0,6 x 30	14 400		
800 x 0,4 x 18	5 760		
Daily litres delivered	20 160	(1)	(R/w)
x Price (2,60)	= R52 416	(1)	(c)
Less Costs			
Base cost 20 160 x 2,00	= (40 320)	(1)	(c)
Power feed cows (800 x 0,6) x 10	(4 800)	(1)	(R/w)
	7 296		

Project income

Daily litres (20 160 x 2,30)	46 368	(1)	(c)
Management fee (300 000 x 0,05)	15 000	(1)	(R/w)
Costs per above	(45 120)	(1)	(c)
Daily income differential	8 952	(1)	(R/w)
		8	

(f) Lawn grass income



- Second bullet regarding quantity not correctly applied.
- Probabilities not applied correctly.
- Net income not correctly calculated.
- Calculations again often on a line by line basis wasting time.

Total parts sold = 6 x 3 + 6 x 1	= 24	(1)
December	3	
Expected (historical)	m^2	
P 0,2 x 240 000	48 000	(½)
N 0,65 x 300 000	195 000	(½)
H 0,15 x 330 000	49 500	(½)
	292 500	
December 3/24	36 563	(½)
Net income (6 – 2 – 1) = 3/m^2		(1)
Total income (R)	109 689	(1)
Market forecast	m^2	
P 0,45 x 240 000	108 000	(½)
N 0,50 x 300 000	150 000	(½)
H 0,05 x 330 000	16 500	(½)
	274 500	
December (3/24)	34 313	(½)
Total income (x3/m^2) (R)	102 939	(1)
		8

QUESTION 14

TUNING TONIES; FELINE PIXELS; TEDDY **SUGGESTED SOLUTION**

(Source: Test 1 2013 MAC4862)

PART 1

(a) *Break-even in sales value*

	Manual		**Automated**		
	R	**%**	**R**	**%**	
Fixed costs:					
Equipment lease paid	55 350		154 980		
Maintenance contract	18 750		28 800		
Rent paid for premises	29 800		29 800		
	103 900		**213 580**		(1)r/w (½ each)
Fees	420 000	100,0	420 000	100,0	
Variable costs	**211 000**	50,2	**115 380**	27,5	(1)r/w (½ each)
Labour	108 000		63 000		
Parts bought	69 000		38 880		
Variable overheads	34 000		13 500		
Contribution margin	209 000	**49,8**	304 620	**72,5**	(1) for % (½ each)r/w
Fixed costs	103 900		213 580		
Net income	105 100		91 040		
Fixed costs	103 900		213 580		
Contribution margin ratio	49,8%		72,5%		
Break-even in sales value =	**208 635**	=	**294 593**		(2)r/w (1 each) OR (1)c (½ each)
				Total	**(5)**

(b) *Margin of safety as a percentage of sales*

	Manual	**Automated**	
	R	**R**	
Total current sales	420 000	420 000	
Break-even sales	208 635	294 593	
Margin of safety in rand value	**211 365**	**125 407**	(3) r/w (1½ each)
Margin of safety as a % of current sales	**50,3%**	**29,9%**	(2) r/w (1 each) OR (1) c (½ each) for formula application

(c) *Evaluate the two tuning methods and advise on what method to choose*

- The company's **optimal level** of tuning is where they are indifferent between the manual and automated methods (**net income is the same** for both methods). (1)
- If the fees are expected to **increase** above R420 000 in future, then it is probably better to **automate** the tuning of the cars. (1)
  The reason is that the **contribution ratio is higher** and its profits will therefore increase more rapidly as sales increase. (1)
- If the fees are expected to **drop** below R420 000 from time to time, then it is probably better to do the tuning of the cars **manually**. (1)
  The reason for this is the **lower fixed costs** that result in a lower break-even point and a **higher margin of safety**. (1)
- The manual method will **not incur losses as quickly as the automated method** in periods of sharply declining fees. Due to its **lower contribution margin**, the manual method will not lose contribution as rapidly as the automated method when fees are lost. (1) (1)
- The manual method will result in **greater stability** in net income and will be more protected from losses during bad years, **but at the cost of lower net income in good years**. (1)
- Consequential mark for choosing a method **with reason**. (1)c

Total 9
Maximum (5)

PART 2

(d) *Traditional absorption costing based on direct labour hours*

Total direct labour hours are budgeted as follows:

(2 x 550 Product A units) + (2 x 110 Product B units) = 1 320 direct labour hours.
This yields the following application rate:

R42 000 budgeted machine setup cost
1 320 budget direct labour hours = **R31,82** application rate per labour hour (½)r/w

	Product A	**Product B**
Total machine setup-related cost applied to each product line	**R35 000** (½)r/w (R31,82 x 1 100 hrs)	**R7 000** (½)r/w (R31,82 x 220 hrs)
Machine setup-related cost per product unit (or R31,82 x 2hrs)	**R63,64** (½)r/w (R35 000 ÷ 550 units)	**R63,64** (½)r/w (R7 000 ÷ 110 units)

Total (2½)
Maximum (2)

Note:

Since Product A is a high-volume product, it picks up 5 times the direct labour hours and thus does the same for machine setup-related cost even though it requires the same number of direct labour hours as does Product B. This gives inaccurate product costs and hinders cost reduction because it gives the wrong signal to management. Conventional costing over costs Product A, the high-volume product and under costs Product B, the low-volume product.

(e) *Activity-based costing (ABC)*

If instead, ABC is used, the cost driver is the budgeted setup hours:

Machine setup hours = (550/110 x 7 hrs) + (110/55 x 17,5)
= (5 runs x 7 hrs per run) + (2 runs x 17,5 hrs per run)
= 35 + 35 = 70 hours

R42 000 budgeted machine setup cost
70 budgeted machine setup hours = **R600** application rate per machine setup hour (½)r/w

	Product A R		**Product B R**	
Total machine setup-related cost applied to each product line (R600 x 35 hrs)	**21 000**	(½)r/w	**21 000**	(½)r/w
Machine setup-related cost per product unit (R21 000 ÷ 550 units) (R21 000 ÷ 110 units)	**38,18**	(½)r/w	**190,91**	(½)r/w
			Total	2½
			Maximum	(2)

Note:
Each product line receives R21 000 machine setup-related cost because each line requires the same amount of engineering attention. However, there are more units of Product A to share this total cost.

(f) **Volume** diversity for Products A and B. (1)r/w

(g) The fixed cost of R42 000 will **remain the same**. However the **allocation** to the two different products **changed**. The question would be if the new allocation (ABC) is so accurate, can the fixed cost be classified as **product specific**. If not the break-even will remain the same. (3)
Or if referred / applied to (d) and (e) above. (1)

Total 4
Maximum (3)

PART 3

(h) *Effect of proposed design changes*

Impact of reduction in decorations

Contribution per unit:
Fixed cost per unit: R4 050 000 / 300 000 = **R13,50** (½) r/w
Contribution per unit – Fixed cost per unit (R13,50) = Net profit per unit (R30)
Therefore contribution per unit = **R43,50** (½) r/w

	R	
Loss of contribution from 10% sales drop (300 000 x 10% x R43,50)	(1 305 000)	(1) c
Production cost saving (300 000 x 90% x R6)	1 620 000	(1) r/w
Net gain from decoration reduction	315 000	

Plastic eyes

The drop in sales volume as a result of the reduction in decorations is also applicable to the evaluation of the proposed changes in the type of eyes and change in filling.

Glass eyes required for production = 270 000 x 2 = 540 000 (½) r/w
Cost per glass eye = R180 / 100 = R1,80 per eye (½) r/w
Cost per plastic eye = R140 / 100 = R1,40 per eye (½) r/w

	R		**Alternative R**	
Glass eye input required to allow for 5% input loss (540 000/0,95 = 568 421 -> 568 500 x R1,80)	1 023 300	(2)r/w	1 023 158	(1)r/w
Plastic eye input required to allow for 10% input loss (540 000 / 0,90 x R1,40)	(840 000)	(1)r/w	(840 000)	(1)r/w
Net saving from plastic eyes	183 300		183 158	

Scrap material for filling

	R	
Cost of synthetic filling (270 000 / 2 000 x R720)	97 200	(1) r/w
Additional shredding cost (270 000 x R0,50)	(135 000)	(1) r/w
Disposal cost saved	12 000	(1) r/w
Net increase in cost due to using scrap material	(25 800)	

Total effect

	R		**Alternative R**	
Net gain from decoration reduction	315 000		315 000	
Net saving from plastic eyes	183 300		183 158	
Net increase in cost due to using scrap material	(25 800)		(25 800)	
Overall net **increase** in annual profit	472 500	(1) c	472 358	(1)c

The proposals will **exceed** the annual target profit increase of R250 000. (1) c

If the proposal to use scrap material for filling is not implemented, annual profit can be increased even further.

	Total	12½
	Maximum	(12)

(i) *Percentage reduction in sales*

	R	
Additional contribution from all three changes	6,583	(2) c
(1 620 000 + 183 300 – 25 800) / 270 000		(1) c
Existing contribution	43,50	
Revised contribution per unit	50,083	

Number of bears required to give the same contribution as prior to the changes:
(R43,50 x 300 000) / R50,083 per unit
= R13 050 000 / R50,083
= 260 568 bears (1) c
Therefore the reduction in sales required to leave net profit unchanged
= (300 000 – 260 568) / 300 000
= 39 432 / 300 000
= 13,1% (1) c

MARKERS' COMMENTS:

PART 1

Part (a) Break-even

Unfortunately many students failed to correctly classify costs into fixed and variable costs which should be basic at post-graduate level. Several students assumed that labour is a fixed cost despite the indication in the question that it is variable ("paid on hourly basis"). In some cases students did not know the break-even formula for calculating the break-even sales in Rand value. Some students got confused as the number of units was not given and they wanted to calculate the break-even units using contribution per unit. This lack of knowledge is alarming as this was an easy question.

Part (b) Margin of safety

For part (b) the answer in (a) was required. Students were penalized if they did not know the margin of safety formula. The rounding was also a bit of a concern.

Part (c) Evaluation of methods

Students generally failed to discuss the impact should the fees increase or decrease in future and instead only focused on the current fees level.

PART 2

Part (d) Traditional costing

Students did relatively well in this question. In some cases students failed to work out the cost per unit and only gave the total fixed cost allocated to each product line or alternatively only calculated the cost per unit but not the total fixed cost. This cost them one mark. This oversight also occurred in part (e).

Part (e) ABC

Quite a few students struggled with the ABC calculations. They did not always correctly calculate the number of runs in order to calculate the number of machine setup hours. If the correct machine setup hours were calculated, they were not always correctly applied (e.g. multiplying or dividing the hours by the number of units produced). Students should do a reasonability check on their final answer. The fixed costs allocated to Product A and Product B should still total R42 000.

Part (g) ABC's impact on break-even

Not a lot of students recognized that the total fixed cost remained the same. The allocation just changed. It was alarming that a lot of students talked about how the contribution would change.

PART 3

Part (h) Effect of proposed changes

Part (h) was not answered very well. In general many students seemed to:

- not read the information and question in detail;
- not understand the nature of fixed costs (in total and per unit) and the impact on net profit per unit when production volume changes; and
- not be able to calculate the quantity of material issued from stores using the loss (damage) percentage and the quantity used in the final product.

Several students approached the question by trying to calculate and evaluate each proposal on its own instead of **combining** the proposals. Furthermore the order of evaluations was a problem. The decorations proposal should be evaluated **first** followed by the plastic eyes and scrap material proposals (i.e. students used 300 000 units instead of 270 000 units for the plastic eyes and scrap material evaluations). This was clearly indicated in the Required section.

Many students incorrectly used the net profit per unit (R30) and multiplied by the revised volume (270 000 units). Fixed costs do not normally change as volume changes. Hence the name **fixed** costs. Only the total variable costs (and sales) change proportionately as volume changes. Therefore, total contribution changes as volume changes (i.e. Sales less Variable costs). It is therefore incorrect to multiply the net profit per unit by the new volume since the net profit per unit includes fixed cost per unit. Only **contribution per unit** can be multiplied by the new volume to get a correct answer. At the new volume level the fixed cost **per unit** will change (and therefore the net profit per unit) but the total fixed cost remains the same.

Another frequent problem was that students forgot that each bear required two eyes (not one). The impact of the increase in damaged eyes was also very often calculated incorrectly as 5% or 10% of the eyes **used on the bears** while the question stated that the percentages of damage were of **"the quantity issued from stores"**. This means that the quantity of eyes used on the bears was already **after** the 5% or 10% loss (i.e. students should divide by 95% or 90%, not multiply by 5% or 10%).

Part (i) Percentage reduction in sales

Students who did attempt this section struggled with it. They frequently calculated the reduction in sales in Rand value instead of in units.

QUESTION 15

MADIBSA **SUGGESTED SOLUTION**

(Source: Extract from Test 1 2013 MAC4861)

MAC4861 only:

PART 3

(h)



- Know the format of the two different costing methods. The question is quite simple and you can score good marks if you know the format.
- Important - calculate the number of units.
- Know and understand the difference between *applied, actual and budgeted overheads*. A clear example is given in the Tut letter.
- In the exam this will probably be more difficult and interlinked with standard costing.

MADIBSA (PTY) LTD STATEMENT OF COMPREHENSIVE INCOME FOR THE YEAR 2013 FOR PRODUCT WINNIE

	Total R	
Sales (47 500 x R90)	4 275 000	
Manufacturing cost	3 144500	
Direct material(49 000 x R30)	1 470 000	(½)r/w
Package cost (49 000/20 xR24)	58 800	(½)r/w
Direct labour (49 000 x R15)	735 000	(½)r/w
Variable overheads(49 000 x R10)	490 000	(½)r/w
Fixed manufacturing overheads (49 000 x (R1 000k/2/50k))	490 000	(1)r/w
Closing stock (R3 243 800/49 000 x **1 500**)	(99 300)	
(1)c (1)r/w		
	1 130 500	
Under recovery(R490k – R500k)	(10 000)	(1)r/w
Expenditure variance (R500k - R480k)	20 000	(1)r/w
Gross profit	1,140,500	
Commission (R4 275 000 x 4%)	(171,000)	(1)r/w
Net profit	969 500	

Total (8)

(i)

- Do not try to do the whole Statement of Comprehensive Income all over again.
- Instead, calculate the fixed cost in the closing inventory.
- Understand the concept that fixed cost is transferred to the Balance sheet with the absorption costing system therefore the profit will decrease with the direct costing method.

Profit from (h)	969,500	
Fixed cost in inventory	(15,000)	(1)r/w
Inventory with variable costing	84,300	
Inventory with absorption costing	(99,300)	
Profit with variable	954,500	(1)c if profit decreased

Total (2)

(j)

This was also to test students on the budget versus actual concept.

Product information is needed for **decision making** purposes for example pricing decisions. Decisions cannot be **delayed** until actual data are available. (1)

Seasonal fluctuations in volume can cause that different rates are used which is not necessarily a true reflection of a typical month. (1)

Total (2)

(k)

Very easy marks!

There will be **no difference** between the income reported under direct costing and income reported under absorption costing. The reason is that there will be **no inventories** on hand and therefore no way to shift fixed overhead cost between periods under absorption costing. JIT greatly enhances the clarity of statements prepared by the absorption costing method.

Total (2)

(l)

- Note the capacity constraints given in the question.
- No overtime is currently allowed and the budget is based on manufacturing time required.
- Therefore only 50 000 units can be used in the calculation.
- Be aware of scenarios like this in future questions
-

Product Winnie

Sales	(**R85,50** x 50,000)	4 275 000	(1)r/w
Manufacturing cost	(30+15+10+1.20+10) x **50,000**	3 310 000	(1)r/w
Commission	(85,50 x 50 000) x 4%	171 000	
Net profit		794 000	(1)r/w

Total (3)

QUESTION 16

P&S LTD; SAVE LTD **SUGGESTED SOLUTION**

(Source: Test 2 2013 MAC4862)

PART A

(a)	(i)	**Material price variance**	**F/(u)**	**F/(u)**		
		P: 1 100 000 (3,8 – 3,6)	(220 000)			
		O: 725 000 (2,30 – 2,2)	(72 500)			
		L: 830 000 (0,56 – 0,6)	33 200	(259 300)	(1½)	R/W
	(ii)	**Material mix variance**				
		Issued: - P: [1 000 000 – 1 056 000 (w1)]x3,60	201 600			
		- O: [790 000 – 739 200 (w1)] x 2,20	(111 760)			
		- L: [850 000 – 844 800 (w1)] x 0,6	(3 120)	86 720	(1½)	R/W
	(iii)	**Material yield variance**				
		5 934 720 (w2) – 5 901 000 (w3)	(33 720)		(1)	R/W
		[P = R180 000F, O = R121 000A, L = R6000A] (½ mark for each variance) Max 1 for usage variance				
	(iv)	**Labour rate variance**				
		53 800 (21,50 – 21,00 (w4))	26 900 U		(1)	R/W
	(v)	**Labour idle time variance**				
		[(53 800 x 21 – (52 000(w5) x 21)]	37 800 U		(2)	
		Or				
		[(53 800 x 96%) – 49 920] x 21,875				
	(vi)	**Labour efficiency variance**				
		(49 920 x 21,875) – (210 000 x 0,24 x 21,875)	10 500 F		(1)	R/W
		Or				
		[(52 000 – (0,25 x 210 000)] x 21				

Workings:

1. Total issued = 1 000 000 + 790 000 + 850 000 = 2 640 000

2 640 000 x 5/12,5	1 056 000
2 640 000 x 3,5/12,5	739 200
2 640 000 x 4/12,5	844 800

2. (1 056 000 x 3,6) + (739 200 x 2,2) + (844 800 x 0,6) 5 934 720

3. 210 000 x 28,10 (Standard cost of material) 5 901 000

4. R21,875 x 96% = R21 (Standard clock rate – normal hours)

5. 49 920 ÷ 96% = 52 000

Some students calculated **unit variances**, which **are not acceptable** as the variances should relate to the costs in the management accounts and income statements. Variances are used in the reconciliation of budget profit to actual profit.

Most students struggled with the concept of idle time on labour efficiency, students were given the productive (work) rate and had to work backwards to get the clock rate first in order to carry out the calculations.

Other problems encountered included:

- Students not stating clearly whether the variance was favourable or unfavourable (adverse).
- Incorrect formula used and mathematical errors.
- Students doing various (random) calculations but not actually calculating the variances
- The distinction between actual and budgeted overhead and the level of the budget.
- Spending too much time on the variance calculation and not completing all sections of the test.

(b) Journal entries

Materials

Dt Production WIP (½)	5 901 000		(½)R/W
Ct Material usage variance (½)		53 000	(½)R/W
Ct Inventory/ stores ledger ctrl acc (½)			
– P (1 000 000 x 3,6)		3 600 000	(½)R/W
– O (790 000 x 2,2)		1 738 000	(½)R/W
– L (850 000 x 0,6)		510 000	(½)R/W

Labour

Dt Wages control Acc (½)	1 102 500		(½)R/W
Ct Bank (½)		1 102 500	(½)R/W

The materials entries were recorded poorly in terms of the description given to the accounts and taking variances into account in the journal itself.

(c) **Maximisation of contribution**

Contribution per unit

	LD R	**SO R**	**AP R**		
Selling price	200	100	300		
Direct material	60	40	172,5		
Direct labour	20	10	35		
Factory OH (40%)	10	5	17,5		
Selling and admin (33,3%)	15	7,5	22,5		
Contribution	95	37,5	52,5		(1½) R/W
Constraints				**Total**	
Mixing-hours needed	25 000	4 500	9 500	39 000	(1½)
Available				(15 000)	
Shortage				24 000	(½)
P & F-hours needed	12 500	2 250	28 500	43 250	(1½)
Available				(30 000)	
Shortage				13 250	(½)
Contribution per limiting factor					
Mixing	190	150	210		(1)
Priority	**2**	**3**	**1**		(½)
P & F	380	300	70		(1)
Priority	**1**	**2**	**3**		(½)

Max contribution = 95*LD* + 37.5*SO* + 52.5*AP* ……(equation 1) (1) C
Mixing constrain = 0.5*LD* + 0.25*SO* + 0.25*AP* ……...(equation 2) (1) C
P& F constrain = 0.25*LD* + 0.125*SO* + 0.75*AP* ……...(equation 3) (1) C

The **linear programming method** should be used to solve the equations. (1)

Max 7

An overwhelming number of students battled with the concept of dealing with capacity constraints. The lack thereof resulted in many students failing to calculate the contribution per limiting factor correctly and hence making a correct decision. And instead they based their ranking on the respective product contribution per unit.

(d) **Incremental outsourcing**
If the calculation is done on a one year basis - <u>no marks</u>

	0	1	2	3	4	
	R'000	**R'000**	**R'000**	**R'000**	**R'000**	
Contract payments		(12 500)	(12 500)	(12 500)		(1) – all yrs
Salary saving		12 840	13 739	14 701		
Staff (480 x 25)		12 000	12 840	13 739		(1) – 25 R/W
Salary increase (12 000 x 7%)		840				(1) C 7% year 1appl.
(12 840 x 7%)			899			
(13 739 x 7%)				962		(1) 14 701 R/W
New mgr		(750)	(803)	(859)		
New mgr		(750)	(750)	(803)		(1)R/W
Salary increase (750 x 7%)			(53)			Year 2
(803 x 7%)				(56)		(1)R/W Year 3
ICT overhead (1 440 x 60% x 25/30)		720	720	720		(1)R/W
Rent (sunk)		-	-	-		(1)R/W
Subletting (upfront)		240	360	480		(1)R/W –corr yr
Pension top-up	(1 000)					(1)R/W
Sale of equipment	1 500					(1)R/W
Relevant cash flow	500	550	1 516	2 542		
Tax at 28%		793	(154)	(424)	(712)	(1) 28% C Appl. (1) next year C
Tax on year 0 at 28%						
Loss on equipment (5 000 x 2/3 – 1 500)	(1 833)					(1) loss R/W
Pension	<u>(1 000)</u>					(1) R/W
	(2 833)					
NCF's	500	1 343	1 362	2 118	(712)	(1) C
Discount at 8%	1,0	0,926	0,857	0,794	0,735	<u>(1)</u> 8%app R/W
					Max	<u>15</u>
NPV = + 4 069 000						(1) C

The company should implement the outsource project as it yields a positive NPV. (1)
[Alternative see next page]



Students evaluated the projects on a one year basis, and they could not identify, given the information (long term, etc.) in the scenario, the need to use the capital appraisal techniques.

(e) **Other issues**

- Contract price after 3 years, what then – negotiate further. (1)
- If above not secured, loss of ICT staff detrimental (1)
- Will there be an improvement in service (1)
- Performance measures for PCC must be quantified (1)
- PM's above must be in service contract (1)
- Will the pension top-up be tax deductible – negotiate with SARS if necessary (1)
- Reliability of PCC (1)
- Can the existing equipment be redeployed / used by PCC / or sold at better price (1)
- Will the 60% / 40% split for overhead cost remain when using PCC (1)

Max 5

[ALTERNATIVE SOLUTION] Total Approach

	0 R'000	1 R'000	2 R'000	3 R'000	4 R'000	
Not contracting						
Salary (30 x 480)		(14 400)	(15 408)	(16 486)		(½) – yr 1
Increases: @ (7% x 14 400)		(1 008)				(1) 7% year 1appl.
(7% x 15 408)			(1 079)			
(7% x 16 486)				(1 154)		(1)
Overhead @ 60% x 1 440		(864)	(864)	(864)		(½)
Rental – sunk		-	-	-		(1)
Cash flow		(16 272)	(17 351)	(18 504)		
Tax saving @28%		-	4 556	4 858	5 181	(1)@28%
		(16 272)	(12 795)	(13 646)	5 181	
Subcontract						
Sub-contract		(12 500)	(12 500)	(12 500)		(1)
Salary 5 x 480		(2 400)	(2 568)	(2 748)		(½)
Increases: @ (7% x 2 400)		(168)				
(7% x 2 568)			(180)			
(7% x 2 748)				(192)		
New employee		(750)	(750)	(803)		(1)
Increases: @ (7% x 750)			(53)			(1)Year 2
(7% x 803)				(56)		
Overhead @ 60% x 1 440 x 5/30 [Accept = 173]		(144)	(144)	(144)		(½)
Rental sublet		240	360	480		(1) upfront year 1
Sale of equipment	1 500					(1)
Pension pmt	(1 000)					(1)
Cash flows	500	(15 722)	(15 835)	(15 963)		
Tax at 28%		793	4 402	4 434	4 470	(1)next yr
Loss on equipment						(1)
(5 000 x 2/3 – 1 500)	(1 833)					(1)
Pension pmt	(1 000)					
	(2 833)					
Net CF	500	(14 929)	(11 433)	(11 529)	4 470	(1)C
Incremental change	500	1 343	1 362	2 117	(711)	(1)8% app
Discount @ 8%	1,0	0,926	0,857	0,794	0,735	

NPV = + 4 069 000 Max 15

General comments: Students in an overwhelming number struggled when it comes to discussion questions

QUESTION 17

P&S LTD **SUGGESTED SOLUTION**

(Source: Test 2 2013 MAC4861)

PART A

(a)	(i)	**Material price variance**	**F/(u)**	**F/(u)**		
		P: 1 100 000 (3,8 – 3,6)	(220 000)			
		O: 725 000 (2,30 – 2,2)	(72 500)			
		L: 830 000 (0,56 – 0,6)	33 200 F	(259 300)	(1½)	R/W
	(ii)	**Material quantity variance**				
		Issued: - P: 1 000 000 x 3,6 - O: 790 000 x 2,2 - L: 850 000 x 0,6	5 848 000		(1)	R/W
		Allowed: 210 000 x 28,10 (st cost)	5 901 000			
			53 000 F		(½)	
	(iii)	**Labour rate variance**				
		53 800 (21,50 – 21(w2))	26 900 U		(1)	R/W
	(iv)	**Labour idle time variance**				
		[(53 800 x 21) – (52 000(w1) x 21)]	37 800 U		(1)	R/W
		OR				
		[(53 800x96%) – 49 920] R21,875				
	(v)	**Labour efficiency**				
		(49 920 x 21.875) – (210 000 x 0,24 x 21,875)	10 500F		(1)	R/W
		OR				
		[52 000 – (0,25 x 210 000)] R21				
	(vi)	**Fixed overhead expenditure**				
		$(1\,689\,200 - \frac{10\,320\,000}{6})$	30 800 F		(1)	R/W
	(vii)	**Fixed overhead volume variance**				
		1 720 000 – (210 000 x 0,24 x 35)	44 000 F		(1)	R/W

(viii) **Container price variance**

(R5,10 – R510) x 212 000	0	(1)	R/W

(ix) **Container quantity variance**

(212 000 – 210 000) x 5,10	(10 200)	(1)	R/W

Workings:

1. 21,875 x 0,96 = R21

2. 49 920 ÷ 0,96 = 52 000

Many students have excelled in the calculation of material price variances, although some could not classify and show the correct calculations into Adverse or Favourable. A pocket of students used the actual units purchased in the calculation of Material usage variance, even though the units used in production were given. Some students calculated **unit variances**, which **are not acceptable** as the variances should relate to the costs in the management accounts and income statements. Variances are used in the reconciliation of budget profit to actual profit.

Most students struggled with the concept of idle time on labour efficiency, students were given the productive (work) rate and had to work backwards to get the clock rate first in order to carry out the calculations.

Other problems encountered included:

- Students not stating clearly whether the variance was favourable or unfavourable (adverse).
- Incorrect formula used and mathematical errors.
- Students doing various (random) calculations but not actually calculating the variances
- The distinction between actual and budgeted overhead and the level of the budget.
- Spending too much time on the variance calculation and not completing all sections of the test.

(b) **Mark-up percentage**

Sales price	R90,00		
Full cost	61,73		
Profit	28,27	(1)	R/W
Mark-up on 61,73	45,8%	(1)	R/W

(c) **Standard cost review**

- Variances can be reviewed quickly (investigated) (1)
- Corrections as result of changes in price/components or variances can be effected (1)
- Pricing can be considered once above done (1)
- Targets for sales/profit/returns can be adjusted accordingly (1)

Max 3

(d) **Break-even for SOAP division**

Fixed costs:

Factory overhead (60% x R1 385 000)	R 831 000	(½)	R/W
Sales and admin (2/3 x R2 400 000)	1 600 000	(½)	R/W
	R2 431 000		

	LD	SO	AP	Total		
Sales price	200	100	300			
Variables:						
Material	60	40	172,5			
Lab our	20	10	35			
Factory OH	10	5	17,5			
S&A	15	7,5	22,5			
Contribution	95	37,5	52,5		(3)	R/W
Demand	50 000	18 000	38 000			
Total contribution	4 750 000	675 000	1 995 000	7 420 000	(1)	R/W
Total sales ('000)	10 000	1 800	11 400	23 200	(1)	R/W
Contribution %				31,98%	(1)	C
			OR	32%		

BEP = $\frac{2\,431\,000}{0{,}32}$

= R7 596 875 (7 600 971) (1) C

8

(e) **Maximise contribution**

Contribution per unit

	LD R	SO R	AP R	
Selling price	200	100	300	
Direct material	60	40	172,5	
Direct labour	20	10	35	
Factory OH (40%)	10	5	17,5	
Selling and admin (33,3%)	15	7,5	22,5	
Contribution	95	37,5	52,5	(1½)

Constraints				**Total**	
Mixing-hours needed	25 000	4 500	9 500	3 900	(1½)
Available				(15 000)	
Shortage				24 000	(½)
P & F-hours needed	12 500	2 250	28 500	43 250	(1½)
Available				(30 000)	
				13 250	(½)
Contribution per limiting factor					
Mixing	190	150	210		(1) C
Priority	2	3	1		(½)C
P & F	380	300	70		(1) C
	1	2	3		(½)C
				Max	7

The linear programming method should be used to solve the two equations. (1)

8

(f)

Performance evaluation	**FP**	**Soap**	
Contribution margin	35 660 / 101 180	43 100 / 132 320	
	= 35,2%	= 32,6%1	(2)
Operating margin	26 100 / 101 180	29 600 / 132 320	
	= 25,8%	22,4%	(2)
ROI	26 100 / 145 900 – 2 030	29 600 / 158 750 – 17 340	
	= 18,1%	20,9%	(2)
RI	= 26 100 – 16% x 145 800) = 26 100 – 23 344 = 2 756	29 600 – (16% x 158 750) 29 600 – 25 400 = 4 200	(2)

- FP performs better ito margin, but Soap does better ito use of assets (1)
- Which measures are used for performance evaluation (1)
- Peer information to compare against (1)
- Comparison against prior years to determine change
- Valid comment ito performance

Max (1)

Max 10

MARKERS' Comments:

General:

Students in an overwhelming number struggled when it comes to discussion questions.

Part (a) Standard costing – variance calculations
Part (b) Mark-up percentage

Some students could not show an understanding of the mark-up concept, instead it was confused with the profit margin percentage (GP/Sales) calculation. Below is a formulae explanation of what mark-up entails:

- Cost x (Mark-up + 1) = Sale price
  - solved for Mark-up = (Sale price / Cost) – 1
  - solved for Mark-up = (Sale price − Cost) / Cost
  - MU = GP/Cost

Part (c) Standard costing
Students in the majority scored less than the allocated marks; this was due to the students not relating the benefits of reviewing every 6 months to the main uses of standard cost, which are:

- Cost control
- Inventory valuation
- Budgeting and planning
- Decision making purposes
- Provides a challenging target that individuals are motivated to achieve

Part (d) Break-even point
The most common error in the calculation of the break-even point for the company was that students tried to calculate the break-even point per product and not the whole division. This was predominantly done using the fixed costs per product and dividing by the contribution margin % (or contribution per unit) per product, instead of dividing the total fixed costs for the division by the contribution margin % for the whole division. A further error that a pocket of students did was showing a calculation of break-even point in units (FC/contribution(R)) and not in Rands as required.

Part (e) contribution maximisation
An overwhelming number of students battled with the concept of dealing with capacity constraints. The lack thereof resulted in many students failing to calculate the contribution per limiting factor correctly and hence making a correct decision. Instead they based their ranking on the respective product contribution per unit.

Part (f) divisional performance evaluation
Students generally performed well in this question. What many, of those who failed to calculate correctly, students did was to imply the text book definition of the capital or assets used to calculate the capital charge for RI (residual income) and not **as per the company policy (as stated in the scenario**). Where a company has a way of calculating a ratio and it has been given to you, remember **the company policy takes precedence**.

QUESTION 18

SOUTH LTD

(Source: Test 1 2012 MAC4861 and MAC4862)

R/W = right or wrong; C = consequential

SUGGESTED SOLUTION **40 marks**

Part (a) – MAC4861 and MAC4862: Calculate the direct labour variances for the 4-week period.

Remember that the labour variances (as well as material and variable overhead variances) should not be calculated using the original budget but rather the flexed budget (i.e. standard quantity based on actual production and not budgeted production).

i) Wage rate variance	= (SP – AP) x AH = (SP x AH) – (AP x AH)	
	= (R20 x **55** x **4** weeks x **40** hrs) – **R177 400**	(2) (½ each) R/W
	= R176 000 – R177 400	
	= R1 400 **Adverse**	(½) for Adverse C
	NB: Deduct 1 mark if "Favourable" or did not indicate Fav. or Adverse	

Alternative:	= (SP – AP) x AH = (R20 – (**R177 400** / **55** x **4** weeks x **40** hrs)) x (55 x 4 weeks x 40 hrs)	(2) (½ each) R/W
	= (R20 – R20,1591) x 8 800 hrs = R1 400 Adverse	(½) for Adverse C

ii) Labour efficiency	= (SH – AH) x SP	
SH (Standard hours)	= (**10 800** x **0,4** hrs) + (**8 200** x **0,56** hrs)	(2) (½ each) R/W
	= 4 320 + 4 592	
	= 8 912 hrs	
AH (Actual hours)	= 55 workers x 4 weeks x 40 hrs	
	= **8 800** hrs	(1) R/W
Variance	= (8 912 – 8 800) x R20	
	= 112 hrs x R20 [Or: R178 240 – R176 000]	
	= R2 240 **Favourable**	(½) for Fav. C
	NB: Deduct 1 mark if "Adverse / Unfav." or did not indicate Fav. or Adverse	

Part (b) - MAC4861 and MAC4862: Calculate the standard purchase price for material Bravo for Super Clean for the 4-week period.

Material price variance = (SP – AP) x AQ

= (AQ x SP) – (AQ x AP)

R4 176 F (given) = (**14 460** x SP) – **R26 190** (1½) (½ each) R/W

SP = $\frac{\text{R4 176 + R26 190}}{\text{14 460 litres}}$

= R30 366 ÷ 14 460 litres

= R2,10 per litre (½) R/W

Alternative:	= (SP – AP) x AQ	
R4 176 F (given)	= (SP – (R26 190 / 14 460) x 14 460	
R4 176 F	= (SP – 1,8112) x 14 460	
R4 176 F	= (SP x 14 460) – (1,8112 x 14 460)	
R4 176 F	= (SP x **14 460**) – **R26 190**	(1½) (½ each) R/W
SP	= $\frac{\text{R4 176 + R26 190}}{\text{14 460 litres}}$	
	= R2,10 per litre	(½) R/W

Part (c) - MAC4861 and MAC4862: Calculate the standard usage of material Bravo per litre of Super Clean produced.

Material usage variance = (SQ – AQ) x SP = (SQ x SP) – (AQ x SP)

C R/W (1) (½ each)

R3 227,70 A (given) = (**R2,10** x SQ) – (**10 285** x R2,10)

- R3 227,70 A = (R2,10 x SQ) – R21 598,50

R2,10 x SQ = R21 598,50 – R3 227,70

R2,10 x SQ = R18 370,80

SQ = R18 370,80 ÷ R2,10

= **8 748** litres (1) R/W

Note that SQ = Actual production (10 800 litres) x Standard usage

Therefore 10 800 x Standard usage = 8 748

Standard usage = 8 748 **÷10 800** (1) R/W

= 0,81 litre per litre of Super Clean

Part (d) - MAC4861 and MAC4862: Calculate the break-even point sales (Rand) for South Ltd based on the budgeted revenue mix.



The most common error in the calculation of the break-even point for the company was that students tried to calculate the break-even point *per product* and then add them together. This was mostly done using the fixed costs *per product* and dividing by the contribution margin % (or contribution per unit) *per product* instead of dividing the *total* fixed costs for the company by the contribution margin % for the whole company.

Remember that the *weighted average* contribution (and selling price) per unit must be weighted based on the **sales volume** (i.e. number of units sold, not produced) while the weighted average contribution margin % must be weighted based on the total sales revenue. Using the ratios of contributions *per unit* or selling prices *per unit* is incorrect because it does not take into account that the products are not sold in the same quantity. A weighted average does *not* mean merely adding the contributions per unit together or dividing by two.

		Super Clean	**Quick Clean**	**Total**	
1	Specific fixed costs per unit (given)	4,90	3,50		
2	General fixed costs per unit (given)	4,60	4,60		
3	Production & Sales volume (given)	143 000	88 400	231 400	
4	Total specific fixed costs (1 x 3)	**700 700**	**309 400**	1 010 100	(2) (1 each) R/W
5	Total general fixed costs (2 x 3)	**657 800**	**406 640**	1 064 440	(2) (1 each) R/W
6	Total fixed costs per unit (1 + 2)	**9,50**	**8,10**		(2) (1 each) R/W
7	Total fixed costs (6 x 3)	**1 358 500**	**716 040**	2 074 540	(2) (1 each) R/W
8	Unit Contribution	13,10	6,30		
9	Total Contribution (8 x 3)	**1 873 300**	**556 920**	2 430 220	(2) (1 each) C
10	Unit selling price (given)	29,20	13,50		
11	Total sales revenue (10 x 3)	**4 175 600**	**1 193 400**	5 369 000	(2) (1 each) R/W

- Contribution margin % = Total contribution / Total sales revenue
  = 2 430 220 / 5 369 000 **Or** R10,5022 / R23,2022 (See below) (1) C
  = 45,26%

- Break-even point (sales value) = Total fixed costs / Contribution margin %
  = (1 010 100 + 1 064 440) / 45,26% (1) C
  = 2 074 540 / 45,26%
  = R4 583 208,62

ALTERNATIVE 1:

- Fixed costs, contribution and sales revenue – Same as above.
- Weighted average contribution per unit = Total contribution / Sales volume
  = R2 430 220 / 231 400
  = R10,50 per litre

  OR = (R13,10 x 143 000 / 231 400) + (R6,30 x 88 400 /231 400)
  = (R13,10 x 61,8%) + (R6,30 x 38,2%)
  = R8,096 + R2,407
  = R10,50 per litre

Note: **Cannot** use the total contributions per unit (i.e. R13,10 + R6,30 = R19,40 or R19,40/2 = R9,70) or contribution per unit ratio (i.e. R13,10 / R19,40 & R6,30 / R19,40 = 67,5% & 32,5%)

- Break-even point (units) = Total fixed costs / Average contribution per unit
  = R2 074 540 / R10,50 (1) C
  = 197 532,96 = 197 533 litres

- Weighted average Selling price per unit = Total sales revenue / Sales volume
  = R5 369 000 / 231 400
  = R23,20

  OR = (R29,20 x 143 000 / 231 400) + (R13,50 x 88 400 / 231 400)
  = (R29,20 x 61,8%) + (R13,50 x 38,2%)
  = R18,045 + R5,157
  = R23,202 per litre

- Break-even point (sales value) = Break-even units x Average selling price
  = 197 533 x R23,202 (1) C
  = R4 583 161 (rounding difference to above)

  OR

  Super Clean = 197 533 x 143 000 / 231 400
  = 197 533 x 61,8%
  = 122 071 litres

  Quick Clean = 197 533 x 88 400 / 231 400
  = 197 533 x 38,2%
  = 75 462 litres

  Total sales value = (122 071 x R29,20) + (75 432 x R13,50)
  = R3 564 473 + R1 018 332
  = R4 582 805 (rounding difference to above)

The following less commonly used variations are also possible:

ALTERNATIVE 2:

- Fixed costs, contribution and sales revenue – Same as above.
- Weighted average contribution Margin

Contribution margins per product:

Super Clean = R13,10 / R29,20 OR R1 873 300 / R4 175 600
= 0,448630

Quick Clean = R6,30 / R13,50 OR R556 920 / R1 193 400
= 0,466666

Sales revenue ratio:

Super Clean = R4 175 600 / R5 369 000
= 0,777724

Quick Clean = R1 193 400 / R5 369 000
= 0,222276

Contribution margins weighted per sales revenue ratio =(0,448630 x 0,777724) + (0,466666 x 0,222276)
= 0,348910 + 0,103729
= 0,4526

Note: **Cannot** use the unit SP ratio (i.e. R29,20 / R42,70 & R13,50 / R42,70 = 68,4% & 31,6%)

- Break-even point (sales value) = Total fixed costs / Contribution margin %
= R2 074 540 / 45,26%
= R4 583 208,62

ALTERNATIVE 3:

- Fixed costs – Same as above.
- Products per batch 143 000 litres : 88 400 litres
1,6176 : 1
- Average contribution per batch = (R13,10 x 1,6176) + (R6,30 x 1)
= R21,19 + R6,30
= R27,49
- Break-even point (batches) = Total fixed costs / Average contribution per batch
= R2 074 540 / R27,49
= 75 462 batches
- Break-even litres per product

Super Clean = 75 462 x 1,6176 litres per batch
= 122 071 litres

Quick Clean = 75 462 x 1 litre per batch
= 75 462 litres

- Break-even sales value = (122 071 x R29,20) + (75 432 x R13,50)
= R3 564 473 + R1 018 332
= R4 582 805 (rounding difference to above)

Note: Calculating total company break-even sales (Rand) using total contribution, total sales and total contribution margin % is the quickest method. This saves time and reduces the risk of errors.

Part (e) - MAC4861 and MAC4862: Calculate the number of litres of Quick Clean at the break-even point determined in (d) above.

Do the calculation in total first and then apportion the litres to Quick Clean based on sales volume.

NB: Zero marks if worked with only one product's (Quick Clean's) fixed cost and contribution

- Weighted average contribution per unit = Total contribution / Sales volume

  = R2 430 220 / 231 400 (1) C

  = R10,50 per litre

Must apply weighting.

Must not use individual contribution or average contribution (i.e. ÷ 2)

- Break-even point (units) = Total fixed costs / Average contribution per unit

  = R2 074 540 / R10,50 (1) C

  = 197 532,96 = 197 533 litres

The planned sales mix for Quick Clean that was used to calculate the break-even point is

88 400 / 231 400. Therefore the number of units of Quick Clean at break-even point is:

88 400 / 231 400 x 197 533 = 75 462 litres

(1) C for apportioning litres to Quick Clean

ALTERNATIVE:

- Total break-even sales (Rand) as calculated above: R4 583 208,62
- Sales revenue ratio for Quick Clean as calculated in Alternative 2 above: 0,222276
- Quick Clean break-even sales value = R4 583 208,62 x 0,222276

  = R1 018 737,28
- Quick Clean break-even units = R1 018 737,28 / R13,50

  = 75 462 litres

Part (f) - MAC4861 only: Calculate the cost driver rate per activity pool using ABC.

	Material Purchase and Storage	Setting up	Production	Quality Control
Amount (Rands)	52 000	195 000	645 000	118 100
Cost Drivers				
Orders	125			
Set up hours		300		
Machine hours			3 029	
Inspections				29
Amount / Cost Drivers	416	650	212,94	4 072,41
	(1) R/W	(1) R/W	(1) C	(1) C

Litres of Super Clean	143 000
Litres of Quick Clean	88 400
Production-units Super Clean (143 000/200)	715 x 3 hours = 2 145 machine hours
Production-units Quick Clean (88 400/200)	442 x 2 hours = 884 machine hours
Total machine hours (2 145 + 884)	3 029 (1) R/W
Machine hours Super (715 x 3)	2 145
Inspections (2 145 /100)	21 (rounded down) (½) R/W
Machine hours Quick (442 x 2)	884
Inspections (884/100)	8 (rounded down) (½) R/W

MAX 6

Part (g) - MAC4861 only: Calculate the product specific overhead per litre, for Super Clean and Quick Clean using ABC

Marks were easily obtainable by using the **cost driver rate.** The concept of total overheads not changing, irrespective of what method is used were missed by a lot of students. The overhead per litre might differ, which was the whole concept of the question.

Super	**R**	
Material purchase and storage (416 x 100)	41 600	(½) R/W
Setting up of machines (650 x 200)	130 000	(½) R/W
Production (**212,94 (C)** x **2 145 R/W**) (½ each)	456 756	(1) C
Quality control (**4 072,41 (C)** x **21 R/W**) (½ each)	85 521	(1) C
Total	713 877	
Litres (Must use correct product quantity)	143 000	
Per Litre (713 877 / **143 000**)	4,99	(1) C

Quick		
Material purchase and storage (416 x 25)	10 400	(½) R/W
Setting up of machines (650 x 100)	65 000	(½) R/W
Production (**212,94 (C)** x **884 (R/W)**) (½ each)	188 239	(1) C
Quality control (**4 072,41 (C)** x **8**) (**R/W**) (½ each)	32 579	(1) C
Total	296 218	
Litres (Must use correct product quantity)	88 400	
Per Litre (296 218 / **88 400**)	3,35	(1) C

Max 7

Part (h) - MAC4861 only: Impact of ABC approach

There is a significant ***change*** in the product specific fixed overhead cost for both products. Super Clean experienced an ***increase*** **and Quick Clean on the other hand a** ***decrease*** **in costs**. (2) R/W

When the ABC method is used, Super Clean which makes more use of the **resources** is allocated more overheads compared to the current allocation. (1)R/W

ABC is a more accurate method for allocating overheads as it assigns costs more accurately. It will thus give the company a better insight on how each product is performing. (1) R/W

Max 3

Part (f) - MAC4862 only: Comment on the viability of Quick Clean.

Students focused calculations on a *per unit* approach instead of looking at the total Rand values involved which is more appropriate for management to determine the impact of a product on the company. Remember to consider Quick Clean's contribution to specific fixed costs and general fixed costs separately.

At the forecast sales volume the profit and contribution for Quick Clean are as follows:

	R		**OR**	**Per Unit R**	
Contribution to all fixed costs (per (d))	556 920			6,30	
Less specific fixed costs	(309 400)			(3,50)	
Contribution to general fixed costs	247 520	(1) C		2,80	
Less share of general fixed costs	(406 640)			(4,60)	
Net Loss	(159 120)	(1) C		(1,80)	(1) C

- Quick Clean provides a contribution of R247 520 towards general fixed costs and, unless savings in general fixed costs in excess of R247 520 can be made if Quick Clean is abandoned, it is still viable to produce Quick Clean. (2)
- The company should investigate whether a greater contribution than R247 520 can be generated from the resources. If this is not possible the company should continue production of Quick Clean. (2)
- If the company ceases production of Quick Clean the net loss for Quick Clean's unused capacity will be R406 640, an increase of R247 520 from the current R159 120. It will lose a contribution R247 520 and total profits will decline by R247 520. (2)

Max 6

Alternative: If students misinterpreted the required to mean that production of Super Clean will be discontinued and that a discussion of the viability of Quick Clean as the sole product was required:

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At the forecast sales volume the profit and contribution for Quick Clean are as follows:

	R		OR	Per Unit R	
Contribution to all fixed costs (per (d))	556 920			6,30	
Less specific fixed costs	(309 400)			(3,50)	
Contribution to general fixed costs	247 520	(1) C		2,80	
Less **all** general fixed costs	(1 064 640)			(12,04)	
Net Loss	(817 120)	(1) C		(9,24)	(1) C

- Quick Clean provides a contribution of R247 520 towards general fixed costs which is insufficient. (1)
- The break-even point with Quick Clean as the sole product of the company will increase to 218 070 litres ((R309 400 + R657 800 + R406 640) / R6,30). The company should investigate whether this is achievable. (1)
- The company should investigate whether a contribution to general fixed costs of R817 120 can be generated from the spare capacity. If this is not possible the company should investigate reducing the capacity in order to reduce costs.

Max (6)

Part (g) - MAC4862 only: Calculate the minimum selling price **per litre** of special cleaning liquid that South will have to ask to break even on the once-off order. **Briefly comment** on the choice of relevant costs.

Students could easily earn several marks in part (g).

Please note:

- Antibact on hand resulted in an opportunity *saving* which had to be deducted from the costs
- There was NOT an opportunity cost due to lost sales
- The Antibact on hand was not sufficient for the order and additional purchases were required
- The main problem encountered by students doing the calculations on a per unit basis was that they did not take into account that the Antibact purchased was only 1 000 litres. They therefore included R4,70 in the cost instead of R0,235 (i.e. R4 700 / 20 000).

		R	R / unit	
Variable costs (excluding Antibact)	20 000 @ R5,40	108 000	5,40	(1) R/W
Antibact – On hand (Calc. 1)	4 000 @ (R0,50)	(2 000)	(0,10)	(1) R/W
Antibact – Puchase (Calc. 1)	1 000 @ R4,70	4 700	0,235	(1) R/W
Fixed costs:				
Product specific	20 000 @ 3,90	78 000	3,90	(1) R/W
General		–		
Temporary staff member		15 500	0,775	(1) R/W
Total costs		204 200	10,21	
Minimum price per litre	204 200 ÷ **20 000**	R10,21		(1) C

Calc. 1: Antibact

Special order needs 20 000 x 0,25 = **5 000** litres Antibact (1) R/W

4 000 litres on hand, therefore purchase 5 000 – 4 000 = **1 000** litres Antibact (1) R/W

Alternative:

4 000 litres on hand is enough for 4 000 ÷ 0,25 = **16 000** litres finished product (1) R/W

Purchase additional Antibact for 20 000 – 16 000 = 4 000 litres finished product

Therefore purchase 4 000 x 0,25 = **1 000** litres Antibact (1) R/W

Max for calculations (7)

Comments:

NB: No mark for comment that **variable costs** are relevant.
No mark for general comment that cost of Antibact is relevant. Must specify which cost relating to Antibact is relevant or irrelevant.

- The **original purchase price** (book value) of Antibact is a **past or sunk cost** and is therefore irrelevant. (1)
- The **cost of disposal** of Antibact is an **opportunity saving** and therefore **relevant**. (1)
- There is not sufficient stock of Antibact on hand for the whole order therefore **additional litres** of Antibact must be **purchased** for which the **future purchase price** is the relevant cost. (1)
- The **general fixed costs** are **not relevant** since the special order will not impact these costs. They are not likely to change in the short term. (1)
- The **specific fixed costs** are **relevant** to the special order. (1)
- The **hiring of a temporary staff member** is as a result of the order, therefore **relevant**. (1)

Max for comments (3)
Max (10)

QUESTION 19

MR JOHNSON **SUGGESTED SOLUTION**

(Source: Test 2 2012 MAC4861)

PART A

(a) **The relevant costs of the contract are as follows:**

	R'000	R'000	
Salary of Ms Q Aplon		80	(1)
Supervision cost[a]		400	(1)
Cost of artisans[a]		640	(1)
Cost of equipment[b]		120	(½)
Material costs: [c]			
B (10 x R1 200) + (90 x R1 200)	120		(1)
L (100 x R360)	36		(½)
U (10 x R2 400)	24		(½)
E (10 x R1 200) + (10 x R1 200)	24		(1)
O (500 x R80)	40		(½)
D (100 x R800) + (200 x R800)	240	484	(1)
Other direct expenses		260	(1)
Owner's opportunity cost[d]		113,6	(2)
		2 097,6	
Less Savings on maintenance work[e]		(60)	(2)
Minimum contract price		**2 037,6**	

OR

Notes

Bookkeeping cost is incidental to the contract. (1)

[a] The costs given in the question **include** apportioned fixed overheads which are not a relevant cost. (1)
Therefore **R40 000** should be deducted from the supervision cost (10/110% x R440 000 = R40 000) and **R32 000** (10/110* R352 000 = R32 000) from each of the artisan's costs. (2)

[b] The **historical cost** of the equipment is a sunk cost (not relevant). (1)
Since the existing equipment is useful to Johnson only for contract work, it is assumed that the equipment would have been sold if the contract were not accepted. (1)

Therefore the relevant cost of using the equipment is the **scrap value at the end of the contract term R120 000.** (1)

[c] Material B: It is assumed that the usage of the 10 units in stock will be **replaced** in the coming year **at R1 200** and be **used on property maintenance.** (2)
The remaining quantities will be replaced at the **current purchase price of R1 200.** (1)

Material L: This material is used **regularly in the business**. (1)
The 100 units issued from stock for the contract will be **replaced at R360 per unit**. (1)

Material U: This material is **purchased specially for the contract**. (1)

Material E: 10 Units will need to be purchased. The 10 in stock will have no value. (2)
OR
The stocks of material E have no alternative use within the business and can neither be used on the contract nor sold as these are **obsolete.** Hence, the current purchase price **(R1200)** is the relevant cost as all stocks required will be obtained at this amount.

Material O: It is assumed that the material taken from stock for this contract will be **replaced at the current purchase price**. This material is used **regularly in the business.** (1)

Material D: 200 Units will need to be purchased. 100 units are in stock and have no cost effect. (1)
OR
Similarly to material E, material D cannot be used on the contract or sold due to its obsolescence to and therefore, the current purchase price relevant cost for the stocks required (**R800**).
Note – (**a pleasure cruiser is a luxury yacht that would exude exquisite quality**)

Other direct expenses are specific to the contract. (1)

d Johnson will incur expenses of R486 400 to maintain the existing business while he earns R600 000 on the one year appointment. (1)
The opportunity cost of accepting the contract is therefore R113 600. (1)

e It is assumed that the contract could be completed and the maintenance programme carried out during the period in which the supervisor and artisans are employed (one year). (1)
It is also assumed that the supervision and artisans will be employed for one year only (term of the contract). (1)

The building lease is a fixed cost and therefore irrelevant. (1)

A further assumption is that the lowest quotation will be accepted. The net effect will be a saving of R60 000 (saving on R100 000 vs R40 000 cost of material used in maintenance) (1)
Conclusion: **Price incorrect**, should be **R2 037 600** (2)

Memorandum format (1)

Maximum 25

(b) **Mark-up percentage**

A process of setting a mark-up percentage has to correspond with established custom, competitive position and likely product demand. (1)
A firm can command a **higher mark-up** for a product that has a **higher demand** as **mark-ups are related to the demand for a product.** (2)
Luxury goods (such as the Blue Blood to be built) with a low sales turnover may **attract high profit margins** whereas **non-luxury goods** with a high sales turnover (such as commodities) **may attract low profit margins.** (2)

Maximum 3

Limitations of cost-plus pricing

Cost-plus pricing **ignores demand** as the mark-up added may bear **no relationship to the price-demand relationship.** (2)
Calculation effect (1)
Cost-plus pricing will only ensure that costs will be met, and the target profits earned, if the sales volume is equal to, or more than, the activity level that was used to estimate total unit costs. (1)

When **sales demand falls below the budgeted activity level** (level used to set the fixed cost recovery rate), the total sales revenue may be **insufficient to cover the total fixed costs.** (2)

Maximum	3
Total	6
Total Part A	31

PART B

(c) **Machining capacity**

If 1 extra machine hour is obtained, the constraints **M + 2R = 700 and 2,5M + 2R = 1000** will still be binding and the new optimal plan can be solved as follows:

M + 2R = 701 (revised machining constraint)
2,5M + 2R = 1 000 (unchanged assembly constraint)

M = 199,33,therefore Mac is reduced by 0,67units (200 – 199,33) (1)
R = 250,83, Ruz is increased by 0,83 units (250 – 250,83) (1)

This will change contribution as follows:

Increase in contribution from Ruz (0,83 * R100)	R83,00	(1)
Decrease in contribution from Mac (0,67 * R80)	(R53,60)	(1)
Net increase in contribution	**R29,40**	
Therefore, the value of of an independent marginal increase in machine capacity is R29,40	R29,40	(1)**C**

Assembly capacity

With an extra hour in assembly capacity, the new optimal plan can be solved as follows:

M + 2R = 700 (unchanged machining constraint)
2,5M + 2R = 1 001 (revised assembly constraint)

M = 200,67, therefore Mac is increased by 0,67 units (200-200,67)		(1)
R = 249,67, Ruz is decreased by 0,33 units (250-249,67)		(1)
This will change contribution as follows:		
Increase in contribution from Mac (0,67 * R80)	R53,60	(1)
Decrease in contribution from Ruz (0,33 * R100)	(R33,00)	(1)
Net increase in contribution	**R20,60**	
Therefore, the value of an independent marginal increase in assembly capacity is R20,60	R20,60	(1)**C**
	Maximum	6

(d) **Principal assumptions underlying linear programming**

(i)	The objective of the firm is to maximise short-term contribution.	(1)
(ii)	Divisibility of products (products can be produced in fractions of units).	(1)
(iii)	Divisibility of resources (supplies of resources may only be available in specified multiples).	(2)
(iv)	Linearity over the whole output range for costs, revenues and quantity of resources used.	(1)
(v)	All of the available opportunities for the use of the resources have been included in the linear programming.	(1)
	Maximum	3
	Total	9
	TOTAL	**40**

MARKERS' COMMENTS:

General

Test 2 covered cost accounting topics. The concept of relevant costing, pricing and linear programming as well as the assumptions were tested. Approximately 2 600 students wrote the test with an average mark achieved of 23%. This indicates that students have not allocated sufficient time to do examples for management accounting, as the theory as well as a practical application of theory was required, which could only be achieved after much study and attempting many questions and analysing the suggested solution.

PART A (Marks 31)

(a): Critical Review of appropriateness and accuracy of Contract Price

Most students focused on the calculations and so lost 50% of the available marks as a result of not critically discussing the calculation of Junior Mercury.

The key definition of a relevant cost was clearly missed as well as the application of the definition to the costs incurred by Mr Johnson.

Basic mathematical calculations proved too difficult. An example of one was the Supervisor cost of R440 000. As this cost included an apportionment of fixed overheads of 10% of labour costs the calculation should be 10/110*R440 000=R40 000 which must be deducted from R440 000 to give a cost of R400 000. Most students did the calculation as R440 000*90%=R396 000.

The definition clearly refers to cash flows but many students included depreciation as a cost.

Many errors made were as a result of not reading the question with understanding. Material E and D were obsolete which means they were to be replaced at current market prices while many students calculated a cost of the 10 units in stock at no cost. Reading the question for interpretation is vital and is the reason many student fail to score marks.

(b): Basis for a Mark-Up Percentage; Limitations of Cost-Plus Pricing

The concept of supply and demand was completely overlooked by almost all students.

The consequential effect on margins of luxury versus non-luxury goods was also not identified.

PART B (Marks 9)

(c): Calculation of an independent Marginal Increase in available capacity: Machining & Assembly

Many students did not even attempt this section which was part of the reason for the low pass percentage rate. The basic constraint equation was problematic as well as the basic calculation of the contribution of R80 for Mac and R100 for Ruz.

(d): Principal assumptions underlying the Linear Programming calculation

Again many students did not even attempt this section which again was part of the reason for the low pass percentage rate. A 100% theory which students were either familiar with or not.

QUESTION 20

KNYSNA SPECIALIST SUPPLIERS **SUGGESTED SOLUTION**

(Source: Test 3 2012 MAC4862)

(a) **Segmental performance**

	Total	EC R'000	Shipping R'000	SC R'000		
Turnover	10 816,0	3 276,0	2 340,0	5 200,0		
VC (apply % of variable production, marketing + distribution costs)		1 638,0	1 240,2	3 224,0		
	4 713,8	1 638,0	1 099,8	1 976,0	(1½)	R/W
FC	2 587,0	1 024,4	624,0	938,6		
Profit	2 126,8	613,6	475,8	1 037,4	(1½)	R/W
WACC charge		360,0	310,0	550,0	(1)	C
(ii) RI	906,8	253,6	165,8	487,4	(1½)	R/W
(i) % PM on sales (613,6 ÷ 3 276,etc)		18,73	20,33	19,95	(1½)	R/W
Assets held						
Working capital (T/O ÷ times)(3 276 ÷ 3, etc)		1 092,0	1 170,0	1 300,0	(1½)	R/W
NCA		708,0	380,0	1 450,0	(1)	R/W
		1 800,0	1 550,0	2 750,0		
WACC charge						
@ 20%		360,0	310,0	550,0	(1½)	C
∴ *Shipping* generates the highest segmental profit margin and					(1)	
Southern Cape generates the highest residual income					(1)	
					13	

(b) **Breakeven for KSS**

Segmental contribution per (a)		4 317,8	(½)	C
Central contribution (284 – 45,8 – 36)		202,2	(½)	R/W
		4 520,0		
Total sales (10 816 + 284)		11 100,0	(1)	R/W
Contribution %		40,72%	(1)	C
Fixed costs = 2 587,0 + 330,0 + 514,5 + 210,0			(2)	R/W
	=	3 641,5		
BE	=	3 641,5		
		,4072		
	= R8 942 780		(1)	C
			6	

(c) **Transfer price policy**

- At cost there is *no compensation* for the segment supplying the products. (1)
- This (transfer) will *benefit the inefficient* segment not planning properly and *disadvantage the efficient*. (2)
- This may lead to consistent *poor planning* as there is a back-up resource. (1)
- It may also lead to *discontent* amongst the segmental managers. (1)

Max 3

(d) **Business risk**

- Availability of products
  - Fresh produce may not be available when required by clients. (1)
  - Specialised meat products need to be ordered in advance – not available on ad hoc basis. (1)
- Impact of *economy on tourists/locals* who may *visit restaurants* and its impact on business/locals who may *use catering facilities*. (3)
- Management of working capital
  - Part of working capital (fresh produce inventory) subject to fast deterioration. (1)
  - Restaurant/catering businesses notorious for failure – how quickly is *debt collected*, who decides on accepting a new client in respect of *credit criteria*. (2)
- Decisions taken by operations managers in respect of new business margins may be contradictory to those based on residual income. (1)
- Any other valid comment – max 1. (1)

Max 6

Relevant to industry/scenario

(e) **Tax liability**

	R		
Total contribution per (b)	4 520,0	(½)	C
Total fixed costs per (b)	(3 641,5)	(½)	C
Profit for the year	878,5		
Bonus liability / expense in respect of 2012		(2)	R/W
25% x (253,6 + 165,8 + 487,4 – 100,8)	(201,5)	(1)	Or C if calc
Taxable income	677,0		
Tax at 28%	189,56	(½)	C
Paid provisionally	(150,00)	(½)	R/W
Tax liability in respect of 30/04/12	(39,56)	(1)	R/W
	Max	5	

(f) **Discontinuation decision**

Relevancy issues:	**R**	
Current cash loss (30/04/12)(62 200 – 35 300)	26 900	(1)
Closure cash flows:		
Sale of equipment	40 000	(1)
Equipment profit – no tax: loss for year	-	(1)
Lease contract R90 000 x 25%	(22 500)	(1)
Retrenchment Employee costs 152 000 + 63 000 – 39 060 = 175 940		(1)
÷ 52 x [(3 x 2) + 2]	(27 068)	(1)
Other closure costs	(3 700)	(1)
	(13 268)	

Other factors to consider

- Time saved by Mr Te Water (R36 000 in respect of 10%) may be used more productively. (1)
- Avoidance of major player threat in future (not quantifiable). (1)
- Current operational costs exceed costs of discontinuance. (1)

∴ Mr Te Water should close this business. (1)

Max 7

QUESTION 21

ROYAL JAMS (PTY) LTD **SUGGESTED SOLUTION**

(Source: Test 1 2011 TOE408W (MAC4862)

Part (a): Prepare the **actual** statement of comprehensive income for the year ended 28 February 2011 using machine hours as the basis to allocate overheads to inventory.

Actual	**Apricot** Kg	**Marmalade** Kg	**Total** Kg	
Opening inventory	15 000	10 000	25 000	
Produced	445 000	487 000	932 000	
Sold	(442 000)	(486 000)	(928 000)	
Closing inventory	18 000	11 000	29 000	(1) R/W

Budgeted fixed overhead rate

R 8 460 000
(67 500 hrs + 144 000 hrs)

= R 8 460 000
211 500 hrs

= R40/hour (1) R/W

Budgeted machine hours per kg

Apricot

67 500 hours
450 000 kg = 0,15 hrs/kg (1) R/W

Marmalade

144 000 hours
480 000 kg = 0,3 hrs/kg (1) R/W

Alternative: (Since budgeted machine hours per kilogram = actual machine hours per kilogram, we could have used actual instead of budget.)

Actual machine hours per kg

Apricot

66 750 hours
445 000 kg = 0,15 hrs/kg (1) R/W

Marmalade

146 100 hours
487 000 kg = 0,3 hrs/kg (1) R/W

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Statement of comprehensive income for the year ended 28 February 2011

	Apricot R	Marmalade R	Total R	
Turnover			20 393 000	
A: R7 480 000/440 000kg x 442 000kg = R17,00 x 442 000kg	7 514 000			(1) R/W
M: R12 852 500/485 000kg x 486 000kg = R26,50 x 486 000kg		12 879 000		(1) R/W
Less: Cost of sales	(6 559 490)	(9 862 800)	(16 422 290)	
Opening inventory: given	(210 000)	(200 000)	(410 000)	(1) R/W
Materials			(6 570 250)	
A: R3 541 500/450 000kg x 445 000kg = R7,87 x 445 000kg	(3 502 150)			(1) R/W
M: R3 024 000/480 000kg x 487 000kg = R6,30 x 487 000kg		(3 068 100)		(1) R/W
Direct labour			(1 419 000)	
A: R450 000/450 000kg x 445 000kg = R1,00 x 445 000kg	(445 000)			(1) R/W
M: R960 000/480 000kg x 487 000kg = R2,00 x 487 000kg		(974 000)		(1) R/W
Fixed overheads			(8 514 000)	
A: R40/hr x 0,15 hr/kg x 445 000kg = R6,00 x 445 000kg (Alt: R40/hr x 66 750 actual hrs) *	(2 670 000)			(1) C
M: R40/hr x 0,3 hr/kg x 487 000kg = R12,00 x 487 000kg (Alt: R40/hr x 146 100 actual hrs) *		(5 844 000)		(1) C
Note: Must use actual kg of 445 000 & 487 000				
Closing inventory			490 960	
A: (R3 502 150+445 000+2 670 000) / 445 000kg x 18 000kg	267 660			(1) C
M: (R3 068 100+974 000+5 844 000) / 487 000kg x 11 000kg		223 300		(1) C
Note: Must include all 3 components (material, labour and overheads). Must divide by actual production kg (445 000 & 487 000). Must multiply by closing inventory kg (18 000 & 11 000).				
	954 510	3 016 200	3 970 710	

Over recovery of fixed manufacturing overheads = R8 514 000 – R8 460 000	54 000	(1) C
Note: 1 mark if said over recovery but amount wrong		
Expenditure variance = R8 560 000 – R8 460 000	(100 000)	(1) R/W
Gross profit	3 924 710	
Less: Selling & distribution costs = (R20 393 000 x 1%) + R820 000 = R203 930 + R820 000	(1 023 930)	(1) C
Less: Other administrative costs (R550 000 x 1,05)	(577 500)	(1) R/W
Profit before interest and tax	2 323 280	

* Alternative fixed overheads: The actual hours could have been used since budgeted machine hours per kilogram = actual machine hours per kilogram (given in question).

Alternative closing inventory:	**R**	**R**	
Materials			
A: R3 541 500/450 000kg	7,87		
M: R3 024 000/480 000kg		6,30	
Direct labour			
A: R450 000/450 000kg	1,00		
M: R960 000/480 000kg		2,00	
Fixed overheads			
A: R40/hr x 0,15 hr/kg	6,00		
M: R40/hr x 0,3 hr/kg		12,00	
Cost per kg	14,87	20,30	
Closing inventory: A: R14,87 x 18 000kg	R267 660		(1) C
Closing inventory: M: R20,30 x 11 000kg		R223 300	(1) C
Note: Must multiply by closing inventory kg (18 000 & 11 000).			

Total (19)

Part (b): Calculate the total **actual** costs attributable to each type of jam using activity based costing (ABC) and the available information.

Activity	**Apricot**	**Marmalade**	**Total**	
No. of batches (NB: Budget kg):			372 batches	
A: **450 000**kg / 2 500kg per batch	180 batches			(½) R/W
M: **480 000**kg / 2 500kg per batch		192 batches		(½) R/W
Soaking:			3 075 hrs	
A: 180 batches x 3,75 hrs/batch	675 hrs			(½) C
M: 192 batches x 12,5 hrs/batch		2 400 hrs		(½) C
Boiling:			1 470 hrs	
A: 450 000kg / 100kg x 0,06 hrs/100kg	270 hrs			(½) C
M: 480 000kg / 100kg x 0,25 hrs/100kg		1 200 hrs		(½) C
Packaging:			744 hrs	
A: 180 batches x 2,0 hrs/batch	360 hrs			(½) C
M: 192 batches x 2,0 hrs/batch		384 hrs		(½) C
Activity	**Apricot R**	**Marmalade R**	**Total R**	
Soaking:			1 845 000	
A: 675 hrs / 3 075 hrs x R1 845 000	405 000			(½) C
M: 2 400 hrs / 3 075 hrs x R1 845 000		1 440 000		(½) C
Boiling:			5 320 000	
A: 270 hrs / 1 470 hrs x **R5 320 000**	977 143			(½) C
M: 1 200 hrs / 1 470 hrs x **R5 320 000**		4 342 857		(½) C
Note: Higher than budget cost (R100 000) is due to boiler, therefore added to boiling cost. Must use actual cost, not budget.				
Packaging:			1 395 000	
A: 360 hrs / 744 hrs x R1 395 000	675 000			(½) C
M: 384 hrs / 744 hrs x R1 395 000		720 000		(½) C
Total	2 057 143	6 502 857	8 560 000	



Note: Budgeted production kilograms should be used to calculate the number of batches, not actual kilograms. If actual kilograms were used to calculate the number of batches and all the subsequent driver calculations were done correctly as presented below, 2 marks can be awarded.
Bonus 2 marks can be awarded if student identified that the difference between budget and actual kilograms produced will result in a variance that must be shown separately.

Activity	Apricot	Marmalade	Total	
No. of batches (Actual kg): A: **445 000**kg / 2 500kg per batch M: **487 000**kg / 2 500kg per batch	178 batches	195 batches	373 batches	
Soaking: A: 178 batches x 3,75 hrs/batch M: 195 batches x 12,5 hrs/batch	667,5 hrs	2 437,5 hrs	3105 hrs	
Boiling: A: 445 000kg / 100kg x 0,06 hrs/100kg M: 487 000kg / 100kg x 0,25 hrs/100kg	267 hrs	1 217,5 hrs	1 484,5 hrs	
Packaging: A: 178 batches x 2,0 hrs/batch M: 195 batches x 2,0 hrs/batch	356 hrs	390 hrs	746 hrs	

Activity	Apricot R	Marmalade R	Total R	
Soaking: A: 667,5 hrs / 3 105 hrs x R1 845 000 M: 2 437,5 hrs / 3 105 hrs x R1 845 000	396 630	1 448 370	1 845 000	
Boiling: A: 267 hrs / 1 484,5 hrs x **R5 320 000** M: 1 217,5 hrs / 1 484,5 hrs x **R5 320 000** Note: Higher than budget cost (R100 000) is due to boiler, therefore added to boiling cost. Must use actual cost, not budget.	956 847	4 363 153	5 320 000	
Packaging: A: 356 hrs / 746 hrs x R1 395 000 M: 390 hrs / 746 hrs x R1 395 000	665 710	729 290	1 395 000	
Total	2 019 187	6 540 813	8 560 000	

Total (7)

Part (c): Calculate the budgeted break-even jars (units) for the year ended 28 February 2011.

Fixed costs	**R**	
Fixed manufacturing overheads	8 460 000	(½) R/W
Fixed selling & distribution costs: R1 053 325 – (R20 332 500 x 1%)	850 000	(1) R/W
Other administrative costs	550 000	(½) R/W
Total fixed costs	9 860 000	

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Contribution	**Apricot R**	**Marmalade R**	
Selling price			
A: R7 480 000 / 440 000kg	17,00		(½) R/W
M: R12 852 500 / 485 000kg		26,50	(½) R/W
Less: Variable costs			
Materials			
A: R3 541 500 / 450 000kg	(7,87)		(½) R/W
M: R3 024 000 / 480 000kg		(6,30)	(½) R/W
Direct labour			
A: R450 000 / 450 000kg	(1,00)		(½) R/W
M: R960 000 / 480 000kg		(2,00)	(½) R/W
Variable selling & distribution costs			
A: 1% x R17,00	(0,17)		(½) R/W
M: 1% x R26,50		(0,27)	(½) R/W
Contribution per kg	7,96	17,93	

Weighted contribution based on revenue mix:

Note: Must use budgeted kg, not actual kg or rand value.

= ((440 000 / 925 000) x R7,96) + ((485 000 / 925 000) x R17,93) (1) C
= (47,568% x R7,96) + (52,432% x R17,93)
= R3,79 + R9,40
= R13,19 per kg

Alternative:
= ((440 000 x R7,96) + (485 000 x R17,93)) / 925 000 (1) C
= (R3 502 400 + R8 696 050) / 925 000
= R12 198 450 / 925 000
= R13,19 per kg

Break-even number of kilograms:

= Fixed costs / Weighted contribution per kg
= R9 860 000 / R13,19 per kg (1) C
= 747 536,01 kg

Jars:
= 747 536,01 / 0,5kg
= 1 495 072,02 jars (½) C
= 1 495 073 jars (round-up) (½) C

Max (8)

Part (d)(i): Discuss what impact the reduced production in the coming year and the costs of the overhaul will have on the fixed manufacturing overhead rate.

- The allocation of fixed production overheads to the costs of conversion is based on the **normal capacity** of the production facilities. Normal capacity is the production **expected to be achieved on average over a number of periods or seasons** under normal circumstances, **taking into account the loss of capacity resulting from planned maintenance.** (IAS 2) (2)
- The **average production expected over a five year period** (including a year with lower production due to the overhaul of the boiler) should be used to calculate the overhead rate. (1)
- Normal capacity will be lower than the budgeted production for the 2011 year (lower production and machine hours) and will result in a **higher overhead rate**. (1)
- The cost of the overhaul should be **recognised in the carrying amount** of the vacuum boiler when that cost is incurred if the recognition criteria are met. (IAS 16) (1)
- The costs capitalised will be **depreciated** over five years and will therefore **increase the fixed manufacturing overheads and the overhead rate**. (1)

Max (4)

Part (d)(ii): Discuss what impact this will have on the fixed manufacturing overheads (Rand) and rate in the flexible budget for the year ended 29 February 2012.

- The flexed fixed cost **does not change** from the budget (i.e. remains the same). (1)
- The overhead rate in the flexed budget will be calculated based on the following:
    - **Budgeted fixed costs** (unchanged from original budget), and (1)
    - **Normal capacity** (instead of 2012 budget). (1)

Max (2)

QUESTION 22

SA HEATING COMPANY (PTY) LTD **SUGGESTED SOLUTION**

(Source: Test 2 2010 DIPAC26 (MAC4861))

R/W – Right/Wrong mark C – Consequential mark

(a) Calculate the budgeted breakeven point in Rand

Fixed costs	SH R'000	CHS R'000	Total R'000	
Labour: (120 000 x R57) + (180 x R10 500)	6 840	1 890	8 730	(1) R/W (or ½ each)
FOH: (120 000 x R19) + (180 x R10 500)	2 280	1 890	4 170	(1) R/W (or ½ each)
Maintenance			3 000	(½) R/W
Administrative			4 000	
Finance cost			1 050	(½) R/W
			20 950	

It is good practice to list each fixed cost item separately – the examiner can then easily follow the process.

Contribution	SH R'000	CHS R'000	Total R'000	
Sales (120 000 x R240); (180 x R100 000)	28 800	18 000	46 800	(1) R/W (or ½ each)
Less:				
Materials(120 000 x R95); (180 x R42 000)	(11 400)	(7 560)	(18 960)	(1) R/W (or ½ each)
VOH (120 000 x R19); (180 x R7 000)	(2 280)	(1 260)	(3 540)	(1) R/W (or ½ each)
Commission (5% x R28 800); (7% x R18 000)	(1 440)	(1 260)	(2 700)	(1) R/W (or ½ each)
Product contribution	13 680	7 920	21 600	

Alternative: Could have calculated contribution per unit.

Sales quantity	120 000	180		
	R	R		
Sales price	240	100 000		
Materials	(95)	(42 000)		(1) R/W (or ½ each)
VOH	(19)	(7 000)		(1) R/W (or ½ each)
Commission (5% x R240); (7% x R100 000)	(12)	(7 000)		(1) R/W (or ½ each)
Contribution per unit	114	44 000		
	R'000	**R'000**	**R'000**	
Sales (120 000 x R240); (180 x R100 000)	28 800	18 000	46 800	(1) R/W (or ½ each)
Product contribution * (120 000 x R114); (180 x R44 000)	13 680	7 920	21 600	

Product contribution	21 600	
Finance costs (R1 590 000 – R1 050 000)	(540)	(1) R/W
Company contribution	21 060	
Contribution % (R21 060 / R46 800 x 100/1)	45%	(1) C

The required was for the company's breakeven in Rand (= total sales). The individual product numbers were not required. Refer below

Breakeven			20 950 000		
Note: Denominator must be contribution %			0,45		
		=	R46 555 556		
	or	=	R46 556 000	(1)	C
			Max	9	

(b) Allocate maintenance and administrative costs using the step method or repeated distribution method

Step method, also known as the specified order of closing method or sequential method
(Refer to Drury, pg 70 – 71)

More administrative costs are absorbed by the maintenance section than vice versa, this indicates the first (allocation) step.

	SH	**CHS**	**Main-tenance**	**Admin**	
Annual cost				4 000 000	
Allocated (50%; 40%; 10%)	2 000 000	1 600 000	400 000	(4 000 000)	(2) R/W
Annual cost			3 000 000	-	
			3 400 000		
Allocated	1 700 000	1 700 000	(3 400 000)	49/49 'equal' to 50/50	(2) R/W
Note: Must use R3.4m and not allocate anything to Admin					
	3 700 000	3 300 000	-		
Units manufactured	120 000	180			
Maintenance and admin cost per unit	30,83	18 333,33			(1) C
					5

OR

Repeated distribution method
(Refer to Drury, pg 68 – 69)

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	SH	CHS	Main-tenance	Admin	
Annual costs			3 000 000	4 000 000	
Allocate Admin (50%; 40%; 10%)	2 000 000	1 600 000	400 000	(4 000 000)	(2) R/W
			3 400 000	-	
Allocate Maintenance (49%; 49%; 2%)	1 666 000	1 666 000	(3 400 000)	68 000	(2) R/W
Note: Must use R3.4m					
			-	68 000	
Allocate Admin (50%; 40%; 10%)	34 000	27 200	6 800	(68 000)	
			6 800	-	
Allocate Maintenance (49%; 49%; 2%)	3 332	3 332	(6 800)	136	
			-	136	
Allocate Admin (50%; 40%; 10%)	68	54	14	(136)	
			14	-	
Allocate Maintenance	7	7	(14)	-	
Total	3 703 407	3 296 593	-	-	
Units manufactured	120 000	180			
Maintenance and admin cost per unit	30,86	18 314,41			(1) C
					5

(c) Allocate maintenance and administrative costs using the ABC method

Follow the ABC steps:

(1) quantify the activities,
(2) allocate costs to activities
(3) obtain cost per activity and then
(4) unit cost

Note: The 2 marks (R/W) for each row are for the Rand value for SH and CHS (1 each)

	No. of Activities	%	Cost R'000	SH R'000	CHS R'000	
Maintenance: R3m						
Scheduled visits (6 + 4)	10	40	1 200	720	480	(2) R/W (or 1 each)
Call-out visits (15 + 5)	20	30	900	675	225	(2) R/W (or 1 each)
Rework hours (1 700 + 100)	1 800	30	900	850	50	(2) R/W (or 1 each)
Admin: R4m						
Orders (150 + 150)	300	10	400	200	200	(2) R/W (or 1 each)
Drs hours (1 350 + 450)	1 800	30	1 200	900	300	(2) R/W (or 1 each)
HR trans (1 800 + 200)	2 000	40	1 600	1 440	160	(2) R/W (or 1 each)
General trans (400 + 100)	500	20	800	640	160	(2) R/W (or 1 each)
				5 425	1 575	
Units manufactured				120 000	180	
Maintenance and admin cost per unit (R)				45,21	8 750	(2) C (or 1 each)
					Max	14

Why would an exercise such as in (b) and (c) be done?
To establish whether different cost allocation methods yield consistent results. If not, establish why. Costs form the base of pricing, margins and decisions, and it is important to get the cost as accurate as possible.

(d) Calculate the value of closing inventory on 28 February 2014 based on the existing costing practice

Both quantities and values are important and should be disclosed, even when in IS.

	SH Units	CHS Units	
Opening inventory	6 000	0	
Production	120 000	180	
	126 000	180	
Sales	(115 000)	(170)	
Closing inventory	11 000	10	(1) R/W

Regardless of the information given, try to put quantities in the 'standard' format as mistakes can easily occur and impact on the reminder of the question.

Value:	SH R	CHS R		
Materials	95	42 000		
Labour	57	10 500		
VOH	19	7 000		
FOH	19	10 500		
Value per unit	190	70 000	(1)	R/W
Inventory value (R)	2 090 000	700 000		
Total (R)		2 790 000	(1)	C
			3	

(e) Discuss the validity of the current maintenance costing policy

Inventory (and production) should include all production costs and be valued as close as possible to actual costs.

- Maintenance is directly related to production and should be included in the product cost – policy not valid. (1)
- Rework on defective items should not be included in above, but be disclose separately for quality and control purposes. (2)
- As the maintenance is currently treated as a period cost (not included in inventory), the annual profit is understated. (2)
- Understatement of profit may have funding implications – bank review. (1)

• The product costs should be	**CH R**	**CHS R**		
Per (d)	190,00	70 000,00	(1)	C
Add per (b) R1 700 000 ÷ 120 000 units	14,17	-	(1)	
R1 700 000 ÷ 180 units	-	9 444,44	(1)	
	204,17	79 444,44		
OR				
Per (d)	190,00	70 000,00	(1)	C
Add per (c) (R720' + R675') ÷ 120 000 units	11,63	-	(1)	
(R480' + R225') ÷ 180 units		3 916,67	(1)	
	201,63	73 916,67		

- Gross profit margins are overstated and may lead to incorrect performance reward. (2)

Max 6

(f) Impact of lower actual sales on breakeven and profitability

- There is no impact on breakeven as a unit contribution or percentage is used which do not change. (2)
- The safety margin will decline as less sales are generated by both products. (2)
- Profitability will decline as less turnover must cover all the costs which remained close to budget. (2)

Max 3

QUESTION 23

MALEMA SPORTS LTD **SUGGESTED SOLUTION**

(Source: Test 2 2011 TOE408W (MAC4862))

(a) **Calculation of the selling price** **R**

Raw materials	100 cubic metres @ R824	=	82 400	(1)
25 cubic metres	@ R946	=	23 650	(1)
Varnish (1 000 litre *R16)		=	16 000	
Workings: 1 000 batches will consume 125 cubic metres				
Labour	2 450 hrs @ R8/hr	=	19 600	(1)
	(2,45hrs * 1 000 batches)			
Variable overheads	(R13* 1 000 batches)	=	13 000	(1)
Supervisor's overtime	25 hrs* R44	=	1 100	(1)
			155 750	(1)
With mark up	(R155 750* 120%)	=	186 900	(1) **C**

- Only the salvage amount of the material on hand is relevant. The original cost (book value) is a sunk cost (already incurred) and is thus not relevant. (1)
- The varnish is relevant as it is a standard item used in several products. Since no new purchase price is mentioned, it is assumed that varnish still retails at R16/litre.
- The company will employ temporary staff to work in the project and therefore, labour cost is relevant. (1)
- Variable overheads are specific to the project, thus relevant. (1)
- The basic monthly salary (including fringe benefits) of supervisor will not be impacted by the project and therefore irrelevant in calculating the selling price. (1)
- The supervisor will work overtime in order to supervise the project, so relevant. (1)
- Fixed costs (these include depreciation) are not relevant since the project will not impact these costs. (1)
- Also, since the company is operating at 45% of production capacity, fixed costs are not likely to change in the short term. (1)

14

(b) **Discussion of the minimum price**

- The minimum price should be R155 750 for the 10 000 bats. (1)**C**
- The mark up would be added to recover some fixed costs as the company is below the break-even point (it is not generating profit). (2)
- The company is keen to obtain this project. The company's name on the bats could induce future players to use its bats normal-sized and thereby generate much needed sales. (1)
- Therefore, the company will benefit from making no profit on this order. (1)

Max 4

(c) **Discuss relevant cost implications**

- The revenue from contracts or a range of products should in the long term exceed their full cost in order to derive profit. (1)
- The selling price based on relevant or incremental cost is lower than the one based on full cost. (1)
- The lower price will create expectations from other customers if they hear of the price. (1)

3

(d) **Calculate RI**

Minimum required profit = R6,25m x 19%
= R1,1875m (1)

Forecast profit:
Contribution = R5,5m x 38%
= R2,09 m (1)

Profit (loss) = Contribution – FC
= R2,09 – R3,0 – R0,86
= (R1,77) (1)

∴ They will NOT achieve RI (1)

4

(e) **Discuss treatment of research expenditure**

- Research expenditure which will/should generate future income. (1)
- MSL should employ EVA as a financial performance tool in this case. A long term decision. (1)
- Above implies that the P/L be adjusted by the R860 000 and that the amount be capitalised and written off over an appropriate period. (2)
- The EVA approach does not penalize decision-makers for taking long-term decisions to benefit the company. (1)

Max 4

(f) **Calculate the standard deviation**

Outcome (r)	Probability	Expected Value	r – EV	$(r-EV)^2$	xP	Expected σ
5 000	0,15	750	(4 850)	23 522 500	0,15	3 528 375
8 000	0,40	3 200	(1 850)	3 422 500	0,40	1 369 000
12 000	0,35	4 200	2 150	4 622 500	0,35	1 617 875
17 000	0,10	1 700	7 150	51 122 500	0,10	5 112 250
	1,00	9 850				11 627 500
		(1)	(1)			(1)

σ^2 = 11 627 500
σ = 3 409,91 (1)

Risk is indicated by the coefficient of variation. (1)

$$CV = \frac{\sigma}{EV}$$

$$= \frac{3\,409{,}91}{9\,850}$$

$$= 0{,}346 \quad (1)$$

Above shows low risk as variation and standard deviation are low. (1)

7

(g) **List discontinuance factors**

- Rent: will less space be required and/or can contract be amended. (1)
- Can space be sublet? (1)
- Allocated costs will remain unchanged. (1)
- Will any of the other fixed costs listed decrease. (1)
- Does this production deliver contribution towards fixed costs? (1)
- Can equipment be sold? (1)
- Costs associated with decommissioning of the hockey stick production line. (1)
- Can affected workers be retrained or will they be made redundant. (1)

Max 4

QUESTION 24

HARARI LTD AND SHELENI LTD **SUGGESTED SOLUTION**

(Source: Test 2 2011 DIPAC26 (MAC4861))

PART A – HARARI LTD **[20]**

(a) **Calculate the following variances:** **(7)**

(i) Direct material price

Formulae	(SP –AP) AQ	
Variance	(6,00 – 6,22) 9 000	
	2 000 Adverse	(1)

(ii) Direct material usage

Formulae	(SQA –AQ) SP	
Variance	(8 600 – 9 000) 6,00	
	2 400 Adverse	(1)

(iii) Direct labour rate

Formulae	(SR –AR) AH	
Variance	(7,00 – 7,13) 4 600	
	600 Adverse	(1)

(iv) Labour idle time

Formulae	Idle Hours x Std Rate/Hr	
Variance	600 X 7	
	4 200 Adverse	(1)

(v) Direct labour efficiency

Formulae	(SHA – AHW) SR	
Variance	(4 300 – 4 000) 7,00	
	2 100 Favourable	(1)

(vi) Fixed overhead expenditure

Formulae	Budget - Actual	
Variance	36 000 – 35 000	
	1 000 Favourable	(1)

(vii) Fixed overhead volume

Formulae	(SHA - BH) SR	
Variance	(4 300 – 4 000) 9,00	
	2 700 Favourable	(1)

NOTE:

Where the description of the variance is incorrect (e.g. Favourable instead of Adverse), the answer should be marked wrong even if the figure is correct.

(b) **Explain the meaning and suggest one potential cause of each of the following variances:** (4)

(i) **Direct labour rate;**

- This measures the effect of paying actual labour hours at a different rate from standard. (1)
- In this case the variance is R600 adverse and could be due to **overtime** working at premium rates, using higher grade **more expensive** labour or possibly as a result of a **recent wage settlement**. (1)

(ii) **Labour idle time;**

- This measures the cost (at standard rate) of having to pay wages for time at work although no actual work is being done (1)
- It is usually the result of a **production stoppage** and could be caused by a machine breakdown, running out of material, a shortage of customer orders or a labour dispute elsewhere in the business. (1)

(c) **Transfer price from Harari Ltd to Ghweri Ltd**

Lost sales (400 units)

			R	
Sales price			40	(½)
Commission (5%)			(2)	(½)
Direct material			(12)	(½)
Direct labour			(7)	(½)
Contribution per unit			19	Total **[2]**
Total contribution (x 400)			7 600	(1)
Production costs				
Normal:				
Direct material	400 x 12		4 800	(½)
Direct labour	400 x 7		2 800	(½)
Additional material	400 x 1,40		560	(½)
Overtime				
Direct material	400 x 12		4 800	(½)
Direct labour	400 x 7 x 1,30		3 640	(1)
Additional material	400 x 1,40		560	(½)
			17 160	(½)
Total costs to be recovered		= 7 600 + 17 160		
		= 24 760		(1)
Cost per unit (÷ 800)		= 30,95		(1)
				9

PART B - SHELENI LTD **[20]**

(a) Determine the production mix that will maximise profit in month 4 and calculate the resulting profit.

Production Mix:

	X R/litre	Y R/litre	Z R/litre	
Selling price	100	120	120	
Less Variable Costs:				
Direct materials	20	16	21	
Direct labour (R12 per hour)	18	24	27	
Other direct expenses	–	3	–	
Variable overhead	12	16	18	
Contribution	50	61	54	(1,5)
Labour hours per unit (Working 1)	1,50	2,00	2,25	(1,5)
Contribution per labour hour	R33,33	R30,50	R24,00	(1,5)
Rank	1	2	3	(1,5)

Production Plan:

	Chemical	Litres	Contribution R	Hours	
Minimum requirements:					
	X	15,00	750,00	22,50	(1)
	Y	15,00	915,00	30,00	(1)
	Z	15,00	810,00	33,75	(1)
Further units:					
	X	135,00	6 750,00	202,50	(1)
	Y	25,00	1 525,00	50,00	(1)
	Z	16,11	869,94	36,25	(1)
Total contribution			11 619,94	375,00	(1)
Fixed overheads (Working 2)			1 760,00		(1)
Profit			9 859,94		(1)
Maximum					**12**

Working 1:

- For chemical X = R18 per litre ÷ R12 per hour = 1,50hrs/l, etc.

Working 2:

- Fixed overhead absorption rate, based on chemical X = R6 per litre ÷ 1,50 hours per litre = R4 per hour.
- Total fixed overhead = 440 budgeted hours x R4 per hour = R1 760

(b) **Using the above information calculate how much chemical Z (if any) should be purchased by Sheleni Ltd from the overseas supplier in month 4.**

Current Production		**R per month**	
Make current amount of Z:			
31,11 litres x (R21 + R27 + R18)	=	2 053,26	(½)
Extra fixed costs	=	1 000,00	(½)
Sub –Total		3 053,26	
Import current amount of :			
31,11 litres x R100	=	(3 111,00)	(½)
Net benefit of in house production		R57,74	(½)

Decision:

- It is better to continue to produce product Z, rather than buy in. (½)

Demand shortfall

As selling price exceeds the buy in cost it is worthwhile to buy in the extra litres required in month 4. (½)
Extra contribution (60 – 31,11) x (R120 – R100) = R577,80 (½)

Conclusion

Sheleni Ltd should buy in 28,89 litres of chemical Z in month 4. (½)

(c) Briefly explain FOUR factors, other than costs or selling price, which should be taken into consideration when deciding whether to subcontract the manufacture of chemical Z.

- **Before subcontracting the production of Z, Sheleni Ltd should consider the following points:**
- Is the quality of imported Z up to requirement? (1)
- Will delivery be as required? (1)
- Whether the supplier has adequate financial resources to meet the delivery? (1)
- Will its customers be happy if they find out that an overseas supplier is producing Z for sale under Sheleni's brand? (1)
- The business standing (reputation) of the supplier (the implication(s) of associating with this supplier on Sheleni Ltd). (1)
- By importing Z, Sheleni Ltd is effectively exporting pollution to the overseas country. There is an ethical consideration to be made. (1)
- Will the overseas supplier eventually turn into a competitor? (1)
- How long will the contract be for, the supplier may not be required once the labour shortage ends? (1)

Maximum 4

Note: only four factors were required.

QUESTION 25

LEKKER RUSKS (PTY) LTD **SUGGESTED SOLUTION**

(Source: Test 1 2010 TOE408W (MAC4862))

PART A

CALCULATIONS

	Buttermilk (units)	Muesli (units)	
Budgeted production			
Opening stock	100 000	55 000	
Production	**1 480 000**	**675 000**	(1)
Closing stock	(80 000)	(30 000)	
Sold	1 500 000	700 000	
Actual production			
Opening stock	100 000	55 000	
Production	**1 380 000**	**575 000**	(1)
Closing stock	(80 000)	(30 000)	
Sold	1 400 000	600 000	

Budgeted Fixed overhead rate

R45 000 000
1 000 000 hrs = R45 / hour (1)

Budgeted labour rate

R25 000 000
1 000 000 hrs = R25 / hour (1)

Hours:

Buttermilk

450 000 hrs
1 480 000 units = 0,304 hours per unit (1)

Muesli

550 000 hrs
675 000 units = 0,815 hours per unit (1)



Note that in this question there are two types of fixed manufacturing overheads – general fixed manufacturing overheads and fixed labour. The absorption costing principles are applied to each separately. Use budgeted numbers to calculate overhead rates. Normal capacity of 1 000 000 hours per year was given.

Income statement for the year ended 28 February 2010

			Buttermilk	Muesli	Total	
Sales					88 800 000	(1)
B: (1 400 000 x R39)			54 600 000			
M: (600 000 x R57)				34 200 000		
Cost of sales			(46 174 000)	(40 752 250)	(86 926 250)	
Opening stock					3 485 000	(1)
B: Given			2 000 000			
M: Given				1 485 000		
Production:						
Materials					26 151 000	(1)
B: (R12,70 x 1 380 000)			17 526 000			
M: (R15,00 x 575 000)				8 625 000		
Labour (Budgeted rate x Actual production)					22 203 625	
B: (R25 x 0,304 hr x 1 380 000 units)			10 488 000			(1)
M: (R25 x 0,815 hr x 575 000 units)				11 715 625		(1)
1 mark if production numbers						
Overheads (Budgeted rate x Actual production)					39 966 525	
B: (R45 x 0,304 hr x 1 380 000 units)			18 878 400			(1)
M: (R45 x 0,815 hr x 575 000 units)				21 088 125		(1)
1 mark if production numbers						
		R				
Closing stock:						
B: Material	R12,70 x 80 000 =	1 016 000				
Labour	0,304 x R25 x 80 000=	608 000				
Overheads	0,304 x 45 x 80 000=	1 094 400				
1 mark for 80 000 if value X		2 718 400	(2 718 400)			(2)
M: Material	R15 x 30 000 =	450 000				
Labour	R25 x 0,815 x 30 000=	611 250				
Overheads	R45 x 0,815 x 30 000=	1 100 250				
1 mark for 30 000 if value X		2 161 500		(2 161 500)	(4 879 900)	(2)
			8 426 000	(6 552 250)	1 873 750	

Over / Under recovery of overheads

Budgeted		**R**		
Overheads		45 000 000		
Labour		25 000 000		
		70 000 000		(1)
Recovered				
Per above:	Labour	22 203 625		
	Overheads	39 966 525		(1)
		62 170 150		
Under recovery (R62 170 150 – R70 000 000)			**(7 829 850)**	(1)

Expenditure variance

Budget (R45 000 000 + R25 000 000)	70 000 000		
Actual (R44 000 000 + R24 000 000)	68 000 000		
	2 000 000	**2 000 000**	(1)
Admin salaries		(777 000)	(½)
Sales and distribution costs		(8 000 000)	(½)
Net loss		(12 733 100)	
		Max	20



Over / under recovery of overheads: Always compare the budgeted overhead rate with what was recovered.
Expenditure variance: Compare budgeted total amount with actual total amount.

PART B

ABC

Quality control

Buttermilk		**Muesli**		
Note: Use production units to calculate number of batches:				
1 380 000 units / 10 000 =	138 batches	575 000 units / 10 000 =	58 batches	(1)
Total = 138 + 58 = 196 batches				
138/196 x R2 100 000 =	R1 478 571,43	58 / 196 x R2 100 000	R621 428,57	(1)

Materials handling

Buttermilk		**Muesli**		
138 x 3 =	414 set-ups	58 x 4 =	232 set-ups	(1)
Total =414 + 232 = 646 set-ups				
414/646 x R21 300 000 =	R13 650 464,40	232/646 x R21 300 000 =	R7 649 535,60	(1)

Baking

Buttermilk		**Muesli**		
1 380 000 units / 100 x 1	13 800 hrs	575 000/100 x 1	5 750 hrs	(1)
Total = 13 800 + 5 750 = 19 550 hrs				
13 800/19 550 x R10 500 000	R7 411 764,71	5 750/19 550 x R10 500 000	R3 088 235,29	(1)

Packaging

Buttermilk		**Muesli**		
1 380 000 units / 100 x 0,1	1 380 hrs	575 000 units / 100 x 0,1	575 hrs	(1)
Total = 1 380 + 575 = 1 955 hrs				
1 380 / 1 955 x R4 600 000 =	R3 247 058,82	575/1 955 x R4 600 000	R1 352 941,18	(1)

Non identifiable

If you are given a portion of overheads that don't have a cost driver you just use hours as you did in part (a) for absorption rates.

Buttermilk			**Muesli**		
R5 500 000 x	460 000 / 1 015 000	R2 492 610,84	R5 500 000 x 555 000 / 1 015 000	3 007 389,16	(2)
Total		28 280 472,20		R15 719 529,80	
Per unit / 1 380 000		R20,49	Per unit / 575 000	R27,34	(2)
1 mark if unit value (X) calculated					
					12

Remember to use production to calculate the number of batches produced. The batch size of 10 000 boxes was given in the question**.**

PART C

REPORT

TO: Mr Terrry
FROM: CTA Student
SUBJECT: Profitability of Buttermilk and Muesli rusks
DATE: 20 June 2010

Overheads rate per unit		**Buttermilk R**	**Muesli R**	
Traditional				
Buttermilk	460 000 hrs / 1 380 000 units x R45 /hr	15,00		(1)
Muesli	555 000 hrs / 575 000 units x R45 /hr		43,44	(1)
ABC				
		20,49	27,34	(1)
			Mark through	

Buttermilk

Where fixed production overheads are allocated on a traditional basis the product's overheads are R15,00 per unit. However when the ABC method is used the product's overheads per unit has increased to R20,49.

Muesli

Where fixed production overheads are allocated on the traditional basis the product incurs overheads of R43,44 whereas if the ABC method is used the overheads per unit decreases to R27,34.

The traditional costing method allocates overheads based on labour hours. Therefore, Muesli which uses more hours for production (0,96 hours per unit for Muesli compared to 0,33 hours per unit for Buttermilk) is allocated more overheads than Buttermilk.

However, when the ABC method is used Buttermilk which makes more use of the resources is allocated more overheads. (1)

ABC is a more accurate method for allocating overheads as it assigns costs more accurately. It will thus give the company a better insight on how each product is performing. (1)

Max 4

PART D



Remember to use budgeted figures, not actual figures. Also remember to split your fixed and variable costs correctly e.g. selling and distribution costs in this test.

Fixed costs = R45 000 000 + R25 000 000 + R777 000 + R3 000 000

= R73 777 000 (1)

Sales of 1 350 000 boxes Buttermilk (B) and 450 000 boxes Muesli (M)

3B + 1 M = 1 bundle

Variable selling and distribution costs = R5 280 000 → R2 640 000

(50/50 split) → R2 640 000

	R2 640 000		R2 640 000	
Buttermilk	= 1 500 000	Muesli	= 700 000	
	= R1,76 /box		= R3,77 /box	(½)

Marginal profit per unit	**Total value**	**B**	**M**	
	R	**R**	**R**	
Selling price	76 725 000	38,50	55,00	
Material (1 350 000 x R12) + (450 000 x R15,50)	(23 175 000)	(12,00)	(15,50)	
Selling and distribution	(5 280 000)	(1,76)	(3,77)	
	48 270 000	24,74	35,73	(1)
Number of units per bundle	62,913%	3	1	

Contribution in bundle = (3 x R24,74) + (1 x R35,73)
= R109,95 (½)

Break even = R73 777 000 / R109,95/bundle = R73 777 000 / 0,62913

= 671 005 bundles (½)

Break-even sales = (671 005 x 3) + (671 005 x 1)
= 2 013 015B + 671 005M (½)

Sales B = 2 013 015 x R38,50
= R77 501 078 (½)

Sales M = 671 005 x R55,00
= R36 905 275 (½)

Total sales = R114 406 353 R117 268 291 (½)

Max 4

MARKERS COMMENTS

Guidance for section (a)

Many students did not calculate the over or under recovery of fixed overheads at all or calculated it incorrectly (using actual figures).
The format of the income statement was also incorrect. Students did not always show opening stock, production and closing stock when calculating cost of sales.When a question asks for an absorption costing income statement we are usually trying to test the following:

- Whether you know what absorption costing means i.e. that you are supposed to included fixed overheads and labour in opening stock, production and closing stock.
- Calculation of under/over recovery of fixed overheads and labour.

Fixed costs variance

This is made up of the following:

- Volume variance (over/under recovery of fixed overheads and labour)
- Expenditure variance

Over or under recovery of fixed overheads and labour

This is calculation done by MANAGEMENT ACCOUNTANTS to ascertain the volume variance of units produced. When a question asks for the preparation of an income statement on an absorption costing basis; remember to do the following:

- Calculate absorption rate using the following equation: Budgeted overheads/ budgeted activity. (In this test it was calculated as R45 per hour for overheads and R25 per hour for labour).
- Include fixed overheads and labour, **on a per unit basis**, in opening stock, production and closing stock. The labour and overheads included in <u>production</u> are the absorbed/recovered overheads. (Please note that if you are given hours you must first calculate how many hours you *expect* to use to complete 1 unit. This was calculated as 0,304 hours for buttermilk and 0,815 hours for muesli).
- Compare the labour and overheads per the budget (R45 m + R25 m) to absorbed/recovered overheads. This difference is the over/under recovery of fixed overheads/labour.
- Ignore ACTUAL OVERHEADS/LABOUR for this calculation.

<u>How does it work?</u>

- When you include overheads in your income statement according to production; you use ACTUAL PRODUCTION UNITS. (1 380 000 buttermilk and 575 000 muesli).
- When you compare the budget to the absorbed you will have a difference due to the difference in production units i.e. volume difference.
- This volume difference is the over or under recovery of overheads/labour.

<u>How do I know whether I have an over/under recovery and how should I treat it in the income statement?</u>

- Always remember that fixed overheads/labour are a fixed cost and should not change. So, in a perfect world your budgeted overheads/labour would equal to your actual absorbed overheads/labour per the income statement. If we budgeted to spend R5 000 on overheads then R5 000 should have been debited to the income statement. This is not the case because per above we include the overheads on a per unit basis. So if we produce more units that the budget we end up debiting more costs to the income statement.
- Think in terms of financial accounting. During the year you have debited a certain amount of overheads and labour to the income statement (absorbed/recovered).
- At the end of the year you compare what was debited to the income statement to your budget.
- If your budget (R70 000 000) is MORE than absorbed (R62 170 150) then you have debited too little to the income statement. This is an under recovery. (R7 829 850) This under recovery should be debited to the income statement. (Treat like an expense.)
- If your budget is less than absorbed you have debited too much to the income statement. This is an over recovery. This over recovery should be credited to the income statement. (Treat it like an income.)

<u>Expenditure variance</u>

This is a separate calculation and is NOT an over/under recovery of fixed overheads/labour.
Expenditure variance = Actual overheads expense – budgeted overheads expense

QUESTION 26

ANTIHISTAMINES; CYCLE ADDICTS; ABC **SUGGESTED SOLUTION**

(Source: Test 1 2010 DIPAC26 (MAC4861))

PART A

Raw materials	**Units (kg)**		
Opening stock	-	Given	(½)
Purchased	15 000 000	R60 000 000 / R4 per kg	(1)
Closing stock	250 000	Given	(½)
Production	14 750 000		
Finished goods			
Opening stock	3 300 000	R21 450 000 / R6,50	(1)
Put in production	11 800 000	14 750 000kg / 1,25kg	(1)
Sold	15 000 000	R142 500 000 / R9,50	(1)
Closing stock	100 000		

Value of closing stock

	R	
Raw materials = 250 000kg x R4	1 000 000	(2)
Finished goods = 100 000kg x R6,65	665 000	(1)
	1 665 000	

Income statement of Antihistamines (Pty) Ltd for the year ended 31 December 2009

	R'000	
Sales	142 500,0	(1)
Less: Cost of sales		
Opening inventory	21 450,0	(1)
Materials	60 000,0	(1)
Depreciation – Manufacturing	7 500,0	(1)
Overheads – Fixed	4 794,2	(1)
Overheads – Variable	2 380,4	(1)
Packaging	500,0	(1)
Wages	15 995,0	(1)
Closing inventory	(1 665,0)	
Cost of sales	110 954,6	

Gross profit		31 545,4	
Delivery costs		500,0	(1)
Depreciation	– Administrative	250,0	(1)
Interest paid		320,0	(1)
Salaries	– Administrative	4 090,5	(1)
	– Sales	2 166,0	(1)
Travelling		7 000,0	(1)
Profit		17 218,9	
			Max 20



Students did not use the correct lay out for the income statement. Production costs should be grouped together before gross profit. Non-production costs should be included after gross profit before the net profit is calculated. Note that the question did not require the under/over recovery of fixed overheads.

PART B AND C

1. (c) (3)
2. (b) (3)
3. (e) (3)
4. (a) (3)
5. (a) (4)
6. (d) (4)

1 (c) In order to calculate the breakeven point we need to determine the fixed and variable costs:
Using the high-low method the variable cost is determined as follows:

= (R45 000 000 - R35 000 000)/(7 000 units - 5 000 units)

= R10 000 000/2 000 units

= R5 000 per unit (1)

The fixed cost is then:
= (R35 000 000 – (R5 000 x 5 000 units))
= R35 000 000 – R25 000 000
= R10 000 000 (1)

The break-even point is then calculated as follows:

Profit = Sales - variable costs - fixed costs
0 = 10 000x - 5 000x - 10 000 000
10 000 000 = 5 000x
x = 2 000 units (1)

(a) This is merely the maximum capacity available. At this point the company will make a profit of [(R10 000 - R5 000) x 7 000 units - R10 000 000] =R 25 000 000.

This is much higher than the break-even point where profit = 0.

(b) This number is higher than the expected sales and the maximum capacity available and is therefore not feasible.

(d) This is the expected sales for the next year, not the point at which the company would not make any profits or suffer losses (i.e. Break-even point).

(e) Not applicable as a correct answer is available.

2 (b) The profit based on the expected sales of 6 000 units, may be calculated as follows:

	R	
Contribution ([R10 000 - R5 000] x 6 000 units)	**30,000,000**	(1)
Less: Total fixed cost	**10,000,000**	(1)
Profit	**20,000,000**	(1)

(a) The amount represents the fixed cost, not a profit, and as the calculation above shows, a profit is made, not a loss.

(c) This is the total contribution for the product, but the fixed cost is incorrectly ignored. To calculate profit, fixed cost should be deducted from the contribution obtained.

(d) This answer merely represents the total sales amount if 6 000 units are sold.

(e) Not applicable as a correct answer is available.

3 (e) To calculate the required number of units to sell in order to generate a profit of R30 000 000, we use the formula:

$$\textbf{Number of units} = \frac{\textbf{(Fixed cost + required profit)}}{\textbf{Contribution per unit}}$$

$$= \frac{\textbf{(R10 000 000 + R30 000 000)}}{\textbf{R5 000}} \quad (2)$$

$$= \textbf{8 000 units}$$

However the maximum capacity is currently 7 000 units, and therefore it is not currently possible for the company to earn a profit of R 30 000 000. (1)

(a) This is the expected sales for the next year, and as calculate above would lead to a profit of R20 000 000

(b) 2 000 bicycles must be sold in order to break even. No profit or loss would be incurred at this level, and it cannot possibly be the correct number of units to generate a high profit of R30 000 000.

(c) This is the correct number of units to produce in order to achieve a profit of R30 000 000.
However, the company cannot manufacture this number of units as its maximum capacity is currently 7 000 units per annum.

(d) This is merely the maximum capacity available. At this point the company will make a profit of (R10 000 - R5 000) x 7 000 units - R10 000 000 = R25 000 000. This does not achieve the profit of R30 000 000.

4 (a) The calculation of margin of safety is:

(Expected sales – Break-even sales) / Expected sales x 100
= (6 000 units - 2 000 units)/ 6 000 units x 100 (2)
= 66,67% (1)

(b) This is merely the difference between the maximum quantity that may be produced (7 000 units) and the lowest quantity given for the cost calculation (5 000 units), divided by the 7 000 units maximum capacity. The answer doesn't make any sense and it's a complete misapplication of the formula.

(c) The answer is incorrect as the formula is applied incorrectly. Instead of dividing by the expected sales, the break-even sales level is used, i.e. (6 000 units - 2 000 units)/ 2 000 units.

(d) This answer is incorrect as the margin of safety may be calculated as 66,67%, which means that the sales may drop from current levels with 66,67% before a loss will be suffered.

(e) Not applicable as a correct answer is available.

5 (a) The total overhead cost of product B based on labour hours is calculated as follows:

$$\frac{\textbf{6 400 000 hours}}{\textbf{8 000 000 hours}} \textbf{ x R2 000 000} \quad (1)$$

= R1 600 000 (2)

(b) This is the overhead cost PER UNIT of product B (i.e. R1 600 000/60 000 units).
This is incorrect as the question asks for TOTAL overhead cost.

(c) This would be the cost if the number of units were incorrectly used to allocate

the total overhead costs to the two products. It is wrong as the question clearly asked that the labour hours be used as an allocation basis.

(d) This is the cost incorrectly calculated (R 2 000 000 / 80 000 units) - the allocation should have been done over the number of labour hours, not the units produced.

(e) Not applicable as a correct answer is available. (1)

6 (d) Machine related cost: R480 000 x 80 000 machine hrs/ 100 000 machine hrs

= R384 000 (1)

Quality control: R920 000 x 1 200 inspections/ 2 000 inspections

= R552 000 (1)

Set up costs: R600 000 x 500 setups / 300 setups

= R360 000 (1)

Total = R384 000 + R552 000 + R360 000
= R1 296 000

Per unit = R1 296 000 / 60 000 units
= R21,60

The question asks for the overhead cost PER UNIT.

(a) This is the cost per unit according to the traditional costing system, using labour hours as the allocation base.

(b) This is the cost per unit according to the traditional costing system, using the number of units produced as the allocation base.

(c) The total cost of R2 000 000 is divided by the 60 000 units of Product B produced. This is incorrect as no cost would then be allocated to Product A, which is clearly wrong as Product A also causes the incurrence of overhead costs.

(e) Not applicable as a correct answer is available. (1)



Part B and C tested basic principles of cost-volume-profit analysis and activity-based costing.

QUESTION 27

ZIMPOPO LTD **SUGGESTED SOLUTION**

(Source: Test 2 2010 TOE408W (MAC4862))

PART A

(a) The fixed overhead (volume) efficiency variance indicates an absorption costing system and consequently unit profit.

(i) Actual sales volume

Budgeted profit per unit	= 42 000 / 15 000	
	= R2,80	(1)
Sales volume variance	= R8 400	
∴ Variance (units)	= R8 400 ÷ R2,80	
	= 3 000 units (A)	(1)
∴ Sales	= 15 000 (budget – 3 000 (var))	
	= 12 000 units	(1)



Follow the logic: the sale volume variance is based on a budgeted profit rate and the difference between budgeted and actual quantities.

(ii) Actual quantity of materials used for production in April

Budgeted (std) price	= R45 000 ÷ 7 500 kg's	
	= R6 / kg	(½)
Budgeted (std) quantity	= 7 500 kg / 15 000 units	
	= 0,5 kg / unit	(½)
For output (std)	= 15 500 x 0,5	
	= 7 750 kg	(½)
Variance	= R1 500 ÷ 6	
	= 250 kg (adv)	(½)
∴ Usage	= 7 750 + 250	
	= 8 000 kg	(1)



As the variance was unfavourable, it follows that the usage must be higher than the standard allowed.

(iii) Actual cost of materials in April

Budgeted cost	= 10 000 kg x R6	
	= R60 000	(1)
Variance	= R10 000 F	
∴ Paid (actual cost)	= R50 000	(1)
Or R50 000 ÷ 10 000 kg	= R5 / kg	

(iv) Actual labour hours worked in April

Budgeted (std) rate	= R45 000 ÷ 11 250 hours	
	= R4 / hour	(½)
Budgeted (std) time	= 11 250 hours / 15 000 units	
	= 0,75 hours / unit	(½)
For output allowed	= 15 500 x 0,75	
	= 11 625 hours	(½)
Variance	= 1 500 ÷ 4	
	= 375	(½)
∴ Actual hours	= 12 000 hours	(1)



The costs for materials and labour are the same and the relationship is 1,5 (labour) to 1 (materials); refer standard usage (11 250 : 7 500) and rates. One could therefore simply take 8 000 x 1,5 = 12 000.

(v) Value of inventory on 30 April 20X0

Raw materials	= (0 + 10 000 kg – 8 000 kg) x R6/kg	
	= R12 000	(1)



Materials valued at standard, no opening inventory.

Completed units

Standard sales price	= R210 000 ÷ 15 000	
	= R14 per unit	(½)
Budget (std) profit	= R42 000 ÷ 15 000	
	= R2,80 per unit	(½)
∴ Standard cost	= R11,20 per unit	(½)

(OR: Std cost: R45' + R45' + R24' + R54' = R168' / 15' units = R11,20 per unit)

	= 4 500 completed units	
Quantity = 1 000 + 15 500 – 12 000		
∴ Value	= R11,20 x 4 500	
	= R50 400	(½)
∴ Total value	= R12 000 + 50 400	
	= R62 400	(1)

PART B

(b)(i) Contribution approach to assess whether product L should be ceased:

Current: **Product**	**L** **R'000**	**I** **R'000**	**M** **R'000**	**Total** **R'000**	
Sales (given)	1 500	1 440	880		(½)
Materials (given)	(500)	(480)	(240)		(½)
Variable labour [C1]	(240)	(192)	(96)		(1½)
VOH [C3]	(250)	(120)	(40)		(1½)
Current contribution	510	648	504	1 662,0	
Alternative: Per unit	**R**	**R**	**R**		
Sales price	15,00	12,00	11,00		
Materials	(5,00)	(4,00)	(3,00)		
Variable labour [C1]	(2,40)	(1,60)	(1,20)		
VOH [C3]	(2,50)	(1,00)	(0,50)		
Current contribution	5,10	5,40	6,30		
After:		**R'000**	**R'000**		
Sales [C4]		1 440	1 710		(1)
Materials (R240'/80' = R3/unit x 180' = R540')		(480)	(540)		(1)
Variable labour [C5]		(192)	(259,2)		(1)
VOH (R40'/80' = R0,5/unit x 180' = R90')		(120)	(90)		(1)
New contribution	-	648	820,8	1 468,8	
Thus **reduction** in contribution:				(193,2)	
Alternative: Per unit		**R**	**R**		
Sales price (R) [C4]		12,00	9,50		
Materials		(4,00)	(3,00)		
Variable labour [C5]		(1,60)	(1,44)		
VOH		(1,00)	(0,50)		
New contribution	-	5,40	4,56		

(b) (ii) Residual Income approach to assess whether product L should be ceased:

Current	**R'000**	
Total contribution (per (b)(i))	1 662	(1)
Less: Fixed costs		
Fixed labour (L: 160 + I: 128 + M:64) [C2]	(352)	(1)
Fixed overhead (given)	(1 200)	(1)
Current profit (OR: L: (50') + I: R40' + M: R120') (given)	110	

∴ RI = R110 000 – (R675 000 x 15%) (1)
= R110 000 – R101 250
= R8 750 (1)

After	**R'000**	
Total contribution (per (b)(i))	1 468,80	(1)
Less: Fixed costs		
Fixed labour (I: 128 + M: 128 [C6])	(256,00)	(1)
Fixed overhead (L: 320 [C8] + I: 480 [C7] + M: 320 [C8])	(1 120,00)	(1)
New profit / (loss)	(92,80)	(1)

∴ Production of product L should not be ceased as both the contribution and the RI will decline due to the lost contribution (R510 000) of product L. (1)

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Remember to compare the **division's** performance before and after the termination and not only that of product L. Students often used profit instead of contribution.

(c) Transfer price

Currently only 80 000 units of product M are produced and sold while the maximum capacity is 100 000 units. Therefore, there is spare capacity of 20 000 units. The spare capacity resulting from the termination of product L will be used to manufacture an additional 100 000 units of product M which will all be sold at the new reduced price. The order from Gweru division for 40 000 units can therefore be partially supplied from the spare capacity of 20 000 units while the other 20 000 units will have to be sacrificed from existing sales resulting in opportunity cost.

	R	
Offered price	7,00	
Less: Variable cost (3,00 + 1,44 + 0,50) (per (b)(i))	(4,94)	
Per unit contribution	2,06	(1)
∴ Effect on Musina division is an additional profit of R82 400 (R2,06 x 40 000 units)	R82 400	(1)
Less: Opportunity cost (contribution lost) (R4,56 x 20 000)	91 200	(1)
	(8 800)	(1)

∴ Offer is not acceptable due to overall loss

4

(d) List the five primary types of transfer prices that companies can use.

- Marginal cost (½)
- Full cost (½)
- Cost plus a mark-up (½)
- Negotiated price (½)
- Market based price (½)
- Plus only if **all** given (½)

3

Calculations – Part B:

1. Variable labour (L) = R400 000 x 60%
   = R240 000
   Per unit = R240 000 ÷ 100 000 units
   = R2,40

2. Fixed labour (L) = R400 000 x 40%
   = R160 000

3. Variable overhead (L) = R650 000 – R400 000 [C7]
   = R250 000

   Per unit = R250 000 ÷ 100 000 units
   = R2,50

4. Sales price (M) = R880 000 ÷ 80 000 units = R11 per unit
   = R11,00 – R1,50
   = R9,50

   Total = R9,50 x (80 000 + 100 000)
   = R1 710 000

5. Variable labour (M) = R96 000 ÷ 80 000 = R1,20 per unit
   = R1,20 x 1,2 (20% increase)
   = R1,44

   Total = R1,44 x 180 000
   = R259 200

6. Fixed labour (M) = 160 000 x 40%
   = R64 000 currently

   New cost = R64 000 x 2 (double)
   = R128 000

7. Fixed admin overhead (before)

$$L = \frac{100}{300} \times R1\,200\,000$$
$$= R400\,000$$

$$I = \frac{120}{300} \times R1\,200\,000$$
$$= R480\,000$$

$$M = \frac{80}{300} \times R1\,200\,000$$
$$= R320\,000$$

8. Fixed OH (L) [C7] = R400 000 x 20%
= R80 000

∴ Left (after ceasing prod L) = R400 000 – R80 000 = R320 000

QUESTION 28

KISUZU LTD; BW LTD **SUGGESTED SOLUTION**

(Source: Test 2 2010 DIPAC26 (MAC4861))

PART A

(a) (i) **Material price variance**

4 255 000 – 230 000 x 18
= 4 255 000 – 4 140 000 = 115 000 U (1)

It is not necessary to calculate a cost per kilogram, as the variance should have a total Rand format. You are reminded that variances should be clearly indicated as favourable or unfavourable.

(ii) **Material usage variance**

18 x (230 000 m^2 – 110 000 x 2 m^2)
= 18 x (230 000 – 220 000) = 180 000 U (1)

The quantity used is equal to that purchased as there was no raw material inventory on 30/4. This quantity should be compared to the standard for the **output**, ie 110 000 shirts. More used, therefore an unfavourable variance.

(iii) **Labour rate variance**

(1 550 000 – 80 000) – 105 000 x 14 (1)
= 1 470 000 – 1 470 000 = 0 (1)

The payment in respect of the long-serving bonuses should be excluded from the labour cost as this does not relate to 'normal' work. The 0 amount indicates that workers were paid the budgeted rate.

(iv) **Labour efficiency variance**

14 x (105 000 – 110 000 x 1)
= 14 x 5 000 hrs = 70 000 F (1)

(v) **VPOH expenditure variance**

410 000 – (110 000 x 0,25 machine hours x 16)
= 410 000 – 440 000 = 30 000 F (1)

(vi) **FPOH absorbed**

110 000 x 12	= 1 320 000	(½)
Standard (budget)		
= 100 000 x 12	= 1 200 000	(½)
Over absorbed	120 000	(1)



Produced more than budget, thus over. 'Absorbed' means each product (shirt) is allocated this cost.

(vii) **Sales commission**

Budgeted %	= 6/120	
	= 5%	(1)
Actual sales	= 60 000 x 120 + 24 000 x 115	
	= 9 960 000	(1)
∴ Commission paid	= 9 960 000 x 5%	
	= 498 000	(1)

(b) **Special order request**

	R	
Sales price	90,00	(½)
Materials	(36,00)	(½)
Labour	(14,00)	(½)
VPOH	(4,00)	(½)
Embroidery and packaging	(16,00)	(½)
Commission	-	(½)
Fixed costs, not relevant	-	(½)
Contribution per shirt	20,00	(½)
∴ Additional contribution (x 15 000)	R300 000	(1)
		5

Motivation

- Sales are well below budget, sitting with huge inventory. (1)
- Sale will contribute towards fixed costs. (1)
- Balls Ltd will pay immediately, low risk of bad debt and improved cash flow. (2)
- May compromise future sales at normal prices. (1)
- Will impact performance return (ROI/RI). (1)

∴ The special order should be accepted. (1)

Max 4

(c) **Residual income**

	Current R'000		Additional R'000	
Sales	9 960,0		1 350,0	(2)
Manufacturing cost – COS (4 255 + 1 550 + 410 + 1 191,5) ÷ 110 000 x 84 000 = 7 406 ÷ 110 x 84	(5 655,9)	(7 406÷110x15)	(1 009,9)	(4)
Embroidery			(240,0)	(1)
Selling costs	(498,0)		-	(½)
Fixed admin	(1 901,0)		-	(½)
Profit	1 905,1		100,1	
WACC charge 16% x R15m	(2 400)		-	(1)
	(494,9)		100,1	(1)

∴ Kisuzu Ltd has not reached their RI target, but the additional sale will contribute towards it (making the RI loss smaller) (2)

12

The WACC charge is the return expected of management, in this case 16%. Although Kisuzu Ltd is profitable, it does not reach this return.

PART B

(d)

		R		
A	200 x 44	8 800	Regular use	(2)
B	600 x 25	15 000	No use – resale	(2)
	800 x 33	26 400	Replacement	(2)
C	400 x 16	6 400	Alternative use better than sale	(2)
D	300 x 10	3 000	No use – resale	(2)
		59 600		
			Max	8

1 mark per calculation and 1 mark for correct motivation.

Commentary on Test. Use the commentary to 'fix' the areas where you may have made a similar mistake.

PART A (a) (i) – (vi)

Most students calculated the value of the price rate variances correctly. The quantity or efficiency variances on the other hand were mostly incorrect as students did not apply the flexed budget principle of linking the standard quantity allowed and rate for the actual output, in this case the production of 110 000 shirts. Furthermore, where the type of variance is shown incorrectly, e.g., favourable where it is unfavourable, no mark can be awarded as the incorrect type indicates that the student does not understand the underlying principles. Variances should also not be calculated in unit (1kg or metre) terms, as this will not indicate clearly value lost or gained. The long term bonus payment of R80 000 included in the labour period cost does not relate to the production of the period and should be excluded from the evaluation process. Students should note that the term 'absorbed' (vi) refers to the difference between the original budget and the actual production: - 100 000 vs. 110 000.

Very few students understood the concept of commission in section (vii). The % - commission (commission paid on sales) is required before a forecast can be made.

In section **(b)** students often used the existing sales price of R120 per unit rather than the offered R90. The motivation required referred to the additional sales from the special order and not the cost components.

Students had a reasonable idea of what was required for section **(c)**, the layout was however problematic. Students are therefore referred to the suggested solution to review the process and layout.

PART B

Relevant costs were done quite well by those students that attempted this section. Students should note that all test/examination questions would contain some sections that are more doable than others. It is good practice to find those (or the ones which suit you best) and do them first.

QUESTION 29

EFICAZ (PTY) LTD ('EFICAZ') **SUGGESTED SOLUTIONS** (UNISA)

(Source: SAICA ITC June 2013 Paper 4 Question 2 adapted)

REFLECTION:

Reading:

- Growth required
- Capacities given – check limits for both
- Target PM 30%
- Options: different contributions, times (especially machine?) quantity, plant requirements
- Overtime (?)
- Non-commitment vs 3 year contract
- FOH recovery rates
- Options exclusionary
- Pricing with recovery rate

Required:

- Calculate and evaluate; justify exclusions (likely to be)
- Non-financial issues
- Pricing including existing OH
- Merits and pitfalls (separate)
- DLH as rate base, impact?

(a) Calculate and evaluate which of the two products will be the most profitable for Eficaz to manufacture in the 2013 financial year.



Note:

1. A relevant costing approach is required: costs that change or have a cash impact as a result of the decision. (The three year alternative is strictly speaking **not** a short-term decision, which would change the rules and method applied). The question's intention was not adequately described/defined.
2. Capacities are specified in the question – check to see whether these have an impact (cash or priority).

	Note	M134 R	IN223 R	
Selling price		120,00	80,00	
Raw material		(40,00)	(15,00)	
Direct labour	2	0,00	(6,60)	
Variable manufacturing overheads		(7,50)	(8,50)	
Fixed manufacturing overheads	3	0,00	0,00	
Machine modification	4	-	(1,30)	
Contribution per unit		72,50	48,60	(1)
Units		35 000	60 000	
Total contribution		R2 537 500	2 916 000	(1)

Notes

1. *Machine capacity*
   Hours to be used (@ 0,5/0,2) 17 500 12 000 (1)
   No capacity constraint (1)

2. *Labour capacity*
   Hours required [35 000 @ 0,60] [60 000 @ 0,5] 21 000 30 000 (1)
   Spare capacity 25 600 (given)

- For product M134 the spare capacity can be utilised and there is no additional cost as labour is already paid anyway – in the SA context labour is a fixed cost unless very specifically classified as variable.

  Although the question specifies 'direct labour', this is not adequate and reality takes preference.

- For product IN223 overtime is required. (1)
  30 000 – 25 600 = 4 400 hours (1)
  Cost 4 400 x 1,5 x R60 = R396 000 (1)
  Cost per unit = R396 000 ÷ 60 000 = R6,60 (1)

3. *Fixed MOH*

   These are not relevant as sunk costs in terms of decision-making (1)

4. *Modification costs*

   This is a relevant cost as cash flow will take place as a result of the decision. (1)
   Manufacturing over 3 years = 60 000 + 60 000 x 1,1 + 60 000 x $(1,1)^2$ = 198 600 (1)
   Note: The estimation (growth) will overrule the contract of 60 000 x 3 = 180 000
   Costs = R308 180 – 50 000 = R258 180 (1)
   Cost per unit = R258 180 ÷ 198 600 = R1,30 (1)

Recommendation
Based on the above, Eficaz should pursue the IN223 alternative. (1)

Max 15

Part (b) Identify and discuss the factors to be considered by Eficaz in evaluating whether to pursue the opportunity to supply XYZ Mining or ABC Industrial



Illustrate your logic and argumentative power by using descriptive headings such as operations (business), financial. In this case there are specific clients, which should also be covered.

Customer due diligence

- Determine the creditworthiness of ABC and XYZ (1)
- Look into the reputation and creditability of ABC and XYZ (1)

Industry analysis

- These products are to be used by the mining industry which means it is likely to be subject to higher safety requirements with a concomitant higher risk of damages. (1)
- The mining industry is more cyclical and therefore, forecast units could vary significantly year on year. (1)
- Eficaz has no BBBEE credentials, which could be a critical factor in supplying mines. (1)

Financial implications

- It will require a working capital investment. (1)
- If the forecasts [costs (raw materials, labour variable OH's) or demand] are not accurate, this could have serious implications. (1)
- The target gross profit margin will not be reached (1)
- This will lead to a decline in overall margin, although overall earnings will increase. (1)

Specific customer considerations

- The fact that ABC is prepared to enter into a three-year contract, provides more comfort with regard to the on-going supply of product. (1)
- If Eficaz is restricted in only supplying IN223 to ABC, it cannot leverage production capability through economies of scale by producing more of this product. (1)
- What are ABC's and XYZ's stance towards future price escalations? (1)

Manufacturing considerations

- Will manufacturing the alternatives restrict future growth? (1)
- Eficaz may need to hire more labour given that overtime is required to meet ABC order. On-going overtime is not sustainable, particularly if the existing business grows. (1)

Other considerations

- Eficaz should prepare a capital budget spanning 3 – 5 years to estimate NPV and IRR (1)
- It would determine whether there are any other profitable opportunities which can be pursued. (1)
- Any other valid comment – limited to a max of 1

Maximum	15
Communication skills – logical argument; clarity of expression	2
Total	17

(c) **Calculate the selling prices Eficaz should charge for the M134 and IN223 products if it used a machine hour recovery rate methodology and the recovery rate is R80 per hour.**



Note:

- The increased machine capacity (part (a)) implies that the 100 000 hours available become normal capacity and a budget rate should be used. Deviance from this capacity will be accounted for by under/or over recovery of the FMOH.

	Note	M134 R	IN223 R	
Raw materials		40,00	15,00	
Direct labour* [6,60 + 25,60]		36,00	32,20*	(2c)
Variable MOH		7,50	8,50	
Fixed MOH	1	6,50	2,60	
Machine mods (per (a))		-	1,30	
Manufacturing costs		90,00	59,60	
Machine hour recovery	2	40,00	16,00	(2)
Selling price		130,00	75,60	

Notes

1. **Fixed MOH**

Allocated on basis of **normal capacity**
Capacities:
M134 = 81 600 + 17 500 (per a) = 99 100 (1c)
IN223 = 81 600 + 12 000 (per a) = 93 600 (1c)

∴ R1 300 000 ÷ 100 000 hours = R13 per hour (2)
∴ M134 = R13 x 0,5 = R6,50 (1)

IN223 = R13 x 0,2 = R2,60 (1)



In terms of the capacities in (a), actual being equal to *estimates*, both alternatives will cause a small under recovery of FOH.
Remember the purpose of using a budgeted rate is to enable the management accountant to produce quick and consistent results to be used in decision-making.

2. **Machine hour recovery**

M134 = R80 x 0,5 = R40,00 (1)
IN223 = R80 x 0,2 = R16,00 (1)



Machine hour recovery basis is a (crude) mechanism to obtain a specific return on equipment used in a business, where the equipment is the driver of the income stream eg an X-ray machine.

Part (d)

Discuss the potential merits and pitfalls of adopting a selling price strategy of adding a machine hour recovery rate to manufacturing costs per unit

Potential pitfalls

- If the R80 per machine hour is an **arbitrary** amount, the base rate will have a profound impact on pricing. (1)
- The end result (selling price) may be unrelated to *market prices*, which may result in a loss of revenue. (1)
- The question is whether machine hours are the correct allocation basis and whether direct labour hours may not be more *appropriate*? (1)
- The recovery rate *limits* the potential *profitability* per product, while target costing may result in higher contributions. (1)
- The pricing strategy may be contrary to *competitors' strategies*, which could result in confusing messages being sent to the customer base. (1)

Merits

- It promotes a greater focus on *cost efficiencies* rather than passing on cost inefficiencies to customers. The only way to improve profitability is to *improve* machine efficiency or lower costs. (2)
- It results in a *more fair allocation* of desired profitability to individual products. (1)
- Changes are more appropriately allocated for *value added* (manufacturing time and expertise) than simply marking up higher input costs. (1)
- May be a cost-effective way to obtain desired returns. (1)

Maximum 8
Communication skills – logical argument 1
Total 9

Part (e)

Estimate the allocation of fixed manufacturing overheads to the M124 and IN223 products during the 2013 financial year based on direct labour hours as opposed to machine hours

	M134	**IN223**	
Planned direct lab our hours (normal time) before new products (132 000 – 25 600)	106 400	106 400	(1)
Labour hours – new products (normal time)	21 000	25 600	(1)
Total planned direct lab our hours	127 400	132 000	
Use budgeted hours	132 000		
Fixed manufacturing overheads per labour hour (R1 300 000 ÷ 132 000)	R9,85	R9,85	(1)
Allocation to M134 & IN223 (9,85 x 0,60); (9,85 x 0,50)	R5,91	R4,93	(2)



Note:
M134 Overhead recovery
In the above case there will be an under recovery of overhead of (132 000 - 127 400) 4 600 x R9,85 = R45 210 which can either be written off or be allocated to the products. It will increase the cost for M134 by 21 c (45 310 ÷ 127 400 x 0,6)

Maximum 5

QUESTION 30

AERO AFRICA LTD **SUGGESTED SOLUTION**

(Source: SAICA ITC January 2013, Paper 2, Part A adapted)

REFLECTION:

Reading:

- Director's policy – wife of audit partner
- Listing issues: directors appointment, pricing, special resolutions?
- Letter of resignation issues
- Composition of board – knowledge?
- One-way flights – returns? Capacity utilised & cancellations → 730
- Mix and 'yld' in respect of revenue
- Cargo revenue 'by-product' see FC
- Subsistence in respect of cancellations?
- Meal costs – variances
- Allocated and apportioned FC
- No hedging

Required:

- Fuel variances in detail
- BE on actual ratio between products
- Concerns re post-listing CG

This solution consists of two parts.

PART A **Marks**

Part (a) Calculate variances for the year ended 31 December 2012

	Budget	**Actual**	**Difference**	
Business class				
Number of passengers	17 568	19 656		(½)
Average fare	R27 000	R28 250		
Catering cost (R180 x 2) and (R200 x 2)	(R360)	(R400)		(½)
Sales margin	R26 640	R27 850		(1)
Economy class				
Number of passengers	108 336	110 656		(½)
Average fare	R3 400	R3 200		
Catering cost (R70 x 2)	(R140)	(R140)		(½)
Sales margin	R3 260	R3 060		(1)

Margin is determined as the net income per product per passenger, the catering cost is therefore deducted from the sales price. The same principle will thus apply for accommodation, health, tourism, education, etc. where the measurement refers to an individual.

Sale margin price variance [(ASP – SSP) x AV]		R2 438 800	
Business class [19 656 x (R28 250-R27 000)]	Fav.	R24 570 000	(1)
Economy class [110 656 x (R3 200-R3 400)]	Unfav.	-R22 131 200	(1)

Budget – Passengers

Business class [732 flights x 24 passengers]	17 568	13,95%
Economy class [732 flights x 148 passengers]	108 336	86,05%
	125 904	100,00%

Actual passenger volumes in budgeted quantities

Business class [13,95% x (19 656+110 656)]	18 183	(1)
Economy class [86,05% x (19 656+110 656)]	112 129	(1)
	130 312	

Sales margin mix variance [(ASQ-ASQ in budget proportions) x Std margin] R34 438 740

Business class [(19 656 - 18 183) x R26 640]	Fav.	R39 240 720	(1)
Economy class [(110 656 - 112 129) x R3 260]	Unfav.	-R4 801 980	(1)
130 312 130 312			

Sales margin quantity variance [ASQ in budget prop.-Budget SQ) x Std margin] R28 748 780

Business class [(18 183 - 17 568) x R26 640]	Fav.	R16 383 600	(1)
Economy class [(112 129 - 108 336) x R3 260]	Fav.	R12 365 180	(1)
130 312 125 904			

Classical mix and yield variances: move from actual split to (actual in terms of budget ratio) to budget ratio.

Jet fuel (Note: 1 000 litres = 1 tonne)

Jet fuel price variance

		R6 000	(1)
Actual fuel cost per tonne (R200 163 600/33 360,6t)			
[(AP-BP) x AQ] = [(R6 000 - R4 654) x 33 360,6t]	Unfav.	-R44 903 368	(1)

Jet fuel consumption variance [(AQ-BQ) x BP]

Option 1 (budgeted not flexed)

Budgeted fuel per year (tonnes) (4,8 t x 10 hrs x 732)		35 136	(1)
[(33 360,6 t - 35 136 t) x R4 654]	Fav.	R8 262 711	(1)
Analysed as follows (**Alternative 1**)	Fav.	R8 262 711	
Saving due to four cancelled flights			
(4,8 t x 10 hrs x 4 flights x R4 654)		R893 568	(1)
Saving due to less flying time			
(4,8 t x 0,25 hrs x 728 flights x R4 654)		R4 065 734	(1)
Saving due to more efficient fuel usage			
(((4,8 t x 9,75 hrs x 728 flights) – 33 360,6 t) x R4 654)		R3 303 409	(1)
Or (((4,8 t – 4,7 t) x 9,75 hrs x 728 flights) x R4 654)			
OR: Analysed as follows (**Alternative 2**)	Fav.	R8 262 711	
Saving due to four cancelled flights			
(4,8 t x 10 hrs x 4 flights x R4 654)		R893 568	(1)
Saving due to less flying time			
(4,7 t x 0,25 hrs x 728 flights x R4 654)		R3 981 032	(1)
Saving due to more efficient fuel usage			
(((4,8 t – 4,7 t) x 10 hrs x 728 flights) x R4 654)		R3 388 112	(1)

Option 2 (budget flexed)

Standard fuel per year (tonnes) – flexed quantity		34 944	(1)
(4,8 t x 10 hrs x 728 flights)			
[(33 360,6 t - 34 944 t) x R4 654]	Fav.	R7 369 144	(1)
Analysed as follows (**Alternative 1**)	Fav.	R7 369 144	
Saving due to four cancelled flights		R -	(1)
Saving due to less flying time			
(4,8 t x 0,25 hrs x 728 flights x R4 654)		R4 065 734	(1)
Saving due to more efficient fuel usage			
(((4,8 t x 9,75 hrs x 728 flights) – 33 360,6 t) x R4 654)		R3 303 409	(1)
Or (((4,8 t – 4,7 t) x 9,75 hrs x 728 flights) x R4 654)			
OR: Analysed as follows (**Alternative 2**)	Fav.	R7 369 144	
Saving due to four cancelled flights		R -	(1)
Saving due to less flying time			
(4,7 t x 0,25 hrs x 728 flights x R4 654)		R3 981 032	(1)
Saving due to more efficient fuel usage			
(((4,8 t – 4,7 t) x 10 hrs x 728 flights) x R4 654)		R3 388 112	(1)

The consumption (quantity) variance has a number of components, all detailed in the question information. Note the 'soft skill' marks awarded.

(Rounding differences possible on above)	
Identification of 2 out of 3 categories above	(1)
Correct identification of Fav/Unfav variances	(1)
Available	**21**
Maximum	**20**

Part (b) Calculate the number of business class and economy class passengers that AA needed to transport in 2012 to break even

Note: *Only penalise candidates once if they have used budgeted costs.*

	Business Class	**Economy Class**	**Total**	
Actual number of passengers	19 656	110 656	130 312	
Alternative 1: Relative number of passengers (per batch)	1,0000	5,63		(1)
Alternative 2: Relative number of passengers (%)	15,08%	84,92%		

BC passengers were 19 656 and EC 110 656, the ratio is thus 1: 5,629 (110 656 ÷ 19 656)

Average fare	R28 250	R3 200	
(Less) catering cost	R (400)	R (140)	(1)
Contribution per passenger	R27 850	R3 060	(1c)

Fixed costs (*651 232 304*)	R 705 718 080	(1c)
Accommodation and refreshments (14 625 360)	15 331 680	(½)
Jet fuel (163 522 944)	200 163 600	(½)
Navigation (18 666 000)	19 091 800	(½)
Aircraft lease costs (42 140 000)	42 570 000	(½)
Electronic data costs (16 278 000)	16 456 000	(½)
Allocated overheads (415 600 000)	431 905 000	(½)
Cargo revenue (-19 600 000)	(19 800 000)	(1)

The cost of the meal is the only cost directly linked to a passenger and consequently variable (also part (a)). The revenue from cargo is by-product income and thus set-off against fixed costs.

MAC4861
MAC4862

Breakeven number of passengers:

Alternative 1

= Fixed costs / Contribution per batch			
= R705 718 080/[(R27 850 x 1) + (R3 060 x 5,63)]			
= 15 655,56 batches			(1c)
Passengers = 15 655,56 x 1 and 15 655,56 x 5,63	15 655,56	88 140,80	(1c)
Round to whole number (passengers)	15 656	88 141	(1c)

Alternative 2

= Fixed costs / Weighted avg. contribution per unit			
= 705 718 080/[(15,08% x27 850)+(84,92% x3 060)]			
= 103 807,56 passengers in total			
= 103 807,56 x 15,08% and 103 807,56 x 84,92%	15 654,18	88 153,38	
Round to whole number (passengers)	15 655	88 154	

Available	11
Maximum	10

Part (c) Analyse and discuss the importance of business class passengers to the profitability of the JNB-GRU route

Total breakeven revenue:		
(Business: (15 656 x R28 250) + Economy: (88 141 x R3 200))	R724 333 200	(1)
Total breakeven contribution:		
(Business: (15 656 x R27 850) + Economy: (88 141 x R3 060))	R705 731 060	(1)
Business class passenger contribution/total contribution FY2012 (19 656 x R27 850) / ((19 656 x R27 850) + (110 656 x R3 060))	61,8%	(1)
Or: Business class passenger revenue/total revenue FY2012 (19 656 x R28 250) / R909 381 200	61,1%	
Maximum contribution from economy class passengers in FY2012 186 seats x 728 x R3 060 [contribution calculated in (b)]	R414 348 480	(1)
Maximum contribution from business class passengers in FY2012 36 seats x 728 x R27 850 [contribution calculated in (b)]	R729 892 800	(1)
Margin of safety: (19 565 – 15 656) /19 656	20%	(1)

Comments

- Fixed overheads totalled R705,7m in FY2012. If the economy class was 100% full at all times, this would have only contributed R414,3m to overheads. (1)
- On the other hand, if business class was 100% full at all times, this would have more than covered fixed overheads, given that maximum contribution equates to R729,9m. (1)
- Business class passengers are much higher margin customers, making 9,1 times (27 850 /3 060) the contribution per passenger compared to economy class passengers. (1)
- Space configuration allocates 186 seats to economy and 36 to business class (5,17 times ratio). Yet on a contribution basis, business class passengers contribute 9,1 times more profits. (1)

- The business class contribution was 61,8% of the total contribution in FY2012, a vital contribution to overall profitability. (1)
- The breakeven revenue was R723,7m in FY2012, which was 79,6% (R724,3m / R909,3812m) of passenger revenue. (1)
- It follows that margin for error is very tight (20% margin of safety) and makes business class passengers' contribution even more critical. (1)
- Another key issue is the occupancy rates of business class passengers. It is important to understand the demand for business class seats, for example: if certain flights are regularly sold out for business class passengers, it may make sense to add more business class seats and sacrifice economy class seats. (1) (1)



The discussion of which product is more critical should centre on which product renders the higher contribution. As a check, you can test the actuals: BC 98,584% and EC 95,625%

Available	16
Maximum	**8**
Communication skills – clarity of expression	**1**
Total for part (c)	**9**

Part (d) Explain how AA could hedge against jet fuel price increases

- **Option contracts with banks** to limit the maximum fuel price it will have to pay. The option contract could stipulate either the jet fuel price or Brent crude price. The option premium will represent the maximum cost if fuel prices decline, but will provide a valuable cap on future prices for the duration of the derivative contract. (1) (1)
- To reduce the hedging costs a **collar** could be purchased. This will entail purchasing a call option and selling a put option on jet fuel prices. In this way the option premiums paid and received will offset one another reducing the overall hedge cost, and also reduce the variability in which the jet fuel price will fluctuate. (1) (1)

Forward contract

- The company should enter into **forward cover contracts** to hedge the US$/ZAR exchange rate. This will not fully hedge against jet fuel price changes but will provide some protection against a strengthening US$, which will indirectly translate into higher fuel prices. (1) (1)
- The company should enter into **supply contracts** with jet fuel suppliers to lock in prices, which would provide certainty as to costs for next 3, 6 or 12 months. (1) (1)
- Purchase **oil futures,** where the buyer agrees to buy an underlying asset (jet fuel) at a future date at a price agreed today. (1) (1)

Natural hedges (offsetting costs with income in foreign currency)

- For example: The sale of tickets to foreigners in their currency (i.e.: dollars, euros). These funds should be kept in foreign accounts in order to pay for fuel in a foreign currency (1) (1)

- Use of **swaps**, where prices are fixed with a supplier but there are actual payments made at the spot rate. Adjustments are then made to decrease/increase the actual price paid in the spot transaction. (1) (1)

- AA should purchase **jet fuel for future use when prices are favourable**. AA will need to finance these inventory holdings but will have certainty regarding fuel costs. (1) (1)



Candidates were expected to identify the hedge **and** how it could be applied in the context of the question.

Available	**16**
Maximum	**6**
Total for part (d)	**6**

Part (e) Identify and describe four major risks faced by AA with respect to the JNB-GRU route

- AA has **high fixed costs,** which means there is a substantial risk of making losses if passenger revenue is insufficient to cover expenses. In FY2012 the breakeven revenue was ~80% (724 333 200 / 909 381 200) of actual revenue which indicates a high risk of future losses (high operating leverage). (1) (1)

- AA is the **only African airline** which offers a flight from South Africa to South America. In the event that other airlines obtain routes to South America, passenger numbers could decline due to **increasing competition**. This could have a significant impact on the profitability and sustainability of the JNB-GRU route. (1) (1)

- **Jet fuel prices** are volatile and represent >26% (200 163 600 / 257 941 350 + 490 931 000) of total operating costs. In the event that fuel prices increase significantly, AA may not be able to increase airfares to cover cost increases, with the result that it would incur operating losses. (1) (1)

- Possibility of AA losing the **single landing slot** to JNB-GRU route due to poor business practices or other uncontrollable factors (e.g.: Brazilian government intervention). Or the possibility that the agreement is not renewed. (1) (1)

- The **transportation of cargo** on the same flights as passengers on this route opens AA up to legal/regulatory risk. (1)

- **Political risk, as emerging market countries** may exhibit political instability. This could reduce the appeal of the country as a business and leisure destination, having a significant impact on the profitability of the JNB-GRU route. (1) (1)

- AA only has **landing rights for one flight per day**. Based on the key assumptions for budget purposes, it will take 4 days to clear the back log of passengers if a flight was cancelled (strikes, weather conditions etc.). AA may lose business if potential passengers do not have confidence in the airline's ability to deliver when flights are cancelled. (1) (1)

The key risks focus on the business environment and the major cash costs. Note that the allocated cost is not a direct cash cost but dependent on the allocation method used. Communication skills will focus on identifying the relevant key risks and coherent explanation thereof.

Available	**13**
Maximum	**8**
Communication skills – logical argument; clarity of expression	**2**
Total for part (e)	**10**

PART B

Part (f) Discuss, with reasons, any concerns you have about the current (i.e. post-listing) corporate governance arrangements of AA

Note the structure of the solution, starting at board appointment and closing with consequences of non-compliance. Communication skills will focus on the structure and the key issues addressed.

1.	**Appointment of directors**	
1.1.	There is no chairman of the Board of Directors. The King Code requires the Board to be chaired by an independent non-executive director.	(1) (½)
1.2.	With the exception of Mr Barclay and Alexia Viljoen, the directors who have been appointed appear to lack the necessary competence and experience required of individuals to serve on the Board – this would be the case for both Kingston and Luhle Jacobs (relevant experience) and Messrs Abbott and Callan (age and experience).	(1) (1)
1.3.	King Code (2.84) requires that directors do not hold more directorships than is reasonable for them to exercise due care, skill and diligence. Given that Mr Barclay holds various directorships and given his age, this could represent a potential risk for AA.	(½) (1)
1.4.	According to King Code (2.18) the Board should be of sufficient size, diversity and demographic make-up to be effective. AA needs to find independent, suitably skilled and demographically acceptable directors. It can be argued that the Board at present is not representative in terms of gender and a risk exists that it is also not representative in terms of race.	(½) (1)
1.5.	In terms of principle 2.17 of the King Code, the Board and not one of the major shareholders, i.e. Mark Brooney, as is the case at AA, should appoint the chief executive officer.	(½) (1)
1.6.	The Board should comprise a balance of power, with a majority of non-executive directors. The majority of the non- executive directors should be independent (principle 2.18). However:	 (½)
1.6.1.	Six directors have been appointed, three executive, and three non-executives – i.e. there is not a majority of non- executives.	(1)
1.6.2.	The risk exists that none of the directors appointed to the Board can be considered independent, as they have been appointed by the Jacobs's (i.e. by the CEO and COO) OR that the individual could have been an employee of AA or that the employee works for a material customer of AA.	(1)

1.6.3.	The desirability of appointing a husband and wife (familiarity threat to objectivity) into two of the three senior executive posts in AA is questionable – as this will undermine effective control and governance (incl. segregation of duties and management oversight) and would be a succession planning	(1)
1.7.	In terms of principle 2.19 of the King Code, directors should be appointed through a formal process – however, with the approval of Mark Brooney, the Jacobs' nominated the directors	(½) (1)
1.8.	In terms of section 66(4) of the Companies Act, at least 50% of the directors must be elected by shareholders – the appointments have been made without the involvement of all the shareholders.	(½) (1)
1.8.1.	Marc Brooney can only have a right to vote on the appointment of directors in accordance with his share- holding. A 40% shareholding will not give him the ability to appoint the majority of the Board of directors. The other shareholders' approval should therefore have been sought in the appointment of	(1)
1.8.2.	In addition, the JSE does not permit a minority shareholding to exercise negative control as a result of any contract or agreement.	(1)
2.	**Remuneration of directors**	
2.1.	There is reason to believe that principle 2.25 of the King Code (that companies should remunerate directors and executives fairly and responsibly) has not been applied appropriately by AA. For example –	(½)
2.1.1.	All directors (including the non-executives) will receive share options on listing and annually thereafter - in terms of recommended practice of King Code 2.25.4 non-executive directors should not receive share options, but rather a base fee and attendance fees for meetings.	(1)
2.1.2.	Principle 2.25.147 (King), states that companies should adopt remuneration policies and practices that create value for the company over the long-term. Given that 3-year appointment terms apply at AA and that directors negotiate large termination payments if not reappointed indicates a shortcoming in the corporate governance of AA.	(½) (1)
2.1.3.	This indicates a lack of a remuneration policy aligned with the strategy of the company and linked to individual performance (a recommended practice per the King Code).	(1)
2.1.4.	A special resolution every two years in terms of which shareholders is required to approve directors' remuneration (in terms of section 66 of the Companies Act) – the shareholders need to approve the directors' remuneration.	(1)
2.1.5.	Principle 2.27.186 (King Code) stipulates that the remuneration policy be discussed at the annual general meeting with shareholders being able to express their views on said policy.	(1)
2.1.6.	Principle 2.25.150 (King), states that the remuneration committee should assist the Board in setting and administering remuneration policies across the company, including for directors.	(1)
2.1.7.	Principle 2.25.165 (King Code) – directors should not be paid large (balloon) severance packages. Hence the negotiation of large severance benefits being negotiated in advance creates an ethical dilemma.	(½) (1)
3.	**Succession planning and rotation of directors**	
3.1.	While appointing directors on three-year contracts is acceptable, the previous practice of replacing virtually all directors on the termination of contracts is of concern. In terms of recommended practice 2.17.5 of the King Code, the Board should ensure that there is succession planning for the CEO and other senior executives and officers.	(½) (1)
3.2.	Recommended practice 2.18.6 of the King Code states that at least one third of the non-executive directors should rotate every year (independence of the non-execs threatened). If all directors' contracts terminate after three years then no rotation policy exists at AA.	(½) (1)

4.	**Board Committees**	
4.1.	In terms of recommended practice 2.23.6 of the King Code the company should also have established a risk committee (in addition to the audit and remuneration committees) – there is no risk committee at present.	(½) (1)
4.2.	Moreover, in terms of recommended practice 2.19.1 of the King Code, the company should have established a nomination committee to assist with the process of identifying suitable Board members – which does not appear to have happened (e.g. the Board members were 'nominated' by the Mark Brooney and the Jacob's).	(½) (1)
4.3.	In terms of Regulation 43 of the Companies Regulations, 2011, every listed public company (such as AA) should establish a social and ethics committee within 12 months of the effective date of this regulation – however, no such committee appears to have been established.	(½) (1)
4.4.	Given that there are no independent directors on the Board –	
4.4.1.	The remuneration committee will not be adequately constituted by at least the majority of independent non-executive directors – as recommended by par. 2.23.7 of the King Code.	(1)
4.4.2.	The audit committee should be comprised exclusively of independent directors (minimum of three) – such an arrangement would be contrary to par. 3.2 of the King Code.	(1)
4.4.3.	Should any of the current independent directors be elected as chairperson of the Board, the audit committee would no longer comply with the aforementioned requirement since the chairperson may not be a member of the audit committee.	(1)
4.5.	Regulation 42 prescribes that at least one-third of the members of an audit committee must have academic qualifications, experience in economics, law, corporate governance, finance, accounting, commerce, industry, public affairs or human resource management – this could be an issue for AA (only Mr Barclay potentially has these skills).	(½) (1)
4.6.	The audit committee, although established, is not effective as it appears to have played no role in the reappointment of United Auditors (in terms of King).	(1)
4.6.1.	The Committee is required to nominate the external auditor for appointment and approve the terms of engagement (per section 94 of Companies Act). According to part 3.9 of the King Code, the committee is responsible for recommending the appointment of external auditors to the Board – this does not appear to have happened..	(½) (1) (½) (1)
4.7.	This committee is required to monitor and report on the independence of the external auditor – however, there are significant threats to the auditor's independence which do not appear to have been addressed (King 3.9.3).	(½) (1)
4.7.1.	In terms of the SAICA Code of Professional Conduct (ET 290.139), in a situation where a key audit partner joins the audit client that is a public interest entity (such as AA), independence would be compromised given that the client has not yet issued audited financial statements covering a period of not less than 12 Months (cooling off period is required).	(1)
4.8.	In terms of King 3.9.4 the audit committee is required to define a policy for non-audit services by the auditor and approve contracts for the external audit firm to render non- audit services – however, the outsourcing of the internal audit services appears to have been done without reference to the audit committee (it was done in terms of a Board resolution).	(½) (1)
4.9.	In terms of section 290.200 of the SAICA Code of Professional Conduct, the appointment of United Auditors to render internal audit services is of concern as the Code states that for a public interest entity (such as AA), the services shall exclude controls and systems relating to financial reporting.	(½) (1)

4.10.	King requires that the internal audit function be independent of management. Given that the internal audit function has been outsourced to United Inc. (small firm) where Mr Jacobs (now the CEO of AA) was a partner, would result in the independence of the internal audit function being questioned (familiarity).	(½) (1)
4.11.	The committee is required to review the quality and effectiveness of the external audit process – however, the competence of the external audit firm to perform the engagement is in doubt (King Code, par. 3.9.6). It is doubtful whether three partners and 30 trainees (capacity) will satisfy the ISQC 1 skills and competence requirements given that AA is a fairly large listed client, with complex operations.	(½) (1)
4.12.	The audit committee needs to ensure that the JSE approves the appointment of the audit firm – given United Inc.'s limited resources, this is debatable.	(1) Bonus
5.	In terms of the Companies Act (section 90), the auditor must be appointed at every annual general meeting of the company (i.e. by the shareholders). Only if no such appointment was made, can the Board of directors proceed to make an appointment. The fact that the appointment was made without the knowledge of the shareholders is therefore of concern.	(½) (1)
6.	The manner in which Alexia Viljoen's resignation was communicated gives rise to the following concerns:	
6.1.	The communication was done by way of a press conference rather than a SENS announcement – which is inconsistent with the JSE Listings Requirements.	(1) Bonus
6.2.	It is doubtful whether the communication satisfies principle 8.5 of the King Code which requires transparent and effective communication with stakeholders in order to build and maintain their confidence – specifically, it appears that the reason cited for Alexia's resignation ('to pursue personal interests') is at odds with her recent acceptance of the appointment. The information was selectively released.	(½) (1)
7.	Given the directors' conduct generally, there is reason to believe that they are acting contrary to their statutory fiduciary duties. For example:	(1)
7.1.	In terms of section 76(2) of the Companies Act, a director must not use the position as a director to gain an advantage for him/herself – yet directors negotiate large payments on the termination of their contracts, and they participate in the rigging of tender processes.	(½) (1)
7.2.	In terms of section 76(3) of the Companies Act, directors must exercise the powers and functions of a director in good faith, in the best interests of the company and with a reasonable degree of care and skill – however, failing to focus on the employees (in an industry where skills are in short supply (e.g. pilots and engineers)) and participating the rigging of tenders, is not consistent with the statutory requirements.	(½) (1)
7.3.	Principle 2.14 (King) requires the board and directors of the company to act in the best interests of the company, which appears not to be happening given the illegal tendering process and large severance packages paid.	(½) (1)
8.	It is doubtful whether the executive directors who 'secretly' received incentives from the recipients of tenders disclosed their interests in the contracts prior to entering into these contracts in the manner required by section 75 of the Companies Act.	(1)
8.1.	By conducting themselves in this manner, the directors would also not be complying with the following principles of the King Code:	
8.1.1.	The Board should ensure that the company's ethics are managed effectively (principle 1.3). According to the King Code, 'Good corporate governance requires that the Board takes responsibility for building and sustaining an ethical corporate culture in the company' the action breaches this requirement since the Board is the problem.	(½) (1)

8.1.2.	The Board should ensure that the company complies with applicable laws (principle 6.1) – such as the Prevention and Combatting of Corrupt Activities Act (PRECCA). Alternatively: principle 2.9 also requires compliance	(1) (1) Bonus
9.	The fact that there is poor corporate governance in place especially from the leadership could result in fraud in other parts of the company. This is more likely when employees are dissatisfied.	(1)
10.	Now that Alexia Viljoen has resigned, AA is operating without a financial director. King (principle 2.18.5) requires the company to have a CEO and chief financial officer indicating non-compliance with King.	(½) (1)
11.	There is doubt as to whether or not a company secretary has been appointed which results in non-compliance with the Companies Act.	(1)
12.	Given the fact that the directors have contravened the Companies Act, the directors need to be aware that they can be held personally liable in such instances where breaches in complying with the Companies Act have taken place.	(½) (1)
13.	From the perspective of the external auditors of AA, they would need to consider the requirements of the Audit Profession Act in respect of the tender rigging qualifying as a reportable irregularity which they would need to report on.	(1)
13.1.	An unlawful act has occurred through the tender rigging as well as non-compliance with various statues (refer above).	(1)
13.2.	This act has been committed by management (the Board of Directors)	(1)
13.3.	Material breach of fiduciary duty (Section 76 – best interest of company) has occurred OR Material financial loss (tender rigging and severance packages) has occurred for the company OR Fraudulent act (tender rigging)	(1)
14.	Given the relevant concerns identified one can conclude that AA is not a responsible corporate citizen.	(1)
15.	Any other valid relevant point (limited to)	(1)
	Available	**75½**
	Maximum	**43**
	Communication skills – clarity of expression; logical argumentation	**2**
	Total for Part (f)	**45**

QUESTION 31

EXPERIENCE AFRICA HOLDINGS **SUGGESTED SOLUTION**

(Source: QI 2011 Paper 3)

PART A (only PART B applicable for Management Accounting)

(a) Audit of Experience Africa under the new Companies Act of 2008, and whether its auditors are in compliance therewith

Company	*Experience Africa is a public company and accordingly must be audited – thus they comply with the Act (section 30 2(a)).*	*(1)*
Auditors	*The same individual may not serve as designated auditor of a company for more than five years (section 92 (2)).*	*(1)*
	As Mr Bob Dlamini is the designated auditor for the past six years, he is contravening the Act.	*(1)*
Note: He is also contravening ISA 220 and the Code of Conduct requiring a five year rotation.		*(1)*
	Maximum	*3*

(b) Discuss any ethical, corporate governance and statutory issues arising from EA SA entering into a new business venture (Beyond 2010) with Interstate as was presented to and approved by the board of directors of EA Holdings Ltd.

SAICA Code of Conduct

Sonny Ndlovu is a CA (SA) and must comply with SAICA's Code of Conduct.

Accordingly he is bound by the ethical principles of integrity, objectivity, professional competence and due care, confidentiality and professional behavior. (1)

Professional behavior imposes an obligation on professional accountants to comply with relevant laws and regulations and avoid any action that may bring discredit to the profession. (1)

Thus Sonny has personal financial interests in the contract as Interstate Bus Tours are owned by his wife. He did not declare his interest or refrain from taking part in the awarding of the contact to her company, thus contravening the Companies Act (section 75) and SAICA Code of Conduct. (1)

Members must also always act with honesty (*and integrity*) and straightforwardness. Non disclosure of his personal interest in the contract, and non-compliance with the Companies Act, may bring the profession in disrepute. (1)

This transaction brings a self-interest threat to objectivity and the threat is significant as his wife is a major shareholder. (1)

Corporate Governance (King III)

The board should provide effective leadership based on an ethical foundation.

This is characterized by the ethical values of responsibility, accountability, fairness and transparency (Principle 1.1). (1)

The board should ensure the company complies with applicable laws and consider compliance with non-binding rules, codes and standards (Principle 2.9). (1)

The board and its directors should act in the best interests of the company (Principle 2.14). (1)

Thus the transaction with Interstate Bus Tours appear not to be in the best interest of the company as they will provide the services below cost, thus potentially resulting in losses for the company. (1)

This situation is further aggrieved as Sonny's Ndlovu has a personal financial interest in the contract which he failed to disclose, and voted on as a board member. (1)

Companies Act

Personal financial interest in the contract by Sonny (section 75)

Interstate Bus Tours are owned by Sonny Ndlovu's (financial director) wife, a related person i.t.o. section 2. As the transaction was approved by the board on 15 September he had a personal financial **interest in the transaction** and should have: (1)

- disclosed the interest and its general nature thereof before the matter was considered; (1)
- disclosed to the meeting any material information relating thereto; (1)
- disclosed any observations or pertinent insights thereto; (1)
- must have left the meeting after the disclosure of his interest; (1)
- may not have taken part in the consideration (vote) on the matter; (1)
- forms part of the quorum of the meeting.

He must also not execute any documents on behalf of the board regarding the matter, unless requested by the board to do so. (1)

Available 7
Interest maximum 4

A decision, transaction or agreement in which a director has a personal financial interest is valid if it was approved by the board (after the interest have been disclosed, etc.); or (1)

If it has been ratified by the shareholders. (1)

Thus the transaction appears not to be valid, but is, however, not void (section 22 *ultra vires*), (special resolution). (1)

Possible non-compliance with IFRS as a result of an onerous contract not correctly accounted for (contracts below cost), resulting in non-compliance with the Companies Act. (1)

Standards of directors conduct (section 76)

A director of a company must: not use the position of director, or information obtained while acting as a director, to gain an advantage for him/herself or another person other than the company or wholly-owned subsidiary; and (1)

A director must exercise the powers and perform the functions of director in good faith and in the best interest of the company; and (1)

with the degree of care, skill and experience that may be reasonably expected of a like person in a similar position. (1)

If a director does not comply with the above he/she can be held liable for costs, losses or damage sustained i.t.o. the common law. (1)

Reckless conduct (section 22)

If the company is making losses by charging less (R4 000) than the cost to the company (which is highly possible), the directors might have also acted recklessly and negligently and can be held liable for loss, damages or costs sustained by the company. (1)

Available 22
Maximum 19

(c) Discuss whether the board of directors of EA Holdings Ltd complies with the *King Code of Governance Principles for South Africa, 2009* (King III), regarding the appointment of Ms Player- Carter as chairman.

King III requires the board to elect a chairman who is an independent non-executive director, and (1)
The CEO should also not fulfill the role of chairman of the board (Principle 2.16). (1)

Where the board appoints a chair who is not independent, they should appoint a lead independent director (Principle 2.16.3). (1)

The board should ensure the company complies with applicable laws and consider compliance with non-binding rules, codes and standards (Principle 2.9). This includes the JSE's listing requirements. (1)

King III is based on the Apply or Explain principle. (1)

The Board of Directors could therefore conclude that an interpretation of the governance principle which is different from the recommended practice contained in the King Report could still lead to good corporate governance. (1)

Different application of the principle will however only result in compliance should the Board disclose the reasons for not adopting the recommended practice in the integrated report. (1)

It is clear that the Board has no intention to disclose the fact that the recommended practice has not been adopted. (1)

It should also be noted that the more established the recommended practice become, the more likely a court in considering whether directors have acted in the best interest of the company, will regard the King Code as the benchmark for the required standard of care. (1)

Response of auditors – Dlamini and Partners Inc.

The auditor has a duty to respond, as there is non-compliance with the King Report, which is a listings requirement. This will include: (1)

Communicating with those charged with governance (ISA 260), namely the audit committee and possibly the board regarding the non-compliance with King and the Listing requirements. (1)

Consider reporting a Reportable Irregularity to IRBA as a result of non-compliance with JSE requirements and King III. (1)

Consider the auditor's reporting responsibilities in terms of ISA 720 regarding other information, such as corporate governance reporting and inconsistencies or misstatements therein. (1)

Response in terms of ISA 240:
Considerations of increasing risk assessment at overall financial statement level and response accordingly through incorporating an element of unpredictability, professional scepticism, involve more senior and experienced staff members. Less reliance on management representation letters. (1)

Available 14
Maximum 7

(d) Discuss the fraud risk factors and their potential impact on the audit report arising from the new business venture between Interstate and EASA.

Fraud Risk Factors

Risk of material misstatement: Correctness of the accounting treatment: related party transactions (IAS 24)

The auditor should consider whether the transaction is correctly disclosed as a related party transaction; as (1)
Mr Sonny Ndlovu is a director of the company and the contract is with his wife's company (the party is a close member of the family of a key management member; para 9(e)); (1)
If the transaction is not correctly disclosed i.t.o. IAS 24, the company is not complying with the accounting standards and are also contravening the company's act (section 29 (a)); (1)

Considerations i.t.o. ISA 240: Fraud and Error

The auditor should consider if the transaction is at arm's length and not detrimental to the company's financial position (of a fraudulent nature). (1)

The fact that the transaction price is below cost to the company seems to indicate the possibility that it is not at arm's length, and given the fact that it is with Sonny's wife, it could be seen as prejudicing the minorities. (1)

It can also be argued that the director (sonny) are not acting in the best interest of the company, and thus are in breach of his fiduciary duties. (1)

Considerations i.t.o. ISA260: Communication to those charged with Governance

The auditor should consider his reporting responsibilities i.t.o ISA 260 "Communication to those charged with governance". (1)

In the above circumstances he should report the company's non-compliance to the Board and fully document this. (1)

Considerations i.t.o. ISA720: Auditors responsibility relating to other information issued with the financial statements

He should also consider his responsibility i.t.o ISA 720 "Auditors responsibility relating to other information issued with the financial statements" and whether the corporate governance reporting therein is not misleading, inconsistent or misstatement of fact. (1)

Reckless or fraudulent conduct of the business (Companies Act section 22)

If the transaction is not at arm's length, it can be argued that the business is carried on recklessly or with a fraudulent purpose. (1)

Impact on audit report

Reportable Irregularity

The auditor should consider whether the above does not constitute a reportable irregularity that should be reported to IRBA. (1)

Qualification of audit report

The auditor should consider whether the above does not require a qualification of the audit report. (1)

Include a paragraph on other legal and regulatory requirements. (1)

Consideration and qualitative and quantitative nature of the misstatement. The non-compliance with laws is qualitative of nature and therefore material. (1)

Available 14
Maximum 7

PART B

Part (e)

Available capacity – six-night packages (150 x 30%)	*45*		*(½)*
Interstate demand (in six-night packages)	*60*		*(½)*
Normal six-night packages sacrificed	*15*		*(1)*
Total approach			
	Current Status quo	*Interstate Option*	
Revenue (18 384 x 105)	*1 930 320*		*(½)*
[(105 – 15) x 18 384] + [120 x 4 000]		*2 134 560*	*(1)*
Management fees paid to Interstate		*(24 000)*	*(½)*
Accommodation costs (R8 000 x 105)	*(840 000)*		*(½)*
Unchanged as these are fixed in nature		*(840 000)*	*(1)*
Food & beverages	*(315 000)*		*(½)*
[(3 000/2) x 120] + [3 000 x 90]		*(450 000)*	*(1)*
Cleaning services	*(52 500)*		*(½)*
[(R150/2) x 270] + 52 500		*(72 750)*	*(2)*
Game drives (105 x R900)	*(94 500)*		*(½)*
[R400 x (90 + 120)] + (105 x R500]		*(136 500)*	*(2)*
Aircraft & flight costs	*(157 500)*	*(157 500)*	*(½)*
Hot air-balloon excursions	*(126 000)*		*(½)*
[(90 + 120) x 1 000]		*(210 000)*	*(1)*
Electricity	*(23 100)*	*(31 200)*	*(1)*
Entertainment costs (R2 000 x 8)		*(16 000)*	*(1)*
Operating profit	*321 720*	*196 610*	*(1)*
Difference in operating profit by entering into arrangement with Interstate		*(125 110)*	*(1)*

Or

Total approach		
Revised number of six day packages	*90*	*(1)*
	Interstate option	
Revenue		
Six day packages [90 x R18 384]	*1 654 560*	*(1)*
Interstate [120 x R4 000]	*480 000*	*(½)*
Management fees paid to Interstate	*(24 000)*	*(1)*
Accommodation costs	*(840 000)*	*(1)*
Food & beverages [(3000/2)x120]+[3000x90] = 180 000 + 270 000	*(450 000)*	*(1)*
Cleaning services		
Additional costs = [R150 x 50% x (150 -105) x 6]	*(20 250)*	*(2)*
Current costs [R500 x 105]	*(52 500)*	*(½)*
Game drive costs		
Existing fixed costs [R500 x 105]	*(52 500)*	*(1)*
Variable costs [(120 + 90) x R400]	*(84 000)*	*(2)*
Aircraft & flight costs	*(157 500)*	*(½)*
Hot air-balloon excursions		
[(90+120)x1000]	*(210 000)*	*(2)*
Electricity	*(31 200)*	*(½)*
Entertainment costs	*(16 000)*	*(1)*
Operating profit	*196 610*	*(1)Principle*

MAC4861
MAC4862

Current monthly operating profit [105 x (R18 384 - R15 320)]	*321 720*	*(1)*
Difference in operating profit by entering into arrangement with Interstate	*(125 110)*	*(1)Principal*
	Maximum	*18*

Incremental method

Per guest on standard six day package

	Current Status quo	
Revenue	*18 384*	
Food & beverages	*(3 000)*	
Game drives	*(400)*	*(1)*
Hot air balloon excursions	*(1 200)*	*(1)*
Electricity [(31 200 – 23 100)/270] = R30 per bed night	*(180)*	*(1)*
Contribution per guest on six night packages sacrificed	*13 604*	

Revenue	*480 000*	*(½)*
Management fees	*(24 000)*	*(½)*
Accommodation costs	*0*	*(1)*
Food & beverages	*(180 000)*	*(1)*
Cleaning costs	*(20 250)*	*(2)*
Game drives	*(48 000)*	*(1)*
Aircraft & flight costs	*0*	*(½)*
Hot air-balloon excursions		
Additional costs [120 x R1 000]	*(120 000)*	*(1)*
Overall cost savings from supplier 'rebate'	*18 000*	*(1)*
Electricity	*(10 800)*	*(1)*
Contribution lost: 6 day packages [15 x R13 604]	*(204 060)*	*(½)*
Additional entertainment	*(16 000)*	*(1)*
Incremental impact on monthly operating profit	*(125 110)*	*(1)*
	Available	*18*
	Maximum	*18*

Incremental method				
Only penalize once for incorrect # of bus tour packages				(1)
Revenue [120 x R4 000]			480 000	(½)
Management fees [5% of 480 000]			(24 000)	(½)
Food & beverages [120 x R3 000/2]			(180 000)	(1)
Cleaning costs				
		Guest nights		
	70% occupancy = 105 x 6	630		
	100% occupancy = 150 x 6	900		
	Additional guest nights	270		(1)
	Extra cost = R150/2 x 270		(20 250)	(1)
Game drives [R400 x 120]			(48 000)	(1)
Hot air balloon excursions				
	Additional costs 120 x R1 000		(120 000)	(1)
	Cost savings [90 x R200]		(18 000)	(2)
Electricity				
	100% occupancy	31 200		
	70% occupancy	23 100		
		8 100		
	Total variable cost [8 100/30%]	27 000		(1)
	Additional cost per guest night [27 000/(150 x 6)]	R30		(1)
	Additional costs [R30 x (120 x3)]		(10 800)	(1)
Additional entertainment [R2 000 x 8]			(16 000)	(1)
Fixed costs therefore ignore				
Accommodation				(½)
Game rangers salaries etc				(½)
Contribution lost per guest				
Revenue		18 384		
Food & beverages		(3 000)		
Game drives		(400)		(1)
Hot air balloon excursions		(1 200)		
Electricity [R30 x 6]		(180)		(1)
Contribution per guest		13 604		
Six night packages sacrificed [105 – [150 – (120/2)]		15		(1)
Total contribution lost [15 x R13 604]			(204 060)	(1) Principle
			(125 110)	(1) Principle
Alternate approach to hot air balloon excursions				
Cost included in contribution calc. per guest				
Cost savings per six night guests [105 x R200]		(1 000)	(21 000)	
			Maximum	18

Part (f)

Pricing & contribution

Pricing of 3 day packages versus 6 day packages (R4 000 x 2 versus R18 384) – out of syncronisation. (1)

Impact on pricing of 6 day packages if word gets out re deal between Experience Africa and Interstate. (1)

Contribution per guest night from entering into deal with Interstate versus existing contribution margins. (1)

Interstate guests receive more 'value for money' than current guests (entertainment, hot air balloon excursion despite only staying 3 nights). (1)

High occupancy levels

Are lodges able to operate at 100% occupancy without **declining service levels**? (1)
Impact on **staff morale** of running at 100% occupancy? (1)
Impact on environment of 100% occupancy levels? (2)
Growth in six night packages not feasible given full utilization. (1)
Loyal guests discouraged from returning (high occupancy/poor service levels etc.). (1)
Operating at 100% capacity may disappoint guests (over-bookings, rooms out of service (1)

Changes in cost structures

Need to **employ more staff** (cleaners etc) to cope with high occupancy? (1)
Given ↓ in # of six night packages, **aircraft utilization even less.** (1)
Staff (game rangers etc) may **demand salary increases** due to higher workload. (1)
Any **other cost increases** due to high occupancy? E.g. vehicle maintenance/road maintenance. (1)

Interstate

7 day package costs **R17k yet EA SA only receives R4k?** (1)
Can Interstate **guarantee a minimum # of packages** monthly? (1)
Should have **offered Interstate fewer guest nights** (e.g. 90 instead of 120 packages). Rather retain more six night package guests contributing higher margin. (2)
Risk of **poor service delivery** by Interstate **(bus breakdowns, booking mistakes etc).** (1)
Risk of **Interstate credibility & reputation** negatively affecting EA SA? (1)
Interstate doing bookings will mean they **retain guest data base.** (2)
Conflict of interest – Sonny's wife owns Interstate. (2)
Service level agreement with Interstate? (1)
Credit risk of Interstate/risk of non-payment? Financial position should be investigated. (1)
Risk of **cannibalizing** own guest base? Six day guests opting for bus tours? (1)
Entertainment may **annoy** other guests and/or wild life. (1)
Risk of over-reliance on Interstate. (1)
Duration of agreement with Interstate? (1)
Interstate management fee is high, should have negotiated lower fee. (1)

Alternate plans to boost revenue

Any other options to ↑ **occupancy levels**? E.g. offer 3 day package at lower rates? (1)
Other joint venture options? Another tour bus operator? (1)
EA SA **offer Interstate package directly**? (1)

Decision making

IRR/NPV approach should be followed to evaluate viability. (1)
Compare **contribution per normal guest and Interstate guest/impact on profitability.** (1)
Business plan (risks & key issues) should have been prepared. (1)

Other factors

Veracity of **market research**? Tourism may decline post World Cup. (1)
Impact of **economic conditions** on tourism? (1)
Competitors' offerings & pricing. (1)
Any other valid factor (1)
Maximum 18

Part (g)

Cost plus pricing

- Experience Africa should rather charge what guests will pay/market related rates (target costing). (2)
- Cost-plus pricing does not encourage rigorous focus on cost containment. In fact, it encourages company to pass on cost inefficiencies to customers. (2)
- Capacity utilization has a significant impact on pricing in cost-plus pricing. If company operates at 100% capacity, prices will reduce whilst at 50% occupancy the prices will increase significantly. (2)
- Cost-plus pricing ignores competitive pricing and realities of the market (EA SA should rather charge what guests will pay). (2)

Bundling of services & pricing

- Guests are charged for hot air-balloon excursions irrespective of whether they use this service. Guests not electing to avail themselves of these services may find the overall pricing of package too expensive. (2)
- By including all food & beverages in package costs, Experience Africa may be encouraging guests to over-indulge or take advantage of offering. Perhaps guests should pay for beverages consumed? (2)
- Local guests may be prejudiced by aircraft transfer costs – they pay R1,500 for service they may not use. (2)
- Experience Africa's aircraft fleet has sufficient capacity to cope with 100% occupancy. Given certain guests opting not to use service and the Interstate arrangement, not all guests will be using service. Experience Africa should consider reducing aircraft fleet or outsourcing this service. (2)

Pricing strategy

Focus is on upper end of market, so guests may be **price insensitive.** (2)
Higher prices may signal higher quality and create demand. (2)
Should **benchmark** against competitors pricing and offerings. (2)
Impact of **global economic conditions**? May need to consider ↓ing prices? (2)
Offer **discounts for advance bookings? Last minute specials?** (2)
Any other valid comment (2)
Available 28
Maximum 10

Part (h)

	Option 1	*Option 2*	
Revenue	*17 000*	*17 000*	*(½)*
Management fee	*200*	*200*	*(½)*
Total revenue	*17 200*	*17 200*	
Experience Africa cost	*(4 000)*	*(4 000)*	*(½)*
Accommodation, meals and sight seeing	*(3 500)*	*-*	*(½)*
Outsourced costs	*-*	*(7 500)*	*(½)*
Contribution per guest	*9 700*	*5 700*	*(1)*
Contribution	*56,4%*	*33,14%*	*(1)*
Fixed costs			
Bus costs	*(220 000*		*(1)*
Administration costs	*(60 000)*	*(60 000)*	*(½)*
Breakeven revenue	*(280 000)*	*(60 000)*	
Breakeven # of guests	*28,8*	*10,5*	*(1)*
Estimated monthly operating profit based on 120 guests	*884 000*	*684 000*	*(1)*
Operating profit margin 33.1%	*42,8%*	*33,1%*	
Degree of operating leverage (Contribution margin/operating profit margin)	*1,32*	*1,00*	*(2)*

Recommendation 2

- *The outsourced option is preferable as breakeven revenue is much lower than the other option. This represents a lower risk option which is appropriate for a start up business. The operating leverage is lower however, this reflects a lower level of fixed costs inherent in this option.*

OR

- *The option of delivering the entire service is a more profitable alternative reflected in the much higher contribution % and higher degree of operating leverage. At the expected revenue levels, this option will deliver R200,000 more profit on a monthly basis.*

Available	*13*
Maximum	*13*

Alternative

		Option 1	
Revenue	*2 040 000*	17 000	
Variable costs			
EA SA package fee	*(456 000)*	(3 800)	(½)
Accommodation, meals & sight seeing	*(420 000)*	(3 500)	(1)
Contribution per guest	*1 164 000*	9 700	
Contribution %		57,1%	
Fixed costs			
Bus costs		(220 000)	(1)
Administration costs		(60 000)	(½)
		(280 000)	
Bonus mark: Discussion of fixed & variable component of bus costs			*(1)*
Breakeven revenue *[Alternate calc. 496 495]*		490 722	(1) Principle
Breakeven # of guests		28,9	(1) Principle
Operating leverage (contribution/operating profit) = [1164000/884000]		1,32	(1) Principle
		Option 2	
Revenue	*2 040 000*	17 000	
Variable costs			
EA SA package fee	*(456 000)*	(3 800)	(½)
Outsourced costs	*(900 000)*	(7 500)	(1)
Contribution per guest	*684 000*	5 700	
Fixed costs: Administration costs		(60 000)	(½)
Contribution %		33,5%	
Breakeven revenue *[Alternate calc. 181 053]*		178 947	(1) Principle
Breakeven # of guests		10,5	(1) Principle
Operating leverage [684000/624000]		1,10	(1) Principle
Recommendation – must follow results of calc's and make logical sense - which is the **lowest risk** option; or - which has the **highest potential** to deliver profits			(2)
Capital budgeting approach more appropriate as Opt 1 =			(2)
			Maximum 13

QUESTION 32

EXCEL LEISURE LTD ('EXCEL') **SUGGESTED SOLUTION**

(Source: SAICA QE I 2010, Paper 1, Question 3 adapted)

Amounts in the question exclude VAT, except where indicated.

PART B

Part (c) — Advantages and disadvantages of selling usage of the golf driving range through annual memberships

Advantages

- Excel's cash flows boosted by receiving membership fees in advance
- Membership fees represent a minimum number of buckets annually, an annuity stream. Provides certainty on revenue streams.
- Entrance fees represent a new income stream
- Members pay directly into Excel's bank account, Excel avoids paying cash deposit fees on money banked
- Potentially higher revenue by selling a "fixed number" of buckets to members p.a. These members may not have used the driving range as often as annual membership fee equivalent
- Increased revenue at restaurants through members using facilities more often because of discount

Maximum 8

Disadvantages

- Members fee rate may be lower than revenue that would have been derived if 'members' charged for buckets of golf balls used
- If membership fees are set too high, Excel may lose members and these individuals could feel aggrieved and no longer use Excel's facilities
- Systems and controls could be subject to abuse through non-members claiming to be members and using facilities for free/claiming discounts at restaurant
- Potentially declining revenue at restaurant through members claiming discounts
- Increased administration in maintaining membership database, following up on payments of membership fees
- Members may over-use facilities causing damage to the grass and driving range
- Increased administration and recording of discounts claimed at restaurants

Maximum 8

Part (c) maximum 10

Part (d) – Variable costing statements

	R	
Driving range division		
Revenue	19 500	
- Annual membership fees	6 000	
- Entrance fees excluded as these are once off	-	(1)
- Usage of golf ball revenue	7 200	
- Sale of beverages & snacks	6 300	
Variable costs		
COS: Beverages & snacks	(4 200)	
Replacement of driving range balls	(800)	(1)
Bank charges [(7 200 + 6 300) x 1,5%]	(203)	(2)
Contribution margin	14 297	
Fixed costs	(7 186)	
Salaries & wages	(3 240)	
Maintenance	(2 160)	(1)
Other expenses	(480)	
Depreciation	(440)	
Electricity [20% x 820] plus explanation of why this cost is fixed	(164)	(2)
Water [780 x 90%]	(702)	(2)
Net profit	7 111	(1)

Restaurant division		
Revenue [R43 200 – R430]	**42 770**	(1)
Variable costs	**(22 181)**	
COS	(19 640)	
Linen replacement & cleaning	(1 080)	(1)
Water (R780 - R702)	(78)	(2)
Electricity (R820 - R164)	(656)	(2)
Bank charges [(R42 770 x 60% x 1,5%) + (R42 770 x 40% x 2,0%)]	(727)	(2)
Contribution margin	20 589	
Fixed costs	**(8 150)**	
Salaries & wages	(5 950)	
Depreciation	(1 480)	
Other expenses	(720)	
Net profit	**12 439**	(1)
	Max	16

Part (e) – ROTA

Driving range division		
Net profit	7 111	
Buildings	1 840	
Equipment	2 400	
Land [% apportioned to Range at fair value]	30 000	(2)
Inventories	300	(1)
Total assets	34 540	(1)
Return on total assets	20,6%	(1)
Restaurant division		
Buildings	16 560	
Equipment	4 480	
Inventories	1 500	
Total assets	22 540	
ROTA	55,2%	(1)
	Max	6

Part (f)

Arguments for

- Divisions would incur such operating costs if they were not part of company/group.
- These costs were incurred to support the division's business operating activities
- Divisions should cover their costs plus head office cots for company to be profitable
- Provides a check on Finance division's costs – other divisions will scrutinize carefully if they bear a portion of such costs

Arguments against

- Allocation basis to divisions would be arbitrary – difficult to provide scientific basis for allocation %'s
- These overheads are not controllable by operating divisions

Max 6